

gilmore-lie-groups: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00002	63	■ 0.978	$S\ 0(3)$	$SO(3)$	13. The group $SO(3)$ is three dimensional. Show that every matrix in $SO(3)$ can be
gilmore-lie-groups_F00003	63	■ 1.000	$S\ 0(4)$	$SO(4)$	degeneracy under the Lie group $SO(4)$. All bound states with the same principal
gilmore-lie-groups_F00004	224	■ 0.997	$S\ 0(4,1)$	$SO(4,1)$	This group can be contracted from either $SO(4,1)$ or $SO(3,2)$.
gilmore-lie-groups_F00005	215	■ 0.943	$S\ 0(4,2)$	$SO(4,2)$	Show that the conformal group on space-time is $SO(4,2)$.
gilmore-lie-groups_F00006	15	■ 0.764	$(+,-,\times,\div)$	$(+,-,\times,\div)$	four standard operations of arithmetic $(+,-,\times,\div)$ as well as extraction of the n th
gilmore-lie-groups_F00007	15	■ 0.764	n	n	four standard operations of arithmetic $(+,-,\times,\div)$ as well as extraction of the n th
gilmore-lie-groups_F00008	17	■ 1.000	G	G	if and only if its Galois group G contains a chain of subgroups $G = G_0 \supset G_1 \supset$
gilmore-lie-groups_F00009	17	■ 1.000	$G=G_{\{0\}} \supset G_{\{1\}} \supset$	$G = G_0 \supset G_1 \supset$	if and only if its Galois group G contains a chain of subgroups $G = G_0 \supset G_1 \supset$
gilmore-lie-groups_F00010	17	■ 1.000	$\cdots \supset G_{\{\omega\}}=I$	$\cdots \supset G_{\omega} = I$	$\cdots \supset G_{\omega} = I$ with the properties:
gilmore-lie-groups_F00011	17	■ 1.000	$G_{\{i+1\}}$	G_{i+1}	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00012	17	■ 1.000	$G_{\{i\}}$	G_i	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00013	17	■ 1.000	$G_{\{i\}} / G_{\{i+1\}}$	G_i/G_{i+1}	(ii) each factor group G_i/G_{i+1} is commutative.
gilmore-lie-groups_F00014	17	■ 0.998	$G=\left\{g_{\{1\}},\ g_{\{2\}},\ \ldots\right\}$	$G = \{g_1,g_2,\ldots\}$	A group G is defined as follows. It consists of a set of operations $G = \{g_1, g_2, \ldots\}$,
gilmore-lie-groups_F00015	18	■ 1.000	$g_{\{i\}} \in G,\ g_{\{j\}} \in G$	$g_i \in G, g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in G$.
gilmore-lie-groups_F00016	18	■ 1.000	$g_{\{i\}} \cdot g_{\{j\}} \in G$	$g_i \cdot g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in G$.
gilmore-lie-groups_F00017	18	■ 1.000	$g_{\{i\}} \in G,\ g_{\{j\}} \in G,\ g_{\{k\}} \in G$	$g_i \in G, g_j \in G, g_k \in G$	(ii) Associativity: for all $g_i \in G, g_j \in G, g_k \in G$,
gilmore-lie-groups_F00018	18	■ 0.835	I	I	(iii) Identity: there is a group operation, I (identity operator), with the property that
gilmore-lie-groups_F00019	18	■ 1.000	$g_{\{i\}}$	g_i	(iv) Inverse: every group operation g_i has an inverse (called g_i^{-1}):
gilmore-lie-groups_F00020	18	■ 1.000	$g_{\{i\}}^{-1}$	g_i^{-1}	(iv) Inverse: every group operation g_i has an inverse (called g_i^{-1}):
gilmore-lie-groups_F00021	18	■ 1.000	$z_{\{1\}},\ z_{\{2\}},\ \ldots,\ z_{\{n\}}$	$z_1,z_2,\ldots,z_n$	is the group that permutes the roots $(z_1,z_2,\ldots,z_n)$ among themselves and leaves the
gilmore-lie-groups_F00022	18	■ 0.658	$P_{\{n\}}$	P_n	This group, called the permutation group P_n or the symmetric group S_n , has $n!$
gilmore-lie-groups_F00023	18	■ 0.658	$S_{\{n\}}$	S_n	This group, called the permutation group P_n or the symmetric group S_n , has $n!$
gilmore-lie-groups_F00024	18	■ 0.658	$n!$	$n!$	This group, called the permutation group P_n or the symmetric group S_n , has $n!$

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gilmore-lie-groups_F00025	18	■ 0.996	$n \times n$	$n \times n$	of $n \times n$ matrices. These matrices have one nonzero element, +1, in each row
gilmore-lie-groups_F00026	18	■ 1.000	$6=3!3 \times 3$	$6 = 3! \times 3$	and each column. For example, the $6 = 3! \times 3$ matrices for the permutation
gilmore-lie-groups_F00027	18	■ 1.000	$S_{\{3\}}$	S_3	representation of S_3 are
gilmore-lie-groups_F00028	19	■ 0.998	$z_{\{1\}}$	z_1	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups_F00029	19	■ 0.998	$z_{\{2\}}, z_{\{2\}}$	z_2, z_2	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups_F00030	19	■ 0.999	$z_{\{3\}}$	z_3	z_3 , and z_3 is replaced by z_1
gilmore-lie-groups_F00031	19	■ 0.768	$\times$	$\times$	stitutions in S_n) and $\times$ represents the multiplication operation among the matrices
gilmore-lie-groups_F00032	19	■ 1.000	$G \rightarrow \Gamma(G)$	$G \rightarrow \Gamma(G)$	tion of a group, $G \rightarrow \Gamma(G)$, is that the product of matrices representing two group
gilmore-lie-groups_F00033	19	■ 0.988	$\left(\Gamma(g_i) \times \Gamma(g_j)\right)$	$(\Gamma(g_i) \times \Gamma(g_j))$	operations $(\Gamma(g_i) \times \Gamma(g_j))$ is equal to the matrix representing the product of these
gilmore-lie-groups_F00034	19	■ 0.999	$\left(\Gamma(g_i \cdot g_j)\right)$	$(\Gamma(g_i \cdot g_j))$	operations in the group $(\Gamma(g_i \cdot g_j))$ for all group operations $g_i, g_j \in G$.
gilmore-lie-groups_F00035	19	■ 0.999	$g_i, g_j \in G$	$g_i, g_j \in G$	operations in the group $(\Gamma(g_i \cdot g_j))$ for all group operations $g_i, g_j \in G$.
gilmore-lie-groups_F00036	19	■ 0.874	3×3	3×3	since knowledge of the 3×3 matrix uniquely identifies the original group operation
gilmore-lie-groups_F00037	19	■ 1.000	H	H	A subgroup H of the group G is a subset of group operations in G that is closed
gilmore-lie-groups_F00038	19	■ 1.000	$A_{\{3\}}$	A_3	particular subgroup is denoted A_3 (alternating group). It consists of those oper-
gilmore-lie-groups_F00039	19	■ 1.000	$A_{\{3\}} \subset S_{\{3\}}$	$A_3 \subset S_3$	$A_3 \subset S_3$ and the three two-element subgroups $S_2(ij)$ of S_3 are illustrated in Fig. 1.1.
gilmore-lie-groups_F00040	19	■ 1.000	$S_{\{2\}}(ij)$	$S_2(ij)$	$A_3 \subset S_3$ and the three two-element subgroups $S_2(ij)$ of S_3 are illustrated in Fig. 1.1.
gilmore-lie-groups_F00041	20	■ 0.706	$(123) \cdot (123) = (321)$	$(123) \cdot (123) = (321)$	of operations in this subset do not lie in this subset: $(123) \cdot (123) = (321)$, $(123) \cdot$
gilmore-lie-groups_F00042	20	■ 1.000	$G, H_{\{1\}} \subset G$	$G, H_1 \subset G$	Two subgroups of G , $H_1 \subset G$ and $H_2 \subset G$ are conjugate if there is a group
gilmore-lie-groups_F00043	20	■ 1.000	$H_{\{2\}} \subset G$	$H_2 \subset G$	Two subgroups of G , $H_1 \subset G$ and $H_2 \subset G$ are conjugate if there is a group
gilmore-lie-groups_F00044	20	■ 1.000	$g \in G$	$g \in G$	element $g \in G$ with the property
gilmore-lie-groups_F00045	20	■ 1.000	$S_{\{2\}}(12)$	$S_2(12)$	Example The subgroups $S_2(12)$ and $S_2(13)$ are conjugate in S_3 since
gilmore-lie-groups_F00046	20	■ 1.000	$S_{\{2\}}(13)$	$S_2(13)$	Example The subgroups $S_2(12)$ and $S_2(13)$ are conjugate in S_3 since
gilmore-lie-groups_F00047	20	■ 1.000	$G=S_{\{3\}}$	$G = S_3$	operation in $G = S_3$ serves merely to permute the group operations in A_3 among
gilmore-lie-groups_F00048	20	■ 1.000	$H \subset G$	$H \subset G$	A subgroup $H \subset G$ which is self-conjugate under all operations in G is called
gilmore-lie-groups_F00049	21	■ 1.000	f	f	A mapping f from a group G with group operations $g_1, g_2, \dots$ and group multi-
gilmore-lie-groups_F00050	21	■ 1.000	$g_{\{1\}}, g_{\{2\}}, \dots$	$g_1, g_2, \dots$	A mapping f from a group G with group operations $g_1, g_2, \dots$ and group multi-
gilmore-lie-groups_F00051	21	■ 0.999	$h_{\{1\}}, h_{\{2\}}, \dots$	$h_1, h_2, \dots$	plication $\cdot$ to a group H with group operations $h_1, h_2, \dots$ and group multiplication

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gilmore-lie-groups-F00052	21	■ 1.000	$h_{\{i\}} \in H$	$h_i \in H$	ments in G may map to a single group element in H . Every element $h_i \in H$ has
gilmore-lie-groups-F00053	21	■ 1.000	$g_{\{j\}} \in G$	$g_j \in G$	the same number of inverse images $g_j \in G$. If each group element $h \in H$ has a
gilmore-lie-groups-F00054	21	■ 1.000	$h \in H$	$h \in H$	the same number of inverse images $g_j \in G$. If each group element $h \in H$ has a
gilmore-lie-groups-F00055	21	■ 0.933	$g \in G \left(h_1 = f \left(g_1 \right) \right)$	$g \in G \left(h_1 = f \left(g_1 \right) \right)$	unique inverse image $g \in G \left(h_1 = f \left(g_1 \right) \right)$ and $h_2 = f \left(g_2 \right), h_1 = h_2 \Rightarrow g_1 = g_2$ the
gilmore-lie-groups-F00056	21	■ 0.933	$\left(h_2 = f \left(g_2 \right) \right), h_1 = h_2 \Rightarrow g_1 = g_2$	$h_2 = f \left(g_2 \right), h_1 = h_2 \Rightarrow g_1 = g_2$	unique inverse image $g \in G \left(h_1 = f \left(g_1 \right) \right)$ and $h_2 = f \left(g_2 \right), h_1 = h_2 \Rightarrow g_1 = g_2$ the
gilmore-lie-groups-F00057	21	■ 1.000	$S_{\{2\}}$	S_2	Example The 3:1 mapping f of S_3 onto S_2 given by
gilmore-lie-groups-F00058	22	■ 0.987	h	h	product of an element h in the subgroup H with a group element in a “quotient,”
gilmore-lie-groups-F00059	22	■ 1.000	G / H	G/H	or <i>coset</i> (denoted G/H). A coset is a subset of G . If the <i>order</i> of G is $ G $ (S_3
gilmore-lie-groups-F00060	22	■ 1.000	$ G \left(S_3 \right)$	$ G \left(S_3 \right)$	or <i>coset</i> (denoted G/H). A coset is a subset of G . If the <i>order</i> of G is $ G $ (S_3
gilmore-lie-groups-F00061	22	■ 0.637	$3! = 6$	$3! = 6$	has $3! = 6$ group elements, so the order of S_3 is 6), then the order of G/H is
gilmore-lie-groups-F00062	22	■ 0.866	$ G / H = G / H $	$ G/H = G / H $	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups-F00063	22	■ 0.866	$H = A_{\{3\}} = \{I$	$H = A_3 = \{I$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups-F00064	22	■ 0.866	$\}$	$\}$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups-F00065	22	■ 0.992	$H = S_{\{2\}}(23) = \{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$ we have
gilmore-lie-groups-F00066	22	■ 1.000	$ G / H $	$ G / H $	The choice of the $ G / H $ group elements in the quotient space is not unique. For
gilmore-lie-groups-F00067	22	■ 0.692	$G / H = S_{\{3\}} / A_{\{3\}} = \{I$	$G/H = S_3/A_3 = \{I$	the subgroup A_3 we could equally well have chosen $G/H = S_3/A_3 = \{I, (13)\}$
gilmore-lie-groups-F00068	22	■ 0.997	$\{I, (23)\}$	$\{I, (23)\}$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups-F00069	22	■ 0.997	$S_{\{2\}}(23)$	$S_2(23)$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups-F00070	22	■ 0.997	$G / H = S_{\{3\}} / S_{\{2\}}(23) =$	$G/H = S_3/S_2(23) =$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups-F00071	22	■ 0.556	$\{I, (123), (321)\}$	$\{I, (123), (321)\}$	$\{I, (123), (321)\}$.
gilmore-lie-groups-F00072	22	■ 1.000	$S_{\{3\}} / A_{\{3\}}$	S_3/A_3	G/H . Since A_3 is an invariant subgroup of S_3 , the coset S_3/A_3 is a group, and
gilmore-lie-groups-F00073	22	■ 0.630	$\operatorname{group} G / H: G = G / H \times H$	$\operatorname{group} G/H: G = G/H \times H$	$\operatorname{group} G/H: G = G/H \times H$.
gilmore-lie-groups-F00074	22	■ 1.000	k	k	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups-F00075	22	■ 1.000	$I_{\{k\}}$	I_k	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups-F00076	22	■ 1.000	$z_{\{i\}}$	z_i	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups-F00077	23	■ 0.992	$I_{\{k\}}(k=1,2,\ldots,n)$	$I_k(k=1,2,\ldots,n)$	The n functions $I_k(k=1,2,\ldots,n)$ of the n roots $(z_1, z_2, \ldots, z_n)$ are symmetric:
gilmore-lie-groups-F00078	23	■ 0.992	$\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}} \right)$	$(z_1, z_2, \ldots, z_n)$	The n functions $I_k(k=1,2,\ldots,n)$ of the n roots $(z_1, z_2, \ldots, z_n)$ are symmetric:

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gilmore-lie-groups_ F00079	23	■ 1.000	$f\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$f(z_1, z_2, \dots, z_n)$	any function $f(z_1, z_2, \dots, z_n)$ that is invariant under S_n can be written as a function
gilmore-lie-groups_ F00080	23	■ 1.000	$I_{\{1\}}, I_{\{2\}}, \ldots, I_{\{n\}}$	$I_1, I_2, \dots, I_n$	of the invariants $I_1, I_2, \dots, I_n$. The invariants are easily expressed in terms of
gilmore-lie-groups_ F00081	23	■ 1.000	$G_{\{\omega-1\}} \supset G_{\{\omega\}}=I$	$G_{\omega-1} \supset G_{\omega} = I$	group-subgroup pair in this chain is isolated: $G_{\omega-1} \supset G_{\omega} = I$. The character
gilmore-lie-groups_ F00082	23	■ 1.000	$G_{\{\omega-1\}} / G_{\{\omega\}}=G_{\{\omega-1\}}$	$G_{\omega-1}/G_{\omega} = G_{\omega-1}$	table for the commutative group $G_{\omega-1}/G_{\omega} = G_{\omega-1}$ is constructed. This lists
gilmore-lie-groups_ F00083	23	■ 1.000	$\left G_{\{\omega-1\}}\right / \left G_{\{\omega\}}\right $	$ G_{\omega-1} / G_{\omega} $	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups_ F00084	23	■ 1.000	$G_{\{\omega-1\}}$	$G_{\omega-1}$	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups_ F00085	23	■ 1.000	$G_{\{\omega\}}=I$	$G_{\omega} = I$	(i) symmetric under $G_{\omega} = I$,
gilmore-lie-groups_ F00086	23	■ 1.000	$G_{\{\omega-2\}} \supset G_{\{\omega-1\}}$	$G_{\omega-2} \supset G_{\omega-1}$	Next, the next pair of groups $G_{\omega-2} \supset G_{\omega-1}$ is isolated. Starting from the set of
gilmore-lie-groups_ F00087	23	■ 1.000	$G_{\{\omega-2\}}$	$G_{\omega-2}$	(ii) not all symmetric under $G_{\omega-2}$.
gilmore-lie-groups_ F00088	23	■ 1.000	$G=G_{\{0\}} \supset$	$G = G_0 \supset$	This bootstrap procedure continues until the last group-subgroup pair $G = G_0 \supset$
gilmore-lie-groups_ F00089	23	■ 1.000	$G_{\{1\}}$	G_1	G_1 is treated. At this stage the last set of functions can be solved by radicals.
gilmore-lie-groups_ F00090	24	■ 1.000	$\Gamma^{\{1\}}, \Gamma^{\{2\}}$	Γ^1, Γ^2	ducible representations Γ^1, Γ^2 are
gilmore-lie-groups_ F00091	24	■ 1.000	$r_{\{1\}}-r_{\{2\}}$	$r_1 - r_2$	That is, the function $r_1 - r_2$ is mapped into itself by the identity, and into its negative
gilmore-lie-groups_ F00092	24	■ 0.899	$\left(r_{\{1\}}-r_{\{2\}}\right)$	$(r_1 - r_2)$	As a result, $r_1 - r_2$ is not symmetric under the action of the group S_2 . It transforms
gilmore-lie-groups_ F00093	24	■ 1.000	$\Gamma^{\{2\}}$	Γ^2	under the irreducible representation Γ^2 , not the identity representation Γ^1 .
gilmore-lie-groups_ F00094	24	■ 1.000	$\Gamma^{\{1\}}$	Γ^1	under the irreducible representation Γ^2 , not the identity representation Γ^1 .
gilmore-lie-groups_ F00095	24	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)^{\{2\}}$	$(r_1 - r_2)^2$	Since the square $(r_1 - r_2)^2$ is symmetric (transforms under the identity repre-
gilmore-lie-groups_ F00096	24	■ 0.999	$I_{\{1\}}, I_{\{2\}}$	I_1, I_2	sentation of S_2), it can be expressed in terms of the two invariants I_1, I_2 as follows
gilmore-lie-groups_ F00097	25	■ 1.000	D	D	where D is the discriminant of the quadratic equation. Since $(r_1 - r_2) = \pm\sqrt{D}$,
gilmore-lie-groups_ F00098	25	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)=\pm\sqrt{D}$	$(r_1 - r_2) = \pm\sqrt{D}$	where D is the discriminant of the quadratic equation. Since $r_1 - r_2 = \pm\sqrt{D}$,
gilmore-lie-groups_ F00099	25	■ 1.000	x	x	to move the origin to the mean value of the roots by defining a new variable, x , in
gilmore-lie-groups_ F00100	25	■ 1.000	z	z	terms of z (see Eq. (1.15)) by a Tschirnhaus transformation
gilmore-lie-groups_ F00101	26	■ 1.000	$S_{\{3\}} / A_{\{3\}}=S_{\{2\}}$	$S_3/A_3 = S_2$	subgroups) is satisfied. Since $S_3/A_3 = S_2$ is commutative and $A_3/I = A_3$ is com-
gilmore-lie-groups_ F00102	26	■ 1.000	$A_{\{3\}} / I=A_{\{3\}}$	$A_3/I = A_3$	subgroups) is satisfied. Since $S_3/A_3 = S_2$ is commutative and $A_3/I = A_3$ is com-
gilmore-lie-groups_ F00103	26	■ 1.000	$A_{\{3\}} \supset I$	$A_3 \supset I$	We begin the solution with the last group-subgroup pair in this chain: $A_3 \supset I$.
gilmore-lie-groups_ F00105	35	■ 1.000	ω	ω	Next, we take cube roots of these quantities. These are unique up to a factor of ω
gilmore-lie-groups_ F00110	26	■ 1.000	$(123)^{-1}$	$(123)^{-1}$	For example, the action of $(123)^{-1}$ on v_2 is

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gilmore-lie-groups_F00111	26	■ 1.000	$v_{\{2\}}$	v_2	For example, the action of $(123)^{-1}$ on v_2 is
gilmore-lie-groups_F00112	27	■ 1.000	$v_{\{1\}}$	v_1	Since v_1 is symmetric under both A_3 and S_3 , it can be expressed in terms of the
gilmore-lie-groups_F00113	27	■ 1.000	$v_{\{3\}}$	v_3	The remaining functions, v_2 and v_3 , are symmetric under I but not under A_3 .
gilmore-lie-groups_F00114	27	■ 1.000	$S_{\{3\}} \supset A_{\{3\}}$	$S_3 \supset A_3$	We now proceed to the next group-subgroup pair: $S_3 \supset A_3$. To construct func-
gilmore-lie-groups_F00115	27	■ 1.000	$S_{\{2\}}=S_{\{3\}} / A_{\{3\}}$	$S_2 = S_3/A_3$	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the
gilmore-lie-groups_F00116	27	■ 1.000	$v_{\{2\}}^{\{3\}}$	v_2^3	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the
gilmore-lie-groups_F00117	27	■ 1.000	$v_{\{3\}}^{\{3\}}$	v_3^3	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the
gilmore-lie-groups_F00118	27	■ 1.000	$J_{\{1\}}, J_{\{2\}}$	J_1, J_2	Since J_1, J_2 are symmetric under S_3 , they can be expressed in terms of the invariants
gilmore-lie-groups_F00119	27	■ 0.995	$I_{\{1\}}, I_{\{2\}}, I_{\{3\}}$	I_1, I_2, I_3	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write
gilmore-lie-groups_F00120	27	■ 0.995	$J_{\{1\}}$	J_1	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write
gilmore-lie-groups_F00121	27	■ 0.995	$J_{\{2\}}$	J_2	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write
gilmore-lie-groups_F00122	27	■ 1.000	$I_{\{1\}}=s_{\{1\}}+s_{\{2\}}+$	$I_1 = s_1 + s_2 +$	These relations can be computed, but they simplify considerably if $I_1 = s_1 + s_2 +$
gilmore-lie-groups_F00123	27	■ 1.000	$s_{\{3\}}=0$	$s_3 = 0$	$s_3 = 0$. This can be accomplished by shifting the origin using a Tschirnhaus trans-
gilmore-lie-groups_F00124	28	■ 1.000	$J_{\{1\}}=v_{\{2\}}^{\{3\}}+v_{\{3\}}^{\{3\}}$	$J_1 = v_2^3 + v_3^3$	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'
gilmore-lie-groups_F00125	28	■ 1.000	$J_{\{2\}}=v_{\{2\}}^{\{3\}} v_{\{3\}}^{\{3\}}$	$J_2 = v_2^3 v_3^3$	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'
gilmore-lie-groups_F00126	28	■ 1.000	$I_{\{2\}}^{\{\prime\}}, I_{\{3\}}^{\{\prime\}}$	I_2', I_3'	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'
gilmore-lie-groups_F00127	28	■ 1.000	$v_{\{2\}}^{\{3\}}, v_{\{3\}}^{\{3\}}$	v_2^3, v_3^3	The resolvent quadratic equation whose solution provides v_2^3, v_3^3 is
gilmore-lie-groups_F00128	28	■ 1.000	$s_{\{1\}}, s_{\{2\}}, s_{\{3\}}$	s_1, s_2, s_3	Finally, the roots s_1, s_2, s_3 are linearly related to v_1, v_2, v_3 by
gilmore-lie-groups_F00129	28	■ 1.000	$v_{\{1\}}, v_{\{2\}}, v_{\{3\}}$	v_1, v_2, v_3	Finally, the roots s_1, s_2, s_3 are linearly related to v_1, v_2, v_3 by
gilmore-lie-groups_F00130	29	■ 1.000	$z=z^{\{\prime\}}+\frac{1}{4} I_{\{1\}}$	$z = z' + \frac{1}{4} I_1$	of coordinates through the Tschirnhaus transformation $z = z' + \frac{1}{4} I_1$
gilmore-lie-groups_F00131	29	■ 1.000	$S_{\{4\}}$	S_4	The Galois group is S_4 . This has the subgroup chain shown in Fig. 1.5. The
gilmore-lie-groups_F00132	29	■ 1.000	$A_{\{4\}}$	A_4	alternating group A_4 consists of the twelve group operations that have determinant
gilmore-lie-groups_F00133	29	■ 0.957	$V_{\{4\}}$	V_4	group, Klein four-group) V_4 is $\{I, (12)(34), (13)(24), (14)(23)\}$. The chain
gilmore-lie-groups_F00134	29	■ 0.957	$\{I, (12)(34), (13)(24), (14)(23)\}$	$\{I, (12)(34), (13)(24), (14)(23)\}$	group, Klein four-group) V_4 is $\{I, (12)(34), (13)(24), (14)(23)\}$. The chain
gilmore-lie-groups_F00135	30	■ 1.000	$S_{\{4\}} / A_{\{4\}}=S_{\{2\}}$	$S_4/A_4 = S_2$	(i) A_4 is invariant in S_4 and $S_4/A_4 = S_2$,
gilmore-lie-groups_F00136	30	■ 0.987	$A_{\{4\}} / V_{\{4\}}=C_{\{3\}}=\{I$	$A_4/V_4 = C_3 = \{I$	(ii) V_4 is invariant in A_4 and $A_4/V_4 = C_3 = \{I, (234), (432)\}$,

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gilmore-lie-groups_ F00137	30	■ 0.987	$V_{\{4\}} / I = V_{\{4\}} = \{I$	$V_4/I = V_4 = \{I$	(iii) I is invariant in V_4 and $V_4/I = V_4 = \{I, (12)(34), (13)(24), (14)(23)\}$.
gilmore-lie-groups_ F00143	30	■ 0.971	$w_{\{2\}}, w_{\{3\}}, w_{\{4\}}$	w_2, w_3, w_4	These basis vectors are symmetric under I but the basis vectors w_2, w_3, w_4 are not
gilmore-lie-groups_ F00144	30	■ 1.000	$A_{\{4\}} \supset V_{\{4\}}$	$A_4 \supset V_4$	We now advance to the next group-subgroup pair: $A_4 \supset V_4$. It is a simple matter
gilmore-lie-groups_ F00145	30	■ 1.000	$A_{\{4\}} / V_{\{4\}}$	A_4/V_4	(ii) permuted among themselves by A_4 and the group A_4/V_4 .
gilmore-lie-groups_ F00146	30	■ 0.999	$w_{\{1\}} = I_{\{1\}}$	$w_1 = I_1$	These functions are $w_1 = I_1$ and w_2^2, w_3^2, w_4^2 . In the coordinate system in which the
gilmore-lie-groups_ F00147	30	■ 0.999	$w_{\{2\}}^2, w_{\{3\}}^2, w_{\{4\}}^2$	w_2^2, w_3^2, w_4^2	These functions are $w_1 = I_1$ and w_2^2, w_3^2, w_4^2 . In the coordinate system in which the
gilmore-lie-groups_ F00148	30	■ 1.000	$w_{\{j\}}^2 (j=2,3,4)$	$w_j^2 (j=2,3,4)$	It is clear that the three $w_j^2 (j=2,3,4)$ are permuted among themselves by the
gilmore-lie-groups_ F00149	30	■ 1.000	$C_{\{3\}} = A_{\{4\}} / V_{\{4\}}$	$C_3 = A_4/V_4$	factor group $C_3 = A_4/V_4$, which is a subgroup of the Galois group of a resolvent
gilmore-lie-groups_ F00150	31	■ 1.000	$J_{\{k\}}$	J_k	Since the three J_k are invariant under C_3 , they can be expressed in terms of the
gilmore-lie-groups_ F00151	31	■ 1.000	$C_{\{3\}}$	C_3	Since the three J_k are invariant under C_3 , they can be expressed in terms of the
gilmore-lie-groups_ F00152	31	■ 1.000	$S_{\{4\}} \supset A_{\{4\}}$	$S_4 \supset A_4$	chain: $S_4 \supset A_4$, with $S_4/A_4 = S_2$. The cubic is solved by introducing the resolvent
gilmore-lie-groups_ F00153	31	■ 1.000	$y_{\{2\}}, y_{\{3\}}, y_{\{4\}}$	y_2, y_3, y_4	If the three solutions of the resolvent cubic equation are called y_2, y_3, y_4 , then
gilmore-lie-groups_ F00154	31	■ 1.000	$w_{\{2\}} w_{\{3\}} w_{\{4\}} = 8 I_{\{3\}}^{\prime}$	$w_2 w_3 w_4 = 8 I_3'$	A simple computation shows that $w_2 w_3 w_4 = 8 I_3'$. The signs $\pm \sqrt{y_j}$ are chosen so
gilmore-lie-groups_ F00155	31	■ 1.000	$\pm \sqrt{y_{\{j\}}}$	$\pm \sqrt{y_j}$	A simple computation shows that $w_2 w_3 w_4 = 8 I_3'$. The signs $\pm \sqrt{y_j}$ are chosen so
gilmore-lie-groups_ F00156	31	■ 1.000	$8 I_{\{3\}}^{\prime}$	$8 I_3'$	that their product is $8 I_3'$. The simple linear relation between the roots t_i and the
gilmore-lie-groups_ F00157	31	■ 1.000	$t_{\{i\}}$	t_i	that their product is $8 I_3'$. The simple linear relation between the roots t_i and the
gilmore-lie-groups_ F00158	31	■ 1.000	$I_{\{1\}}$	I_1	invariants I_1 and functions $w_j(I')$ is easily inverted:
gilmore-lie-groups_ F00159	31	■ 1.000	$w_{\{j\}} \left(I^{\prime} \right)$	$w_j(I')$	invariants I_1 and functions $w_j(I')$ is easily inverted:
gilmore-lie-groups_ F00160	31	■ 1.000	$w_{\{j\}}$	w_j	where the w_j are square roots of the solutions of the resolvent cubic equation whose
gilmore-lie-groups_ F00161	31	■ 1.000	$S_{\{5\}}$	S_5	is sufficient to study the structure of its Galois group S_5 . The alternating subgroup
gilmore-lie-groups_ F00162	32	■ 1.000	$A_{\{5\}}$	A_5	A_5 of order 60 is an invariant subgroup. S_5 has no invariant subgroups except A_5
gilmore-lie-groups_ F00163	32	■ 1.000	$S_{\{5\}} / A_{\{5\}} = S_{\{2\}}$	$S_5/A_5 = S_2$	Although $S_5/A_5 = S_2$ is commutative, $A_5/I = A_5$ is not. Therefore the quintic
gilmore-lie-groups_ F00164	32	■ 1.000	$A_{\{5\}} / I = A_{\{5\}}$	$A_5/I = A_5$	Although $S_5/A_5 = S_2$ is commutative, $A_5/I = A_5$ is not. Therefore the quintic
gilmore-lie-groups_ F00165	33	■ 1.000	$t_{\{1\}}^{\prime} = -3, t_{\{2\}}^{\prime} = -2, t_{\{3\}}^{\prime} =$	$t_1' = -3, t_2' = -2, t_3' =$	Next, we introduce linear combinations of the four roots $t_1' = -3, t_2' = -2, t_3' =$
gilmore-lie-groups_ F00166	33	■ 0.961	$1, t_{\{4\}}^{\prime} = 4$	$1, t_4' = 4$	$1, t_4' = 4$
gilmore-lie-groups_ F00167	33	■ 1.000	$y = y^{\prime} + \frac{1}{3} J_{\{1\}} = y^{\prime} + \frac{1}{3} (4 + 16 + 100) = y^{\prime} + 40$	$y = y' + \frac{1}{3} J_1 = y' + \frac{1}{3} (4 + 16 + 100) = y' + 40$	$y = y' + \frac{1}{3} J_1 = y' + \frac{1}{3} (4 + 16 + 100) = y' + 40$. The roots are now

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gilmore-lie-groups_F00168	34	■ 0.990	$v_2^3 + v_3^3, v_2^3 v_3^3$	$v_2^3 + v_3^3, v_2^3 v_3^3$	We can express $v_2^3 + v_3^3, v_2^3 v_3^3$ in terms of J'_2, J'_3 :
gilmore-lie-groups_F00169	34	■ 0.990	$J_2^{\prime}, J_3^{\prime}$	J'_2, J'_3	We can express $v_2^3 + v_3^3, v_2^3 v_3^3$ in terms of J'_2, J'_3 :
gilmore-lie-groups_F00170	34	■ 1.000	$x = x^{\prime} + \frac{1}{2} K_1$	$x = x' + \frac{1}{2} K_1$	A Tschirnhaus transformation $x = x' + \frac{1}{2} K_1$ produces the auxiliary quadratic
gilmore-lie-groups_F00171	35	■ 1.000	y_1, y_2, y_3	y_1, y_2, y_3	The values y_1, y_2, y_3 of the resolvent cubic are complex linear combinations of
gilmore-lie-groups_F00172	35	■ 0.982	v_2, v_3	v_2, v_3	<u>v_2, v_3</u>
gilmore-lie-groups_F00173	35	■ 1.000	$w_2 w_3 w_4 = 8 I_3^{\prime} = 80$	$w_2 w_3 w_4 = 8 I'_3 = 80$	Since <u>$w_2 w_3 w_4 = 8 I'_3 = 80$</u> , an even number of these signs must be negative. The
gilmore-lie-groups_F00174	36	■ 1.000	$S_4 / A_4, A_4 / V_4, V_4$	$S_4 / A_4, A_4 / V_4, V_4$	1. Compute $S_4 / A_4, A_4 / V_4, V_4$ and show that they are commutative.
gilmore-lie-groups_F00175	36	■ 0.986	V_8	V_8	2. Construct the group V_8 with the property $S_4 \supset V_8 \supset V_4$ (see Fig. 1.5). (Hint: include
gilmore-lie-groups_F00176	36	■ 0.986	$S_4 \supset V_8 \supset V_4$	$S_4 \supset V_8 \supset V_4$	2. Construct the group V_8 with the property <u>$S_4 \supset V_8 \supset V_4$</u> (see Fig. 1.5). (Hint: include
gilmore-lie-groups_F00177	36	■ 1.000	$z^3 - 7z + 6 = 0 ((z - 1)(z - 2)(z + 3) = 0)$	$z^3 - 7z + 6 = 0 ((z - 1)(z - 2)(z + 3) = 0)$	3. For the cubic equation <u>$z^3 - 7z + 6 = 0 ((z - 1)(z - 2)(z + 3) = 0)$</u> show
gilmore-lie-groups_F00178	36	■ 1.000	$\left(x - v_2^3\right)\left(x - v_3^3\right) = x^2 - 162 x +$	$(x - v_2^3)(x - v_3^3) = x^2 - 162x +$	Show that the resolvent equation for v_2^3, v_3^3 is $(x - v_2^3)(x - v_3^3) = x^2 - 162x +$
gilmore-lie-groups_F00179	36	■ 0.996	$9261 = 0$	$9261 = 0$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i 30 \sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00180	36	■ 0.996	$v_2^3, v_3^3 = 81 \pm i 30 \sqrt{3}$	$v_2^3, v_3^3 = 81 \pm i 30 \sqrt{3}$	$9261 = 0$. Solve this quadratic to find <u>$v_2^3, v_3^3 = 81 \pm i 30 \sqrt{3}$</u> , so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00181	36	■ 0.996	$v_2, v_3 = \frac{1}{2}(3 \pm$	$v_2, v_3 = \frac{1}{2}(3 \pm$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i 30 \sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00182	36	■ 1.000	$i 5 \sqrt{3}$	$i 5 \sqrt{3}$	$i 5 \sqrt{3}$). Invert Eq. (1.43) to determine the three roots of the original equation:
gilmore-lie-groups_F00183	36	■ 0.543	$x + y, x - y, x y$	$x + y, x - y, x y$	to construct intervals of lengths <u>$x + y, x - y, x y$</u> and x / y using ruler and compass.
gilmore-lie-groups_F00184	36	■ 0.543	x / y	x / y	to construct intervals of lengths $x + y, x - y, x y$ and x / y using ruler and compass.
gilmore-lie-groups_F00185	36	■ 1.000	$\sqrt{x}$	$\sqrt{x}$	It is also possible to construct intervals of length $\sqrt{x}$ by these means. The set of all
gilmore-lie-groups_F00186	36	■ 1.000	$x + i y = (x, y)$	$x + i y = (x, y)$	forms a subset of the numbers <u>$x + i y = (x, y)$</u> in the complex plane. If a number is
gilmore-lie-groups_F00187	36	■ 1.000	K	K	number satisfies an algebraic equation of degree K with integer coefficients, where
gilmore-lie-groups_F00188	36	■ 1.000	$K = 2^n$	$K = 2^n$	$K = 2^n$ must be some power of two.
gilmore-lie-groups_F00189	36	■ 1.000	π	π	a. Square a circle? For the circle of radius 1 the area is π . Squaring a circle means
gilmore-lie-groups_F00190	36	■ 1.000	$x^2 - \pi = 0$	$x^2 - \pi = 0$	finding an interval of length x , where $x^2 - \pi = 0$. This is of degree 2 but π is not
gilmore-lie-groups_F00191	36	■ 1.000	$1^3 = 1$	$1^3 = 1$	b. Double the cube? A cube with edge length 1 has volume $1^3 = 1$. A cube with
gilmore-lie-groups_F00192	36	■ 1.000	$x^3 - 2 = 0$	$x^3 - 2 = 0$	twice the volume has edge length x , where x satisfies $x^3 - 2 = 0$. Although the
gilmore-lie-groups_F00193	36	■ 1.000	$3 \neq 2^n$	$3 \neq 2^n$	coefficients are integers this equation is of degree $3 \neq 2^n$ for any integer n . Argue

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gilmore-lie-groups_F00194	36	■ 1.000	$3 \ \theta$	3θ	c. Trisect an angle? If 3θ is some angle, the trigonometric functions of 3θ and
gilmore-lie-groups_F00195	36	■ 1.000	$\frac{1}{3}(3 \ \theta)=\theta$	$\frac{1}{3}(3\theta) = \theta$	$\frac{1}{3}(3\theta) = \theta$ are related by
gilmore-lie-groups_F00196	37	■ 1.000	$\cos (3 \ \theta)$	$\cos(3\theta)$	Whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00197	37	■ 1.000	$\cos (\theta)$	$\cos(\theta)$	Whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00198	37	■ 1.000	$\left(x^2+a \ x+b\right)(x+c)=0$	$(x^2 + ax + b)(x + c) = 0$	cubic factors into the form $\boxed{x^2 + ax + b}(x + c) = 0$, where a, b, c are rational.
gilmore-lie-groups_F00199	37	■ 1.000	$a, \ b, \ c$	a, b, c	cubic factors into the form $(x^2 + ax + b)(x + c) = 0$, where a, b, c are rational.
gilmore-lie-groups_F00200	37	■ 1.000	$\cos (3 \ \theta)=0, \ c=0$	$\cos(3\theta) = 0, c = 0$	For example, if $\boxed{\cos(3\theta) = 0, c = 0}$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00201	37	■ 1.000	$a=0$	$a = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00202	37	■ 1.000	$b=-3 \ / \ 4$	$b = -3/4$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00203	37	■ 1.000	$\cos (\theta)=0$	$\cos(\theta) = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00204	37	■ 0.986	$\pm \sqrt{3} \ / \ 2$	$\pm\sqrt{3}/2$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2 (+), 3\pi/2 (0)$, or $5\pi/2 (-)$.
gilmore-lie-groups_F00205	37	■ 0.986	$3 \ \theta=\pi \ / \ 2(+), \ 3 \ \pi \ / \ 2(0)$	$3\theta = \pi/2(+), 3\pi/2(0)$	or $\pm\sqrt{3}/2$ for $\boxed{3\theta = \pi/2 (+), 3\pi/2 (0)}$, or $5\pi/2 (-)$.
gilmore-lie-groups_F00206	37	■ 0.986	$5 \ \pi \ / \ 2(-)$	$5\pi/2(-)$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2 (+), 3\pi/2 (0)$, or $5\pi/2 (-)$.
gilmore-lie-groups_F00207	38	■ 1.000	$g_{\{i\}}, \ g_{\{j\}}, \ g_{\{k\}}, \ \ldots$	$g_i, g_j, g_k, \dots$	Definition A set $\boxed{g_i, g_j, g_k, \dots}$ (called group elements or group operations)
gilmore-lie-groups_F00208	38	■ 1.000	$g_{\{i\}} \ \circ g_{\{j\}} \ \in G$	$g_i \circ g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \circ g_j \in G$.
gilmore-lie-groups_F00209	38	■ 1.000	e	e	(iii) Identity: there is an operator e (the identity operation) with the property that for
gilmore-lie-groups_F00210	38	■ 1.000	$g_{\{i\}} \ \in G$	$g_i \in G$	every group operation $\boxed{g_i \in G}$
gilmore-lie-groups_F00211	39	■ 1.000	$2 \ \times \ 2$	2×2	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups_F00212	39	■ 1.000	$S \ L(2 \ ; \ \mathbb{R})$	$SL(2; \mathbb{R})$	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups_F00213	39	■ 1.000	$\alpha, \ \beta, \ \gamma, \ \delta$	$\alpha, \beta, \gamma, \delta$	where $\alpha, \beta, \gamma, \delta$ are real numbers. This set forms a group under matrix multiplica-
gilmore-lie-groups_F00214	39	■ 0.797	A	A	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00215	39	■ 0.797	B	B	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00216	39	■ 0.797	$A \ \circ B=C$	$A \circ B = C$	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00217	39	■ 0.797	$\circ$	$\circ$	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00218	39	■ 0.922	C	C	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups_F00219	39	■ 0.922	$\operatorname{det}(A)=+1$	$\det(A) = +1$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups_F00220	39	■ 0.922	$\operatorname{det}(B)=+1$	$\det(B) = +1$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,

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gilmore-lie-groups_ F00221	39	■ 1.000	$\operatorname{det}(C)=\operatorname{det}(A)\operatorname{det}(B)=+1$	$\det(C) = \det(A) \det(B) = +1$	then $\det(C) = \det(A) \det(B) = +1$.
gilmore-lie-groups_ F00222	39	■ 0.998	$(A \circ B) \circ C$	$(A \circ B) \circ C$	(ii) Associativity: $(A \circ B) \circ C$ and $A \circ (B \circ C)$ are given explicitly by
gilmore-lie-groups_ F00223	39	■ 0.998	$A \circ (B \circ C)$	$A \circ (B \circ C)$	(ii) Associativity: $(A \circ B) \circ C$ and $A \circ (B \circ C)$ are given explicitly by
gilmore-lie-groups_ F00224	39	■ 1.000	$g_i \rightarrow g(x)$	$g_i \rightarrow g(x)$	in the group with a point in some topological space: $g_i \rightarrow g(x)$. In other words, the
gilmore-lie-groups_ F00225	39	■ 1.000	i	i	index i depends on one or more continuous real variables.
gilmore-lie-groups_ F00226	39	■ 1.000	$S^2 \subset R^3: x^2+y^2+z^2=1$	$S^2 \subset R^3: x^2 + y^2 + z^2 = 1$	example, every point on the surface of a unit sphere $S^2 \subset R^3: x^2 + y^2 + z^2 = 1$,
gilmore-lie-groups_ F00227	39	■ 1.000	R^2	R^2	has a neighborhood that looks, over small distances, like a piece of the plane R^2
gilmore-lie-groups_ F00228	39	■ 1.000	S^2	S^2	(see Fig. 2.1). Locally, the two spaces S^2 and R^2 are topologically equivalent but
gilmore-lie-groups_ F00229	39	■ 1.000	M^n	M^n	Definition An n -dimensional differentiable manifold M^n consists of the follow-
gilmore-lie-groups_ F00230	39	■ 1.000	T	T	(i) A topological space T . This includes a collection of open sets U_α (a topology) that
gilmore-lie-groups_ F00231	39	■ 1.000	U_{α}	U_α	(i) A topological space T . This includes a collection of open sets U_α (a topology) that
gilmore-lie-groups_ F00232	39	■ 0.896	$T: \cup_{\alpha} U_{\alpha}=T$	$T: \cup_\alpha U_\alpha = T$	cover $T: \cup_\alpha U_\alpha = T$.
gilmore-lie-groups_ F00233	40	■ 1.000	p	p	Figure 2.1. Every point p on a sphere S^2 is surrounded by an open neighborhood
gilmore-lie-groups_ F00234	40	■ 1.000	ϕ_{α}	ϕ_α	(ii) A collection of charts ϕ_α , with $\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$. Each ϕ_α is a homeomorphism of
gilmore-lie-groups_ F00235	40	■ 1.000	$\phi_{\alpha}(U_{\alpha})=V_{\alpha} \subset R^n$	$\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$	(ii) A collection of charts ϕ_α , with $\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$. Each ϕ_α is a homeomorphism of
gilmore-lie-groups_ F00236	40	■ 1.000	V_{α}	V_α	U_α to V_α .
gilmore-lie-groups_ F00237	40	■ 1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1}: \phi_{\beta}(U_{\alpha} \cap U_{\beta}) \rightarrow \phi_{\alpha}(U_{\alpha} \cap U_{\beta})$	$\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta)$	(iii) Smoothness conditions. The homeomorphisms $\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta)$
gilmore-lie-groups_ F00238	40	■ 0.988	U_{β}	U_β	U_β) of open sets in R^n to open sets in R^n are 1:1, invertible, and differentiable.
gilmore-lie-groups_ F00239	40	■ 0.988	R^n	R^n	U_β) of open sets in R^n to open sets in R^n are 1:1, invertible, and differentiable.
gilmore-lie-groups_ F00240	40	■ 1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1}$	$\phi_\alpha \circ \phi_\beta^{-1}$	functions” $\phi_\alpha \circ \phi_\beta^{-1}$ map $R^n \rightarrow R^n$, all the definitions of elementary multivariable
gilmore-lie-groups_ F00241	40	■ 1.000	$R^n \rightarrow R^n$	$R^n \rightarrow R^n$	functions” $\phi_\alpha \circ \phi_\beta^{-1}$ map $R^n \rightarrow R^n$, all the definitions of elementary multivariable
gilmore-lie-groups_ F00242	40	■ 0.697	C^k	C^k	replaced by other adjectives (C^k , smooth, analytic, . . .) in the definition above.
gilmore-lie-groups_ F00243	40	■ 1.000	$x_1 \rightarrow 0$	$x_1 \rightarrow 0$	lem, as is clear when one asks: “what happens as $x_1 \rightarrow 0$?” We will consider this
gilmore-lie-groups_ F00244	40	■ 1.000	$\times S^1$	$\times S^1$	fold R^2 (plane) $\times S^1$ (circle) (see Fig. 2.2). This is not at all obvious, but will become
gilmore-lie-groups_ F00245	41	■ 0.999	$z^2-x^2-y^2=1$	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$. The rotation matrix is parameterized by a point on a circle. The
gilmore-lie-groups_ F00247	41	■ 0.698	$\left.U_{\alpha}\right)$	$U_\alpha)$	$U_\alpha)$ has a finite subcover: $\cup_\alpha^{\text{finite}} T \subset U_\alpha$.

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gilmore-lie-groups_ F00248	41	■ 0.698	$\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	$\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	U_{α}) has a finite subcover: $\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$.
gilmore-lie-groups_ F00249	41	■ 1.000	$\left(\left x-x^{\prime}\right ^2=\left x_1-x_1^{\prime}\right ^2+\cdots+\left x_n-x_n^{\prime}\right ^2\right)$	$\left(\left x-x^{\prime}\right ^2=\left x_1-x_1^{\prime}\right ^2+\cdots+\left x_n-x_n^{\prime}\right ^2\right)$	In spaces R^n with a Euclidean notion of distance ($ x-x' ^2= x_1-x_1' ^2+\cdots+ x_n-x_n' ^2$), this definition is equivalent to an older definition of compact spaces:
gilmore-lie-groups_ F00250	41	■ 1.000	$\left x_n-x_n^{\prime}\right ^2$	$ x_n-x_n' ^2$	$ x_n-x_n' ^2$), this definition is equivalent to an older definition of compact spaces:
gilmore-lie-groups_ F00251	42	■ 1.000	$\left(g(x), x \in M^n\right)$	$(g(x), x \in M^n)$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups_ F00252	42	■ 1.000	$g(x) \circ g(y)=$	$g(x) \circ g(y) =$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups_ F00253	42	■ 1.000	$g(z)$	$g(z)$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00254	42	■ 1.000	$z \in M^n$	$z \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00255	42	■ 1.000	$x \in M^n$	$x \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00256	42	■ 1.000	$y \in M^n$	$y \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00257	42	■ 1.000	$z=\phi(x, y)$	$z=\phi(x, y)$	through a function $z=\phi(x, y)$.
gilmore-lie-groups_ F00258	42	■ 1.000	$g(x) \circ g(y)=g(z)$	$g(x) \circ g(y)=g(z)$	defined by $g(x) \circ g(y)=g(z)$, is differentiable.
gilmore-lie-groups_ F00259	42	■ 1.000	$y=\psi(x)$	$y=\psi(x)$	(ii) Smoothness of the group inversion map The group inversion map $y=\psi(x)$, defined
gilmore-lie-groups_ F00260	42	■ 1.000	$g(x)^{-1}=g(y)$	$g(x)^{-1}=g(y)$	by $g(x)^{-1}=g(y)$, is differentiable.
gilmore-lie-groups_ F00261	42	■ 1.000	$g(\phi(x, y))$	$g(\phi(x, y))$	$g(\phi(x, y))$
gilmore-lie-groups_ F00262	42	■ 1.000	ϕ	ϕ	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_ F00263	42	■ 1.000	x_1	x_1	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_ F00264	42	■ 1.000	y_1	y_1	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_ F00265	42	■ 1.000	x_2-x_3	x_2-x_3	bounded away from the x_2-x_3 plane $x_1=0$ and the y_2-y_3 plane $y_1=0$. In the
gilmore-lie-groups_ F00266	42	■ 1.000	$x_1=0$	$x_1=0$	bounded away from the x_2-x_3 plane $x_1=0$ and the y_2-y_3 plane $y_1=0$. In the
gilmore-lie-groups_ F00267	42	■ 1.000	y_2-y_3	y_2-y_3	bounded away from the x_2-x_3 plane $x_1=0$ and the y_2-y_3 plane $y_1=0$. In the
gilmore-lie-groups_ F00268	42	■ 1.000	$y_1=0$	$y_1=0$	bounded away from the x_2-x_3 plane $x_1=0$ and the y_2-y_3 plane $y_1=0$. In the
gilmore-lie-groups_ F00269	43	■ 0.983	$\left(y_1, y_2, y_3\right)$	(y_1, y_2, y_3)	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_ F00270	43	■ 0.983	$\left[g\left(x_1, x_2, x_3\right)\right]^{-1}$	$[g\left(x_1, x_2, x_3\right)]^{-1}$	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_ F00271	43	■ 0.998	$\left(z_1, z_2, z_3\right)=(1,0,0)$	$(z_1, z_2, z_3)=(1,0,0)$	$(z_1, z_2, z_3)=(1,0,0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_ F00272	43	■ 0.998	$\left(x_1, x_2, x_3\right)$	(x_1, x_2, x_3)	$(z_1, z_2, z_3)=(1,0,0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_ F00273	43	■ 1.000	$[g(x)]^{-1}=g(y)=g(\psi(x))$	$[g(x)]^{-1}=g(y)=g(\psi(x))$	The inverse mapping $[g(x)]^{-1}=g(y)=g(\psi(x))$ is

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gilmore-lie-groups_ F00274	43	■ 0.964	(x, y, z , θ)	(x, y, z , θ)	The rotation matrix is parameterized by a point on a circle. Two points (x, y, z , θ)
gilmore-lie-groups_ F00275	43	■ 0.882	$(-x, -y, - z , \theta + \pi)$	$(-x, -y, - z , \theta + \pi)$	and $(-x, -y, - z , \theta + \pi)$ map to the same matrix in $SL(2; \mathbb{R})$. The manifold that
gilmore-lie-groups_ F00276	43	■ 1.000	$H^{2+} \times S^1$	$H^{2+} \times S^1$	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_ F00277	43	■ 1.000	H^{2+}	H^{2+}	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_ F00278	43	■ 1.000	$\phi(x, y)$	$\phi(x, y)$	operation that obeys the group axioms. The combinatorial operation $\phi(x, y)$ de-
gilmore-lie-groups_ F00279	43	■ 1.000	$g(x) \circ g(y) = g(z) = g(\phi(x, y))$	$g(x) \circ g(y) = g(z) = g(\phi(x, y))$	fined by $g(x) \circ g(y) = g(z) = g(\phi(x, y))$ is differentiable in both sets of variables.
gilmore-lie-groups_ F00280	44	■ 1.000	$[g(x)]^{-1} =$	$[g(x)]^{-1} =$	In addition, the mapping $y = \psi(x)$ of a group operation to its inverse $[g(x)]^{-1} =$
gilmore-lie-groups_ F00281	44	■ 1.000	$g(y) = g(\psi(x))$	$g(y) = g(\psi(x))$	$g(y) = g(\psi(x))$ is also differentiable.
gilmore-lie-groups_ F00282	44	■ 0.863	$\operatorname{SL}(2; \mathbb{R})$	$SL(2; \mathbb{R})$	1. Construct the analytic mapping $\phi(x, y)$ for the parameterization of $SL(2; \mathbb{R})$ illus-
gilmore-lie-groups_ F00283	44	■ 1.000	M	M	3. Convince yourself that every matrix M in the group $SL(n; \mathbb{R})$ can be written as the
gilmore-lie-groups_ F00284	44	■ 1.000	$SL(n; \mathbb{R})$	$SL(n; \mathbb{R})$	3. Convince yourself that every matrix M in the group $SL(n; \mathbb{R})$ can be written as the
gilmore-lie-groups_ F00285	44	■ 1.000	S	S	product of an $n \times n$ real symmetric unimodular matrix S and an orthogonal matrix
gilmore-lie-groups_ F00286	44	■ 0.755	O	O	O in $SO(n)$: $M = SO$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_ F00287	44	■ 0.755	$SO(n): M = SO$	$SO(n): M = SO$	O in $SO(n)$: $M = SO$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_ F00288	44	■ 1.000	$S = (MM^t)^{1/2}$	$S = (MM^t)^{1/2}$	$S = (MM^t)^{1/2}$ and $O = S^{-1}M$. How do you compute the square root of a matrix?
gilmore-lie-groups_ F00289	44	■ 1.000	$O = S^{-1}M$	$O = S^{-1}M$	$S = (MM^t)^{1/2}$ and $O = S^{-1}M$. How do you compute the square root of a matrix?
gilmore-lie-groups_ F00290	44	■ 1.000	$(x, y, z) \rightarrow (x', y', z')$	$(x, y, z) \rightarrow (x', y', z')$	4. Construct the most general linear transformation $(x, y, z) \rightarrow (x', y', z')$ that leaves
gilmore-lie-groups_ F00291	44	■ 1.000	M_1	M_1	where the real symmetric matrices M_1 and M_2 satisfy
gilmore-lie-groups_ F00292	44	■ 1.000	M_2	M_2	where the real symmetric matrices M_1 and M_2 satisfy
gilmore-lie-groups_ F00293	44	■ 0.995	$[SO(1, 1)]$	$[SO(1, 1)]$	5. Construct the group of linear transformations $[SO(1, 1)]$ that leaves invariant the
gilmore-lie-groups_ F00294	44	■ 0.972	$(ct)^2 - x^2$	$(ct)^2 - x^2$	quantity $(ct)^2 - x^2$. Compare this with the group of linear transformations $[SO(2)]$
gilmore-lie-groups_ F00295	44	■ 0.972	$SO(2)$	$SO(2)$	quantity $(ct)^2 - x^2$. Compare this with the group of linear transformations $[SO(2)]$
gilmore-lie-groups_ F00296	44	■ 0.999	$x^2 + y^2$	$x^2 + y^2$	that leaves invariant the radius of the circle $x^2 + y^2$. (This comparison involves map-
gilmore-lie-groups_ F00297	45	■ 1.000	$(ct)^2 - x^2 - y^2 - z^2$	$(ct)^2 - x^2 - y^2 - z^2$	$(ct)^2 - x^2 - y^2 - z^2$. This is the Lorentz group $O(3, 1)$. Four disconnected mani-
gilmore-lie-groups_ F00298	45	■ 1.000	$O(3, 1)$	$O(3, 1)$	$(ct)^2 - x^2 - y^2 - z^2$. This is the Lorentz group $O(3, 1)$. Four disconnected mani-
gilmore-lie-groups_ F00299	45	■ 0.535	$SL(2; \mathbb{C})$	$SL(2; \mathbb{C})$	7. The group of 2×2 complex matrices with determinant +1 is named $SL(2; \mathbb{C})$. Ma-
gilmore-lie-groups_ F00300	45	■ 0.788	$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$	$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$	trices in this group have the structure $\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$, where $\alpha, \beta, \gamma, \delta$ are complex numbers

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gilmore-lie-groups_ F00301	45	■ 0.987	$\alpha \delta - \beta \gamma = 1$	$\alpha \delta - \beta \gamma = 1$	and $\alpha \delta - \beta \gamma = 1$. Define the matrix X by
gilmore-lie-groups_ F00302	45	■ 0.987	x	X	and $\alpha \delta - \beta \gamma = 1$. Define the matrix X by
gilmore-lie-groups_ F00303	45	■ 1.000	$\mathbf{x}$	$\mathbf{x}$	where $\mathbf{x}$ is the three vector $\mathbf{x} = (x, y, z)$ and $\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$ are the
gilmore-lie-groups_ F00304	45	■ 1.000	$\mathbf{x} = (x, y, z)$	$\mathbf{x} = (x, y, z)$	where $\mathbf{x}$ is the three vector $\mathbf{x} = (x, y, z)$ and $\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$ are the
gilmore-lie-groups_ F00305	45	■ 1.000	$\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$	$\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$	where $\mathbf{x}$ is the three vector $\mathbf{x} = (x, y, z)$ and $\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$ are the
gilmore-lie-groups_ F00306	45	■ 1.000	$X^\dagger \equiv (X^t)^* = X$	$X^\dagger \equiv (X^t)^* = X$	a. Show that X is hermitian: $X^\dagger \equiv (X^t)^* = X$.
gilmore-lie-groups_ F00307	45	■ 1.000	$g \in SL(2; \mathbb{C})$	$g \in SL(2; \mathbb{C})$	c. If $g \in SL(2; \mathbb{C})$, show that $g^\dagger X g = X' = H(x', y', z', ct')$.
gilmore-lie-groups_ F00308	45	■ 1.000	$g^\dagger X g = X' = H(x', y', z', ct')$	$g^\dagger X g = X' = H(x', y', z', ct')$	c. If $g \in SL(2; \mathbb{C})$, show that $g^\dagger X g = X' = H(x', y', z', ct')$.
gilmore-lie-groups_ F00309	45	■ 0.964	(x', y', z', ct')	(x', y', z', ct')	d. How are the new space-time coordinates (x', y', z', ct') related to the original
gilmore-lie-groups_ F00310	45	■ 0.918	(x, y, z, ct)	(x, y, z, ct)	coordinates (x, y, z, ct) ? (They are linearly related by coefficients that are bilinear
gilmore-lie-groups_ F00311	45	■ 1.000	g	g	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^\dagger$.)
gilmore-lie-groups_ F00312	45	■ 1.000	$\alpha^*, \beta^*, \gamma^*, \delta^*$	$\alpha^*, \beta^*, \gamma^*, \delta^*$	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^\dagger$.)
gilmore-lie-groups_ F00313	45	■ 1.000	$g^\dagger$	$g^\dagger$	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^\dagger$.)
gilmore-lie-groups_ F00314	45	■ 0.990	$t' = t$	$t' = t$	e. Find the subgroup of $SL(2; \mathbb{C})$ that leaves $t' = t$. (It is $SU(2) \subset SL(2; \mathbb{C})$).
gilmore-lie-groups_ F00315	45	■ 0.990	$SU(2) \subset SL(2; \mathbb{C})$	$SU(2) \subset SL(2; \mathbb{C})$	e. Find the subgroup of $SL(2; \mathbb{C})$ that leaves $t' = t$. (It is $SU(2) \subset SL(2; \mathbb{C})$).
gilmore-lie-groups_ F00316	45	■ 0.952	$g = kh$	$g = kh$	f. For any $g \in SL(2; \mathbb{C})$ write $g = kh$, where $h \in SU(2)$, $h^\dagger = h^{-1}$, h has the
gilmore-lie-groups_ F00317	45	■ 0.952	$h \in SU(2), h^\dagger = h^{-1}, h$	$h \in SU(2), h^\dagger = h^{-1}, h$	f. For any $g \in SL(2; \mathbb{C})$ write $g = kh$, where $h \in SU(2)$, $h^\dagger = h^{-1}$, h has the
gilmore-lie-groups_ F00318	45	■ 0.461	$h = \exp\left(\frac{i}{2} \sigma \cdot \theta\right)$	$h = \exp\left(\frac{i}{2} \sigma \cdot \theta\right)$	form $h = \exp(\frac{i}{2} \sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2)$, $k^\dagger = k^{+1}$, k has the form
gilmore-lie-groups_ F00319	45	■ 0.461	$k \in SL(2; \mathbb{C})/SU(2), k^\dagger = k^{+1}, k$	$k \in SL(2; \mathbb{C})/SU(2), k^\dagger = k^{+1}, k$	form $h = \exp(\frac{i}{2} \sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2)$, $k^\dagger = k^{+1}$, k has the form
gilmore-lie-groups_ F00320	45	■ 1.000	$k = \exp\left(\frac{1}{2} \sigma \cdot \mathbf{b}\right)$	$k = \exp\left(\frac{1}{2} \sigma \cdot \mathbf{b}\right)$	$k = \exp(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F00321	45	■ 1.000	$\mathbf{b}$	$\mathbf{b}$	$k = \exp(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F00322	45	■ 1.000	θ	θ	$k = \exp(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F00323	45	■ 0.899	$k^\dagger H(x, y, z, ct) k = H(x', y', z', ct')$	$k^\dagger H(x, y, z, ct) k = H(x', y', z', ct')$	and $\mathbf{b}$ are real. Construct $k^\dagger H(x, y, z, ct) k = H(x', y', z', ct')$. If this is too diffi-
gilmore-lie-groups_ F00324	45	■ 0.893	$\mathbf{b} = (0, 0, b)$	$\mathbf{b} = (0, 0, b)$	cult, choose $\mathbf{b}$ along the z -axis, $\mathbf{b} = (0, 0, b)$.
gilmore-lie-groups_ F00325	45	■ 0.458	$k(b')$	$k(b')$	h. Applying $k(b')$ after applying $k(b)$ results in (a) $k(b' + b)$, (b) two successive
gilmore-lie-groups_ F00326	45	■ 0.458	$k(b)$	$k(b)$	h. Applying $k(b')$ after applying $k(b)$ results in (a) $k(b' + b)$, (b) two successive
gilmore-lie-groups_ F00327	45	■ 0.458	$(a) k(b' + b)$	$(a) k(b' + b)$	h. Applying $k(b')$ after applying $k(b)$ results in (a) $k(b' + b)$, (b) two successive

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gilmore-lie-groups_F00328	45	■ 0.981	$\mathbf{b}^{\prime}$	$\mathbf{b}'$	i. If $\mathbf{b}$ and $\mathbf{b}'$ are not colinear, $k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$. Compute $\mathbf{b}'', \theta$. The angle θ is
gilmore-lie-groups_F00329	45	■ 0.981	$k(\mathbf{b}^{\prime})k(\mathbf{b})=k(\mathbf{b}'')h(\theta)$	$k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$	i. If $\mathbf{b}$ and $\mathbf{b}'$ are not colinear, $k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$. Compute $\mathbf{b}'', \theta$. The angle θ is
gilmore-lie-groups_F00330	45	■ 0.981	$\mathbf{b}^{\prime \prime}, \theta$	$\mathbf{b}'', \theta$	i. If $\mathbf{b}$ and $\mathbf{b}'$ are not colinear, $k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$. Compute $\mathbf{b}'', \theta$. The angle θ is
gilmore-lie-groups_F00331	45	■ 1.000	$\theta \rightarrow \theta^{\prime}=\theta+k+f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$, where k is a real number, $0 \leq k < 2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_F00332	45	■ 1.000	$\theta \leq k < 2 \pi$	$0 \leq k < 2\pi$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$, where k is a real number, $0 \leq k < 2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_F00333	45	■ 1.000	$f(\theta)$	$f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$, where k is a real number, $0 \leq k < 2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_F00334	45	■ 0.991	$f(\theta+2 \pi)=f(\theta)$	$f(\theta+2\pi) = f(\theta)$	$f(\theta+2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_F00335	45	■ 0.991	1: 1	1 : 1	$f(\theta+2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_F00336	45	■ 1.000	$f(\theta): d f(\theta) / d \theta>-1$	$f(\theta): df(\theta)/d\theta > -1$	on $f(\theta): df(\theta)/d\theta > -1$ everywhere. Does this set of transformations form a group?
gilmore-lie-groups_F00337	46	■ 0.769	(a, b, c, d)	(a, b, c, d)	9. Rational fractional transformations (a, b, c, d) map points on the real line (real pro-
gilmore-lie-groups_F00338	46	■ 1.000	$R P^{\{1\}}$	RP^1	jective line RP^1) to the real line as follows:
gilmore-lie-groups_F00339	46	■ 1.000	$(\lambda a, \lambda b, \lambda c, \lambda d)=\lambda(a, b, c, d)(\lambda \neq 0)$	$(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d)(\lambda \neq 0)$	The transformations (a, b, c, d) and $(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d)(\lambda \neq 0)$ generate
gilmore-lie-groups_F00340	46	■ 1.000	A, B, C, D	A, B, C, D	values of A, B, C, D .
gilmore-lie-groups_F00341	46	■ 0.668	$(\lambda, 0, 0, \lambda)$	$(\lambda, 0, 0, \lambda)$	b. Show that the transformations $(\lambda, 0, 0, \lambda)$ map x to itself.
gilmore-lie-groups_F00342	46	■ 0.999	$x^{\prime} \rightarrow x$	$x' \rightarrow x$	c. Construct the inverse transformation $x' \rightarrow x$, and show that it is $\lambda(d,-b,-c, a)$
gilmore-lie-groups_F00343	46	■ 0.999	$\lambda(d,-b,-c, a)$	$\lambda(d,-b,-c, a)$	c. Construct the inverse transformation $x' \rightarrow x$, and show that it is $\lambda(d,-b,-c, a)$
gilmore-lie-groups_F00344	46	■ 1.000	$\lambda \neq 0$	$\lambda \neq 0$	provided $\lambda \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups_F00345	46	■ 1.000	$D=a d-b c \neq 0$	$D = ad - bc \neq 0$	provided $\lambda \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups_F00346	46	■ 1.000	$x^{\prime}=(a, b, c, d) x=\lambda(a, b, c, d) x$	$x' = (a, b, c, d)x = \lambda(a, b, c, d)x$	d. Show that the transformation degeneracy $x' = (a, b, c, d)x = \lambda(a, b, c, d)x$ can
gilmore-lie-groups_F00347	46	■ 1.000	a, b, c, d	a, b, c, d	be lifted by requiring that the four parameters a, b, c, d describing these transfor-
gilmore-lie-groups_F00348	46	■ 1.000	$D=a d-b c=1$	$D = ad - bc = 1$	mations satisfy the constraint $D = ad - bc = 1$.
gilmore-lie-groups_F00349	46	■ 0.980	(y, z)	(y, z)	e. It is useful to introduce homogeneous coordinates (y, z) and define the real projec-
gilmore-lie-groups_F00350	46	■ 1.000	$x=y / z$	$x = y/z$	tive coordinate x as the ratio of these homogeneous coordinates: $x = y/z$. If the
gilmore-lie-groups_F00351	46	■ 0.477	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3	distinct points x_1, x_2, x_3 on the real line to the three standard positions $(0, 1, \infty)$,
gilmore-lie-groups_F00352	46	■ 0.477	∞	∞	distinct points x_1, x_2, x_3 on the real line to the three standard positions $(0, 1, \infty)$,
gilmore-lie-groups_F00353	46	■ 0.996	$D=1$	$D = 1$	What matrix in $SL(2; \mathbb{R})$ describes this mapping? (Careful of the condition $D = 1$.)
gilmore-lie-groups_F00354	46	■ 1.000	$\left(x_{\{1\}}^{\prime}, x_{\{2\}}^{\prime}, x_{\{3\}}^{\prime}\right)$	(x'_1, x'_2, x'_3)	points (x_1, x_2, x_3) to any other triple of distinct points (x'_1, x'_2, x'_3) .
gilmore-lie-groups_F00355	46	■ 1.000	$R P^{\{n\}}$	RP^n	10. The real projective space RP^n is the space of all straight lines through the origin

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gilmore-lie-groups_ F08356	46	■ 0.987	R^{n+1}	R^{n+1}	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_ F08357	46	■ 0.987	$SL(n+1; \mathbb{R}) \operatorname{maps} x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	$SL(n+1; \mathbb{R}) \operatorname{maps} x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	in R^{n+1} . The group $SL(n+1; \mathbb{R}) \operatorname{maps} x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_ F08358	46	■ 0.987	$x^{\prime} \in R^{n+1}$	$x' \in R^{n+1}$	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_ F08359	46	■ 1.000	$x^{\prime} \neq 0 \rightarrow x \neq 0$	$x' \neq 0 \leftrightarrow x \neq 0$	with $x' \neq 0 \leftrightarrow x \neq 0$ and $x' = 0 \leftrightarrow x = 0$. A straight line through the origin contains
gilmore-lie-groups_ F08360	46	■ 1.000	$x^{\prime}=0 \rightarrow x=0$	$x' = 0 \leftrightarrow x = 0$	with $x' \neq 0 \leftrightarrow x \neq 0$ and $x' = 0 \leftrightarrow x = 0$. A straight line through the origin contains
gilmore-lie-groups_ F08361	46	■ 1.000	$x \neq 0$	$x \neq 0$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_ F08362	46	■ 1.000	$y \neq 0$	$y \neq 0$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_ F08363	46	■ 1.000	$y=\lambda x$	$y = \lambda x$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_ F08364	46	■ 1.000	y	y	factor can always be chosen so that y is in the unit sphere in R^{n+1} : $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups_ F08365	46	■ 1.000	$R^{n+1}: y \in S^n \subset R^{n+1}$	$R^{n+1}: y \in S^n \subset R^{n+1}$	factor can always be chosen so that y is in the unit sphere in R^{n+1} : $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups_ F08366	46	■ 1.000	λ	λ	In fact, two values of λ can be chosen: $\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_ F08367	46	■ 1.000	$\lambda= \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$	$\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$	In fact, two values of λ can be chosen: $\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_ F08368	46	■ 1.000	R^3	R^3	In fact, two values of λ can be chosen: $\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_ F08369	46	■ 0.983	(x, y, z)	(x, y, z)	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups_ F08370	46	■ 0.983	$(X, Y)=$	$(X, Y) =$	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups_ F08371	46	■ 1.000	$(x/z, y/z)$	$(x/z, y/z)$	$(x/z, y/z)$ if $z \neq 0$. Straight lines through the origin of R^3 are mapped to straight lines
gilmore-lie-groups_ F08372	46	■ 1.000	$z \neq 0$	$z \neq 0$	$(x/z, y/z)$ if $z \neq 0$. Straight lines through the origin of R^3 are mapped to straight lines
gilmore-lie-groups_ F08373	46	■ 1.000	$x \rightarrow x^{\prime}=M x, M \in S L(3 ; \mathbb{R})$	$x \rightarrow x' = Mx, M \in SL(3; \mathbb{R})$	in R^3 by $x \rightarrow x' = Mx, M \in SL(3; \mathbb{R})$. Show that the homogeneous coordinates
gilmore-lie-groups_ F08374	47	■ 1.000	$x^{\prime}$	x'	representing the two lines containing x and x' are related by the linear fractional
gilmore-lie-groups_ F08375	47	■ 1.000	$R P^n \rightarrow R P^n$	$RP^n \rightarrow RP^n$	Generalize for linear fractional transformations $RP^n \rightarrow RP^n$.
gilmore-lie-groups_ F08376	47	■ 0.989	$S L(2 ; \mathbb{R}) / S O(2) \simeq\left[\begin{array}{cc} z+x & y \\ y & z-x \end{array}\right]$	$SL(2; \mathbb{R})/SO(2) \simeq \begin{bmatrix} z+x & y \\ y & z-x \end{bmatrix}$	11. The hyperbolic two-space $SL(2; \mathbb{R})/SO(2) \simeq [\frac{z+x}{y} \frac{y}{z-x}]$ consists of the algebraic sub-
gilmore-lie-groups_ F08377	47	■ 1.000	$x=r \cos \phi, y=r \sin \phi$	$x = r \cos \phi, y = r \sin \phi$	b. Introduce polar coordinates $x = r \cos \phi$, $y = r \sin \phi$, and show
gilmore-lie-groups_ F08378	47	■ 0.633	$S O(3) / S O(2)$	$SO(3)/SO(2)$	d. Repeat this calculation for $SO(3)/SO(2)$. This space is a sphere $S^2 \subset R^3$: the
gilmore-lie-groups_ F08379	47	■ 0.633	$S^2 \subset R^3$	$S^2 \subset R^3$	d. Repeat this calculation for $SO(3)/SO(2)$. This space is a sphere $S^2 \subset R^3$: the
gilmore-lie-groups_ F08380	47	■ 1.000	$z^2+\left(x^2+y^2\right)=1$	$z^2 + (x^2 + y^2) = 1$	algebraic manifold in R^3 that satisfies $z^2 + (x^2 + y^2) = 1$ and inherits the metric
gilmore-lie-groups_ F08381	47	■ 1.000	$d s^2=d z^2+\left(d x^2+d y^2\right)$	$ds^2 = dz^2 + (dx^2 + dy^2)$	$ds^2 = dz^2 + (dx^2 + dy^2)$ from this Euclidean space. Show that the metric and

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gilmore-lie-groups_ F08382	47	■ 1.000	H^2	H^2	measure on S^2 are obtained from the results above for H^2 by the substitutions
gilmore-lie-groups_ F08383	47	■ 1.000	$1+r^2 \rightarrow 1-r^2$	$1+r^2 \rightarrow 1-r^2$	$1+r^2 \rightarrow 1-r^2$. Show that the disk $0 \leq r \leq 1, 0 \leq \phi \leq 2\pi$ maps onto the up-
gilmore-lie-groups_ F08384	47	■ 1.000	$0 \leq r \leq 1, 0 \leq \phi \leq 2\pi$	$0 \leq r \leq 1, 0 \leq \phi \leq 2\pi$	$1+r^2 \rightarrow 1-r^2$. Show that the disk $0 < r < 1, 0 < \phi < 2\pi$ maps onto the up-
gilmore-lie-groups_ F08385	47	■ 1.000	$r=0$	$r=0$	per hemisphere of the sphere, with $r=0$ mapping to the north pole and $r=1$
gilmore-lie-groups_ F08386	47	■ 1.000	$r=1$	$r=1$	per hemisphere of the sphere, with $r=0$ mapping to the north pole and $r=1$
gilmore-lie-groups_ F08387	47	■ 1.000	$\phi=0$	$\phi=0$	equator along the longitude $\phi=0$ is $s = \int_0^1 dr/\sqrt{1-r^2} = \pi/2$ and the volume of
gilmore-lie-groups_ F08388	47	■ 1.000	$s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$	$s = \int_0^1 dr/\sqrt{1-r^2} = \pi/2$	equator along the longitude $\phi=0$ is $s = \int_0^1 dr/\sqrt{1-r^2} = \pi/2$ and the volume of
gilmore-lie-groups_ F08389	47	■ 1.000	$V=\int_{r=0}^1 \int_{\phi=0}^{2\pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2\pi} d \phi=$	$V = \int_{r=0}^1 \int_{\phi=0}^{2\pi} dV(r, \phi) = \int_0^1 r dr/\sqrt{1-r^2} \int_0^{2\pi} d\phi =$	the hemisphere surface is $V = \int_{r=0}^1 \int_{\phi=0}^{2\pi} dV(r, \phi) = \int_0^1 r dr/\sqrt{1-r^2} \int_0^{2\pi} d\phi =$
gilmore-lie-groups_ F08390	47	■ 1.000	2π	2π	2π .
gilmore-lie-groups_ F08391	48	■ 0.969	$GL(n ; \mathbb{F})$	$GL(n; \mathbb{F})$	typically encountered. These include the general linear groups $GL(n; \mathbb{F})$
gilmore-lie-groups_ F08392	48	■ 1.000	$\mathbb{F}$	$\mathbb{F}$	of nonsingular $n \times n$ matrices over the fields $\mathbb{F}$ of real numbers, complex
gilmore-lie-groups_ F08393	48	■ 1.000	$H_1 \subset G$	$H_1 \subset G$	Theorem If $H_1 \subset G$ and $H_2 \subset G$ are subgroups of G then their intersection
gilmore-lie-groups_ F08394	48	■ 1.000	$H_{12}=H_1 \cap H_2$	$H_{12} = H_1 \cap H_2$	$H_{12} \equiv H_1 \cap H_2$ is a subgroup of G .
gilmore-lie-groups_ F08395	48	■ 1.000	$H_1 \cap H_2$	$H_1 \cap H_2$	Proof Verify that the four group axioms are satisfied for all operations in $H_1 \cap H_2$.
gilmore-lie-groups_ F08396	48	■ 0.950	H_1	H_1	Example If H_1 is the two-dimensional subgroup of $SL(2; \mathbb{R})$ described in (3.1)
gilmore-lie-groups_ F08397	48	■ 0.817	H_2	H_2	above and H_2 is the one-dimensional subgroup of 2×2 orthogonal matrices
gilmore-lie-groups_ F08398	49	■ 1.000	$\pm I_2$	$\pm I_2$	discrete group operations $\pm I_2$.
gilmore-lie-groups_ F08399	49	■ 0.950	$(\mathbb{F}=\mathbb{R})$	$(\mathbb{F} = \mathbb{R})$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups_ F08400	49	■ 0.950	$\mathbb{F}=\mathbb{C}$	$\mathbb{F} = \mathbb{C}$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups_ F08401	49	■ 0.950	$\mathbb{F}=\mathbb{Q}$	$\mathbb{F} = \mathbb{Q}$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups_ F08402	49	■ 1.000	$1, \mathcal{I}, J, K$	$1, \mathcal{I}, J, K$	The four basis vectors $1, \mathcal{I}, J, K$ for this map are related to the four Pauli spin
gilmore-lie-groups_ F08403	49	■ 0.900	$[\operatorname{det}(\mathrm{M})=+1]$	$[\det(M) = +1]$	$\det(M) \equiv +1$ are imposed (12).
gilmore-lie-groups_ F08404	49	■ 0.999	$GL(1 ; \mathbb{Q})$	$GL(1; \mathbb{Q})$	the real, complex, or quaternion fields. The group $GL(1; \mathbb{Q})$ consists of 1×1
gilmore-lie-groups_ F08405	49	■ 0.999	1×1	1×1	the real, complex, or quaternion fields. The group $GL(1; \mathbb{Q})$ consists of 1×1
gilmore-lie-groups_ F08406	50	■ 1.000	A_i^j	A_i^j	The determinant of an $n \times n$ matrix A with matrix elements A_i^j is defined by
gilmore-lie-groups_ F08407	50	■ 1.000	$\epsilon^{i_1 i_2 \cdots i_n}$	$\epsilon^{i_1 i_2 \cdots i_n}$	Here $\epsilon^{i_1 i_2 \cdots i_n}$ and its covariant version are the Levi–Civita symbols: +1 for an even

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gilmore-lie-groups_ F00408	50	0.719	$1, 2, \ldots, n; -1$	$1, 2, \dots, n; -1$	permutation of the integers $\boxed{1, 2, \dots, n; -1}$ for an odd permutation; and 0 if two
gilmore-lie-groups_ F00409	50	1.000	$i_{\{*\}}$	i_*	or more values of the indices i_* are equal. With this definition there is no difficulty
gilmore-lie-groups_ F00410	50	1.000	$U\ T(p, q)$	$UT(p, q)$	2. $UT(p, q)$. Upper triangular groups. The $n \times n$ ($n = p + q$) matrix is parti-
gilmore-lie-groups_ F00411	50	1.000	$n \times n (n=p+q)$	$n \times n (n = p + q)$	2. $UT(p, q)$. Upper triangular groups. The $n \times n$ ($n = p + q$) matrix is parti-
gilmore-lie-groups_ F00412	50	1.000	$U\ T(1, 1)$	$UT(1, 1)$	Example The action of transformations in $UT(1, 1)$ on the plane R^2 is as
gilmore-lie-groups_ F00413	50	0.991	$y=0$	$y = 0$	The x -axis $y = 0$ remains invariant. It is an invariant subspace ($y = 0 \rightarrow y' = 0$),
gilmore-lie-groups_ F00414	50	0.991	$y=0 \rightarrow y^{\prime}=0$	$y = 0 \rightarrow y' = 0$	The x -axis $y = 0$ remains invariant. It is an invariant subspace ($\boxed{y = 0 \rightarrow y' = 0}$),
gilmore-lie-groups_ F00415	50	1.000	$x=0$	$x = 0$	mapped into itself by all group operations in $UT(1, 1)$. The y -axis $x = 0$ is not
gilmore-lie-groups_ F00416	50	1.000	$V_{\{p\}} \oplus V_{\{q\}}$	$V_p \oplus V_q$	invariant. More generally, if $UT(p, q)$ acts on the direct sum vector space $\boxed{V_p \oplus V_q}$,
gilmore-lie-groups_ F00417	50	1.000	$V_{\{q\}}$	V_q	the subspace V_q is invariant while V_p is not. For lower triangular matrices reverse
gilmore-lie-groups_ F00418	50	1.000	$V_{\{p\}}$	V_p	the subspace V_q is invariant while V_p is not. For lower triangular matrices reverse
gilmore-lie-groups_ F00419	50	1.000	q	q	p and q .
gilmore-lie-groups_ F00420	51	1.000	$H\ T(p, q)$	$HT(p, q)$	3. $HT(p, q)$. This is a subgroup of $UT(p, q)$ obtained by imposing the additional
gilmore-lie-groups_ F00421	51	1.000	$H\ T(1, 1) \left(m_{22}=1 \right)$	$HT(1, 1) (m_{22} = 1)$	Example Affine transformations in $HT(1, 1) (m_{22} = 1)$ act on the x -axis by $x \rightarrow$
gilmore-lie-groups_ F00422	51	1.000	$x \rightarrow$	$x \rightarrow$	Example Affine transformations in $HT(1, 1) (m_{22} = 1)$ act on the x -axis by $x \rightarrow$
gilmore-lie-groups_ F00423	51	0.997	$x^{\prime}=a\ x+b$	$x' = ax + b$	$x' = ax + b$:
gilmore-lie-groups_ F00424	51	0.999	$U\ T(p, q, r)$	$UT(p, q, r)$	4. $UT(p, q, r)$. This matrix group consists of upper triangular matrices that are the
gilmore-lie-groups_ F00425	51	1.000	$U\ T(p, q+r) \cap U\ T(p+q, r)$	$UT(p, q + r) \cap UT(p + q, r)$	intersection of the matrix groups $UT(p, q + r) \cap UT(p + q, r)$.
gilmore-lie-groups_ F00426	52	0.460	$S\ U(1, 1)$	$SU(1, 1)$	where the 2×2 matrix $SU(1, 1)$ is defined below in (3.30). Matrix groups with the
gilmore-lie-groups_ F00427	52	1.000	$\operatorname{Sol}(n)=U\ T(1, 1, 1, \ldots, 1)$	$Sol(n) = UT(1, 1, 1, \dots, 1)$	5. $Sol(n) = UT(1, 1, 1, \dots, 1)$. Solvable groups are strictly upper triangular.
gilmore-lie-groups_ F00428	52	0.971	$U\ T(1, 1, 1)$	$UT(1, 1, 1)$	Example We consider the subgroup of 3×3 matrices in $UT(1, 1, 1)$ of the
gilmore-lie-groups_ F00429	52	1.000	$\hat{n}=a^{\dagger} a$	$\hat{n} = a^\dagger a$	of the photon number operator ($\boxed{\hat{n} = a^\dagger a}$), the creation ($a^\dagger$) and annihilation (a)
gilmore-lie-groups_ F00430	52	1.000	$a^{\dagger}$	$a^\dagger$	of the photon number operator ($\hat{n} = a^\dagger a$), the creation ($a^\dagger$) and annihilation (a)
gilmore-lie-groups_ F00431	52	1.000	a	a	of the photon number operator ($\hat{n} = a^\dagger a$), the creation ($a^\dagger$) and annihilation (a)
gilmore-lie-groups_ F00432	52	0.994	$I=a\ a^{\dagger}-a^{\dagger} a=\left[a, a^{\dagger}\right]$	$I = aa^\dagger - a^\dagger a = [a, a^\dagger]$	operators, and their commutator ($\boxed{I = aa^\dagger - a^\dagger a = [a, a^\dagger]}$). We will use this iden-
gilmore-lie-groups_ F00433	52	1.000	$\operatorname{Nil}(n)$	$Nil(n)$	6. $Nil(n)$. Nilpotent groups are subgroups of $Sol(n)$ whose diagonal matrix
gilmore-lie-groups_ F00434	52	1.000	$\operatorname{Sol}(n)$	$Sol(n)$	6. $Nil(n)$. Nilpotent groups are subgroups of $Sol(n)$ whose diagonal matrix

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gilmore-lie-groups_ F00435	52	■ 0.998	$\operatorname{Nil}(3)$	$\operatorname{Nil}(3)$	Example Matrices in $\operatorname{Nil}(3)$ of the form
gilmore-lie-groups_ F00436	52	■ 0.976	$a^{\dagger}, a, I$	$a^{\dagger}, a, I$	are closely related to the photon creation and annihilation operators ($a^{\dagger}, a, I$) and
gilmore-lie-groups_ F00437	52	■ 0.716	$(p$	$(p$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_ F00438	52	■ 0.716	$q)$	$q)$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_ F00439	52	■ 0.716	$[p, q] = \hbar / i$	$[p, q] = \hbar/i$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_ F00440	52	■ 1.000	$\langle p \mid q \rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / \hbar}$	$\langle p \mid q \rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / \hbar}$	The set of change of basis transformations $\langle p \mid q \rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / \hbar}$ encountered in
gilmore-lie-groups_ F00441	52	■ 0.995	$A(p, q)$	$A(p, q)$	7. $A(p, q)$. This group consists of matrices that are the sum of an identity matrix
gilmore-lie-groups_ F00442	52	■ 0.998	(p, q)	(p, q)	and the upper right-hand off-diagonal block of a (p, q) blocked matrix. Its matrix
gilmore-lie-groups_ F00443	53	■ 1.000	$AB = BA$	$AB = BA$	This group is abelian or commutative: $AB = BA$ for all elements (matrices) in this
gilmore-lie-groups_ F00444	53	■ 1.000	$A(1, 1)$	$A(1, 1)$	Example We consider the translation subgroup $A(1, 1)$ of the affine group of
gilmore-lie-groups_ F00445	53	■ 0.798	$x \rightarrow x' = x + a$	$x \rightarrow x' = x + a$	transformations of the x -axis (3.9): $x \rightarrow x' = x + a$. Successive transformations
gilmore-lie-groups_ F00446	53	■ 1.000	$M^{\dagger} G M = G$	$M^{\dagger} G M = G$	all satisfy the bilinear or quadratic constraint condition $M^{\dagger} G M = G$. If G is sym-
gilmore-lie-groups_ F00447	53	■ 0.679	$G = I_n$	$G = I_n$	metric positive-definite we can set $G = I_n$ (8). If G is nonsingular and symmetric
gilmore-lie-groups_ F00448	53	■ 0.573	$G = I_{p, q}$	$G = I_{p, q}$	but indefinite we can set $G = I_{p, q}$ (9). If G is nonsingular and antisymmetric, we
gilmore-lie-groups_ F00449	53	■ 1.000	I_n	I_n	which we can take as I_n
gilmore-lie-groups_ F00450	54	■ 0.936	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	terize the matrix groups $GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.
gilmore-lie-groups_ F00451	54	■ 1.000	C^2	C^2	two-dimensional linear vector space C^2 (of spin states, for example), and quaternion
gilmore-lie-groups_ F00452	54	■ 1.000	$\mathbb{Q}$	$\mathbb{Q}$	over $\mathbb{Q}$
gilmore-lie-groups_ F00453	54	■ 0.808	$SU(1; \mathbb{Q})$	$SU(1; \mathbb{Q})$	The group $SU(1; \mathbb{Q})$ is the subgroup of $GL(1; \mathbb{Q})$ (3.5) subject to the condition
gilmore-lie-groups_ F00454	54	■ 1.000	R^4	R^4	in R^4
gilmore-lie-groups_ F00455	54	■ 0.992	$G = I_{p, q}, p + q = n$	$G = I_{p, q}, p + q = n$	$G = I_{p, q}, p + q = n$. This is a diagonal matrix with p elements $+1$ and q ele-
gilmore-lie-groups_ F00456	54	■ 1.000	$p \neq 0, q \neq 0$	$p \neq 0, q \neq 0$	The manifolds that parameterize these groups are noncompact when $p \neq 0, q \neq 0$.
gilmore-lie-groups_ F00457	54	■ 1.000	$x^2 + y^2 + z^2 - (ct)^2$	$x^2 + y^2 + z^2 - (ct)^2$	Example The Lorentz group preserves the invariant $x^2 + y^2 + z^2 - (ct)^2$ and is
gilmore-lie-groups_ F00458	55	■ 0.999	$N \times N$	$N \times N$	10. Antisymmetric metric-preserving groups The metric G is an $N \times N$ non-
gilmore-lie-groups_ F00459	55	■ 1.000	$\operatorname{det}(G) = \operatorname{det}(\left(G^t\right)) = \operatorname{det}(-G) = (-1)^N \operatorname{det}(G), N$	$\operatorname{det}(G) = \operatorname{det}(G^t) = \operatorname{det}(-G) = (-1)^N \operatorname{det}(G), N$	Since $\operatorname{det}(G) = \operatorname{det}(G^t) = \operatorname{det}(-G) = (-1)^N \operatorname{det}(G), N$ must be even: $N = 2n$. The
gilmore-lie-groups_ F00460	55	■ 1.000	$N = 2n$	$N = 2n$	Since $\operatorname{det}(G) = \operatorname{det}(G^t) = \operatorname{det}(-G) = (-1)^N \operatorname{det}(G), N$ must be even: $N = 2n$. The

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gilmore-lie-groups_ F00461	55	■ 1.000	$i \sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	This consists of n copies of the matrix $i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ along the diagonal. Symplectic
gilmore-lie-groups_ F00462	55	■ 0.990	$\operatorname{Sp}(2n; \mathbb{R})$	$\operatorname{Sp}(2n; \mathbb{R})$	transformations in $\operatorname{Sp}(2n; \mathbb{R})$ leave invariant the form of the classical hamiltonian
gilmore-lie-groups_ F00463	55	■ 0.935	$\operatorname{Sp}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$	Example The symplectic group $\operatorname{Sp}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$ satisfies the constraint
gilmore-lie-groups_ F00464	55	■ 0.510	$ad - bc = +1$	$ad - bc = +1$	The constraint is $ad - bc = +1$. Thus, $\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$.
gilmore-lie-groups_ F00465	55	■ 0.510	$\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$	The constraint is $ad - bc = +1$. Thus, $\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$.
gilmore-lie-groups_ F00466	55	■ 1.000	$MGM^t = G$	$MGM^t = G$	Partitioning M into blocks and imposing the condition $MGM^t = G$, we find
gilmore-lie-groups_ F00467	56	■ 0.999	$E(3)$	$E(3)$	Euclidean group $E(3)$ whose action on the coordinates (x, y, z) of a point in
gilmore-lie-groups_ F00468	56	■ 0.607	$\mathbf{t}$	$\mathbf{t}$	That is, the coordinates are rotated by the matrix A and translated by the vector $\mathbf{t}$. By
gilmore-lie-groups_ F00469	56	■ 0.792	$A \in \operatorname{SO}(3, 1), AI_{3,1}A^t = I_{3,1}$	$A \in \operatorname{SO}(3, 1), AI_{3,1}A^t = I_{3,1}$	($A \in \operatorname{SO}(3, 1), AI_{3,1}A^t = I_{3,1}$ (3.17)) and space-time displacements is isomorphic
gilmore-lie-groups_ F00470	56	■ 0.608	$\mathbf{v}$	$\mathbf{v}$	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00471	56	■ 0.608	$(\mathbf{t})$	$(\mathbf{t})$	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00472	56	■ 0.608	(t_4)	(t_4)	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00473	57	■ 0.991	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \operatorname{SL}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \operatorname{SL}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$	trices $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \operatorname{SL}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$ satisfies the constraint $ad - bc = +1$, which
gilmore-lie-groups_ F00474	57	■ 0.998	$U(1, 1)$	$U(1, 1)$	taking the intersection of $U(1, 1)$ with $\operatorname{SL}(2; \mathbb{C})$:
gilmore-lie-groups_ F00475	57	■ 1.000	$a^*a - b^*b = +1$	$a^*a - b^*b = +1$	where $a^*a - b^*b = +1$.
gilmore-lie-groups_ F00476	57	■ 1.000	$U(n)$	$U(n)$	The unitary group $U(n)$ consists of $n \times n$ complex matrices that obey the constraint
gilmore-lie-groups_ F00477	57	■ 1.000	$U^{\dagger}U = I_n$	$U^{\dagger}U = I_n$	$U^{\dagger}U = I_n$. For some purposes it is useful to represent this group as a group of real
gilmore-lie-groups_ F00478	57	■ 1.000	$2n \times 2n$	$2n \times 2n$	is a real $2n \times 2n$ matrix M . This matrix inherits the constraint that comes with the
gilmore-lie-groups_ F00479	57	■ 1.000	$M^tM = I_{2n}$	$M^tM = I_{2n}$	unitary group, $U^{\dagger}U = I_n$. This constraint now appears in the form $M^tM = I_{2n}$.
gilmore-lie-groups_ F00480	57	■ 1.000	$\dagger$	$\dagger$	We have been able to replace $\dagger$ by t since the matrices are real, and must replace
gilmore-lie-groups_ F00481	57	■ 1.000	t	t	We have been able to replace $\dagger$ by t since the matrices are real, and must replace
gilmore-lie-groups_ F00482	57	■ 1.000	I_{2n}	I_{2n}	I_n by I_{2n} since the matrices are $2n \times 2n$. In other words, the matrices M obey the
gilmore-lie-groups_ F00483	57	■ 0.712	$\operatorname{SO}(2n)$	$\operatorname{SO}(2n)$	an <i>orthogonal</i> representation of the <i>unitary</i> group. It is a subgroup of $\operatorname{SO}(2n)$. This
gilmore-lie-groups_ F00484	57	■ 1.000	$\operatorname{OU}(2n)$	$\operatorname{OU}(2n)$	matrix group is called $\operatorname{OU}(2n)$. Symbolically,
gilmore-lie-groups_ F00485	58	■ 0.999	$U(n; \mathbb{Q}) = \operatorname{Sp}(n)$	$U(n; \mathbb{Q}) = \operatorname{Sp}(n)$	For the metric-preserving quaternion group $U(n; \mathbb{Q}) = \operatorname{Sp}(n)$ whose matrices obey
gilmore-lie-groups_ F00486	58	■ 1.000	$M^{\dagger}M = I_{2n}$	$M^{\dagger}M = I_{2n}$	constraint in the form $M^{\dagger}M = I_{2n}$. In other words, the matrices M obey the condi-
gilmore-lie-groups_ F00487	58	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	tion that determines unitary groups (over $\mathbb{C}$). This group of $2n \times 2n$ matrices forms

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gilmore-lie-groups_ F00488	58	■ 0.945	$\operatorname{SU}(2n)$	$SU(2n)$	a <i>unitary</i> representation of the <i>symplectic</i> group. It is a subgroup of $SU(2n)$. This
gilmore-lie-groups_ F00489	58	■ 1.000	$U \operatorname{Sp}(2n)$	$U Sp(2n)$	matrix group is called $U Sp(2n)$. Symbolically,
gilmore-lie-groups_ F00490	58	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	because they lack the geometric structure contributed by the continuous fields $\mathbb{R}$,
gilmore-lie-groups_ F00491	58	■ 1.000	$GL(n; \mathbb{Z})$	$GL(n; \mathbb{Z})$	(i) $GL(n; \mathbb{Z})$: if $m \in GL(n; \mathbb{Z})$, $\det(m) = \pm 1$.
gilmore-lie-groups_ F00492	58	■ 1.000	$m \in GL(n; \mathbb{Z}), \operatorname{det}(m) = \pm 1$	$m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$	(i) $GL(n; \mathbb{Z})$: if $m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$.
gilmore-lie-groups_ F00493	58	■ 1.000	$SL(n; \mathbb{Z})$	$SL(n; \mathbb{Z})$	(ii) $SL(n; \mathbb{Z})$: if $m \in SL(n; \mathbb{Z})$, $\det(m) = +1$.
gilmore-lie-groups_ F00494	58	■ 1.000	$m \in SL(n; \mathbb{Z}), \operatorname{det}(m) = +1$	$m \in SL(n; \mathbb{Z}), \det(m) = +1$	(ii) $SL(n; \mathbb{Z})$: if $m \in SL(n; \mathbb{Z}), \det(m) = +1$.
gilmore-lie-groups_ F00495	58	■ 1.000	$PSL(n; \mathbb{Z}), n$	$PSL(n; \mathbb{Z}), n$	(iii) $PSL(n; \mathbb{Z}), n$ even: $PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$.
gilmore-lie-groups_ F00496	58	■ 1.000	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \left\{ I_n, -I_n \right\}$	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$	(iii) $PSL(n; \mathbb{Z}), n$ even: $PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$
gilmore-lie-groups_ F00497	58	■ 0.999	$n=2$	$n = 2$	For $n = 2$ these groups of matrices have the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with a, b, c, d all integers.
gilmore-lie-groups_ F00498	58	■ 0.999	$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	For $n = 2$ these groups of matrices have the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with a, b, c, d all integers.
gilmore-lie-groups_ F00499	58	■ 1.000	$\operatorname{det}(m) = n$	$\det(m) = n$	If $\det(m) = n$, with n an integer, then $\det(m^{-1}) = 1/n$. Since the determinant of
gilmore-lie-groups_ F00500	58	■ 1.000	$\operatorname{det}(m^{-1}) = 1/n$	$\det(m^{-1}) = 1/n$	If $\det(m) = n$, with n an integer, then $\det(m^{-1}) = 1/n$. Since the determinant of
gilmore-lie-groups_ F00501	58	■ 1.000	$\operatorname{det}(m) =$	$\det(m) =$	any matrix composed of integers must be an integer, the condition is that $\det(m) =$
gilmore-lie-groups_ F00502	58	■ 0.714	± 1	± 1	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_ F00503	58	■ 0.714	$GL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_ F00504	58	■ 0.714	$SL(2; \mathbb{Z}) \subset$	$SL(2; \mathbb{Z}) \subset$	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_ F00505	58	■ 1.000	$PSL(2; \mathbb{Z})$	$PSL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$. The modular group $PSL(2; \mathbb{Z})$ is obtained by identifying each pair of
gilmore-lie-groups_ F00506	58	■ 0.906	$SL(2; \mathbb{Z})$	$SL(2; \mathbb{Z})$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_ F00507	58	■ 0.906	$\left[\begin{array}{ll} -a & -b \\ -c & -d \end{array} \right] \simeq \left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_ F00508	59	■ 1.000	$F(n)$	$F(n)$	where $F(n)$ is the n th Fibonacci number, defined recursively by
gilmore-lie-groups_ F00509	59	■ 1.000	$n=1(F(0)=0, F(1)=1)$	$n = 1(F(0) = 0, F(1) = 1)$	and by comparison of initial conditions for $n = 1$ ($F(0) = 0, F(1) = 1$). Many
gilmore-lie-groups_ F00510	59	■ 0.998	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	in the groups $GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$, etc.
gilmore-lie-groups_ F00511	59	■ 0.998	$GL(n; \mathbb{R})$	$GL(n; \mathbb{R})$	with $GL(n; \mathbb{R})$.
gilmore-lie-groups_ F00512	59	■ 1.000	$M^t I_n M = I_n$	$M^t I_n M = I_n$	Imposing quadratic constraints, for example $M^t I_n M = I_n$, generates a sub-
gilmore-lie-groups_ F00513	59	■ 1.000	$O(n; \mathbb{Z})$	$O(n; \mathbb{Z})$	group, $O(n; \mathbb{Z})$, consists of $n \times n$ matrices in which all but one matrix element

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gilmore-lie-groups-F00514	59	■ 1.000	$\operatorname{det}(m)=+1$	$\det(m) = +1$	Finally, the multilinear condition $\det(m) = +1$ defines the unimodular subgroup
gilmore-lie-groups-F00515	60	■ 1.000	$A_{\{n\}}$	A_n	ple, the alternating group A_n consists of unimodular matrices in P_n :
gilmore-lie-groups-F00516	60	■ 0.932	$O(2 ; \mathbb{Z})$	$O(2; \mathbb{Z})$	Example The group $O(2; \mathbb{Z})$ consists of the $8 = 2^2 \times 2!$ matrices
gilmore-lie-groups-F00517	60	■ 0.932	$8=2^{\{2\}} \times 2!$	$8 = 2^2 \times 2!$	Example The group $O(2; \mathbb{Z})$ consists of the $8 = 2^2 \times 2!$ matrices
gilmore-lie-groups-F00518	60	■ 0.974	$O(3 ; \mathbb{Z})$	$O(3; \mathbb{Z})$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups-F00519	60	■ 0.974	$2^{\{3\}} \times 3!=48$	$2^3 \times 3! = 48$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups-F00520	60	■ 0.974	$6=3!$	$6 = 3!$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups-F00521	60	■ 1.000	$A_{\{3\}} \subset S_{\{3\}} \subset O(3 ; \mathbb{Z})$	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	Its alternating subgroup $A_3 \subset S_3 \subset O(3; \mathbb{Z})$ consists of the three matrices with
gilmore-lie-groups-F00522	60	■ 0.997	$G_{\{2\}}, F_{\{4\}}, E_{\{6\}}, E_{\{7\}}, E_{\{8\}}$	G_2, F_4, E_6, E_7, E_8	G_2, F_4, E_6, E_7, E_8 and their real forms (encountered in Chapters 10 and 11), and
gilmore-lie-groups-F00523	60	■ 1.000	$GL(n ; \mathbb{Z}), SL(n ; \mathbb{Z})$	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	integers, $GL(n; \mathbb{Z})$, $SL(n; \mathbb{Z})$, and $PSL(n; \mathbb{Z})$. Matrix groups over finite fields are
gilmore-lie-groups-F00524	60	■ 1.000	$PSL(n ; \mathbb{Z})$	$PSL(n; \mathbb{Z})$	integers, $GL(n; \mathbb{Z})$, $SL(n; \mathbb{Z})$, and $PSL(n; \mathbb{Z})$. Matrix groups over finite fields are
gilmore-lie-groups-F00525	61	■ 1.000	$\mathcal{IJ}=-\mathcal{IK}$	$\mathcal{IJ} = -\mathcal{K}$	where $\mathcal{IJ} = -\mathcal{K}$:
gilmore-lie-groups-F00526	61	■ 0.997	$\sigma_x, \sigma_y, \sigma_z$	$\sigma_x, \sigma_y, \sigma_z$	Here $\sigma_x, \sigma_y, \sigma_z$ are the Pauli spin matrices, and i is the usual square root of -1 . Show
gilmore-lie-groups-F00527	61	■ 0.924	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$	that any pair of the unit quaternions anticommute: i.e., $\mathcal{IJ} + \mathcal{JI} = 0$.
gilmore-lie-groups-F00528	61	■ 1.000	$\mathcal{I}, J, K$	$\mathcal{I}, J, K$	2. Show that the unit quaternions $\mathcal{I}, J, K$ generate a group of order 8 under multi-
gilmore-lie-groups-F00529	61	■ 0.464	$SU(1 ; \mathbb{Q}) \sim SU(2 ; \mathbb{C})$	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	3. Show that $SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$.
gilmore-lie-groups-F00530	61	■ 1.000	$p+q=n$	$p + q = n$	in Example (9), for suitable positive integers p and q , with $p + q = n$.
gilmore-lie-groups-F00531	61	■ 0.996	$SL_{\{i\}}(n ; \mathbb{C})$	$SL_i(n; \mathbb{C})$	6. Show that it is possible to define subgroups $SL_i(n; \mathbb{C})$ of $GL(n; \mathbb{C})$ by the con-
gilmore-lie-groups-F00532	61	■ 0.996	$GL(n ; \mathbb{C})$	$GL(n; \mathbb{C})$	6. Show that it is possible to define subgroups $SL_i(n; \mathbb{C})$ of $GL(n; \mathbb{C})$ by the con-
gilmore-lie-groups-F00533	61	■ 1.000	ϕ, λ, r	ϕ, λ, r	where ϕ, λ, r are real and $r \neq 0$. Show that the dimensions of these three subgroups
gilmore-lie-groups-F00534	61	■ 1.000	$r \neq 0$	$r \neq 0$	where ϕ, λ, r are real and $r \neq 0$. Show that the dimensions of these three subgroups
gilmore-lie-groups-F00535	61	■ 1.000	$2n^{\{2\}}-1$	$2n^2 - 1$	are $2n^2 - 1$ and that $SL_3(n; \mathbb{C})$ is disconnected. It consists of two topologically iden-
gilmore-lie-groups-F00536	61	■ 1.000	$SL_{\{3\}}(n ; \mathbb{C})$	$SL_3(n; \mathbb{C})$	are $2n^2 - 1$ and that $SL_3(n; \mathbb{C})$ is disconnected. It consists of two topologically iden-
gilmore-lie-groups-F00537	61	■ 1.000	$SL(n ; \mathbb{C})=SL_{\{1\}}(n ; \mathbb{C}) \cap SL_{\{2\}}(n ; \mathbb{C})$	$SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$	Show that $SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$. Do these results extend under the field
gilmore-lie-groups-F00538	61	■ 1.000	$\mathbb{C} \rightarrow \mathbb{R}$	$\mathbb{C} \rightarrow \mathbb{R}$	restriction $\mathbb{C} \rightarrow \mathbb{R}$? and the field extension $\mathbb{C} \rightarrow \mathbb{Q}$?
gilmore-lie-groups-F00539	61	■ 1.000	$\mathbb{C} \rightarrow \mathbb{Q}$	$\mathbb{C} \rightarrow \mathbb{Q}$	restriction $\mathbb{C} \rightarrow \mathbb{R}$? and the field extension $\mathbb{C} \rightarrow \mathbb{Q}$?
gilmore-lie-groups-F00540	61	■ 0.936	$\left[\begin{array}{cc} -1 & a \\ 0 & 1 \end{array}\right], a \in \mathbb{R}$	$\left[\begin{array}{cc} -1 & a \\ 0 & 1 \end{array}\right], a \in \mathbb{R}$	7. A subgroup of $UT(1, 1)$ includes matrices of the form $\begin{bmatrix} -1 & a \\ 0 & 1 \end{bmatrix}, a \in \mathbb{R}$. Show that the

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gilmore-lie-groups_F08541	61	■ 1.000	R^1	R^1	underlying group manifold consists of two copies of the real line R^1 . If matrices of
gilmore-lie-groups_F08542	61	■ 1.000	$\left[\begin{array}{cc} 1 & a \\ 0 & -1 \end{array}\right]$	$\begin{bmatrix} 1 & a \\ 0 & -1 \end{bmatrix}$	the form $\begin{bmatrix} 1 & a \\ 0 & -1 \end{bmatrix}$ are also included, then the parameterizing manifold consists of how
gilmore-lie-groups_F08547	84	■ 1.000	$n(n-1) / 2$	$n(n-1)/2$	Show that a useful set of basis vectors (matrices) consists of the $n(n-1)/2$ matrices
gilmore-lie-groups_F08549	62	■ 1.000	$\mathbf{F} = d \mathbf{p} / d t$	$\mathbf{F} = d\mathbf{p}/dt$	9. Newton's equations of motion are $\mathbf{F} = d\mathbf{p}/dt$. In the Lorentz gauge Maxwell's equa-
gilmore-lie-groups_F08551	62	■ 0.814	$S U(2)$	$SU(2)$	10. Show that the group of 2×2 matrices $SU(2)$ is parameterized by two complex num-
gilmore-lie-groups_F08552	62	■ 1.000	$c_1 = a_1 + i b_1$	$c_1 = a_1 + ib_1$	bers $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$, so that
gilmore-lie-groups_F08553	62	■ 1.000	$c_2 = a_2 + i b_2$	$c_2 = a_2 + ib_2$	bers $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$, so that
gilmore-lie-groups_F08554	62	■ 1.000	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	subject to the condition $a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$. Convince yourself (a) that topolog-
gilmore-lie-groups_F08555	62	■ 0.991	$S^3 \subset R^4$	$S^3 \subset R^4$	$S^3 \subset R^4$; and (b) algebraically it is equivalent to $SU(1; \mathbb{Q})$ (cf. (3.16)).
gilmore-lie-groups_F08556	62	■ 1.000	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	subject to the condition $a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$. Identify the parameterizing manifold.
gilmore-lie-groups_F08557	62	■ 1.000	$\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array}\right]$	$\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$	in the form $\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row
gilmore-lie-groups_F08558	62	■ 1.000	$m_{11}^2 + x^2 = 1$	$m_{11}^2 + x^2 = 1$	in the form $\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row
gilmore-lie-groups_F08559	62	■ 1.000	$m_{11} = \pm \sqrt{1-x^2}$	$m_{11} = \pm\sqrt{1-x^2}$	in the form $\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row
gilmore-lie-groups_F08560	63	■ 1.000	$m_{21} m_{11} + m_{22} x = 0$	$m_{21}m_{11} + m_{22}x = 0$	is orthogonal to the first, so that $m_{21}m_{11} + m_{22}x = 0$. As a result, we find
gilmore-lie-groups_F08561	63	■ 0.994	$m_{ij} i \geq j$	$m_{ij}i \geq j$	$m_{ij} i \geq j$ in terms of the three parameters (x, y, z) .
gilmore-lie-groups_F08562	63	■ 0.495	Z_2	Z_2	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .
gilmore-lie-groups_F08563	63	■ 0.495	Z_1	Z_1	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .
gilmore-lie-groups_F08564	63	■ 0.495	(x, y)	(x, y)	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .
gilmore-lie-groups_F08565	63	■ 0.854	$M \in GL(n; \mathbb{Z})$	$M \in GL(n; \mathbb{Z})$	15. If $M \in GL(n; \mathbb{Z})$, show that $\det(M)$ must be ± 1 .
gilmore-lie-groups_F08566	63	■ 0.854	$\operatorname{det}(M)$	$\det(M)$	15. If $M \in GL(n; \mathbb{Z})$, show that $\det(M)$ must be ± 1 .
gilmore-lie-groups_F08567	63	■ 1.000	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	16. Show that the orders of $O(n; \mathbb{Z}) \supset S_n \supset A_n$ are $2^n \times n!$, $n!$, $\frac{1}{2}n!$.
gilmore-lie-groups_F08568	63	■ 1.000	$2^n \times n!, n!, \frac{1}{2} n!$	$2^n \times n!, n!, \frac{1}{2}n!$	16. Show that the orders of $O(n; \mathbb{Z}) \supset S_n \supset A_n$ are $2^n \times n!$, $n!$, $\frac{1}{2}n!$.
gilmore-lie-groups_F08569	63	■ 1.000	$\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$	$\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$	17. Estimate the Fibonacci number $F(n)$ from the eigenvalues $\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$ of the
gilmore-lie-groups_F08570	63	■ 0.999	$F(0)=0, F(1)=1$	$F(0) = 0, F(1) = 1$	(other than $F(0) = 0, F(1) = 1$) are introduced?
gilmore-lie-groups_F08571	63	■ 1.000	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array}\right]$	$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$	$GL(2; \mathbb{Z})$ (for example, $\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$) and other initial conditions.
gilmore-lie-groups_F08572	63	■ 1.000	$ n l m\rangle$	$ nlm\rangle$	19. The energy levels $ nlm\rangle$ of the nonrelativistic hydrogen atom exhibit an n^2 -fold

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gilmore-lie-groups_F00573	63	■ 1.000	$n^{\{2\}}$	n^2	19. The energy levels $ nlm\rangle$ of the nonrelativistic hydrogen atom exhibit an n^2 -fold
gilmore-lie-groups_F00574	63	■ 1.000	$E(n\ l\ m)=-E_{\{0\}} / n^{\{2\}}\backslashleft(E_{\{0\}}=13.6\ \mathrm{eV}\backslashright)$	$E(nlm) = -E_0/n^2$ ($E_0 = 13.6\mathrm{eV}$)	quantum number n have the same energy $E(nlm) = -E_0/n^2$ ($E_0 = 13.6\ \mathrm{eV}$). If the
gilmore-lie-groups_F00575	63	■ 1.000	$S\ 0(4)\ \downarrow S\ 0(3)$	$SO(4) \downarrow SO(3)$	symmetry reduction $SO(4) \downarrow SO(3)$. The representations of $SO(4)$ that enter into
gilmore-lie-groups_F00576	63	■ 0.993	$n(n=1,2,3,\ \ldots)$	$n(n = 1, 2, 3, \dots)$	quantum number $n\ (n = 1, 2, 3, \dots)$. The $SO(4)$ representation with quantum num-
gilmore-lie-groups_F00577	63	■ 0.980	$l,\ l=0,1,2\ \ldots,\ n-1$	$l, l = 0, 1, 2 \dots, n - 1$	number $l\ (=0, 1, 2 \dots, n - 1)$, with $\sum_{l=0}^{l=n-1} (2l + 1) = n^2$. The $SO(3)$ multiplet
gilmore-lie-groups_F00578	63	■ 0.980	$\sum_{l=0}^{l=n-1} (2\ l+1)=n^{\{2\}}$	$\sum_{l=0}^{l=n-1} (2l + 1) = n^2$	number $l, l = 0, 1, 2 \dots, n - 1$, with $\sum_{l=0}^{l=n-1} (2l + 1) = n^2$. The $SO(3)$ multiplet
gilmore-lie-groups_F00579	63	■ 1.000	l	l	with quantum number l is $2l + 1$ -fold degenerate. An empirical hamiltonian with
gilmore-lie-groups_F00580	63	■ 1.000	$2\ l+1$	$2l + 1$	with quantum number l is $2l + 1$ -fold degenerate. An empirical hamiltonian with
gilmore-lie-groups_F00581	63	■ 1.000	$+Z\ e$	$+Ze$	introduced into a Coulomb potential established by a central charge $+Ze$ can be
gilmore-lie-groups_F00582	64	■ 1.000	$\delta=0.28$	$\delta = 0.28$	trum with $\delta=0.28$ provides the filling ordering that accounts for Mendelev's
gilmore-lie-groups_F00583	64	■ 0.608	$(n,\ l)\ \rightarrow 1\ s$	$(n, l) \rightarrow 1s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00584	64	■ 0.608	$2\ s$	$2s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00585	64	■ 0.608	$2\ p$	$2p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00586	64	■ 0.608	$3\ s$	$3s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00587	64	■ 0.608	$3\ p$	$3p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00588	64	■ 0.608	$4\ s$	$4s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00589	64	■ 0.608	$3\ d$	$3d$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00590	64	■ 0.608	$4\ p$	$4p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00591	64	■ 0.957	$5\ s,\ 4\ d,\ 5\ p;\ 6\ s,\ 4\ f,\ 5\ d,\ 6\ p;\ 7\ s,\ 5\ f,\ 6\ d,\ 7\ p;\ 8\ s,\ 6\ f,\ 7\ d,\ 8\ p;\ \ldots$	$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$	$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$.
gilmore-lie-groups_F00592	64	■ 1.000	$G_{\{2\}}$	G_2	21. G_1 and G_2 are two metrics on a real $2n$ -dimensional linear vector space that are
gilmore-lie-groups_F00593	64	■ 1.000	$2\ n$	$2n$	21. G_1 and G_2 are two metrics on a real $2n$ -dimensional linear vector space that are
gilmore-lie-groups_F00594	64	■ 1.000	$M^{\{t\}}\ G_{\{i\}}\ M=G_{\{i\}}$	$M^t G_i M = G_i$	Show that the $2n \times 2n$ matrices M that satisfy the bilinear constraints $M^t G_i M = G_i$
gilmore-lie-groups_F00595	64	■ 1.000	$y^{\{j\}}=x^{\{i\}}\ M_{\{i\}}^{\{j\}}$	$y^j = x^i M_i^{\ j}$	by a linear transformation: $y^j = x^i M_i^{\ j}$. Show that the derivatives are related by the
gilmore-lie-groups_F00596	64	■ 0.990	$f(q,\ p)$	$f(q, p)$	23. The Poisson brackets between two functions $f(q, p)$ and $g(q, p)$ on a classical phase
gilmore-lie-groups_F00597	64	■ 0.990	$g(q,\ p)$	$g(q, p)$	23. The Poisson brackets between two functions $f(q, p)$ and $g(q, p)$ on a classical phase
gilmore-lie-groups_F00598	65	■ 0.666	$(Q,\ P)$	(Q, P)	b. Introduce a new coordinate system (Q, P) , related to the original by a linear
gilmore-lie-groups_F00599	65	■ 1.000	$A^{\{t\}}\ C$	$A^t C$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.

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gilmore-lie-groups_F00600	65	■ 1.000	$B^{\{t\}} D$	$B^t D$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00601	65	■ 1.000	$A^{\{t\}} D - B^{\{t\}} C = I_n$	$A^t D - B^t C = I_n$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00602	65	■ 0.999	$\left[q_{\{j\}}, q_{\{k\}}\right] = \left[p_{\{j\}}, p_{\{k\}}\right] = 0$	$[q_j, q_k] = [p_j, p_k] = 0$	mechanical commutator bracket: $q_j, q_k = p_j, p_k = 0$ and $[q_j, p_k] = i\hbar\delta_{jk}$.
gilmore-lie-groups_F00603	65	■ 0.999	$\left[q_{\{j\}}, p_{\{k\}}\right] = i\hbar\delta_{jk}$	$[q_j, p_k] = i\hbar\delta_{jk}$	mechanical commutator bracket: $[q_j, q_k] = [p_j, p_k] = 0$ and $[q_j, p_k] = i\hbar\delta_{jk}$.
gilmore-lie-groups_F00604	65	■ 1.000	$e^{+i k x}$	e^{+ikx}	with a similar expression for the wavefunction on the right. The exponential e^{+ikx}
gilmore-lie-groups_F00605	65	■ 0.644	m	m	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00606	65	■ 0.644	$(+)$	$(+)$	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00607	65	■ 0.644	$\hbar k$	$\hbar k$	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00608	65	■ 1.000	$E = (\hbar k)^2 / 2 m$	$E = (\hbar k)^2 / 2m$	energy $E = (\hbar k)^2 / 2m$. The complex number A_L is the probability amplitude for
gilmore-lie-groups_F00609	65	■ 1.000	$A_{\{L\}}$	A_L	energy $E = (\hbar k)^2 / 2m$. The complex number A_L is the probability amplitude for
gilmore-lie-groups_F00610	65	■ 1.000	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	momentum in the left-hand region is $\langle \hat{p} \rangle = (A_L ^2 - B_L ^2) \hbar k$, where the operator
gilmore-lie-groups_F00611	65	■ 1.000	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$.
gilmore-lie-groups_F00615	66	■ 1.000	$A_{\{L\}}, A_{\{R\}}$	A_L, A_R	b. Since the Schrödinger equation is second order the four amplitudes $A_L, A_R,$
gilmore-lie-groups_F00616	66	■ 1.000	$B_{\{L\}}, B_{\{R\}}$	B_L, B_R	B_L, B_R are not independent. Only two are independent. Two linear relations exist
gilmore-lie-groups_F00617	66	■ 1.000	E	E	of energy E . Show that $T(E) \in U(1, 1)$.
gilmore-lie-groups_F00618	66	■ 1.000	$T(E) \in U(1, 1)$	$T(E) \in U(1, 1)$	of energy E . Show that $T(E) \in U(1, 1)$.
gilmore-lie-groups_F00619	66	■ 1.000	$T \in SU(1, 1)$	$T \in SU(1, 1)$	c. Show that $T \in SU(1, 1)$ by appropriate choice of phase.
gilmore-lie-groups_F00620	66	■ 1.000	$S \in U(2)$	$S \in U(2)$	a. Invoke conservation of momentum arguments to conclude $S \in U(2)$.
gilmore-lie-groups_F00621	66	■ 1.000	$S(E)$	$S(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_F00622	66	■ 1.000	$T(E)$	$T(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_F00623	66	■ 1.000	$t_{11}(E)$	$t_{11}(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_F00624	66	■ 1.000	$r_{\{j\}} / \left[\left(E - E_{\{j\}} \right) + i \left(\Gamma_{\{j\}} / 2 \right) \right]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$ describe resonances at energy E_j with characteristic de-
gilmore-lie-groups_F00625	66	■ 1.000	$E_{\{j\}}$	E_j	$r_j / [(E - E_j) + i(\Gamma_j/2)]$ describe resonances at energy E_j with characteristic de-
gilmore-lie-groups_F00626	66	■ 1.000	$\Gamma_{\{j\}} / \hbar$	$\Gamma_j / \hbar$	cay time $\Gamma_j / \hbar$.
gilmore-lie-groups_F00627	66	■ 1.000	$V_{\{1\}}$	V_1	26. Two interaction regions V_1 and V_2 on the line are characterized by transfer matrices
gilmore-lie-groups_F00628	66	■ 1.000	$V_{\{2\}}$	V_2	26. Two interaction regions V_1 and V_2 on the line are characterized by transfer matrices
gilmore-lie-groups_F00629	66	■ 1.000	$T_{\{1\}}$	T_1	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region

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gilmore-lie-groups_F00630	66	■ 1.000	$T_{\{2\}}$	T_2	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00631	66	■ 1.000	$S_{\{1\}}$	S_1	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00632	67	■ 1.000	$S_{\{\text{Tot}\}}$	S_{Tot}	c. Show that S_{Tot} is unitary.
gilmore-lie-groups_F00633	67	■ 1.000	$1/\left(1-s_{12}s_{43}\right)$	$1/(1-s_{12}s_{43})$	by expanding the fraction $1/(1-s_{12}s_{43})$ as a geometric sum and interpreting each
gilmore-lie-groups_F00634	67	■ 1.000	$V_{\{1\}}^{\prime}$	V_1'	27. If the potentials V_1 and V_2 are modified to V_1' and V_2' their transfer matrices and
gilmore-lie-groups_F00635	67	■ 1.000	$V_{\{2\}}^{\prime}$	V_2'	27. If the potentials V_1 and V_2 are modified to V_1' and V_2' their transfer matrices and
gilmore-lie-groups_F00636	67	■ 1.000	$T_{\{i\}}(E) \rightarrow T_{\{i\}}^{\prime}(E)$	$T_i(E) \rightarrow T_i'(E)$	their scattering matrices will also be modified $T_i(E) \rightarrow T_i'(E)$ and $S_i(E) \rightarrow S_i'(E)$,
gilmore-lie-groups_F00637	67	■ 1.000	$S_{\{i\}}(E) \rightarrow S_{\{i\}}^{\prime}(E)$	$S_i(E) \rightarrow S_i'(E)$	their scattering matrices will also be modified $T_i(E) \rightarrow T_i'(E)$ and $S_i(E) \rightarrow S_i'(E)$,
gilmore-lie-groups_F00638	67	■ 1.000	$i=1,2$	$i=1,2$	$i=1,2$. It is possible that for some energy E , $S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$. Find the set of
gilmore-lie-groups_F00639	67	■ 1.000	$E, S_{\{\text{Tot}\}}^{\prime}(E)=S_{\{\text{Tot}\}}(E)$	$E, S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$	$i=1,2$. It is possible that for some energy E , $S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$. Find the set of
gilmore-lie-groups_F00640	67	■ 1.000	$S_{\{1\}}^{\prime}(E)$	$S_1'(E)$	all modified scattering matrices $S_1'(E)$ and $S_2'(E)$ with the property that the modi-
gilmore-lie-groups_F00641	67	■ 1.000	$S_{\{2\}}^{\prime}(E)$	$S_2'(E)$	all modified scattering matrices $S_1'(E)$ and $S_2'(E)$ with the property that the modi-
gilmore-lie-groups_F00642	67	■ 0.998	$S_{\{\text{Tot}\}}(E)$	$S_{\text{Tot}}(E)$	fied pair maps into the original total S -matrix $S_{\text{Tot}}(E)$. In fancy terms, find the fiber
gilmore-lie-groups_F00643	67	■ 0.999	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	in <u>$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$</u> . (Hint: if this seems daunting, note that to satisfy
gilmore-lie-groups_F00644	67	■ 0.997	$T_{\{1\}}(E) T_{\{2\}}(E)=T_{\{\text{Tot}\}}(E)=T_{\{1\}}^{\prime}(E) T_{\{2\}}^{\prime}(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$ we can take $T_1'(E) = T_1(E)R$ and $T_2'(E) =$
gilmore-lie-groups_F00645	67	■ 0.997	$T_{\{1\}}^{\prime}(E)=T_{\{1\}}(E) R$	$T_1'(E) = T_1(E)R$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$ we can take $T_1'(E) = T_1(E)R$ and $T_2'(E) =$
gilmore-lie-groups_F00646	67	■ 0.997	$T_{\{2\}}^{\prime}(E)=$	$T_2'(E) =$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$ we can take $T_1'(E) = T_1(E)R$ and $T_2'(E) =$
gilmore-lie-groups_F00647	67	■ 1.000	$R^{-1} T_{\{2\}}(E)$	$R^{-1}T_2(E)$	$R^{-1}T_2(E)$ for any $R \in U(1,1)$. The fiber in $U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$
gilmore-lie-groups_F00648	67	■ 1.000	$R \in U(1,1)$	$R \in U(1,1)$	$R^{-1}T_2(E)$ for any $R \in U(1,1)$. The fiber in $U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$
gilmore-lie-groups_F00649	67	■ 1.000	$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$	$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$	$R^{-1}T_2(E)$ for any $R \in U(1,1)$. The fiber in <u>$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$</u>
gilmore-lie-groups_F00650	67	■ 0.932	$T_{\{\text{Tot}\}}(E)$	$T_{\text{Tot}}(E)$	over $T_{\text{Tot}}(E)$ consists of the matrices $(T_1(E)R, R^{-1}T_2(E))$. Now map this into the
gilmore-lie-groups_F00651	67	■ 0.932	$\left(T_{\{1\}}(E) R, R^{-1} T_{\{2\}}(E)\right)$	$(T_1(E)R, R^{-1}T_2(E))$	over $T_{\text{Tot}}(E)$ consists of the matrices $(T_1(E)R, R^{-1}T_2(E))$. Now map this into the
gilmore-lie-groups_F00652	67	■ 0.850	$\left(S_{\{1\}}^{\prime}(E), S_{\{2\}}^{\prime}(E)\right)$	$(S_1'(E), S_2'(E))$	fiber <u>$(S_1'(E), S_2'(E))$</u> over $S_{\text{Tot}}(E)$.
gilmore-lie-groups_F00653	68	■ 0.987	$n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{k\}}$	$n_1, n_2, \dots, n_k$	with <u>$n_1, n_2, \dots, n_k$</u> leads can be connected together by soldering some of the leads
gilmore-lie-groups_F00654	68	■ 0.718	$k S$	kS	direct sum of k S matrices of sizes $n_j \times n_j$ ($j=1,2,\dots,k$). Through appropriate
gilmore-lie-groups_F00655	68	■ 0.718	$n_{\{j\}} \times n_{\{j\}}(j=1,2, \ldots, k)$	$n_j \times n_j(j=1,2,\dots,k)$	direct sum of k S matrices of sizes <u>$n_j \times n_j(j=1,2,\dots,k)$</u> . Through appropriate
gilmore-lie-groups_F00656	68	■ 1.000	Γ	Γ	The matrix Γ that relates internal outputs to internal inputs describes the topology, or

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gilmore-lie-groups_F00657	68	■ 0.999	$\left[o_e\right]=S_{\text {Network }}\left[i_e\right]$	$[o_e] = S_{\text{Network}} [i_e]$	a. Show that the S matrix that describes the network, defined by $[o_e] = S_{\text{Network}} [i_e]$,
gilmore-lie-groups_F00658	68	■ 1.000	$S_{\text {Newtork }}$	S_{Newtork}	b. Show that S_{Newtork} is unitary: $S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$.
gilmore-lie-groups_F00659	68	■ 1.000	$S_{\text {Newtork }}^{\dagger}=S_{\text {Newtork }}, S_{\text {Newtork }} \subset U(d)$	$S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$	b. Show that S_{Newtork} is unitary: $S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$.
gilmore-lie-groups_F00660	68	■ 1.000	$U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k U\left(n_j\right)$	$U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k U\left(n_j\right)$	$U(\sum_{j=1}^k n_j) \supset \Pi_{j=1}^k U(n_j)$ and a projection to the total network scattering matrix
gilmore-lie-groups_F00661	68	■ 1.000	$U(d)$	$U(d)$	in $U(d)$, where d is the number of the network's external leads. The connectivity is
gilmore-lie-groups_F00662	68	■ 1.000	d	d	in $U(d)$, where d is the number of the network's external leads. The connectivity is
gilmore-lie-groups_F00663	68	■ 0.917	$\Pi_{j=1}^k U\left(n_j\right)$	$\Pi_{j=1}^k U\left(n_j\right)$	determined by the permutation matrix Γ . Determine the fiber in $\Pi_{j=1}^k U(n_j)$ over
gilmore-lie-groups_F00664	68	■ 1.000	U	U	30. All the matrices in this problem are square $n \times n$, with: H hermitian; U unitary; A
gilmore-lie-groups_F00665	70	■ 1.000	b	b	into the neighborhood of group element b by multiplying a , and every element in
gilmore-lie-groups_F00666	70	■ 1.000	$b a^{-1}$	ba^{-1}	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This
gilmore-lie-groups_F00667	70	■ 1.000	$a^{-1} b$	$a^{-1}b$	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This
gilmore-lie-groups_F00668	71	■ 0.953	(a, b, c)	(a, b, c)	done by allowing the parameters (a, b, c) to become infinitesimals and expanding
gilmore-lie-groups_F00669	71	■ 1.000	$\phi(x, y) \rightarrow \phi(x, \theta+\delta y)$	$\phi(x, y) \rightarrow \phi(x, 0 + \delta y)$	arguments, say $\phi(x, y) \rightarrow \phi(x, 0 + \delta y)$. This generates a left-invariant vector field.
gilmore-lie-groups_F00670	72	■ 1.000	ϵ	ϵ	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_F00671	72	■ 1.000	$I+\epsilon X$	$I + \epsilon X$	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_F00672	72	■ 0.744	$X_{\{a\}}$	X_a	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00673	72	■ 0.744	$X_{\{b\}}$	X_b	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00674	72	■ 0.744	$X_{\{c\}}$	X_c	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00675	72	■ 1.000	$X^{\{n\}}$	X^n	With brute force, each of the matrices X^n is computed explicitly, a pattern is
gilmore-lie-groups_F00676	72	■ 1.000	$X^{\{0\}}=I_{\{2\}}, X^{\{1\}}=X$	$X^0 = I_2, X^1 = X$	recognized, and the sum is carried out. The first few powers are $X^0 = I_2, X^1 = X$
gilmore-lie-groups_F00677	72	■ 1.000	$X^{\{2\}}=\theta^2 I_{\{2\}}$	$X^2 = \theta^2 I_2$	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00678	72	■ 1.000	$X^{\{2\}}$	X^2	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00679	72	■ 1.000	$X^{\{3\}}=X^{\{2\}} X^{\{1\}}$	$X^3 = X^2 X^1$	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00680	72	■ 0.991	$X\left(=\theta^2 X\right), X^{\{4\}}$	$X (= \theta^2 X), X^4$	must be proportional to $X (= \theta^2 X)$, X^4 is proportional to the identity, and so on.
gilmore-lie-groups_F00681	72	■ 0.995	$X^{\{0\}}=I_{\{2\}}$	$X^0 = I_2$	equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00682	72	■ 0.995	$X^{\{1\}}=X$	$X^1 = X$	equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.

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gilmore-lie-groups_ F06683	72	■ 1.000	$X^{\{3\}}$	X^3	Similarly, X^3 can be replaced by a linear combination of X^2 and X , and then X^2
gilmore-lie-groups_ F06684	72	■ 0.759	$I_{\{2\}}$	I_2	replaced by I_2 and X . By induction, any function of the 2×2 matrix X can be
gilmore-lie-groups_ F06685	73	■ 1.000	$f_{\{0\}}, f_{\{1\}}$	f_0, f_1	Furthermore, the functions f_0, f_1 are not arbitrary functions of the three parameters
gilmore-lie-groups_ F06686	73	■ 1.000	$\theta^2 = a^2 + bc$	$\theta^2 = a^2 + bc$	X is $\theta^2 = a^2 + bc$. As a result, we know from general and simple considerations
gilmore-lie-groups_ F06687	73	■ 1.000	$f_{\{0\}}(\theta) = 1 + \theta^2/2! + \theta^4/4! + \theta^6/6! + \dots = \cosh \theta$	$f_0(\theta^2) = 1 + \theta^2/2! + \theta^4/4! + \theta^6/6! + \dots = \cosh \theta$	The two functions are $f_0(\theta^2) = 1 + \theta^2/2! + \theta^4/4! + \theta^6/6! + \dots = \cosh \theta$ and
gilmore-lie-groups_ F06688	73	■ 1.000	$f_{\{1\}}(\theta) = 1 + \theta^2/3! + \theta^4/5! + \theta^6/7! + \dots = \sinh(\theta)/\theta$	$f_1(\theta^2) = 1 + \theta^2/3! + \theta^4/5! + \theta^6/7! + \dots = \sinh(\theta)/\theta$	$f_1(\theta^2) = 1 + \theta^2/3! + \theta^4/5! + \theta^6/7! + \dots = \sinh(\theta)/\theta$. These arguments are ap-
gilmore-lie-groups_ F06689	73	■ 1.000	Y	Y	If X and Y are elements in the Lie algebra, then $g_1 = I + \epsilon X$ is an element in
gilmore-lie-groups_ F06690	73	■ 1.000	$g_{\{1\}} = I + \epsilon X$	$g_1 = I + \epsilon X$	If X and Y are elements in the Lie algebra, then $g_1 = I + \epsilon X$ is an element in
gilmore-lie-groups_ F06691	73	■ 1.000	$I + \epsilon \alpha X$	$I + \epsilon \alpha X$	the Lie group near the identity for ϵ sufficiently small. In fact, so also is $I + \epsilon \alpha X$
gilmore-lie-groups_ F06692	74	■ 1.000	α	α	for any real number α . We can form the product
gilmore-lie-groups_ F06693	74	■ 1.000	ϵ, δ	ϵ, δ	If X and Y are in the Lie algebra, then for any ϵ, δ sufficiently small, $g_1(\epsilon) =$
gilmore-lie-groups_ F06694	74	■ 1.000	$g_{\{1\}}(\epsilon) =$	$g_1(\epsilon) =$	If X and Y are in the Lie algebra, then for any ϵ, δ sufficiently small, $g_1(\epsilon) =$
gilmore-lie-groups_ F06695	74	■ 1.000	$\operatorname{EXP}(\epsilon X)$	$\operatorname{EXP}(\epsilon X)$	$\operatorname{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups_ F06696	74	■ 1.000	$g_{\{1\}}(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$	$g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$	$\operatorname{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups_ F06697	74	■ 1.000	$g_{\{2\}}(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$. Expanding the commutator to lowest order nonvanishing
gilmore-lie-groups_ F06698	74	■ 0.996	$g_{\{1\}}(\epsilon) = \operatorname{EXP}(\epsilon X)$	$g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$	Therefore, the commutator of two group elements, $g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$ and $g_2(\delta) =$
gilmore-lie-groups_ F06699	74	■ 0.996	$g_{\{2\}}(\delta) =$	$g_2(\delta) =$	Therefore, the commutator of two group elements, $g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$ and $g_2(\delta) =$
gilmore-lie-groups_ F06700	74	■ 1.000	$\operatorname{EXP}(\delta Y)$	$\operatorname{EXP}(\delta Y)$	$\operatorname{EXP}(\delta Y)$, which is in the group G , requires the commutator of the operators X and
gilmore-lie-groups_ F06701	74	■ 1.000	$Y, [X, Y] = (XY - YX)$	$Y, [X, Y] = (XY - YX)$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups_ F06702	74	■ 1.000	$\mathfrak{g}$	$\mathfrak{g}$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups_ F06703	74	■ 1.000	$g_{\{1\}} H g_{\{1\}}^{-1} \subset H$	$g_1 H g_1^{-1} \subset H$	If H is an invariant subgroup of G , then $g_1 H g_1^{-1} \subset H$. This means that if X is in
gilmore-lie-groups_ F06704	74	■ 1.000	X, Y, Z	X, Y, Z	If X, Y, Z are in the Lie algebra, then the Jacobi identity is satisfied
gilmore-lie-groups_ F06705	74	■ 0.996	$([X, Y] = XY - YX)$	$([X, Y] = XY - YX)$	$([X, Y] = XY - YX)$ and showing that the 12 terms so obtained cancel pairwise.
gilmore-lie-groups_ F06706	74	■ 1.000	XY	XY	for which composition (e.g., XY is well defined, as are all other pairwise products)
gilmore-lie-groups_ F06707	75	■ 1.000	$X \in \mathfrak{g}$	$X \in \mathfrak{g}$	(i) It is a linear vector space under vector addition and scalar multiplication. If $X \in \mathfrak{g}$ and
gilmore-lie-groups_ F06708	75	■ 1.000	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$ then every linear combination of X and Y is in $\mathfrak{g}$:
gilmore-lie-groups_ F06709	75	■ 1.000	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	(iii) The Jacobi identity is satisfied. If $X \in \mathfrak{g}, Y \in \mathfrak{g}$, and $Z \in \mathfrak{g}$, then

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gilmore-lie-groups-F00710	75	■ 1.000	$Z \in \mathfrak{g}$	$Z \in \mathfrak{g}$	(iii) The Jacobi identity is satisfied. If $X \in \mathfrak{g}$, $Y \in \mathfrak{g}$, and $Z \in \mathfrak{g}$, then
gilmore-lie-groups-F00711	75	■ 0.996	$X_{\{1\}}, X_{\{2\}}, \dots, X_{\{n\}}$	$X_1, X_2, \dots, X_n$	$X_1, X_2, \dots, X_n$, then we can ask several additional questions such as: Is there a
gilmore-lie-groups-F00712	75	■ 1.000	$\left(X_{\{i\}}, X_{\{j\}}\right)$	(X_i, X_j)	(X_i, X_j) ? We return to these questions shortly.
gilmore-lie-groups-F00713	76	■ 1.000	$C_{\{i\} \{j\}}^{\{k\}}$	C_{ij}^k	The coefficients C_{ij}^k in this expansion are called structure constants . The structure
gilmore-lie-groups-F00714	76	■ 0.865	$X_{\{a\}}, X_{\{b\}}, X_{\{c\}}$	X_a, X_b, X_c	X_a, X_b, X_c (4.4) in the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ for the Lie group $SL(2; \mathbb{R})$ are, from
gilmore-lie-groups-F00715	76	■ 0.865	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	X_a, X_b, X_c (4.4) in the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ for the Lie group $SL(2; \mathbb{R})$ are, from
gilmore-lie-groups-F00716	76	■ 1.000	Z	Z	change of basis affects the commutator of an arbitrary element Z in the algebra
gilmore-lie-groups-F00717	76	■ 1.000	$R(Z)$	$R(Z)$	The association of a matrix $R(Z)$ with each element of a Lie algebra is called
gilmore-lie-groups-F00718	77	■ 1.000	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	Example The regular representation of the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ is easily con-
gilmore-lie-groups-F00719	77	■ 0.844	a, b	a, b	The rows and columns of this 3×3 matrix are labeled by the indices a, b and c ,
gilmore-lie-groups-F00720	77	■ 0.844	c	c	The rows and columns of this 3×3 matrix are labeled by the indices a, b and c ,
gilmore-lie-groups-F00721	78	■ 0.899	$X_{\{a\}}, X_{\{b\}}$	X_a, X_b	the three generators X_a, X_b , and X_c of $\mathfrak{sl}(2; \mathbb{R})$ to either the nonsemisimple or the
gilmore-lie-groups-F00722	78	■ 0.997	$E(2)$	$E(2)$	$E(2)$ with structure
gilmore-lie-groups-F00723	78	■ 1.000	$P_{\{x\}}, P_{\{y\}}$	P_x, P_y	where the rows and columns are labeled successively by the basis vectors P_x, P_y , and
gilmore-lie-groups-F00724	78	■ 1.000	$L_{\{z\}}$	L_z	L_z . This regular representation has the block diagonal structure of a nonsemisimple
gilmore-lie-groups-F00725	78	■ 0.985	A, B	A, B	very useful inner product when A, B are $p \times q$ matrices is the Hilbert–Schmidt
gilmore-lie-groups-F00726	78	■ 0.985	$p \times q$	$p \times q$	very useful inner product when A, B are $p \times q$ matrices is the Hilbert–Schmidt
gilmore-lie-groups-F00727	79	■ 0.998	$X_{\{i\}}$	X_i	generators that commute with all operators in the Lie algebra (X_i , where $C_{i\star}^* = 0$)
gilmore-lie-groups-F00728	79	■ 0.998	$C_{\{i\} \star}^{\{*\}} = 0$	$C_{i\star}^* = 0$	generators that commute with all operators in the Lie algebra (X_i , where $C_{i\star}^* = 0$)
gilmore-lie-groups-F00729	79	■ 1.000	$C_{\{*\} \star}^{\{i\}} = 0$	$C_{**}^i = 0$	generators (X_i , where $C_{**}^i = 0$).
gilmore-lie-groups-F00730	79	■ 1.000	$V_{\{0\}}$	V_0	The subspace V_0 is a subalgebra of $\mathfrak{g}$. It is the largest nilpotent invariant subal-
gilmore-lie-groups-F00731	79	■ 1.000	$V_{\{-\}}$	V_-	The subspace V_- is also a subalgebra of $\mathfrak{g}$. It consists of compact (a topological
gilmore-lie-groups-F00732	79	■ 1.000	$V_{\{+\}}$	V_+	Finally, the subspace V_+ is not a subalgebra of $\mathfrak{g}$. It consists of noncompact
gilmore-lie-groups-F00733	80	■ 1.000	$X_{\{\pm\}} = X_{\{b\}} \pm X_{\{c\}}$	$X_{\pm} = X_b \pm X_c$	and $X_{\pm} = X_b \pm X_c$. In this basis the metric matrix is
gilmore-lie-groups-F00734	80	■ 1.000	$X_{\{-\}}$	X_-	In this representation it is clear that the operator X_- spans a one-dimensional
gilmore-lie-groups-F00735	80	■ 1.000	$X_{\{a\}}, X_{\{+\}}$	X_a, X_+	compact subalgebra in $\mathfrak{sl}(2; \mathbb{R})$ and the generators X_a, X_+ are noncompact.
gilmore-lie-groups-F00736	81	■ 0.644	$\alpha^{\{1\}}, \alpha^{\{2\}}, \dots, \alpha^{\{n\}}$	$\alpha^1, \alpha^2, \dots, \alpha^n$	ric and measure . We assume the coordinates of the identity are $(\alpha^1, \alpha^2, \dots, \alpha^n)$

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gilmore-lie-groups_ F00737	81	■ 0.840	$\alpha^1+d \alpha^1, \alpha^2+d \alpha^2, \ldots, \alpha^n+d \alpha^n$	$\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n +$	and the coordinates of a point near the identity are $(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n +$
gilmore-lie-groups_ F00738	81	■ 0.866	$d \alpha^n$	$d\alpha^n$	$d\alpha^n$). If $(x^1, x^2, \dots, x^n)$ represents some other group operation, then the point
gilmore-lie-groups_ F00739	81	■ 0.866	$\left(x^1, x^2, \ldots, x^n\right)$	$(x^1, x^2, \dots, x^n)$	$d\alpha^n$). If $(x^1, x^2, \dots, x^n)$ represents some other group operation, then the point
gilmore-lie-groups_ F00740	81	■ 0.831	$\alpha^1+d \alpha^1, \alpha^2+d \alpha^2, \ldots, \alpha^n+d \alpha^n$	$\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n$	$(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$ is mapped to the point $(x^1 + dx^1, x^2 +$
gilmore-lie-groups_ F00741	81	■ 0.831	$x^1+d x^1, x^2+$	$x^1 + dx^1, x^2 +$	$(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$ is mapped to the point $(x^1 + dx^1, x^2 +$
gilmore-lie-groups_ F00742	81	■ 1.000	$d x^2, \ldots, x^n+d x^n$	$dx^2, \dots, x^n + dx^n$	$(dx^2, \dots, x^n + dx^n)$ under left (right) multiplication by the group operation associ-
gilmore-lie-groups_ F00743	81	■ 0.997	$d x$	dx	ated with $(x^1, x^2, \dots, x^n)$. The displacements dx and $d\alpha$ are related by a position-
gilmore-lie-groups_ F00744	81	■ 0.997	$d \alpha$	$d\alpha$	ated with $(x^1, x^2, \dots, x^n)$. The displacements dx and $d\alpha$ are related by a position-
gilmore-lie-groups_ F00745	81	■ 1.000	$d s$	ds	Suppose now that the distance ds between the identity and a point with coordi-
gilmore-lie-groups_ F00746	81	■ 1.000	$\alpha^i+d \alpha^i$	$\alpha^i + d\alpha^i$	nates $\alpha^i + d\alpha^i$ infinitesimally close to the identity is given by
gilmore-lie-groups_ F00747	81	■ 0.671	$x, g_{rs}(x)$	$x, g_{rs}(x)$	Killing inner product. Can we define a metric at $x, g_{rs}(x)$, with the property that
gilmore-lie-groups_ F00748	81	■ 1.000	$x, g(x)$	$x, g(x)$	In order to enforce the invariance condition, the metric at $x, g(x)$, must be related
gilmore-lie-groups_ F00749	82	■ 0.976	$\left(d \alpha^1, d \alpha^2, d \alpha^3\right)$	$(d\alpha^1, d\alpha^2, d\alpha^3)$	The linear relation between the infinitesimals $(d\alpha^1, d\alpha^2, d\alpha^3)$ in the neighborhood
gilmore-lie-groups_ F00750	82	■ 0.999	$(d x, d y, d z)$	(dx, dy, dz)	of the identity and the infinitesimals (dx, dy, dz) in the neighborhood of the group
gilmore-lie-groups_ F00751	83	■ 1.000	$[X, Y]=(X Y-Y X)$	$[X, Y] = (XY - YX)$	tional combinatorial operation, the commutator $[X, Y] = (XY - YX)$, and obey the
gilmore-lie-groups_ F00752	83	■ 1.000	$g_1=(I+\epsilon X), g_1^{-1}=(I+\epsilon X)^{-1}=I-$	$g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I -$	1. Carry out the commutator calculation for $g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I -$
gilmore-lie-groups_ F00753	83	■ 1.000	$\epsilon X+\epsilon^2 X^2-\cdots$	$\epsilon X + \epsilon^2 X^2 - \dots$	$(\epsilon X + \epsilon^2 X^2 - \dots)$, with similar expressions for g_2 , to obtain the same result as in
gilmore-lie-groups_ F00754	83	■ 1.000	g_2	g_2	$\epsilon X + \epsilon^2 X^2 - \dots$, with similar expressions for g_2 , to obtain the same result as in
gilmore-lie-groups_ F00755	83	■ 1.000	$X^{\prime}$	X'	2. The inner product of two vectors X and X' in a linear vector space can be computed
gilmore-lie-groups_ F00756	83	■ 1.000	$(X, X)=2\left(a^2+b c\right)$	$(X, X) = 2(a^2 + bc)$	c. Use the result from Eq. (4.43) that $(X, X) = 2(a^2 + bc)$ to show
gilmore-lie-groups_ F00757	83	■ 1.000	$Y=e^X$	$Y = e^X$	a Lie algebra: $Y = e^X$. Show that “for Y sufficiently close to the identity” the matrix
gilmore-lie-groups_ F00758	83	■ 0.570	λ_i	λ_i	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00759	83	■ 0.570	$0<\lambda_i<+2$	$0 < \lambda_i < +2$	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00760	83	■ 0.570	$Y \in \operatorname{SL}(2 ; \mathbb{R})$	$Y \in \text{SL}(2; \mathbb{R})$	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00761	83	■ 0.570	$\operatorname{tr} Y<-2$	$\text{tr } Y < -2$	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00762	84	■ 1.000	$X \in \mathfrak{sl}(2 ; \mathbb{R})$	$X \in \mathfrak{sl}(2; \mathbb{R})$	expansion does not converge. That is, there is no 2×2 matrix $X \in \mathfrak{sl}(2; \mathbb{R})$ with the
gilmore-lie-groups_ F00763	84	■ 0.637	$\operatorname{tr} e^X<-2$	$\text{tr } e^X < -2$	property $\text{tr } e^X < -2$.

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gilmore-lie-groups_F00764	84	■ 0.952	$\mathbf{L} =$	$\mathbf{L} =$	4. The Lie algebra of $SO(3)$ is spanned by three 3×3 antisymmetric matrices $\mathbf{L} =$
gilmore-lie-groups_F00765	84	■ 1.000	$\left(L_1, L_2, L_3\right) = \left(X_{23}, X_{31}, X_{12}\right)$	$(L_1, L_2, L_3) = (X_{23}, X_{31}, X_{12})$	$(L_1, L_2, L_3) = (X_{23}, X_{31}, X_{12})$, with
gilmore-lie-groups_F00766	84	■ 1.000	$\mathbf{X} = \theta \cdot \mathbf{L}$	$\mathbf{X} = \theta \cdot \mathbf{L}$	where θ is the single invariant that can be constructed from the matrix $\mathbf{X} = \theta \cdot \mathbf{L}$:
gilmore-lie-groups_F00767	84	■ 1.000	$\theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2$	$\theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2$	$\theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2$. Show
gilmore-lie-groups_F00768	84	■ 1.000	$SO(n)$	$SO(n)$	5. The Lie algebra for the matrix group $SO(n)$ consists of antisymmetric $n \times n$ matrices.
gilmore-lie-groups_F00769	84	■ 1.000	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$ with matrix elements $(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$.
gilmore-lie-groups_F00770	84	■ 1.000	$(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$	$(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$ with matrix elements $(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$.
gilmore-lie-groups_F00771	84	■ 1.000	$\mathcal{X}_{ij} = x^i \partial_j - x^j \partial_i$	$\mathcal{X}_{ij} = x^i \partial_j - x^j \partial_i$	b. Show that the operators $\mathcal{X}_{ij} = x^i \partial_j - x^j \partial_i$ satisfy isomorphic commutation rela-
gilmore-lie-groups_F00772	84	■ 1.000	$\mathcal{B}_{ij} =$	$\mathcal{B}_{ij} =$	c. Show that bilinear products of boson creation and annihilation operators $\mathcal{B}_{ij} =$
gilmore-lie-groups_F00773	84	■ 1.000	$b_i^\dagger b_j - b_j^\dagger b_i (1 \leq i \neq j \leq n)$	$b_i^\dagger b_j - b_j^\dagger b_i (1 \leq i \neq j \leq n)$	$b_i^\dagger b_j - b_j^\dagger b_i (1 \leq i \neq j \leq n)$ satisfy isomorphic commutation relations.
gilmore-lie-groups_F00774	84	■ 0.963	$\mathcal{F}_{ij} =$	$\mathcal{F}_{ij} =$	d. Show that bilinear products of fermion creation and annihilation operators $\mathcal{F}_{ij} =$
gilmore-lie-groups_F00775	84	■ 1.000	$f_i^\dagger f_j - f_j^\dagger f_i (1 \leq i \neq j \leq n)$	$f_i^\dagger f_j - f_j^\dagger f_i (1 \leq i \neq j \leq n)$	$f_i^\dagger f_j - f_j^\dagger f_i (1 \leq i \neq j \leq n)$ satisfy isomorphic commutation relations.
gilmore-lie-groups_F00776	84	■ 1.000	D, Y, Z	D, Y, Z	6. The Jacobi identity for operators D, Y, Z (replace X by D in Eq. (4.17)) can be
gilmore-lie-groups_F00777	84	■ 0.998	$\mathfrak{so}(4)$	$\mathfrak{so}(4)$	7. For the matrix Lie algebra $\mathfrak{so}(4)$ the defining matrix representation consists of 4×4
gilmore-lie-groups_F00778	84	■ 0.998	4×4	4×4	7. For the matrix Lie algebra $\mathfrak{so}(4)$ the defining matrix representation consists of 4×4
gilmore-lie-groups_F00779	85	■ 1.000	$a_{ij} X_{ij}$	$a_{ij} X_{ij}$	element $a_{ij} X_{ij}$ in the Lie algebra:
gilmore-lie-groups_F00780	85	■ 1.000	$SO(p, q)$	$SO(p, q)$	stant? How does this result extend to $SO(n)$? to $SO(p, q)$? Is there a simple relation
gilmore-lie-groups_F00781	85	■ 1.000	$\operatorname{EXP}(V_-)$	$\operatorname{EXP}(V_-)$	$\operatorname{EXP}(V_-)$ is multiply periodic. Apply this construction to the noncompact groups
gilmore-lie-groups_F00782	85	■ 0.546	$SO(3, 1)$	$SO(3, 1)$	$SO(3, 1)$ and $SO(2, 2)$. Show $\operatorname{EXP}(V_-)$ is a two-sphere S^2 for $SO(3, 1)$ and a two-
gilmore-lie-groups_F00783	85	■ 0.546	$SO(2, 2)$	$SO(2, 2)$	$SO(3, 1)$ and $SO(2, 2)$. Show $\operatorname{EXP}(V_-)$ is a two-sphere S^2 for $SO(3, 1)$ and a two-
gilmore-lie-groups_F00784	85	■ 0.982	T^2	T^2	torus T^2 for $SO(2, 2)$.
gilmore-lie-groups_F00785	85	■ 0.752	$e^{\theta \cdot \mathbf{L}} \in SO(3)$	$e^{\theta \cdot \mathbf{L}} \in SO(3)$	operator $e^{\theta \cdot \mathbf{L}} \in SO(3)$. From this construct the invariant density $\rho(\theta)$ and the metric
gilmore-lie-groups_F00786	85	■ 0.752	$\rho(\theta)$	$\rho(\theta)$	operator $e^{\theta \cdot \mathbf{L}} \in SO(3)$. From this construct the invariant density $\rho(\theta)$ and the metric
gilmore-lie-groups_F00787	85	■ 1.000	$g_{\mu\nu}(\theta)$	$g_{\mu\nu}(\theta)$	tensor $g_{\mu\nu}(\theta)$. Give a reason for the strange behavior (singularities) that these invariant
gilmore-lie-groups_F00788	85	■ 1.000	$dV(x) = \ g(x)\ ^{1/2} d^n x$	$dV(x) = \ g(x)\ ^{1/2} d^n x$	standard result of Riemannian geometry that $dV(x) = \ g(x)\ ^{1/2} d^n x$. Discuss the
gilmore-lie-groups_F00789	85	■ 0.623	$(\mathbf{x}, \mathbf{x})$	$(\mathbf{x}, \mathbf{x})$	12. An inner product $(\mathbf{x}, \mathbf{x})$ is imposed on a real n -dimensional linear vector space.
gilmore-lie-groups_F00790	85	■ 1.000	$g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$	$g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$	It is represented by a real symmetric nonsingular $n \times n$ matrix $g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$, where

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gilmore-lie-groups_F00791	85	■ 1.000	$\mathbf{x}=\mathbf{e}_i x^i$	$\mathbf{x} = \mathbf{e}_i x^i$	$\mathbf{x} = \mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.
gilmore-lie-groups_F00792	85	■ 1.000	$g^{\{r\ s\}}$	g^{rs}	$\mathbf{x} = \mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.
gilmore-lie-groups_F00793	85	■ 1.000	$\mathbf{y}=G\ \mathbf{x},(\mathbf{y},\mathbf{y})=(\mathbf{x},\mathbf{x})$	$\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$	a. Lie group G preserves inner products. If $\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$. Show $G^t g G = g$.
gilmore-lie-groups_F00794	85	■ 1.000	$G^{\{t\}}\ g\ G=g$	$G^t g G = g$	a. Lie group G preserves inner products. If $\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$. Show $G^t g G = g$.
gilmore-lie-groups_F00795	85	■ 1.000	$H^{\{t\}}\ g+g\ H=0$	$H^t g + g H = 0$	b. Show the Lie algebra H of G satisfies $H^t g + g H = 0$.
gilmore-lie-groups_F00796	85	■ 1.000	$X_{\{r\ s\}}=g_{\{r\ t\}}\ x^{\{t\}}\ \partial_{\{s\}}-g_{\{s\ t\}}\ x^{\{t\}}\ \partial_{\{r\}}$	$X_{rs} = g_{rt} x^t \partial_s - g_{st} x^t \partial_r$	c. Show that the infinitesimal generators of G are $X_{rs} = g_{rt} x^t \partial_s - g_{st} x^t \partial_r$.
gilmore-lie-groups_F00797	86	■ 1.000	$X_{\{r\ s\}}$	X_{rs}	d. Show that the operators X_{rs} satisfy commutation relations
gilmore-lie-groups_F00798	86	■ 0.887	$M=S\ O$	$M = SO$	13. Every real unimodular 2×2 matrix M can be written in the form $M = SO$, where S
gilmore-lie-groups_F00808	98	■ 0.986	$A^{\{t\}}=-A$	$A^t = -A$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00809	86	■ 0.999	$M\ M^{\{t\}}=S^{\{2\}}=e^{\{2\}\ \Sigma}$	$MM^t = S^2 = e^{2\Sigma}$	a. Show that $MM^t = S^2 = e^{2\Sigma}$.
gilmore-lie-groups_F00810	86	■ 1.000	$O=S^{\{-1\}}\ M=e^{\{-\Sigma\}}\ M$	$O = S^{-1} M = e^{-\Sigma} M$	b. Show $O = S^{-1} M = e^{-\Sigma} M$
gilmore-lie-groups_F00811	86	■ 1.000	Σ	Σ	c. Write S as a power series expansion in Σ .
gilmore-lie-groups_F00812	86	■ 1.000	$S-I_{\{2\}}$	$S - I_2$	d. Write Σ as a power series expansion in $S - I_2$.
gilmore-lie-groups_F00813	86	■ 1.000	$M=H\ U$	$M = HU$	14. Extend the result of the previous problem to complex $n \times n$ matrices $M = HU$, with
gilmore-lie-groups_F00814	86	■ 1.000	$H^{\{\dagger\}}=H^{\{+1\}}$	$H^\dagger = H^{+1}$	M arbitrary but nonsingular, $H^\dagger = H^{+1}$ hermitian and $U^\dagger = U^{-1}$ unitary.
gilmore-lie-groups_F00815	86	■ 1.000	$U^{\{\dagger\}}=U^{\{-1\}}$	$U^\dagger = U^{-1}$	M arbitrary but nonsingular, $H^\dagger = H^{+1}$ hermitian and $U^\dagger = U^{-1}$ unitary.
gilmore-lie-groups_F00816	86	■ 1.000	$\hbar k_{\{L\}}$	$\hbar k_L$	or right with momentum $\hbar k_L$ or $-\hbar k_R$, has the form (Gilmore, 2004)
gilmore-lie-groups_F00817	86	■ 1.000	$-\hbar k_{\{R\}}$	$-\hbar k_R$	or right with momentum $\hbar k_L$ or $-\hbar k_R$, has the form (Gilmore, 2004)
gilmore-lie-groups_F00818	86	■ 1.000	$m_{\{i\ j\}}$	m_{ij}	The real quantities m_{ij} are the four matrix elements of a group operation in $SL(2; \mathbb{R})$.
gilmore-lie-groups_F00819	86	■ 1.000	$\hbar k_{\{L\}}=\hbar k_{\{R\}}$	$\hbar k_L = \hbar k_R$	They are energy dependent. By appropriate choice of $\hbar k_L = \hbar k_R$ and the matrix
gilmore-lie-groups_F00820	86	■ 0.679	$\operatorname{SU}(1,1)$	$SU(1,1)$	Show that these three matrices span the Lie algebra of the group $SU(1, 1)$.
gilmore-lie-groups_F00821	87	■ 1.000	$\kappa_{\{R\}}$	κ_R	The parameters κ_R and κ_L describe the decay length of the exponentially decaying
gilmore-lie-groups_F00822	87	■ 1.000	$\kappa_{\{L\}}$	κ_L	The parameters κ_R and κ_L describe the decay length of the exponentially decaying
gilmore-lie-groups_F00823	87	■ 1.000	$\kappa_{\{L\}}=\kappa_{\{R\}}$	$\kappa_L = \kappa_R$	energy dependent. By appropriate choice of $\kappa_L = \kappa_R$ and the matrix elements m_{ij} ,
gilmore-lie-groups_F00824	87	■ 1.000	$k_{\{*\}}=\sqrt{2\ m\left(E-V_{\{*\}}\right)\ /\ \hbar^2}$	$k_* = \sqrt{2m(E - V_*)/\hbar^2}$	(Hint: $k_* = \sqrt{2m(E - V_*)/\hbar^2}$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$ for $E < V_*$,
gilmore-lie-groups_F00825	87	■ 1.000	$E>V_{\{*\}}$	$E > V_*$	(Hint: $k_* = \sqrt{2m(E - V_*)/\hbar^2}$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$ for $E < V_*$,
gilmore-lie-groups_F00826	87	■ 1.000	$\kappa_{\{*\}}=\sqrt{2\ m\left(V_{\{*\}}-E\right)\ /\ \hbar^2}$	$\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$	(Hint: $k_* = \sqrt{2m(E - V_*)/\hbar^2}$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$ for $E < V_*$,

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gilmore-lie-groups_ F08827	87	■ 1.000	$E < V_*$	$E < V_*$	(Hint: $k_* = \sqrt{2m(E - V_*)/\hbar^2}$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$ for $E < V_*$,
gilmore-lie-groups_ F08828	87	■ 0.978	$*=L, R$	$* = L, R$	$\textcolor{red}{*} \equiv \textcolor{red}{L, R}.$)
gilmore-lie-groups_ F08829	88	■ 0.998	$\operatorname{det}(M) \neq 0$	$\det(M) \neq 0$	groups $GL(n; \mathbb{F})$ have no defining condition (the only condition is $\det(M) \neq 0$),
gilmore-lie-groups_ F08830	88	■ 1.000	$I + \epsilon A$	$I + \epsilon A$	the constraint to matrices infinitesimally close to the identity: $I + \epsilon A$. The matrices
gilmore-lie-groups_ F08831	88	■ 1.000	$\mathfrak{gl}(n; \mathbb{F})$	$\mathfrak{gl}(n; \mathbb{F})$	1. $\mathfrak{gl}(n; \mathbb{F})$. This algebra consists of arbitrary $n \times n$ matrices over the field $\mathbb{F}$.
gilmore-lie-groups_ F08832	89	■ 1.000	$\mathfrak{ut}(p, q)$	$\mathfrak{ut}(p, q)$	2. $\mathfrak{ut}(p, q)$. Upper triangular algebras. The matrix algebra has the same structure
gilmore-lie-groups_ F08833	89	■ 0.998	$\mathfrak{ht}(p, q)$	$\mathfrak{ht}(p, q)$	3. $\mathfrak{ht}(p, q)$. The algebra for this class of groups is defined by the condition
gilmore-lie-groups_ F08834	90	■ 1.000	$(a, b) = (1, 0)$	$(a, b) = (1, 0)$	The identity is at $\textcolor{red}{(a, b) = (1, 0)}$. Its algebra is spanned by the two operators
gilmore-lie-groups_ F08835	90	■ 0.999	$\left[X_a, X_b\right] = X_b$	$[X_a, X_b] = X_b$	The commutation relations are given by $[X_a, X_b] = X_b$.
gilmore-lie-groups_ F08836	90	■ 1.000	$\mathfrak{ut}(p, q, r)$	$\mathfrak{ut}(p, q, r)$	4. $\mathfrak{ut}(p, q, r)$. This matrix algebra is identical in structure to the parent group.
gilmore-lie-groups_ F08837	90	■ 0.999	$\mathfrak{ut}(1, 2, 1)$	$\mathfrak{ut}(1, 2, 1)$	Example A very useful six-parameter subalgebra of $\mathfrak{ut}(1, 2, 1)$ is given by
gilmore-lie-groups_ F08838	90	■ 1.000	j	j	algebra are summarized in the table below. The operator in the i th row and j th
gilmore-lie-groups_ F08839	90	■ 1.000	$\left[X_i, X_j\right]$	$[X_i, X_j]$	column is $[X_i, X_j]$.
gilmore-lie-groups_ F08845	91	■ 1.000	X_{δ}	X_{δ}	matrix X_{δ} representing the operator $I = [a, a^{\dagger}]$ is not diagonal. It need not be, as
gilmore-lie-groups_ F08854	91	■ 1.000	$a^{\dagger} a^{\dagger}$	$a^{\dagger} a^{\dagger}$	and annihilation operators $a^{\dagger} a^{\dagger}$ and aa , single-photon creation and annihilation
gilmore-lie-groups_ F08855	91	■ 1.000	$a a$	aa	and annihilation operators $a^{\dagger} a^{\dagger}$ and aa , single-photon creation and annihilation
gilmore-lie-groups_ F08863	90	■ 1.000	$\hat{n} = a^{\dagger} a + \frac{1}{2}$	$\hat{n} = a^{\dagger} a + \frac{1}{2}$	and the algebra of the photon energy operator $\textcolor{red}{\hat{n} = a^{\dagger} a + \frac{1}{2}}$, two-photon creation
gilmore-lie-groups_ F08864	91	■ 0.996	$I = [a, a^{\dagger}]$	$I = [a, a^{\dagger}]$	operators $a^{\dagger}$ and a , and the identity operator $I = [a, a^{\dagger}]$. We observe that the 4×4
gilmore-lie-groups_ F08865	91	■ 0.887	$\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$	$\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$	5. $\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$. This matrix algebra is also identical in structure to
gilmore-lie-groups_ F08866	91	■ 1.000	$\mathfrak{ut}(1, 1, 1)$	$\mathfrak{ut}(1, 1, 1)$	its parent group. A very useful four-parameter subalgebra of $\mathfrak{ut}(1, 1, 1)$ is given by
gilmore-lie-groups_ F08867	91	■ 0.936	$\mathfrak{nil}(n)$	$\mathfrak{nil}(n)$	6. $\mathfrak{nil}(n)$. Nilpotent matrices have an upper triangular structure, with +1 along the
gilmore-lie-groups_ F08868	91	■ 1.000	$\eta = 0$	$\eta = 0$	This is easily seen by setting $\eta = 0$ in (5.9). As a result, a number of difficult
gilmore-lie-groups_ F08869	91	■ 1.000	$\mathfrak{a}(p, q)$	$\mathfrak{a}(p, q)$	7. $\mathfrak{a}(p, q)$. The matrix algebra for the commutative group of Example (7) in Chap-
gilmore-lie-groups_ F08870	92	■ 0.998	$O(3)$	$O(3)$	antihermitian matrices. The Lie algebras for the groups $O(3)$ and $U(2)$ are
gilmore-lie-groups_ F08871	92	■ 0.998	$U(2)$	$U(2)$	antihermitian matrices. The Lie algebras for the groups $O(3)$ and $U(2)$ are
gilmore-lie-groups_ F08872	92	■ 0.728	σ_{μ}	σ_{μ}	The four 2×2 matrices σ_{μ} are called Pauli spin matrices .
gilmore-lie-groups_ F08873	92	■ 1.000	$\mathfrak{so}(2, 1)$	$\mathfrak{so}(2, 1)$	example, the algebra $\mathfrak{so}(2, 1)$ for the Lorentz group in the plane is

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gilmore-lie-groups_ F08874	93	■ 1.000	$\mathfrak{so}(3,1)$	$\mathfrak{so}(3,1)$	By an identical argument the matrix Lie algebra for the Lorentz group $\mathfrak{so}(3, 1)$ is
gilmore-lie-groups_ F08875	93	■ 1.000	$M=-G^{-1} M^{\dagger} G$	$M = -G^{-1} M^{\dagger} G$	Since G is nonsingular, $M = -G^{-1} M^{\dagger} G$ and
gilmore-lie-groups_ F08876	93	■ 1.000	$(p+q) \times (p+q)$	$(p+q) \times (p+q)$	In the case that the $(p+q) \times (p+q)$ matrix G has singular block diagonal struc-
gilmore-lie-groups_ F08877	93	■ 0.983	$\left[\begin{smallmatrix} g & 0 \\ 0 & 0 \end{smallmatrix}\right]$	$\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$	ture $\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$ with $\det(g) \neq 0$, this constraint reduces to
gilmore-lie-groups_ F08878	93	■ 0.983	$\operatorname{det}(g) \neq 0$	$\det(g) \neq 0$	ture $\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$ with $\det(g) \neq 0$, this constraint reduces to
gilmore-lie-groups_ F08879	93	■ 1.000	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1, 1, 1, 0)$ the Lie algebra is
gilmore-lie-groups_ F08880	94	■ 1.000	θ_i	θ_i	Here the parameters θ_i describe rotations about the i th coordinate axis and the
gilmore-lie-groups_ F08881	94	■ 1.000	$s_{\{4\}}$	s_4	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4} t$. If s_4 is set to zero
gilmore-lie-groups_ F08882	94	■ 1.000	$t^{\{\prime\}}=e^{s_{\{4\}}} t$	$t' = e^{s_4} t$	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4} t$. If s_4 is set to zero
gilmore-lie-groups_ F08883	94	■ 1.000	$E(3)=I \ S \ 0(3)$	$E(3) = ISO(3)$	Euclidean group $E(3) \equiv ISO(3)$ (inhomogeneous rotation group in R^3).
gilmore-lie-groups_ F08884	94	■ 0.981	$I \ S \ 0(3,1) (3.26)$	$ISO(3,1)(3.26)$	The matrix Lie algebra for the Poincaré group $ISO(3, 1)$ (3.26) is obtained by
gilmore-lie-groups_ F08885	94	■ 0.837	5×5	5×5	similar arguments using a singular 5×5 metric $G = \operatorname{diag}(1, 1, 1, -1, 0)$. The Lie
gilmore-lie-groups_ F08886	94	■ 0.837	$G=\operatorname{diag}(1,1,1,-1,0)$	$G = \operatorname{diag}(1, 1, 1, -1, 0)$	similar arguments using a singular 5×5 metric $G = \operatorname{diag}(1, 1, 1, -1, 0)$. The Lie
gilmore-lie-groups_ F08887	94	■ 0.988	$I \ S \ 0(3,1)$	$ISO(3,1)$	“contraction” (cf., Chapter 13) from the Lie algebra of $ISO(3, 1)$:
gilmore-lie-groups_ F08888	94	■ 1.000	$\mathfrak{su}(n)$	$\mathfrak{su}(n)$	Lie algebras. The important algebra $\mathfrak{su}(n)$ is obtained from the intersection of $\mathfrak{u}(n)$
gilmore-lie-groups_ F08889	94	■ 1.000	$\mathfrak{u}(n)$	$\mathfrak{u}(n)$	Lie algebras. The important algebra $\mathfrak{su}(n)$ is obtained from the intersection of $\mathfrak{u}(n)$
gilmore-lie-groups_ F08890	94	■ 1.000	$\mathfrak{sl}(n; \mathbb{C})$	$\mathfrak{sl}(n; \mathbb{C})$	and $\mathfrak{sl}(n; \mathbb{C})$ (cf. (5.14)). For example
gilmore-lie-groups_ F08891	95	■ 1.000	$M_{\{i \ j\}(i < j)} (M \in \mathfrak{u}(n))$	$M_{ij}(i < j) (M \in \mathfrak{u}(n))$	each of the $n(n-1)/2$ complex matrix elements M_{ij} ($i < j$) ($M \in \mathfrak{u}(n)$) above the
gilmore-lie-groups_ F08892	95	■ 0.588	$M_{\{i \ i\}}$	M_{ii}	M_{ii} by a real 2×2 matrix representing an imaginary complex number ($a = 0$,
gilmore-lie-groups_ F08893	95	■ 0.761	$M_{\{i \ j\}}$	M_{ij}	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_ F08894	95	■ 0.761	$(i > j)$	$(i > j)$	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_ F08895	95	■ 0.999	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	$2n \times 2n$ matrix with the property $\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$. The dimension of $\mathfrak{ou}(2n)$
gilmore-lie-groups_ F08896	95	■ 0.999	$\mathfrak{o} \ \mathfrak{u}(2n)$	$\mathfrak{ou}(2n)$	$2n \times 2n$ matrix with the property $\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$. The dimension of $\mathfrak{ou}(2n)$
gilmore-lie-groups_ F08897	95	■ 0.640	$\mathcal{u}(n): 2 \times n(n-1) / 2+1 \times n=n^2$	$\square(n): 2 \times n(n-1)/2 + 1 \times n = n^2$	is the dimension of $\mathfrak{u}(n): 2 \times n(n-1)/2 + 1 \times n = n^2$.
gilmore-lie-groups_ F08898	95	■ 0.914	$S \ p(n)$	$Sp(n)$	The Lie algebra of $Sp(n)$ consists of $n \times n$ antihermitian matrices M over $\mathbb{Q}$:
gilmore-lie-groups_ F08899	95	■ 0.791	$\operatorname{Sp}(n)$	$\operatorname{Sp}(n)$	tained from the Lie algebra of $Sp(n)$ by replacing each of the $n(n-1)/2$ quaternion
gilmore-lie-groups_ F09900	95	■ 0.712	$M_{\{i \ j\}(i < j)} (M \in \mathfrak{sp}(n))$	$M_{ij}(i < j) (M \in \mathfrak{sp}(n))$	matrix elements M_{ij} ($i < j$) ($M \in \mathfrak{sp}(n)$) above the diagonal of M by a 2×2 com-

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gilmore-lie-groups-F00901	95	■ 0.998	$q_{\{0\}}=0, \ q_{\{i\}}$	$q_0 = 0, q_i$	matrix representing an imaginary quaternion ($q_0 = 0, q_i$ arbitrary in Eq. (3.4)). The
gilmore-lie-groups-F00902	95	■ 0.615	$i>j$	$i > j$	matrix elements M_{ij} below the diagonal ($i > j$) are obtained from the antihermitic-
gilmore-lie-groups-F00903	95	■ 1.000	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$ $\subset \mathfrak{u}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups-F00904	95	■ 1.000	$\mathfrak{u}(2n)$	$\mathfrak{usp}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups-F00905	95	■ 1.000	$\mathfrak{sp}(n)$	$\mathfrak{sp}(n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups-F00906	95	■ 1.000	$4 \times n(n-1) / 2 + n \times 3 = 2n(2n+1) / 2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$.
gilmore-lie-groups-F00907	96	■ 0.999	$(X, X)=0 \Rightarrow X=$	$(X, X) = 0 \Rightarrow X =$	is usually very useful. This inner product is positive–definite: $(X, X) = 0 \Rightarrow X =$
gilmore-lie-groups-F00908	96	■ 0.949	$X, X^{\prime}$	X, X'	0. For example, for the algebra $\mathfrak{so}(2, 1)$ (Eq. (5.18)), if X, X' are two 3×3 matrices
gilmore-lie-groups-F00909	96	■ 0.811	(X, Y)	(X, Y)	Lie algebra. The Cartan–Killing inner product (X, Y) is specifically defined as the
gilmore-lie-groups-F00910	97	■ 1.000	$\exp(\theta X)$	$\exp(\theta X)$	In the compact case, the group element $\exp(\theta X)$ periodically returns to the identity
gilmore-lie-groups-F00911	97	■ 1.000	$-\pi \leq \theta \leq +\pi$	$-\pi \leq \theta \leq +\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups-F00912	97	■ 1.000	$-\pi$	$-\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups-F00913	97	■ 1.000	$+\pi$	$+\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups-F00914	97	■ 1.000	$-\infty < \theta < +\infty$	$-\infty < \theta < +\infty$	the noncompact case the group is parameterized by the entire line $-\infty < \theta < +\infty$.
gilmore-lie-groups-F00915	97	■ 1.000	S^1	S^1	The underlying manifolds for the two groups are the circle S^1 and the line R^1 .
gilmore-lie-groups-F00916	97	■ 1.000	R^k	R^k	the entire subspace in the Lie algebra, R^k , where k is the number of noncompact
gilmore-lie-groups-F00917	97	■ 1.000	$G=I_{\{n\}}, \ I_{\{p, \ q\}}$	$G = I_n, I_{p,q}$	the metric-preserving groups ($G = I_n, I_{p,q}$, nonsingular antisymmetric, general
gilmore-lie-groups-F00918	97	■ 0.989	$(A, A)=$	$(A, A) =$	Show that in this matrix Lie algebra an inner product can be defined by $(A, A) =$
gilmore-lie-groups-F00919	97	■ 1.000	$\operatorname{tr}(A)^2=a^2+c^2$	$\operatorname{tr}(A)^2 = a^2 + c^2$	$\operatorname{tr}(A)^2 = a^2 + c^2$.
gilmore-lie-groups-F00920	97	■ 1.000	$\mathfrak{ut}(1,1)$	$\mathfrak{ut}(1, 1)$	2. Show that the regular representation of the matrix Lie algebra $\mathfrak{ut}(1, 1)$ given in Prob-
gilmore-lie-groups-F00921	98	■ 1.000	$(A, A)=\operatorname{tr} R(A)^2=(a-c)^2$	$(A, A) = \operatorname{tr} R(A)^2 = (a - c)^2$	inner product in the regular representation is $(A, A) = \operatorname{tr} R(A)^2 = (a - c)^2$. The inner
gilmore-lie-groups-F00922	98	■ 0.998	$X_{\{a\}}+X_{\{c\}}$	$X_a + X_c$	product in the regular representation suggests that the linear combination $X_a + X_c$
gilmore-lie-groups-F00923	98	■ 1.000	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$ $\mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	3. Write down the algebra inclusions $\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$ explicitly in terms
gilmore-lie-groups-F00924	98	■ 1.000	$X_{\{\eta\}}, X_{\{r\}}, X_{\{l\}}, X_{\{\delta\}}$	$X_\eta, X_r, X_l, X_\delta$	the matrices $X_\eta, X_r, X_l, X_\delta$ defined in (5.9). Show that the two Lie algebras are
gilmore-lie-groups-F00925	98	■ 1.000	$\hat{E}=\left(a^{\dagger} a+\frac{1}{2}\right) \hbar \omega \rightarrow a^{\dagger} a+\frac{1}{2}$	$\hat{E} = \left(a^\dagger a + \frac{1}{2}\right) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$	is $\hat{E} = (a^\dagger a + \frac{1}{2}) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$ for $\hbar \omega = 1$.
gilmore-lie-groups-F00926	98	■ 1.000	$\hbar \omega=1$	$\hbar \omega = 1$	is $\hat{E} = (a^\dagger a + \frac{1}{2}) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$ for $\hbar \omega = 1$.
gilmore-lie-groups-F00927	98	■ 1.000	$X, X^{\prime} \in \mathfrak{h}$	$X, X' \in \mathfrak{h}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.

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gilmore-lie-groups_F00928	98	■ 1.000	$Y, Y^{\prime} \in \mathfrak{p}$	$Y, Y' \in \mathfrak{p}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00929	98	■ 1.000	$\left[X, X^{\prime}\right] \in \mathfrak{h},[X, Y] \in \mathfrak{p},\left[Y, Y^{\prime}\right] \in \mathfrak{h}$	$[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00930	98	■ 1.000	$\mathfrak{h}$	$\mathfrak{h}$	Conclude that the subspace $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$. Is $\mathfrak{p}$ a subalgebra (under what
gilmore-lie-groups_F00931	98	■ 1.000	$\mathfrak{p}$	$\mathfrak{p}$	Conclude that the subspace $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$. Is $\mathfrak{p}$ a subalgebra (under what
gilmore-lie-groups_F00932	98	■ 0.995	$X=\left[\begin{array}{cc} A & 0 \\ 0 & D \end{array}\right] \in \mathfrak{h}$	$X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$	If $X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$, then
gilmore-lie-groups_F00933	98	■ 0.995	$Y=\left[\begin{array}{cc} 0 & B \\ C & 0 \end{array}\right] \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$	If $X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$, then
gilmore-lie-groups_F00934	98	■ 1.000	$(X, Y)=0$	$(X, Y) = 0$	Show that X and Y are orthogonal under this inner product: $(X, Y) = 0$.
gilmore-lie-groups_F00935	98	■ 0.970	$\mathfrak{so}(p, q)$	$\mathfrak{so}(p, q)$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00936	98	■ 0.970	$\left[\begin{array}{cc} A & B \\ B^t & C \end{array}\right]$	$\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00937	98	■ 0.970	$p \times p$	$p \times p$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00938	98	■ 0.970	$q \times q$	$q \times q$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00939	98	■ 0.986	$C^t=-C$	$C^t = -C$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00940	98	■ 0.986	$X=\left[\begin{array}{cc} A & 0 \\ 0 & C \end{array}\right] \in \mathfrak{h}$	$X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00941	98	■ 0.986	$Y=\left[\begin{array}{cc} 0 & B \\ B^t & 0 \end{array}\right] \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00942	98	■ 0.988	$(X, X) \leq 0, \quad(X, X)=0 \Rightarrow X=0$	$(X, X) \leq 0, \quad (X, X) = 0 \Rightarrow X = 0$	• $(X, X) \leq 0, \quad (X, X) = 0 \Rightarrow X = 0$
gilmore-lie-groups_F00943	98	■ 1.000	$(Y, Y) \geq 0, \quad(Y, Y)=0 \Rightarrow Y=0$	$(Y, Y) \geq 0, \quad (Y, Y) = 0 \Rightarrow Y = 0$	• $(Y, Y) \geq 0, \quad (Y, Y) = 0 \Rightarrow Y = 0$
gilmore-lie-groups_F00944	99	■ 0.818	$\mathfrak{su}(p, q)$	$\mathfrak{su}(p, q)$	8. The Lie algebra $\mathfrak{su}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^\dagger & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00945	99	■ 0.818	$\left[\begin{array}{c} A \\ B^\dagger \\ C \end{array}\right]$	$\begin{bmatrix} A \\ B^\dagger \\ C \end{bmatrix}$	8. The Lie algebra $\mathfrak{su}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^\dagger & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00946	99	■ 0.999	$A^\dagger=-A, \quad C^\dagger=-C$	$A^\dagger = -A, C^\dagger = -C$	A and C satisfy $A^\dagger = -A, C^\dagger = -C$, and $\text{tr}(A + C) = 0$. If $X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and
gilmore-lie-groups_F00947	99	■ 0.999	$\text{tr}(A+C)=0$	$\text{tr}(A + C) = 0$	A and C satisfy $A^\dagger = -A, C^\dagger = -C$, and $\text{tr}(A + C) = 0$. If $X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and
gilmore-lie-groups_F00948	99	■ 1.000	$Y=\left[\begin{array}{cc} 0 & B \\ B^\dagger & 0 \end{array}\right] \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^\dagger & 0 \end{bmatrix} \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^\dagger & 0 \end{bmatrix} \in \mathfrak{p}$, then show once again that
gilmore-lie-groups_F00949	99	■ 1.000	$(X, X)=0 \Rightarrow X=0$	$(X, X) = 0 \Rightarrow X = 0$	Show that $(X, X) = 0 \Rightarrow X = 0$ and $(Y, Y) = 0 \Rightarrow Y = 0$.
gilmore-lie-groups_F00950	99	■ 1.000	$(Y, Y)=0 \Rightarrow Y=0$	$(Y, Y) = 0 \Rightarrow Y = 0$	Show that $(X, X) = 0 \Rightarrow X = 0$ and $(Y, Y) = 0 \Rightarrow Y = 0$.
gilmore-lie-groups_F00951	99	■ 1.000	$\mathfrak{sl}(n ; \mathbb{R})$	$\mathfrak{sl}(n; \mathbb{R})$	9. The Lie algebra for $\mathfrak{sl}(n; \mathbb{R})$ has a decomposition in terms of real antisymmetric and
gilmore-lie-groups_F00952	99	■ 1.000	$B^t=B$	$B^t = B$	traceless symmetric matrices $A^t = -A$ and $B^t = B$ with $\text{tr } B = 0$:

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gilmore-lie-groups_F00953	99	■ 1.000	$\operatorname{tr} B=0$	$\operatorname{tr} B = 0$	traceless symmetric matrices $A^t = -A$ and $B^t = B$ with $\operatorname{tr} B = 0$:
gilmore-lie-groups_F00954	99	■ 1.000	$(A, A)=0 \Rightarrow A=0$	$(A, A) = 0 \Rightarrow A = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(B, B) = 0 \Rightarrow B = 0$.
gilmore-lie-groups_F00955	99	■ 1.000	$(B, B)=0 \Rightarrow B=0$	$(B, B) = 0 \Rightarrow B = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(B, B) = 0 \Rightarrow B = 0$.
gilmore-lie-groups_F00956	99	■ 1.000	$A^{\dagger}=-A$	$A^{\dagger} = -A$	matrices $A^{\dagger} = -A$ and traceless hermitian matrices $H^{\dagger} = H$:
gilmore-lie-groups_F00957	99	■ 1.000	$H^{\dagger}=H$	$H^{\dagger} = H$	matrices $A^{\dagger} = -A$ and traceless hermitian matrices $H^{\dagger} = H$:
gilmore-lie-groups_F00958	99	■ 1.000	$(H, H)=0 \Rightarrow H=0$	$(H, H) = 0 \Rightarrow H = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(H, H) = 0 \Rightarrow H = 0$.
gilmore-lie-groups_F00959	99	■ 1.000	$\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$	11. Assume that $\mathfrak{g}$ is a Lie algebra with a Cartan decomposition $\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$, with commu-
gilmore-lie-groups_F00960	100	■ 1.000	$\mathfrak{g}'=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}'$	$\mathfrak{g}' = \mathfrak{h} + i\mathfrak{p} = \mathfrak{h} + \mathfrak{p}'$	defined by $\mathfrak{g}' = \mathfrak{h} + i\mathfrak{p} = \mathfrak{h} + \mathfrak{p}'$ then
gilmore-lie-groups_F00961	100	■ 1.000	$\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$	$\mathfrak{g} = \mathfrak{sl}(n; \mathbb{Q})$	nion algebra $\mathfrak{g} = \mathfrak{sl}(n; \mathbb{Q})$ with respect to the subalgebra $\mathfrak{h} = \mathfrak{sl}(n; \mathbb{C})$.
gilmore-lie-groups_F00962	100	■ 1.000	$\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$	$\mathfrak{h} = \mathfrak{sl}(n; \mathbb{C})$	nion algebra $\mathfrak{g} = \mathfrak{sl}(n; \mathbb{Q})$ with respect to the subalgebra $\mathfrak{h} = \mathfrak{sl}(n; \mathbb{C})$.
gilmore-lie-groups_F00963	210	■ 1.000	μ	μ	The contravariant index μ can be lowered with the metric tensor, which is the
gilmore-lie-groups_F00964	126	■ 1.000	σ	σ	representation $\mathbf{J} \rightarrow \frac{1}{2}\sigma$, where σ are the Pauli spin matrices (5.14):
gilmore-lie-groups_F00969	100	■ 1.000	$\mathfrak{g}=A$	$\mathfrak{g} = A$	14. Assume that $\mathfrak{g} = A$, where A is a Lie algebra of $n \times n$ matrices on which the in-
gilmore-lie-groups_F00970	100	■ 1.000	$\operatorname{tr} A^2 \leq 0, =0 \Rightarrow A=0$	$\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$	ner product is negative-definite: $\operatorname{tr} A^2 < 0, = 0 \Rightarrow A = 0$. Then show that $\operatorname{EXP}(tA)$
gilmore-lie-groups_F00971	100	■ 1.000	$\operatorname{EXP}(tA)$	$\operatorname{EXP}(tA)$	ner product is negative-definite: $\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$. Then show that $\operatorname{EXP}(tA)$
gilmore-lie-groups_F00972	100	■ 1.000	t	t	returns to any neighborhood of the identity I_n if t becomes large enough. If the eigen-
gilmore-lie-groups_F00973	100	■ 1.000	$\lambda_i=\gamma n_i, n_i$	$\lambda_i = \gamma n_i, n_i$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or
gilmore-lie-groups_F00974	100	■ 1.000	$\gamma \neq 0$	$\gamma \neq 0$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or
gilmore-lie-groups_F00975	100	■ 1.000	t_0	t_0	irrational), $\operatorname{EXP}(tA)$ returns periodically to I_n . What is t_0 , the minimum period in t ?
gilmore-lie-groups_F00976	100	■ 0.979	$\mathfrak{so}(3)$	$\mathfrak{so}(3)$	15. Use the parameterization of $\mathfrak{so}(3)$ given in Problem 3.14. Show that the differentials
gilmore-lie-groups_F00977	100	■ 1.000	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$ in the neighborhood of the identity by
gilmore-lie-groups_F00978	100	■ 1.000	$g(x, y, z)$	$g(x, y, z)$	metric tensor $g(x, y, z)$ and the invariant measure $d\mu(x, y, z)$ on the group $SO(3)$
gilmore-lie-groups_F00979	100	■ 1.000	$d\mu(x, y, z)$	$d\mu(x, y, z)$	metric tensor $g(x, y, z)$ and the invariant measure $d\mu(x, y, z)$ on the group $SO(3)$
gilmore-lie-groups_F00980	101	■ 1.000	$\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p}$	$(\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p})$	exist in the algebra $\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p}$,
gilmore-lie-groups_F00981	102	■ 0.987	$b_i^{\dagger}$	$b_i^{\dagger}$	a set of n independent boson creation ($b_i^{\dagger}$) and annihilation (b_j) operators that obey
gilmore-lie-groups_F00982	102	■ 0.987	b_j	b_j	a set of n independent boson creation ($b_i^{\dagger}$) and annihilation (b_j) operators that obey

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gilmore-lie-groups_F00983	102	■ 0.971	$\left[b_{\{i\}}, b_{\{j\}}\right], \left[b_{\{i\}}^{\dagger}, b_{\{j\}}^{\dagger}\right], \left[b_{\{i\}}, I\right], \left[b_{\{j\}}^{\dagger}, I\right]$	$[b_i, b_j], [b_i^\dagger, b_j^\dagger], [b_i, I], [b_j^\dagger, I]$	with all other commutators (e.g., $[b_i, b_j], [b_i^\dagger, b_j^\dagger], [b_i, I], [b_j^\dagger, I]$) equal to zero.
gilmore-lie-groups_F00984	102	■ 1.000	$\mathcal{A}$	$\mathcal{A}$	matrix A the operator $\mathcal{A}$ that is a linear combination of creation and annihilation
gilmore-lie-groups_F00985	103	■ 1.000	$[A, B]=C$	$[A, B] = C$	where $[A, B] = C$. This argument is invertible. An algebra of operators bilinear in
gilmore-lie-groups_F00986	103	■ 0.999	$2n+1$	$2n+1$	Remark The $2n+1$ operators $b_i, b_j^\dagger, I$ ($1 \leq i, j \leq n$) span the Heisenberg
gilmore-lie-groups_F00987	103	■ 0.999	$b_{\{i\}}, b_{\{j\}}^{\dagger}, I(1 \leq i, j \leq n)$	$b_i, b_j^\dagger, I(1 \leq i, j \leq n)$	Remark The $2n+1$ operators $b_i, b_j^\dagger, I$ ($1 \leq i, j \leq n$) span the Heisenberg
gilmore-lie-groups_F00988	103	■ 1.000	$X_{\{i\} \{j\}}$	X_{ij}	Any set of operators X_{ij} that satisfies isomorphic commutation relations
gilmore-lie-groups_F00989	103	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{j\}}$	$b_i^\dagger b_j$	can be used in place of the bilinear combinations $b_i^\dagger b_j$:
gilmore-lie-groups_F00990	103	■ 0.970	$f_{\{i\}}^{\dagger}$	$f_i^\dagger$	creation ($f_i^\dagger$) and annihilation (f_j) operators for n independent modes. These oper-
gilmore-lie-groups_F00991	103	■ 0.970	$f_{\{j\}}$	f_j	creation ($f_i^\dagger$) and annihilation (f_j) operators for n independent modes. These oper-
gilmore-lie-groups_F00992	103	■ 1.000	$\left\{f_{\{i\}}, f_{\{j\}}\right\}, \left\{f_{\{i\}}^{\dagger}, f_{\{j\}}^{\dagger}\right\}$	$\{f_i, f_j\}, \{f_i^\dagger, f_j^\dagger\}$	with all other bilinear anticommutators (e.g., $\{f_i, f_j\}, \{f_i^\dagger, f_j^\dagger\}$) equal to zero. Bi-
gilmore-lie-groups_F00993	104	■ 1.000	$SU(n)$	$SU(n)$	subgroup $SU(n)$. The mapping to a boson operator algebra greatly simplifies the
gilmore-lie-groups_F00994	104	■ 1.000	$\mathcal{L}_1=x_2 \partial_3-x_3 \partial_2, \mathcal{L}_2=x_3 \partial_1-x_1 \partial_3, \mathcal{L}_3=x_1 \partial_2-x_2 \partial_1$	$\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$	where $\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$. The two algebras
gilmore-lie-groups_F00995	104	■ 0.705	$L_{\{i\}}$	L_i	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_F00996	104	■ 0.705	$\mathcal{L}_{\{i\}}$	$\mathcal{L}_i$	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_F00997	105	■ 1.000	$E(2)=ISO(2)$	$E(2) = ISO(2)$	As another example, we treat the Lie algebra for the group $E(2) = ISO(2)$ of
gilmore-lie-groups_F00998	105	■ 1.000	$x-y$	$x-y$	rigid motions (translations and rotations) in the $x-y$ plane, whose matrix algebra
gilmore-lie-groups_F00999	105	■ 1.000	$t_{\{1\}}$	t_1	angle θ and displacements in the x and y directions by t_1 and t_2 . The associated
gilmore-lie-groups_F01000	105	■ 1.000	$t_{\{2\}}$	t_2	angle θ and displacements in the x and y directions by t_1 and t_2 . The associated
gilmore-lie-groups_F01001	105	■ 1.000	$\mathcal{L}_z=x_1 \partial_2-x_2 \partial_1$	$\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$	where $\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$ and $\mathcal{T}_i = \partial_i$. The matrix algebra and operator algebra have
gilmore-lie-groups_F01002	105	■ 1.000	$\mathcal{T}_{\{i\}}=\partial_{\{i\}}$	$\mathcal{T}_i = \partial_i$	where $\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$ and $\mathcal{T}_i = \partial_i$. The matrix algebra and operator algebra have
gilmore-lie-groups_F01003	105	■ 0.867	(a, b)	(a, b)	the real line parameterized by points (a, b) in R^2 as follows
gilmore-lie-groups_F01004	105	■ 1.000	$x(p)$	$x(p)$	is chosen a coordinate, $x(p)$, can be introduced and the function can be written
gilmore-lie-groups_F01005	105	■ 1.000	$S^{\prime}$	S'	If a new coordinate system S' is chosen, the value of the function at p remains
gilmore-lie-groups_F01006	105	■ 1.000	$p, x^{\prime}(p)$	$p, x'(p)$	unchanged but the new coordinate of $p, x'(p)$, is different. Therefore the functions
gilmore-lie-groups_F01007	105	■ 1.000	$f_{\{S\}}$	f_S	f_S and $f_{S'}$ must be different. We ask: how is $f_{S'}$ related to f_S ?

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gilmore-lie-groups_ F01059	109	■ 1.000	$f_{\mathrm{i}}^{\dagger} f_{\mathrm{j}}$	$f_i^{\dagger} f_j$	ators $f_i^{\dagger} f_j$. What is now the dimension of this representation?
gilmore-lie-groups_ F01060	109	■ 1.000	$d, d^{\dagger}$	$d, d^{\dagger}$	5. Construct operators $d, d^{\dagger}$ defined formally from the standard creation and annihilation
gilmore-lie-groups_ F01061	109	■ 0.996	$a, a^{\dagger}$	$a, a^{\dagger}$	operators $a, a^{\dagger}$ as follows:
gilmore-lie-groups_ F01062	110	■ 0.996	$\left[d, d^{\dagger}\right]=1$	$[d, d^{\dagger}] = 1$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01063	110	■ 0.996	$[d, d]=\left[d^{\dagger}, d^{\dagger}\right]=0$	$[d, d] = [d^{\dagger}, d^{\dagger}] = 0$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01064	110	■ 0.996	$A D$	AD	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01065	110	■ 1.000	$B C=1$	$BC = 1$	$BC = 1$.
gilmore-lie-groups_ F01066	110	■ 0.998	$\operatorname{Sp}(2 ; \mathbb{R})=$	$\operatorname{Sp}(2; \mathbb{R}) =$	b. Argue that the commutation relations are invariant under the group $\operatorname{Sp}(2; \mathbb{R}) =$
gilmore-lie-groups_ F01067	110	■ 0.982	$\operatorname{Sp}(2 ; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R})$	c. Show that under $\operatorname{Sp}(2; \mathbb{R})$, linear combinations of the coordinate and differential
gilmore-lie-groups_ F01068	110	■ 1.000	x, ∂	x, ∂	operators x, ∂ preserve the commutation relations. In particular, show that
gilmore-lie-groups_ F01069	110	■ 0.946	$a_{\{1\}}, a_{\{2\}}, \ldots, a_{\{n\}}$	$a_1, a_2, \dots, a_n$	d. Replace a by $(a_1, a_2, \dots, a_n)$ and similarly for $a^{\dagger}$ and their images $d, d^{\dagger}$ under
gilmore-lie-groups_ F01070	110	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{N\}}\right\rangle$	$ n_1, n_2, \dots, n_N\rangle$	and eigenstates $ n_1, n_2, \dots, n_N\rangle$.
gilmore-lie-groups_ F01071	110	■ 1.000	$\hbar \omega(n+$	$\hbar \omega(n+$	a. Show that the degeneracy of the multiplet containing n quanta, with energy $\hbar \omega(n +$
gilmore-lie-groups_ F01072	110	■ 0.531	$\frac{N}{2}$	$\frac{N}{2}$	$\frac{N}{2}$) is $\deg(N, n) = (n + N - 1)!/n!(N - 1)!$. This solution to the Bose–Einstein
gilmore-lie-groups_ F01073	110	■ 0.531	$\operatorname{deg}(N, n)=(n+N-1)! / n !(N-1) !$	$\deg(N, n) = (n + N - 1)!/n!(N - 1)!$	$\frac{N}{2}$) is $\deg(N, n) = (n + N - 1)!/n!(N - 1)!$. This solution to the Bose–Einstein
gilmore-lie-groups_ F01074	110	■ 0.999	$f\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{N\}}\right)$	$f\left(x_1, x_2, \dots, x_N\right)$	Taylor series expansion of a function of N variables: $f(x_1, x_2, \dots, x_N)$.
gilmore-lie-groups_ F01075	110	■ 1.000	$N^{\{2\}}$	N^2	N^2 operators $a_i^{\dagger} a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^{\dagger} a_j] = 0$,
gilmore-lie-groups_ F01076	110	■ 1.000	$\mathfrak{u}(N)$	$\mathfrak{u}(N)$	N^2 operators $a_i^{\dagger} a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^{\dagger} a_j] = 0$,
gilmore-lie-groups_ F01077	110	■ 1.000	$\left[\mathcal{H}, a_{\mathrm{i}}^{\dagger} a_{\mathrm{j}}\right]=0$	$[\mathcal{H}, a_i^{\dagger} a_j] = 0$	N^2 operators $a_i^{\dagger} a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^{\dagger} a_j] = 0$,
gilmore-lie-groups_ F01078	110	■ 1.000	$\mathfrak{s u}(N)$	$\mathfrak{su}(N)$	this algebra is a direct sum of a simple Lie algebra, $\mathfrak{su}(N)$, plus the one-dimensional
gilmore-lie-groups_ F01079	110	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	algebra spanned by $\mathcal{H}$.
gilmore-lie-groups_ F01080	110	■ 1.000	$a_{\{j\}}$	a_j	single creation and annihilation operators $a_i^{\dagger}$ and a_j , as well as their commutator I ,
gilmore-lie-groups_ F01081	110	■ 1.000	$(N+1)^{\{2\}}$	$(N + 1)^2$	the resulting set of operators closes to form an $(N + 1)^2$ dimensional Lie algebra
gilmore-lie-groups_ F01082	110	■ 1.000	$n^{\prime}$	n'	state in a multiplet with n' excitations.
gilmore-lie-groups_ F01083	110	■ 1.000	$R, S, T, U, \ldots$	$R, S, T, U, \dots$	7. The set of matrices $R, S, T, U, \dots$ belong to a Lie algebra of $n \times n$ matrices, $a^{\dagger} =$
gilmore-lie-groups_ F01084	110	■ 1.000	$a^{\dagger}=$	$a^{\dagger} =$	7. The set of matrices $R, S, T, U, \dots$ belong to a Lie algebra of $n \times n$ matrices, $a^{\dagger} =$

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gilmore-lie-groups_F01085	110	■ 0.999	$a_{1}^{\dagger}, a_{2}^{\dagger}, \ldots, a_{n}^{\dagger}$	$a_1^\dagger, a_2^\dagger, \dots, a_n^\dagger$	$(a_1^\dagger, a_2^\dagger, \dots, a_n^\dagger)$ is a row vector of creation operators for n boson modes, and a is its
gilmore-lie-groups_F01086	111	■ 1.000	$\mathcal{R} = a^{\dagger} R$ $a = a_{i}^{\dagger} R_{ij} a_j$	$\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$	adjoint, a column vector of annihilation operators. Define $\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$, and
gilmore-lie-groups_F01087	111	■ 1.000	$\mathcal{S}, T, U, \ldots$	$\mathcal{S}, T, U, \dots$	similarly for $\mathcal{S}, T, U, \dots$.
gilmore-lie-groups_F01088	111	■ 1.000	$[R, S] = T$ $\Leftrightarrow [\mathcal{R}, S] = \mathcal{T}$	$[R, S] = T \Leftrightarrow [\mathcal{R}, S] = \mathcal{T}$	a. $[R, S] = T \Leftrightarrow [\mathcal{R}, S] = \mathcal{T}$
gilmore-lie-groups_F01089	111	■ 1.000	$e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$ $\Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$	$e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$	b. $e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$
gilmore-lie-groups_F01090	111	■ 0.994	$\left[\frac{d}{dx}, e^{-x^2/2} \right] = -x e^{-x^2/2}$	$\left[\frac{d}{dx}, e^{-x^2/2} \right] = -x e^{-x^2/2}$	a. Show $\left[\frac{d}{dx}, e^{-x^2/2} \right] = -x e^{-x^2/2}$.
gilmore-lie-groups_F01091	111	■ 1.000	$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$	$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$	d. Introduce the annihilation operator $a = \frac{1}{\sqrt{2}} (x + \frac{d}{dx})$, define the normalized ground
gilmore-lie-groups_F01092	111	■ 1.000	$\langle x 0 \rangle$	$\langle x 0 \rangle$	state $\langle x 0 \rangle$ by $a \langle x 0 \rangle = 0$. Solve this equation, normalize the solution, and show
gilmore-lie-groups_F01093	111	■ 1.000	$a \langle x 0 \rangle = 0$	$a \langle x 0 \rangle = 0$	state $\langle x 0 \rangle$ by $a \langle x 0 \rangle = 0$. Solve this equation, normalize the solution, and show
gilmore-lie-groups_F01094	111	■ 0.990	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$.
gilmore-lie-groups_F01095	111	■ 1.000	$a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right)$	$a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right)$	e. Introduce the creation operator $a^\dagger = \frac{1}{\sqrt{2}} (x - \frac{d}{dx})$ and show
gilmore-lie-groups_F01096	111	■ 1.000	$\psi_n(x)$	$\psi_n(x)$	where $\psi_n(x)$ is the n th normalized harmonic oscillator eigenstate $\langle x n \rangle =$
gilmore-lie-groups_F01097	111	■ 1.000	$\langle x n \rangle =$	$\langle x n \rangle =$	where $\psi_n(x)$ is the n th normalized harmonic oscillator eigenstate $\langle x n \rangle =$
gilmore-lie-groups_F01098	111	■ 0.999	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x 0 \rangle$	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x 0 \rangle$	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x 0 \rangle$.
gilmore-lie-groups_F01099	111	■ 0.749	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i$	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i$	$(L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i)$ and a quadrupole interaction $(Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i)$.
gilmore-lie-groups_F01100	111	■ 0.749	$Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i$	$Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i$	$(L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i)$ and a quadrupole interaction $(Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i)$.
gilmore-lie-groups_F01101	111	■ 1.000	$i \leq j$	$i \leq j$	for $i \leq j$ the matrix elements are
gilmore-lie-groups_F01102	111	■ 0.999	$\Gamma_{ji}^* = \Gamma_{ij}$	$\Gamma_{ji}^* = \Gamma_{ij}$	with $\Gamma_{ji}^* = \Gamma_{ij}$.
gilmore-lie-groups_F01103	112	■ 0.992	$b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i' \left(b_i^\dagger b_i + \frac{1}{2} \right) +$	$b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i' \left(b_i^\dagger b_i + \frac{1}{2} \right) +$	annihilation operators $b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i' (b_i^\dagger b_i + \frac{1}{2}) +$ constant. Express
gilmore-lie-groups_F01104	112	■ 1.000	$\Gamma(H)$	$\Gamma(H)$	the amplitudes m_{ij} in terms of the eigenvectors of $\Gamma(H)$.
gilmore-lie-groups_F01105	114	■ 1.000	$\operatorname{EXP}(X)$	$\operatorname{EXP}(X)$	Lie algebra onto the entire Lie group through a single mapping of the form $\operatorname{EXP}(X)$.
gilmore-lie-groups_F01106	114	■ 1.000	$\operatorname{Tr} X = 0$	$\operatorname{Tr} X = 0$	Now choose S to diagonalize (7.2). Since $\operatorname{Tr} X = 0$, the eigenvalues λ can only
gilmore-lie-groups_F01107	114	■ 1.000	$\pm \theta$	$\pm \theta$	have the form $\pm \theta$ or $\pm i \theta$ (θ real)
gilmore-lie-groups_F01108	114	■ 1.000	$\pm i \theta$	$\pm i \theta$	have the form $\pm \theta$ or $\pm i \theta$ (θ real)
gilmore-lie-groups_F01109	116	■ 0.997	$a = b = 0$	$a = b = 0$	To see this most easily set $a = b = 0$. Points on the c -axis map onto
gilmore-lie-groups_F01110	116	■ 1.000	$2\pi n$	$2\pi n$	and therefore points separated by $2\pi n$ along the c -axis map onto the same group

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gilmore-lie-groups_ F01111	116	0.651	$S O(2) \subset S L(2 ; R)$	$SO(2) \subset SL(2; R)$	operation in $SO(2) \subset SL(2; R)$. The complementary subspace $(a, b, 0)$ maps onto
gilmore-lie-groups_ F01112	116	0.651	$a, b, 0$	$a, b, 0$	operation in $SO(2) \subset SL(2; R)$. The complementary subspace $(a, b, 0)$ maps onto
gilmore-lie-groups_ F01113	116	1.000	$z=\cosh r \geq 1$	$z = \cosh r \geq 1$	For simplicity we have set $z = \cosh r \geq 1$ and $(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$
gilmore-lie-groups_ F01114	116	1.000	$(x, y)=(b, a) \sinh (r) / r, r^2=a^2+$	$(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$	For simplicity we have set $z = \cosh r \geq 1$ and $(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$
gilmore-lie-groups_ F01115	116	1.000	b^2	b^2	b^2 . We observe that
gilmore-lie-groups_ F01116	116	1.000	H_{2+}^2	H_{2+}^2	which is just the upper sheet of the two-sheeted hyperboloid H_{2+}^2 , shown in
gilmore-lie-groups_ F01119	117	1.000	H_1^2	H_1^2	This equation describes the surface of the single-sheeted hyperboloid H_1^2 , shown
gilmore-lie-groups_ F01120	118	1.000	$c, 0 \leq c < 2 \pi$	$c, 0 \leq c < 2 \pi$	set parameterized by $c, 0 \leq c < 2 \pi$, is the coset. In the theory of Lie groups, cosets
gilmore-lie-groups_ F01121	118	0.999	$S L(2 ; \mathbb{R}) / S O(2)$	$SL(2; \mathbb{R})/SO(2)$	obtained from the coset decomposition $SL(2; \mathbb{R})/SO(2)$, the manifold parameteriz-
gilmore-lie-groups_ F01122	118	0.996	$R^2 \times S^1$	$R^2 \times S^1$	is the direct product $R^2 \times S^1$. A different parameterization (7.13) based on the coset
gilmore-lie-groups_ F01123	118	1.000	$[S L(2 ; \mathbb{R}) / S O(1,1)] \times S O(1,1) \left(S O(1,1) \simeq R^1\right)$	$[SL(2; \mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$	decomposition $[SL(2; \mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$ shows that the
gilmore-lie-groups_ F01124	118	1.000	$R^1 \times S^1$	$R^1 \times S^1$	boloid (equivalent to $R^1 \times S^1$) with R^1 . This product is once again $R^2 \times S^1$.
gilmore-lie-groups_ F01125	118	1.000	$R^N, N=n^2$	$R^N, N = n^2$	the underlying manifold is a subset of $R^N, N = n^2$, that is defined by algebraic
gilmore-lie-groups_ F01126	119	1.000	R^m	R^m	generators is R^m , for suitable m (m is the number of noncompact generators). The
gilmore-lie-groups_ F01127	119	0.943	$S O(2,1)$	$SO(2, 1)$	the groups $SO(2, 1)$ and $SU(1, 1)$ with Lie algebras
gilmore-lie-groups_ F01128	119	0.993	$S O(2,1) / S O(2)$	$SO(2, 1)/SO(2)$	The cosets $SO(2, 1)/SO(2)$ and $SU(1, 1)/U(1)$ are both isomorphic to R^2 and have
gilmore-lie-groups_ F01129	119	0.993	$S U(1,1) / U(1)$	$SU(1, 1)/U(1)$	The cosets $SO(2, 1)/SO(2)$ and $SU(1, 1)/U(1)$ are both isomorphic to R^2 and have
gilmore-lie-groups_ F01130	119	0.969	$U(1)$	$U(1)$	a 1:1 correspondence. The subgroups $SO(2)$ and $U(1)$ have a 2:1 correspondence.
gilmore-lie-groups_ F01131	119	0.969	$2: 1$	$2 : 1$	a 1:1 correspondence. The subgroups $SO(2)$ and $U(1)$ have a 2:1 correspondence.
gilmore-lie-groups_ F01132	119	0.936	$b_{\{3\}}$	b_3	This can be seen by increasing b_3 by 2π and noticing that the 2×2 unitary matrix
gilmore-lie-groups_ F01133	119	1.000	$U \left(b_{\{3\}}+2 \pi\right)=-U \left(b_{\{3\}}\right)$	$U\left(b_3+2 \pi\right)=-U\left(b_3\right)$	in (7.19) goes to its negative: $U\left(b_3+2 \pi\right)=-U\left(b_3\right)$. However, increasing a_3 by
gilmore-lie-groups_ F01134	119	1.000	$a_{\{3\}}$	a_3	in (7.19) goes to its negative: $U\left(b_3+2 \pi\right)=-U\left(b_3\right)$. However, increasing a_3 by
gilmore-lie-groups_ F01135	120	0.984	$U(1) \subset S U(1,1)$	$U(1) \subset SU(1, 1)$	$U(1) \subset SU(1, 1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2, 1)$ the result is
gilmore-lie-groups_ F01136	120	0.984	4π	4π	$U(1) \subset SU(1, 1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2, 1)$ the result is
gilmore-lie-groups_ F01137	120	0.984	$S O(2) \subset S O(2,1)$	$SO(2) \subset SO(2, 1)$	$U(1) \subset SU(1, 1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2, 1)$ the result is
gilmore-lie-groups_ F01138	120	0.811	$\operatorname{SO}(2,1)$	$SO(2, 1)$	2π . Therefore, $SU(1, 1)$ is “twice as large” as $SO(2, 1)$. More formally, there is a
gilmore-lie-groups_ F01139	120	0.992	$2 \rightarrow 1$	$2 \rightarrow 1$	$2 \rightarrow 1$ homomorphism of $SU(1, 1)$ onto $SO(2, 1)$.

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gilmore-lie-groups_F01140	120	■ 1.000	$\sqrt{a_1^2+a_2^2+a_3^2} \leq$	$\sqrt{a_1^2+a_2^2+a_3^2} \leq$	It can be checked that all points in the interior of a sphere of radius $\sqrt{a_1^2+a_2^2+a_3^2} \leq$
gilmore-lie-groups_F01141	120	■ 0.994	$ \mathbf{a} =\pi$	$ \mathbf{a} = \pi$	π) map onto $SO(3)$ provided antipodal points at $ \mathbf{a} = \pi$ are identified
gilmore-lie-groups_F01142	120	■ 0.999	$\operatorname{SU}(2)$	$SU(2)$	with θ the latitude, and ϕ the longitude on a sphere. For $SU(2)$ all points within a
gilmore-lie-groups_F01143	120	■ 0.998	$2\pi\left(\sqrt{b_1^2+b_2^2+b_3^2}<2\pi\right)$	$2\pi\left(\sqrt{b_1^2+b_2^2+b_3^2}<2\pi\right)$	sphere of radius 2π ($\sqrt{b_1^2+b_2^2+b_3^2}<2\pi$) are mapped onto distinct elements of
gilmore-lie-groups_F01144	120	■ 0.986	$-I_2$	$-I_2$	$SU(2)$ and all points at a radius of 2π are mapped onto $-I_2$. There is an easier way
gilmore-lie-groups_F01145	121	■ 0.652	$x^2+y^2+z^2=1$	$x^2+y^2+z^2=1$	(x,y,z) subject to $x^2+y^2+z^2=1$, and $SU(2)/U(1)$, parameterized by the real
gilmore-lie-groups_F01146	121	■ 0.652	$SU(2)/U(1)$	$SU(2)/U(1)$	(x,y,z) subject to $x^2+y^2+z^2=1$, and $SU(2)/U(1)$, parameterized by the real
gilmore-lie-groups_F01147	121	■ 0.989	x',y',z'	x',y',z'	numbers (x',y',z') subject to $x'^2+y'^2+z'^2=1$, are in 1:1 correspondence with
gilmore-lie-groups_F01148	121	■ 0.989	$x'^2+y'^2+z'^2=1$	$x'^2+y'^2+z'^2=1$	numbers (x',y',z') subject to $x'^2+y'^2+z'^2=1$, are in 1:1 correspondence with
gilmore-lie-groups_F01149	121	■ 0.604	$SU(2)\rightarrow SO(3)$	$SU(2)\rightarrow SO(3)$	of the mapping $SU(2)\rightarrow SO(3)$ can be seen from the 2:1 nature of the rotations
gilmore-lie-groups_F01150	121	■ 1.000	$\bar{G}$	$\bar{G}$	group, $\bar{G}$, called the universal covering group . This group is simply connected:
gilmore-lie-groups_F01151	121	■ 1.000	$\bar{G}/D$	$\bar{G}/D$	this simply connected Lie group, or else has the form of a quotient $\bar{G}/D$, where D
gilmore-lie-groups_F01152	121	■ 1.000	$\bar{G}:gd_i=d_ig$	$\bar{G}:gd_i=d_ig$	is a discrete invariant subgroup of $\bar{G}$ whose elements commute with $\bar{G}$: $gd_i=d_ig$
gilmore-lie-groups_F01153	121	■ 1.000	$d_i\in D$	$d_i\in D$	for $d_i\in D$ and $g\in G$. If $\bar{G}$ is compact it is useful to determine the largest such
gilmore-lie-groups_F01154	121	■ 0.863	D_{MAX}	D_{MAX}	subgroup, D_{MAX} , of $\bar{G}$. Then all compact Lie groups with the same Lie algebra as
gilmore-lie-groups_F01155	121	■ 0.979	D_{MAX}	D_{MAX}	$\bar{G}$ are obtained by “dividing” $\bar{G}$ by all possible subgroups of D_{MAX} , as shown in
gilmore-lie-groups_F01157	121	■ 1.000	$G_1=\bar{G}/D_1$	$G_1=\bar{G}/D_1$	G_2 have the same Lie algebra, $G_1=\bar{G}/D_1$ and G_1 is locally isomorphic with $\bar{G}$.
gilmore-lie-groups_F01158	121	■ 0.848	$\bar{G}/D_{\mathrm{MAX}}$	$\bar{G}/D_{\mathrm{MAX}}$	G_1 . If $\bar{G}$ is compact, G_1 and G_2 are also locally isomorphic with $\bar{G}/D_{\mathrm{MAX}}$, which
gilmore-lie-groups_F01161	122	■ 1.000	λI_2	λI_2	λI_2 that obey $\lambda^*\lambda=1$ and $\det(\lambda I_2)=+1$, so that $\lambda=\pm 1$. D is the two-element
gilmore-lie-groups_F01162	122	■ 1.000	$\lambda^*\lambda=1$	$\lambda^*\lambda=1$	λI_2 that obey $\lambda^*\lambda=1$ and $\det(\lambda I_2)=+1$, so that $\lambda=\pm 1$. D is the two-element
gilmore-lie-groups_F01163	122	■ 1.000	$\det(\lambda I_2)=+1$	$\det(\lambda I_2)=+1$	λI_2 that obey $\lambda^*\lambda=1$ and $\det(\lambda I_2)=+1$, so that $\lambda=\pm 1$. D is the two-element
gilmore-lie-groups_F01164	122	■ 1.000	$\lambda=\pm 1.D$	$\lambda=\pm 1.D$	λI_2 that obey $\lambda^*\lambda=1$ and $\det(\lambda I_2)=+1$, so that $\lambda=\pm 1$. D is the two-element
gilmore-lie-groups_F01165	122	■ 1.000	$D=\{I_2,-I_2\}$	$D=\{I_2,-I_2\}$	subgroup $D=\{I_2,-I_2\}$. For the locally isomorphic Lie group $SO(3)$, $D=\lambda I_3$
gilmore-lie-groups_F01166	122	■ 1.000	$SO(3), D=\lambda I_3$	$SO(3), D=\lambda I_3$	subgroup $D=\{I_2,-I_2\}$. For the locally isomorphic Lie group $SO(3)$, $D=\lambda I_3$
gilmore-lie-groups_F01167	122	■ 0.874	$\lambda=+1$	$\lambda=+1$	with $\lambda=+1$. As a result $SU(2)/\{I_2,-I_2\}=SO(3)/I_3=SO(3)$. For each group
gilmore-lie-groups_F01168	122	■ 0.874	$SU(2)/\{I_2,-I_2\}=SO(3)/I_3=SO(3)$	$SU(2)/\{I_2,-I_2\}=SO(3)/I_3=SO(3)$	with $\lambda=+1$. As a result $SU(2)/\{I_2,-I_2\}=SO(3)/I_3=SO(3)$. For each group

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gilmore-lie-groups_ F01169	122	■ 1.000	$\overline{\{SO(2,1)\}}$	$\overline{SO(2,1)} =$	is R^1 , the group of translations of the line. The covering group $\overline{SO(2,1)} =$
gilmore-lie-groups_ F01170	122	■ 1.000	$\overline{\{SU(1,1)\}}$	$\overline{SU(1,1)}$	$\overline{SU(1,1)}$ has no compact subgroup at all. Its underlying group manifold
gilmore-lie-groups_ F01171	122	■ 1.000	$\overline{\{SO(2,1)/SO(2)\}} \times \overline{\{SO(2)\}} = \overline{\{SU(1,1)/U(1)\}} \times \overline{\{U(1)\}} = [SO(2,1)/SO(2)] \times$ $\overline{\{U(1)\}} = [SO(2,1)/SO(2)] \times$	$\overline{SO(2,1)/SO(2)} \times \overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = [SO(2,1)/SO(2)] \times$	is $\boxed{SO(2,1)/SO(2) \times SO(2) = SU(1,1)/U(1) \times U(1) = [SO(2,1)/SO(2)] \times}$
gilmore-lie-groups_ F01172	122	■ 1.000	$\overline{\{SO(2)\}} = \overline{\{SU(1,1)/U(1)\}} \times \overline{\{U(1)\}} = R^2 \times R^1$	$\overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = R^2 \times R^1$	$\boxed{SO(2) = SU(1,1)/U(1) \times U(1) = R^2 \times R^1}$. It is the only group we will encounter
gilmore-lie-groups_ F01173	122	■ 1.000	$\overline{\{SO(2,1)\}} = \overline{\{SU(1,1)\}}$	$\overline{SO(2,1)} = \overline{SU(1,1)}$	in this book that is not a matrix group. The covering group $\overline{SO(2,1)} = \overline{SU(1,1)}$ has
gilmore-lie-groups_ F01174	123	■ 1.000	R_+^2	R_+^2	right half-plane R_+^2 into the group operator
gilmore-lie-groups_ F01175	123	■ 0.974	(w, z)	(w, z)	The second maps a point (w, z) in R^2 into the group under the EXPonential map
gilmore-lie-groups_ F01176	123	■ 1.000	$R_+^2(x > 0, y)$	$R_+^2(x > 0, y)$	We ask: is there some mapping of the half-plane $R_+^2(x > 0, y)$ into $R^2(w, z)$
gilmore-lie-groups_ F01177	123	■ 1.000	$R^2(w, z)$	$R^2(w, z)$	We ask: is there some mapping of the half-plane $R_+^2(x > 0, y)$ into $R^2(w, z)$
gilmore-lie-groups_ F01178	123	■ 0.995	$x > 0$	$x > 0$	These transformations are analytic for $x > 0$.
gilmore-lie-groups_ F01179	124	■ 0.981	$a, a^\dagger, I$	$a, a^\dagger, I$	The single-mode photon operators $a, a^\dagger, I$ obey isomorphic commutation relations
gilmore-lie-groups_ F01180	124	■ 1.000	$l = l' = l'', r = r' = r'', \delta' = \delta + \frac{1}{2}lr = \delta'' + l''r''$	$l = l' = l'', r = r' = r'', \delta' = \delta + \frac{1}{2}lr = \delta'' + l''r''$	We see immediately that $\boxed{= l' = l'', r = r' = r'', \delta' = \delta + \frac{1}{2}lr = \delta'' + l''r''}$, and
gilmore-lie-groups_ F01181	124	■ 1.000	$\delta = 0$	$\delta = 0$	obtain the Heisenberg identity (for $\delta = 0$)
gilmore-lie-groups_ F01182	125	■ 1.000	$\hat{n} = a^\dagger a, a, a^\dagger, I$	$\hat{n} = a^\dagger a, a, a^\dagger, I$	photon operators $\boxed{\hat{n} = a^\dagger a, a, a^\dagger, I}$ under the identification
gilmore-lie-groups_ F01183	125	■ 1.000	$\text{EXP}(\eta a^\dagger a + r a^\dagger + l a)$	$\text{EXP}(\eta a^\dagger a + r a^\dagger + l a)$	If for some reason $\text{EXP}(\eta a^\dagger a + r a^\dagger + l a)$ needed to be rewritten in the more con-
gilmore-lie-groups_ F01184	125	■ 1.000	$\text{EXP}(r' a^\dagger) \text{EXP}(\eta' a^\dagger a + \delta' I) \text{EXP}(l' a)$	$\text{EXP}(r' a^\dagger) \text{EXP}(\eta' a^\dagger a + \delta' I) \text{EXP}(l' a)$	veniently ordered form $\text{EXP}(r' a^\dagger) \text{EXP}(\eta' a^\dagger a + \delta' I) \text{EXP}(l' a)$, then the reparam-
gilmore-lie-groups_ F01185	125	■ 0.948	$e^{l' a^\dagger} 0\rangle = 0\rangle, \langle 0 e^{r' a^\dagger} = \langle 0 $	$e^{l' a^\dagger} 0\rangle = 0\rangle, \langle 0 e^{r' a^\dagger} = \langle 0 $	Since $e^{l' a^\dagger} 0\rangle = 0\rangle, \langle 0 e^{r' a^\dagger} = \langle 0 $ and $e^{\eta' a^\dagger a} 0\rangle = 0\rangle$, the expectation value is
gilmore-lie-groups_ F01186	125	■ 0.948	$e^{\eta' a^\dagger a} 0\rangle = 0\rangle, \langle 0 e^{r' a^\dagger} = \langle 0 $	$e^{\eta' a^\dagger a} 0\rangle = 0\rangle, \langle 0 e^{r' a^\dagger} = \langle 0 $	Since $e^{l' a^\dagger} 0\rangle = 0\rangle, \langle 0 e^{r' a^\dagger} = \langle 0 $ and $e^{\eta' a^\dagger a} 0\rangle = 0\rangle$, the expectation value is
gilmore-lie-groups_ F01187	125	■ 1.000	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	As a final example we treat the Lie algebra $\mathfrak{su}(2)$. First, we show how to compute
gilmore-lie-groups_ F01188	125	■ 1.000	$(j, -j\rangle)$	$(j, -j\rangle)$	$(j, -j\rangle)$
gilmore-lie-groups_ F01189	126	■ 1.000	θ_z'	θ_z'	The only problem that remains is to compute θ_z' as a function of θ . To do this
gilmore-lie-groups_ F01190	126	■ 1.000	$\mathbf{J} \rightarrow \frac{1}{2}\sigma$	$\mathbf{J} \rightarrow \frac{1}{2}\sigma$	representation $\mathbf{J} \rightarrow \frac{1}{2}\sigma$, where σ are the Pauli spin matrices (5.14):
gilmore-lie-groups_ F01191	126	■ 1.000	$\theta_\pm = \theta_1 \pm i\theta_2$	$\theta_\pm = \theta_1 \pm i\theta_2$	where $\boxed{\theta_\pm = \theta_1 \pm i\theta_2}$. Comparison of the two matrices gives immediately
gilmore-lie-groups_ F01192	127	■ 1.000	$e^{\alpha J_+} e^{\beta J_-}$	$e^{\alpha J_+} e^{\beta J_-}$	The operator product $e^{\alpha J_+} e^{\beta J_-}$ is written in normally ordered form $\text{EXP}(\beta' J_-)$

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gilmore-lie-groups_ F01193	127	■ 1.000	$\operatorname{EXP}\left(\beta^{\prime}\right) J_{-}\right)$	$\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$	The operator product $e^{\alpha J_{+}} e^{\beta J_{-}}$ is written in normally ordered form $\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$
gilmore-lie-groups_ F01194	127	■ 0.999	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime}\right) J_{+}\right)$ $\operatorname{EXP}\left(\alpha^{\prime}\right) J_{+}\right)$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$ and the parameters $\alpha^{\prime}, \beta^{\prime}, n^{\prime}$ computed. We find
gilmore-lie-groups_ F01195	127	■ 0.999	$\alpha^{\prime}, \beta^{\prime}, n^{\prime}$	$\alpha^{\prime}, \beta^{\prime}, n^{\prime}$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$ and the parameters $\alpha^{\prime}, \beta^{\prime}, n^{\prime}$ computed. We find
gilmore-lie-groups_ F01196	127	■ 1.000	$(1+\alpha \beta)^{2 j}$	$(1+\alpha \beta)^{2 j}$	By expanding $(1+\alpha \beta)^{2 j}$ and invoking analyticity, we find
gilmore-lie-groups_ F01197	127	■ 1.000	$\mathcal{A}, \mathcal{B}$	$\mathcal{A}, \mathcal{B}$	For example, if operators $\mathcal{A}, \mathcal{B}$ belong to two complementary subspaces in some
gilmore-lie-groups_ F01198	127	■ 0.995	$e^{\mathcal{A}} e^{\mathcal{B}}$	$e^{\mathcal{A}} e^{\mathcal{B}}$	operator Lie algebra $\mathfrak{g}$ then the operator product $e^{\mathcal{A}} e^{\mathcal{B}}$ can be reparameterized as
gilmore-lie-groups_ F01199	127	■ 0.999	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}\left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime}\right)$ $\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}\left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime}\right)$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}\left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime}\right)$ different operators in the same subspaces as $\mathcal{A}, \mathcal{B}$ by
gilmore-lie-groups_ F01200	127	■ 0.999	$\left.\mathcal{A}, \mathcal{B}\right)$	$\mathcal{A}, \mathcal{B})$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}\left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime}\right)$ different operators in the same subspaces as $\mathcal{A}, \mathcal{B}$ by
gilmore-lie-groups_ F01201	127	■ 1.000	$e^{\mathcal{A}} e^{\mathcal{B}}$	$e^{\mathcal{A}} e^{\mathcal{B}}$	(iii) Carrying out the matrix calculations $e^{\mathcal{A}} e^{\mathcal{B}}$ and $e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$,
gilmore-lie-groups_ F01202	127	■ 1.000	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$	(iii) Carrying out the matrix calculations $e^{\mathcal{A}} e^{\mathcal{B}}$ and $e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$,
gilmore-lie-groups_ F01203	127	■ 1.000	$\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$	$\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$	(iv) determining the matrices $\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$ by comparing matrix elements; and
gilmore-lie-groups_ F01204	127	■ 1.000	$\mathcal{A}^{\prime} \leftrightarrow \mathcal{A}^{\prime} \mathcal{B}^{\prime} \leftrightarrow \mathcal{B}^{\prime}$	$\mathcal{A}^{\prime} \leftrightarrow \mathcal{A}^{\prime} \mathcal{B}^{\prime} \leftrightarrow \mathcal{B}^{\prime}$	(v) using the isomorphism $\mathcal{A}^{\prime} \leftrightarrow \mathcal{A}^{\prime} \mathcal{B}^{\prime} \leftrightarrow \mathcal{B}^{\prime}$.
gilmore-lie-groups_ F01205	127	■ 1.000	$(\mathcal{A}, \mathcal{B}) \leftrightarrow$	$(\mathcal{A}, \mathcal{B}) \leftrightarrow$	This procedure will produce a local analytic reparameterization $(\mathcal{A}, \mathcal{B}) \leftrightarrow$
gilmore-lie-groups_ F01206	127	■ 0.998	$(\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$	$(\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$	$(\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$. If the matrix group used to construct this reparameterization is simply
gilmore-lie-groups_ F01207	127	■ 1.000	$\mathcal{A}, \mathcal{B}, \ldots$	$\mathcal{A}, \mathcal{B}, \ldots$	groups of the operator group and the matrix group. When the operators $\mathcal{A}, \mathcal{B}, \ldots$ are
gilmore-lie-groups_ F01208	128	■ 1.000	$A, B, \ldots$	$A, B, \ldots$	related to matrices $A, B, \ldots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A} \leftrightarrow A$,
gilmore-lie-groups_ F01209	128	■ 1.000	$\mathcal{A} \leftrightarrow A$	$\mathcal{A} \leftrightarrow A$	related to matrices $A, B, \ldots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A} \leftrightarrow A$,
gilmore-lie-groups_ F01210	128	■ 1.000	$t+\delta t$	$t+\delta t$	The state of the system at time $t+\delta t$ is related to the state at time t by
gilmore-lie-groups_ F01211	128	■ 1.000	$ \psi\left(t_f\right)\rangle$	$ \psi\left(t_f\right)\rangle$	$ \psi\left(t_f\right)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi\left(t_f\right)\rangle =$
gilmore-lie-groups_ F01212	128	■ 1.000	t_f	t_f	$ \psi\left(t_f\right)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi\left(t_f\right)\rangle =$
gilmore-lie-groups_ F01213	128	■ 1.000	$ \psi\left(t_f\right)\rangle =$	$ \psi\left(t_f\right)\rangle =$	$ \psi\left(t_f\right)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi\left(t_f\right)\rangle =$
gilmore-lie-groups_ F01214	128	■ 0.996	$U\left(t_f, t_i\right) \psi\left(t_i\right)\rangle$	$U\left(t_f, t_i\right) \psi\left(t_i\right)\rangle$	$U\left(t_f, t_i\right) \psi\left(t_i\right)\rangle$. The finite time unitary operator is built up from small dis-
gilmore-lie-groups_ F01215	128	■ 1.000	$H\left(t^{\prime}\right)$	$H\left(t^{\prime}\right)$	Care must be taken with the formal integration in this equation, as in general $H\left(t^{\prime}\right)$
gilmore-lie-groups_ F01216	128	■ 1.000	$H(t), t^{\prime} \neq t$	$H(t), t^{\prime} \neq t$	does not commute with $H(t)$, $t^{\prime} \neq t$. It is for this reason that the symbol “ T ”
gilmore-lie-groups_ F01217	129	■ 1.000	$2 j+1$	$2 j+1$	Example 1. A Hamiltonian acts in a $2 j+1$ dimensional space through a set of
gilmore-lie-groups_ F01218	129	■ 1.000	$ j, m_j\rangle$	$ j, m_j\rangle$	wish to determine the evolution of some particular state $ j, m_j\rangle$. The Hamiltonian

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gilmore-lie-groups_ F01219	129	■ 1.000	$g(t) \in SU(2)$	$g(t) \in SU(2)$	$g(t) \in SU(2)$ evolves, and then construct its unitary representation, than it is to de-
gilmore-lie-groups_ F01220	129	■ 1.000	$(2j+1) \times (2j+1)$	$(2j+1) \times (2j+1)$	termine the time evolution of the $(2j+1) \times (2j+1)$ unitary matrix. Specifically,
gilmore-lie-groups_ F01221	129	■ 1.000	$a(t)$	$a(t)$	the complex coefficients $a(t)$ and $b(t)$ or three equations for the real coefficients of
gilmore-lie-groups_ F01222	129	■ 1.000	$b(t)$	$b(t)$	the complex coefficients $a(t)$ and $b(t)$ or three equations for the real coefficients of
gilmore-lie-groups_ F01223	129	■ 1.000	$\sigma_1, \sigma_2, \sigma_3$	$\sigma_1, \sigma_2, \sigma_3$	the Pauli spin matrices $\sigma_1, \sigma_2, \sigma_3$. These are first order equations and can be solved
gilmore-lie-groups_ F01224	129	■ 1.000	$a(t_i) =$	$a(t_i) =$	by standard integration methods (e.g., RK4). The initial conditions are $a(t_i) =$
gilmore-lie-groups_ F01225	129	■ 0.999	$1, b(t_i) = 0$	$1, b(t_i) = 0$	$1, b(t_i) = 0$. The final 2×2 unitary matrix is determined by $a(t_f), b(t_f)$. This is
gilmore-lie-groups_ F01226	129	■ 0.999	$a(t_f), b(t_f)$	$a(t_f), b(t_f)$	$1, b(t_i) = 0$. The final 2×2 unitary matrix is determined by $a(t_f), b(t_f)$. This is
gilmore-lie-groups_ F01227	129	■ 0.971	$(2j+1) \times$	$(2j+1) \times$	a group operation in $SU(2)$ that can subsequently be mapped into the $(2j+1) \times$
gilmore-lie-groups_ F01228	129	■ 1.000	$(2j+1)$	$(2j+1)$	$(2j+1)$ unitary irreducible representation of this group. At this point the problem
gilmore-lie-groups_ F01229	129	■ 1.000	$ \psi(t_i)\rangle$	$ \psi(t_i)\rangle$	is solved, independent of the initial state $ \psi(t_i)\rangle$.
gilmore-lie-groups_ F01230	130	■ 1.000	$ n\rangle, n=0,1,2, \dots$	$ n\rangle, n=0,1,2, \dots$	space with basis vectors $ n\rangle, n=0,1,2, \dots$. The matrix on the right is a faith-
gilmore-lie-groups_ F01231	130	■ 1.000	$U =$	$U =$	general unitary operator that can be constructed from these operators is $U =$
gilmore-lie-groups_ F01232	130	■ 1.000	$\text{EXP}(i[n(t)a^\dagger a + r(t)a^\dagger + r^*(t)a + d(t)I])$	$\text{EXP}(i[n(t)a^\dagger a + r(t)a^\dagger + r^*(t)a + d(t)I])$	$\text{EXP}(i[n(t)a^\dagger a + r(t)a^\dagger + r^*(t)a + d(t)I])$. This exponential is easy to compute
gilmore-lie-groups_ F01233	130	■ 0.982	(n, r, r^*, d)	(n, r, r^*, d)	tions in four real variables (n, r, r^*, d) that can be integrated by standard methods.
gilmore-lie-groups_ F01234	130	■ 1.000	$d\omega(t)/dt = 0$	$d\omega(t)/dt = 0$	In the case that $d\omega(t)/dt = 0$ the equations simplify considerably, and can almost
gilmore-lie-groups_ F01235	130	■ 1.000	ρ	ρ	density operator ρ
gilmore-lie-groups_ F01236	130	■ 1.000	$\rho = e^{-\beta H}/Z$	$\rho = e^{-\beta H}/Z$	hamiltonian describing the system as $\rho = e^{-\beta H}/Z$, where the normalization con-
gilmore-lie-groups_ F01237	130	■ 1.000	$Z = \text{tr} e^{-\beta H}$	$Z = \text{tr} e^{-\beta H}$	stant, or partition function, is $Z = \text{tr} e^{-\beta H}$ and $\beta = 1/k_B T$, k_B is the Boltzmann
gilmore-lie-groups_ F01238	130	■ 1.000	$\beta = 1/k_B T, k_B$	$\beta = 1/k_B T, k_B$	stant, or partition function, is $Z = \text{tr} e^{-\beta H}$ and $\beta = 1/k_B T$, k_B is the Boltzmann
gilmore-lie-groups_ F01239	131	■ 1.000	$\langle e^\Lambda \rangle$	$\langle e^\Lambda \rangle$	generating functions for these expectation values. To do this we compute $\langle e^\Lambda \rangle$,
gilmore-lie-groups_ F01240	131	■ 1.000	$\Lambda = \lambda \cdot \mathbf{J}$	$\Lambda = \lambda \cdot \mathbf{J}$	where $\Lambda = \lambda \cdot \mathbf{J}$. All symmetric moments can be constructed by taking derivatives
gilmore-lie-groups_ F01241	131	■ 1.000	$H \cdot \Lambda = (H, \Lambda) = \frac{1}{2} \text{tr} H \Lambda$	$H \cdot \Lambda = (H, \Lambda) = \frac{1}{2} \text{tr} H \Lambda$	In these expressions $H \cdot \Lambda = (H, \Lambda) = \frac{1}{2} \text{tr} H \Lambda$, and similarly for $ H = \sqrt{(H, H)}$
gilmore-lie-groups_ F01242	131	■ 1.000	$ H = \sqrt{(H, H)}$	$ H = \sqrt{(H, H)}$	In these expressions $H \cdot \Lambda = (H, \Lambda) = \frac{1}{2} \text{tr} H \Lambda$, and similarly for $ H = \sqrt{(H, H)}$
gilmore-lie-groups_ F01243	131	■ 1.000	$ \Lambda = \sqrt{(\Lambda, \Lambda)}$	$ \Lambda = \sqrt{(\Lambda, \Lambda)}$	and $ \Lambda = \sqrt{(\Lambda, \Lambda)}$.
gilmore-lie-groups_ F01244	131	■ 1.000	2^N	2^N	If N two-level atoms are acting incoherently, the trace over the 2^N states of all

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gilmore-lie-groups_F01245	131	■ 1.000	$2J+1$	$2J+1$	N atoms are acting coherently, there are $2J+1$ states, where $N=2J$. The trace
gilmore-lie-groups_F01246	131	■ 1.000	$N=2J$	$N=2J$	N atoms are acting coherently, there are $2J+1$ states, where $N=2J$. The trace
gilmore-lie-groups_F01247	131	■ 1.000	$\mu(H, \Lambda, T)$	$\mu(H, \Lambda, T)$	where $\mu(H, \Lambda, T)$ is determined from Eq. (7.65). The thermodynamic generating
gilmore-lie-groups_F01248	131	■ 1.000	$\langle J_- \rangle$	$\langle J_- \rangle$	To construct explicit expectation values (e.g., $\langle J_- \rangle$) it is sufficient to differenti-
gilmore-lie-groups_F01249	131	■ 1.000	$\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$	$\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$	ate the generating function (e.g., $\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$) and evaluate the result at $\Lambda=0$.
gilmore-lie-groups_F01250	131	■ 1.000	$\Lambda=0$	$\Lambda=0$	ate the generating function (e.g., $\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$) and evaluate the result at $\Lambda=0$.
gilmore-lie-groups_F01251	131	■ 1.000	$\left. \frac{\partial}{\partial \lambda^*} \log(\langle e^\Lambda \rangle) \right _{\Lambda=0}$	$\left. \frac{\partial}{\partial \lambda^*} \log(\langle e^\Lambda \rangle) \right _{\Lambda=0}$	$\frac{\partial}{\partial \lambda^*} \log(\langle e^\Lambda \rangle) _{\Lambda=0}$.
gilmore-lie-groups_F01252	132	■ 0.999	$\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	$\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	The density operator is $\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$. The generating function for
gilmore-lie-groups_F01253	132	■ 1.000	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$	operator expectation values is $\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$.
gilmore-lie-groups_F01254	132	■ 0.967	$ 0\rangle, 1\rangle, 2\rangle, \dots$	$ 0\rangle, 1\rangle, 2\rangle, \dots$	$ 0\rangle, 1\rangle, 2\rangle, \dots$. It would be insane to attempt to compute this expectation value
gilmore-lie-groups_F01255	132	■ 0.989	H, Λ	H, Λ	(i) Write each of the operators H, Λ in the 3×3 matrix representation M (cf., Eq.
gilmore-lie-groups_F01256	132	■ 0.999	Z_l	Z_l	(iv) Find a similarity transformation, S , that zeroes out Z_l and Z_r :
gilmore-lie-groups_F01257	132	■ 0.999	Z_r	Z_r	(iv) Find a similarity transformation, S , that zeroes out Z_l and Z_r :
gilmore-lie-groups_F01258	133	■ 1.000	$A < 0$	$A < 0$	(vi) Take the trace. Assuming $A < 0$ the sum converges to
gilmore-lie-groups_F01259	133	■ 1.000	$\lambda_n =$	$\lambda_n =$	The generating function for only the creation and annihilation operators ($\lambda_n =$
gilmore-lie-groups_F01260	133	■ 1.000	$d=0$	$d=0$	$d=0$) is considerably simpler.
gilmore-lie-groups_F01261	133	■ 1.000	$D = \text{Id}, G$	$D = \text{Id}, G$	D is a discrete invariant subgroup of $\overline{G}$. If $\overline{D} = \text{Id}, \overline{G}$ is isomorphic to $\overline{G}$, otherwise
gilmore-lie-groups_F01262	133	■ 1.000	λI_n	λI_n	of the identity matrix, λI_n , and is simple to compute. If G_1 and G_2 have isomorphic
gilmore-lie-groups_F01263	134	■ 0.940	$it/\hbar \leftrightarrow 1/k_B T$	$it/\hbar \leftrightarrow 1/k_B T$	(dynamics and statics) are related by a “Wick rotation”: $it/\hbar \leftrightarrow 1/k_B T$. We have
gilmore-lie-groups_F01264	134	■ 1.000	$\phi((x_1, y_1), (x_2, y_2))$	$\phi((x_1, y_1), (x_2, y_2))$	1. Construct the analytic group mapping $\phi((x_1, y_1), (x_2, y_2))$ for the parameterization
gilmore-lie-groups_F01265	134	■ 1.000	$\phi((w_1, z_1), (w_2, z_2))$	$\phi((w_1, z_1), (w_2, z_2))$	(7.24) of the affine group. Construct the mapping $\phi((w_1, z_1), (w_2, z_2))$ for the param-
gilmore-lie-groups_F01266	134	■ 1.000	$a^2 + b^2 - c^2 < 0$	$a^2 + b^2 - c^2 < 0$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_F01267	134	■ 1.000	$SO(2) \subset$	$SO(2) \subset$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_F01268	134	■ 0.999	c_T	c_T	c_T , in the subgroup is 2π but if $a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$), the basic
gilmore-lie-groups_F01269	134	■ 0.999	$a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$)	$a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$)	c_T , in the subgroup is 2π but if $a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$), the basic

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gilmore-lie-groups_F01270	134	■ 0.999	$2 \pi \gamma$	$2\pi\gamma$	repetition period in the c -direction is increased to $2\pi\gamma$, where $\gamma = 1/\sqrt{1 - \beta^2}$ and
gilmore-lie-groups_F01271	134	■ 0.999	$\gamma=1 / \sqrt{1-\beta^2}$	$\gamma = 1/\sqrt{1 - \beta^2}$	repetition period in the c -direction is increased to $2\pi\gamma$, where $\gamma = 1/\sqrt{1 - \beta^2}$ and
gilmore-lie-groups_F01272	134	■ 1.000	$\beta^2=\left(a^2+b^2\right) / c^2$	$\beta^2 = (a^2 + b^2) / c^2$	$\beta^2 = (a^2 + b^2) / c^2$. Compare this renormalization of periodicity with “time dilation.”
gilmore-lie-groups_F01273	134	■ 0.947	$S U(3)$	$SU(3)$	3. Compute the maximal discrete invariant subgroup D_{MAX} of $SU(3)$ and show that it
gilmore-lie-groups_F01274	134	■ 0.954	$\left\{I_3, \lambda I_3, \lambda^2 I_3\right\}$	$\{I_3, \lambda I_3, \lambda^2 I_3\}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_F01275	134	■ 0.954	$\lambda=e^{\left\{2 \pi i / 3\right\}}$	$\lambda = e^{2\pi i/3}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_F01276	134	■ 0.954	$\operatorname{SU}(3) / D_{\text {MAX }}$	$SU(3)/D_{\text{MAX}}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_F01277	134	■ 0.229	$\left[\operatorname{Re} \operatorname{g}\left(\operatorname{mathfrak{su}}(3)\right)\right]$	$[\Re g(\mathfrak{su}(3))]$	the group of real 8×8 matrices $\text{EXP}[\Re g(\mathfrak{su}(3))]$ (“eight-fold way”).
gilmore-lie-groups_F01278	134	■ 1.000	$\epsilon I_n, \epsilon=e^{\left\{2 \pi i / n\right\}}$	$\epsilon I_n, \epsilon = e^{2\pi i/n}$	and show that it is the cyclic group of order n generated by $\epsilon I_n, \epsilon = e^{2\pi i/n}$. What real
gilmore-lie-groups_F01279	134	■ 0.988	$S U(n) / D_{\text {MAX }}$	$SU(n)/D_{\text{MAX}}$	matrix group is $SU(n)/D_{\text{MAX}}$ equivalent to?
gilmore-lie-groups_F01280	135	■ 1.000	$a^{\dagger}, a$	$a^\dagger, a$	6. It is convenient to introduce the creation and annihilation operators $a^\dagger, a$ to study the
gilmore-lie-groups_F01281	135	■ 1.000	$\left\langle n^{\prime} \mid x^k \mid n\right\rangle$	$\langle n' x^k n \rangle$	tor basis, $\langle n' x^k n \rangle$, can be simplified using disentangling theorems. This problem
gilmore-lie-groups_F01282	135	■ 1.000	$e^{\left\{\lambda x\right\}}$	$e^{\lambda x}$	a. The function $e^{\lambda x}$ is a generating function for matrix elements of x^k . Show that
gilmore-lie-groups_F01283	135	■ 1.000	x^k	x^k	a. The function $e^{\lambda x}$ is a generating function for matrix elements of x^k . Show that
gilmore-lie-groups_F01284	135	■ 1.000	$\left[a, a^{\dagger}\right]=I$	$[a, a^\dagger] = I$	ators and their commutator $[a, a^\dagger] = I$ to show
gilmore-lie-groups_F01285	135	■ 0.955	$e^{r a^{\dagger}} e^{\delta I} e^{l a}$	$e^{ra^\dagger} e^{\delta I} e^{la}$	$e^{ra^\dagger} e^{\delta I} e^{la}$ by constructing the matrix product of these three operators:
gilmore-lie-groups_F01286	135	■ 0.999	$\left\langle n^{\prime} \mid x^4 \mid n\right\rangle$	$\langle n' x^4 n \rangle$	e. Use the disentangling theorem in d to compute $\langle n' x^4 n \rangle$. In particular, show
gilmore-lie-groups_F01287	135	■ 1.000	$\left\langle e^{i k x}\right\rangle$	$\langle e^{ikx} \rangle$	potential it is necessary to compute a structure factor $\langle e^{ikx} \rangle$. The expectation value is
gilmore-lie-groups_F01288	136	■ 1.000	$P_n \simeq e^{-n \beta \hbar \omega}$	$P_n \simeq e^{-n\beta\hbar\omega}$	thermal: $P_n \simeq e^{-n\beta\hbar\omega}$. This expectation value can be written in algebraic form as
gilmore-lie-groups_F01289	136	■ 1.000	$k \rightarrow 0$	$k \rightarrow 0$	We concentrate on the numerator, as the denominator is obtained in the limit $k \rightarrow 0$.
gilmore-lie-groups_F01290	136	■ 1.000	$\left\langle e^{i k x}\right\rangle=e^{\delta}$	$\langle e^{ikx} \rangle = e^\delta$	As a result $\langle e^{ikx} \rangle = e^\delta$.
gilmore-lie-groups_F01291	136	■ 0.976	δ	δ	c. Compute δ using 3×3 nonunitary matrix multiplications to carry out multi-
gilmore-lie-groups_F01292	136	■ 1.000	$\infty \times \infty$	$\infty \times \infty$	plications <i>in the group</i> rather than in an $\infty \times \infty$ unitary representation of the
gilmore-lie-groups_F01293	136	■ 0.999	$\omega^{\prime}=\omega$	$\omega' = \omega$	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_F01294	136	■ 0.999	α, β	α, β	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_F01295	136	■ 0.999	γ	γ	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_F01296	136	■ 1.000	$\left[X_i, X_j\right]=\sum_{k=1}^N C_{i j}^k X_k$	$[X_i, X_j] = \sum_{k=1}^N C_{ij}^k X_k$	8. A finite set of operators X_i closes under commutation: $[X_i, X_j] = \sum_{k=1}^N C_{ij}^k X_k$.

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gilmore-lie-groups_F01297	136	■ 1.000	R	R	this set of operators has two representations R and S with the following properties:
gilmore-lie-groups_F01298	136	■ 1.000	$\left(R\left(a^i X_i\right)\right)^{\dagger}=\left(a^i R\left(X_i\right)\right)^{\dagger}=\left(a^i\right)^{*} R^{\dagger}\left(X_i\right)$	$\left(R\left(a^i X_i\right)\right)^{\dagger}=\left(a^i R\left(X_i\right)\right)^{\dagger}=\left(a^i\right)^{*} R^{\dagger}\left(X_i\right)$	R is hermitian: $\left(R\left(a^i X_i\right)\right)^{\dagger}=\left(a^i R\left(X_i\right)\right)^{\dagger}=\left(a^i\right)^{*} R^{\dagger}\left(X_i\right)$
gilmore-lie-groups_F01299	136	■ 1.000	$S\left(a^i X_i\right)=0 \Rightarrow a^i=0$	$S\left(a^i X_i\right)=0 \Rightarrow a^i=0$	S is faithful: $S\left(a^i X_i\right)=0 \Rightarrow a^i=0$.
gilmore-lie-groups_F01300	136	■ 1.000	$\mathcal{H}=R\left(a^i X_i\right)$	$\mathcal{H}=R\left(a^i X_i\right)$	a. It happens frequently that $\mathcal{H}=R\left(a^i X_i\right)$ describes the physics of some quantum
gilmore-lie-groups_F01301	136	■ 0.999	$H_1, H_2, \ldots, H_r \in \mathfrak{g}$	$H_1, H_2, \ldots, H_r \in \mathfrak{g}$	mechanical system. Show that if $H_1, H_2, \ldots, H_r \in \mathfrak{g}$ span a maximal commutative
gilmore-lie-groups_F01302	136	■ 1.000	$\left[H_i, H_j\right]=0, 1 \leq i, j \leq r$	$\left[H_i, H_j\right]=0, 1 \leq i, j \leq r$	subspace, so that $\left[H_i, H_j\right]=0, 1 \leq i, j \leq r$, then the hermitian operators $R\left(H_i\right)$
gilmore-lie-groups_F01303	136	■ 1.000	$R\left(H_i\right)$	$R\left(H_i\right)$	subspace, so that $\left[H_i, H_j\right]=0, 1 \leq i, j \leq r$, then the hermitian operators $R\left(H_i\right)$
gilmore-lie-groups_F01304	136	■ 1.000	$\left[R\left(H_i\right)\right]_{\alpha \beta}=r_{\alpha}(i) \delta_{\alpha \beta}$	$\left[R\left(H_i\right)\right]_{\alpha \beta}=r_{\alpha}(i) \delta_{\alpha \beta}$	representation: $\left[R\left(H_i\right)\right]_{\alpha \beta}=r_{\alpha}(i) \delta_{\alpha \beta}$.
gilmore-lie-groups_F01305	137	■ 1.000	$\left[S\left(H_i\right), S\left(H_j\right)\right]=0$	$\left[S\left(H_i\right), S\left(H_j\right)\right]=0$	b. Show that $\left[S\left(H_i\right), S\left(H_j\right)\right]=0$, but show by example that the r matrices $S\left(H_i\right)$
gilmore-lie-groups_F01306	137	■ 1.000	r	r	b. Show that $\left[S\left(H_i\right), S\left(H_j\right)\right]=0$, but show by example that the r matrices $S\left(H_i\right)$
gilmore-lie-groups_F01307	137	■ 1.000	$S\left(H_i\right)$	$S\left(H_i\right)$	b. Show that $\left[S\left(H_i\right), S\left(H_j\right)\right]=0$, but show by example that the r matrices $S\left(H_i\right)$
gilmore-lie-groups_F01308	137	■ 1.000	$U(t)=R\left(e^{-\frac{i}{\hbar} \mathcal{H}} \mathcal{H} t\right)=e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$	$U(t)=R\left(e^{-\frac{i}{\hbar} \mathcal{H}}\right)=e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$	$U(t)=R\left(e^{-\frac{i}{\hbar} \mathcal{H}}\right)=e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$.
gilmore-lie-groups_F01309	137	■ 1.000	$\rho(T)=e^{-\beta \mathcal{H}} / Z$	$\rho(T)=e^{-\beta \mathcal{H}} / Z$	d. Show that the density operator for thermal expectation values is $\rho(T)=e^{-\beta \mathcal{H}} / Z$
gilmore-lie-groups_F01310	137	■ 0.975	$R\left(e^{-\beta \mathcal{H}}\right) / Z=e^{-\beta R(\mathcal{H})} / Z$	$R\left(e^{-\beta \mathcal{H}}\right) / Z=e^{-\beta R(\mathcal{H})} / Z$	$R\left(e^{-\beta \mathcal{H}}\right) / Z=e^{-\beta R(\mathcal{H})} / Z$. What is Z ?
gilmore-lie-groups_F01311	137	■ 1.000	$U(t)$	$U(t)$	e. Show that the unitary time evolution operator $U(t)$ and the hermitian density
gilmore-lie-groups_F01312	137	■ 1.000	$\rho(T)$	$\rho(T)$	operator $\rho(T)$ are related by a Wick rotation $it / \hbar \leftrightarrow \beta=1 / k_B T$.
gilmore-lie-groups_F01313	137	■ 1.000	$it / \hbar \leftrightarrow \beta=1 / k_B T$	$it / \hbar \leftrightarrow \beta=1 / k_B T$	operator $\rho(T)$ are related by a Wick rotation $it / \hbar \leftrightarrow \beta=1 / k_B T$
gilmore-lie-groups_F01314	137	■ 1.000	$G=e^{\mathfrak{g}}$	$G=e^{\mathfrak{g}}$	g. The operator product in the numerator is in the group $G=e^{\mathfrak{g}}$ or its complex exten-
gilmore-lie-groups_F01315	137	■ 1.000	H_i	H_i	in terms of the operators H_i) the trace can easily be constructed. Show that for
gilmore-lie-groups_F01316	137	■ 1.000	x^i	x^i	x^i sufficiently small it is always possible to construct a similarity transformation
gilmore-lie-groups_F01317	137	■ 1.000	$S=e^{y^k X_k}$	$S=e^{y^k X_k}$	$S=e^{y^k X_k}$ with the property
gilmore-lie-groups_F01318	137	■ 1.000	$e^{x^i X_i}$	$e^{x^i X_i}$	function $e^{x^i X_i}$, and the Wick rotation.
gilmore-lie-groups_F01319	137	■ 1.000	$U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^{*} a\right)}$	$U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^{*} a\right)}$	The unitary transformation has the form $U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^{*} a\right)}$, where $a^{\dagger}$ and a are the
gilmore-lie-groups_F01320	138	■ 0.896	$a^{\dagger} n\rangle=$	$a^{\dagger} n\rangle=$	Hilbert space and out of the nonunitary 3×3 matrix representation), $a^{\dagger} n\rangle=$
gilmore-lie-groups_F01321	138	■ 1.000	$ n+1\rangle \sqrt{n+1}$	$ n+1\rangle \sqrt{n+1}$	$ n+1\rangle \sqrt{n+1}$ to conclude
gilmore-lie-groups_F01322	138	■ 1.000	$\langle\beta \alpha\rangle$	$\langle\beta \alpha\rangle$	e. Compute the inner product $\langle\beta \alpha\rangle$ and show $\langle\alpha \alpha\rangle=1$.
gilmore-lie-groups_F01323	138	■ 1.000	$\langle\alpha \alpha\rangle=1$	$\langle\alpha \alpha\rangle=1$	e. Compute the inner product $\langle\beta \alpha\rangle$ and show $\langle\alpha \alpha\rangle=1$.

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gilmore-lie-groups_F01324	138	■ 1.000	$a \alpha\rangle=\alpha \alpha\rangle$	$a \alpha\rangle=\alpha \alpha\rangle$	f. Show $\boxed{a \alpha\rangle=\alpha \alpha\rangle}$.
gilmore-lie-groups_F01325	138	■ 1.000	$\langle\alpha x \alpha\rangle=\langle\alpha^*+\alpha\rangle/\sqrt{2}$	$\langle\alpha x \alpha\rangle=(\alpha^*+\alpha)/\sqrt{2}$	g. Show $\boxed{\langle\alpha x \alpha\rangle=(\alpha^*+\alpha)/\sqrt{2}}$
gilmore-lie-groups_F01326	138	■ 0.966	$e^{i\phi}J_z\left j\right\rangle=\left j\right\rangle e^{-ij\phi}$	$e^{i\phi}J_z\left j\right\rangle=\left j\right\rangle e^{-ij\phi}$	by a phase angle: $e^{i\phi}J_z\left j\right\rangle=\left j\right\rangle e^{-ij\phi}$. This simply “renormalizes” the fiducial
gilmore-lie-groups_F01327	138	■ 1.000	$e^{i\left(\theta_xJ_x+\theta_yJ_y\right)}$	$e^{i\left(\theta_xJ_x+\theta_yJ_y\right)}$	c. Rewrite $e^{i\left(\theta_xJ_x+\theta_yJ_y\right)}$ in the form $e^{i\alpha_+J_+}e^{i\alpha_zJ_z}e^{i\alpha_-J_-}$ and compute the analytic relation
gilmore-lie-groups_F01328	138	■ 1.000	$e^{i\alpha_+J_+}e^{i\alpha_zJ_z}e^{i\alpha_-J_-}$	$e^{i\alpha_+J_+}e^{i\alpha_zJ_z}e^{i\alpha_-J_-}$	c. Rewrite $e^{i\left(\theta_xJ_x+\theta_yJ_y\right)}$ in the form $e^{i\alpha_+J_+}e^{i\alpha_zJ_z}e^{i\alpha_-J_-}$ and compute the analytic relation
gilmore-lie-groups_F01329	138	■ 0.859	$e^{i\alpha_-J_-}\left j\right\rangle=\left j\right\rangle$	$e^{i\alpha_-J_-}\left j\right\rangle=\left j\right\rangle$	d. Show $\boxed{e^{i\alpha_-J_-}\left j\right\rangle=\left j\right\rangle}$.
gilmore-lie-groups_F01330	138	■ 0.952	$e^{i\alpha_zJ_z}\left j\right\rangle=\left j\right\rangle e^{-ij\alpha_z}$	$e^{i\alpha_zJ_z}\left j\right\rangle=\left j\right\rangle e^{-ij\alpha_z}$	e. Show $\boxed{e^{i\alpha_zJ_z}\left j\right\rangle=\left j\right\rangle e^{-ij\alpha_z}}$.
gilmore-lie-groups_F01331	139	■ 0.815	$J_-^j\left j\right\rangle$	$J_-^j\left j\right\rangle$	g. Show that $J_-^j\left j\right\rangle$ cannot be proportional to $\left j\right\rangle$ because the state $\left j\right\rangle$ is not
gilmore-lie-groups_F01332	139	■ 0.815	$J_-^j\left j\right\rangle$	$J_-^j\left j\right\rangle$	g. Show that $J_-^j\left j\right\rangle$ cannot be proportional to $\left j\right\rangle$ because the state $\left j\right\rangle$ is not
gilmore-lie-groups_F01333	139	■ 0.815	$J_-^j\left j\right\rangle$	$J_-^j\left j\right\rangle$	g. Show that $J_-^j\left j\right\rangle$ cannot be proportional to $\left j\right\rangle$ because the state $\left j\right\rangle$ is not
gilmore-lie-groups_F01334	139	■ 1.000	H_4	H_4	irreducible representations and the harmonic oscillator group H_4 is not compact
gilmore-lie-groups_F01335	139	■ 1.000	$e^{-i\phi}=(\theta_x-i\theta_y)/\theta$	$e^{-i\phi}=(\theta_x-i\theta_y)/\theta$	where $e^{-i\phi}=(\theta_x-i\theta_y)/\theta$, and similarly for θ' (cf., Eq. (7.46)).
gilmore-lie-groups_F01336	139	■ 1.000	θ'	θ'	where $e^{-i\phi}=(\theta_x-i\theta_y)/\theta$, and similarly for θ' (cf., Eq. (7.46)).
gilmore-lie-groups_F01337	139	■ 1.000	$ u\rangle$	$ u\rangle$	sometimes does) represent a charge, and $ u\rangle$ is an eigenfunction. In this toy example,
gilmore-lie-groups_F01338	139	■ 1.000	J_3	J_3	the operators J_3 and J_1 are assumed to belong to the Lie algebra $\mathfrak{su}(2)$ and the equation
gilmore-lie-groups_F01339	139	■ 1.000	$(E'J_3-Z) v\rangle=0$	$(E'J_3-Z) v\rangle=0$	$\boxed{E'J_3-Z} v\rangle=0$, where $E'=\sqrt{E^2+p^2}$ and $ v\rangle=U u\rangle$.
gilmore-lie-groups_F01340	139	■ 1.000	$E'=\sqrt{E^2+p^2}$	$E'=\sqrt{E^2+p^2}$	$(E'J_3-Z) v\rangle=0$, where $E'=\sqrt{E^2+p^2}$ and $ v\rangle=U u\rangle$.
gilmore-lie-groups_F01341	139	■ 1.000	$ v\rangle=U u\rangle$	$ v\rangle=U u\rangle$	$(E'J_3-Z) v\rangle=0$, where $E'=\sqrt{E^2+p^2}$ and $ v\rangle=U u\rangle$.
gilmore-lie-groups_F01342	139	■ 1.000	$E=\pm\sqrt{(Z/m)^2-p^2}$	$E=\pm\sqrt{(Z/m)^2-p^2}$	b. Show that $\boxed{E=\pm\sqrt{(Z/m)^2-p^2}}$.
gilmore-lie-groups_F01343	139	■ 0.871	$(p\rightarrow 0)$	$(p\rightarrow 0)$	c. Compare this spectrum with the unperturbed spectrum $(p\rightarrow 0)$.
gilmore-lie-groups_F01344	139	■ 1.000	j, p, Z	j, p, Z	d. Under what conditions on j, p, Z are these solutions valid?
gilmore-lie-groups_F01345	139	■ 0.986	$ u\rangle=e^{i\theta J_2}\left j\right\rangle$	$ u\rangle=e^{i\theta J_2}\left j\right\rangle$	and show that $\boxed{ u\rangle=e^{i\theta J_2}\left j\right\rangle}$. Compute θ for each E .
gilmore-lie-groups_F01346	139	■ 1.000	$z=\cos(\beta)$	$z=\cos(\beta)$	where $z=\cos(\beta)$. The Wigner matrix elements D_{mn}^j are related to the Jacobi poly-
gilmore-lie-groups_F01347	139	■ 1.000	D_{mn}^j	D_{mn}^j	where $z=\cos(\beta)$. The Wigner matrix elements D_{mn}^j are related to the Jacobi poly-
gilmore-lie-groups_F01348	139	■ 1.000	$j=l$	$j=l$	nomials when $j=l$, where l is an integer.
gilmore-lie-groups_F01349	140	■ 1.000	$I_2\in SU(2)$	$I_2\in SU(2)$	a. Geodesics through $I_2\in SU(2)$ focus at $-I_2$ and geodesics through $I_3\in SO(3)$

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gilmore-lie-groups_F01350	140	■ 1.000	$I_{\{3\}} \in S O(3)$	$I_3 \in SO(3)$	a. Geodesics through $I_2 \in SU(2)$ focus at $-I_2$ and geodesics through $I_3 \in SO(3)$
gilmore-lie-groups_F01351	140	■ 0.835	$I_{\{3\}}$	I_3	focus at I_3 . Conclude that $SU(2)$ is a two-fold covering group of $SO(3)$.
gilmore-lie-groups_F01352	140	■ 0.999	$S U(2) / U(1)(z=1, x=y=0)$	$SU(2)/U(1)(z = 1, x = y = 0)$	b. Geodesics through the “north pole” of $SU(2)/U(1)(z = 1, x = y = 0)$ focus at
gilmore-lie-groups_F01353	140	■ 0.793	$z=-1, x=y=0$	$z = -1, x = y = 0$	its “south pole” $(z = -1, x = y = 0)$ and geodesics through the north pole of
gilmore-lie-groups_F01354	140	■ 0.701	$S O(3) / S O(2)(z=1, x=y=0)$	$SO(3)/SO(2)(z = 1, x = y = 0)$	$SO(3)/SO(2)(z = 1, x = y = 0)$ focus at its south pole $(z = -1, x = y = 0)$.
gilmore-lie-groups_F01355	140	■ 0.701	$(z=-1, x=y=0)$	$(z = -1, x = y = 0)$	$SO(3)/SO(2)(z = 1, x = y = 0)$ focus at its south pole $(z = -1, x = y = 0)$.
gilmore-lie-groups_F01356	140	■ 1.000	$S U(2) / U(1)=S^2=S O(3) / S O(2)$	$SU(2)/U(1) = S^2 = SO(3)/SO(2)$	c. Conclude that $SU(2)/U(1) = S^2 = SO(3)/SO(2)$ and the $2 \rightarrow 1$ nature of the
gilmore-lie-groups_F01357	140	■ 0.986	$S U(2) \searrow S O(3)$	$SU(2) \downarrow SO(3)$	covering $SU(2) \downarrow SO(3)$ is contained in the subgroup of rotations about the z -
gilmore-lie-groups_F01358	140	■ 0.918	$U(1) \searrow S O(2)$	$U(1) \downarrow SO(2)$	axis $U(1) \downarrow SO(2)$:
gilmore-lie-groups_F01359	140	■ 1.000	$e^{2 \pi i k / r} I_{\{n\}}$	$e^{2 \pi i k / r} I_n$	order r , with group elements $e^{2 \pi i k / r} I_n$, with n / r integer. Compute the foci in $SU(n)$.
gilmore-lie-groups_F01360	140	■ 1.000	n / r	n / r	order r , with group elements $e^{2 \pi i k / r} I_n$, with n / r integer. Compute the foci in $SU(n)$.
gilmore-lie-groups_F01361	140	■ 1.000	$e^{2 \pi i k / n} I_{\{n\}}$	$e^{2 \pi i k / n} I_n$	How are the foci related to the group operations of the form $e^{2 \pi i k / n} I_n$?
gilmore-lie-groups_F01362	140	■ 0.999	$\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1 / \lambda \end{array}\right]$	$\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1 / \lambda \end{array}\right]$	14. Show that the matrix $\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1 / \lambda \end{array}\right]$ in $SL(2 ; \mathbb{R})$ cannot be reached by exponentiating
gilmore-lie-groups_F01363	140	■ 1.000	$\lambda > 1$	$\lambda > 1$	any element in the Lie algebra if $\lambda > 1$. Show that it can be reached by following a
gilmore-lie-groups_F01364	140	■ 1.000	$\hbar \omega$	$\hbar \omega$	photon has energy $\hbar \omega$. The hamiltonian is chosen as
gilmore-lie-groups_F01365	140	■ 1.000	$\sigma_z^{(i)}$	$\sigma_z^{(i)}$	The operator $\sigma_z^{(i)}$ describes the two states of atom i and the operator $a^\dagger a$ describes
gilmore-lie-groups_F01366	140	■ 1.000	$a^{\dagger} a$	$a^\dagger a$	The operator $\sigma_z^{(i)}$ describes the two states of atom i and the operator $a^\dagger a$ describes
gilmore-lie-groups_F01367	140	■ 0.999	$\sigma_+^{(j)} \left(\sigma_{\pm}^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	$\sigma_+^{(j)} \left(\sigma_{\pm}^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	the number of photons in the field mode. The operator $\sigma_{\pm}^{(j)} (\sigma_{\pm}^{(j)}) = \frac{1}{2} (\sigma_x^{(j)} \pm i \sigma_y^{(j)})$
gilmore-lie-groups_F01368	140	■ 1.000	$\sigma_-^{(j)} a^\dagger$	$\sigma_-^{(j)} a^\dagger$	operator $\sigma_-^{(j)} a^\dagger$, describes deexcitation of an atom with emission (creation, $a^\dagger$) of a
gilmore-lie-groups_F01369	140	■ 1.000	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	b. If all the atoms behave cooperatively it is possible to replace $\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$,
gilmore-lie-groups_F01370	140	■ 0.999	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$. Show that the operators $J_z, J_{\pm}$ satisfy the usual $\mathfrak{su}(2)$ commuta-
gilmore-lie-groups_F01371	140	■ 0.999	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$. Show that the operators $J_z, J_{\pm}$ satisfy the usual $\mathfrak{su}(2)$ commuta-
gilmore-lie-groups_F01372	141	■ 1.000	$J_{\{z\}} \rightarrow \langle J_z(t) \rangle, J_{\{+\}} \rightarrow \langle J_+(t) \rangle, J_{\{-\}} \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$. Show that this semiclas-
gilmore-lie-groups_F01373	141	■ 1.000	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	of the electromagnetic field: $ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$. Use the disentan-
gilmore-lie-groups_F01374	141	■ 1.000	$\alpha(t)$	$\alpha(t)$	gling theorems to compute the relation between the coherent state parameter $\alpha(t)$

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gilmore-lie-groups_F01375	141	■ 1.000	$\left\langle J_z(t) \right\rangle$	$\langle J_z(t) \rangle$	and the classical driving fields $\langle J_z(t) \rangle$ and $\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$.
gilmore-lie-groups_F01376	141	■ 1.000	$\left\langle J_+(t) \right\rangle = \left\langle J_-(t) \right\rangle^*$	$\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$	and the classical driving fields $\langle J_z(t) \rangle$ and $\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$.
gilmore-lie-groups_F01377	141	■ 1.000	$ \beta\rangle$	$ \beta\rangle$	state $ \beta\rangle$, the state obtained by the action of the classical current is still a coherent
gilmore-lie-groups_F01378	141	■ 1.000	β	β	state. How are the parameters β , describing the initial condition, and α , describing
gilmore-lie-groups_F01379	141	■ 1.000	$a^\dagger \rightarrow \langle a(t) \rangle^*$	$a^\dagger \rightarrow \langle a(t) \rangle^*$	considered classically. Show that this amounts to the substitutions $a^\dagger \rightarrow \langle a(t) \rangle^*$,
gilmore-lie-groups_F01380	141	■ 1.000	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$, and $a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$.
gilmore-lie-groups_F01381	141	■ 1.000	$a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$, and $a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$.
gilmore-lie-groups_F01382	141	■ 1.000	$m = -\frac{1}{2}$	$m = -\frac{1}{2}$	ground state ($m = -\frac{1}{2}$ for each atom, or $M = -J$, $J = N/2$ for the ensemble of
gilmore-lie-groups_F01383	141	■ 1.000	$M = -J$, $J = N/2$	$M = -J$, $J = N/2$	ground state ($m = -\frac{1}{2}$ for each atom, or $M = -J$, $J = N/2$ for the ensemble of
gilmore-lie-groups_F01384	141	■ 0.997	$ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$	$ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$	evolve into a coherent state: $ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$, where $J = N/2$. How are the
gilmore-lie-groups_F01385	141	■ 0.997	$J = N/2$	$J = N/2$	evolve into a coherent state: $ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$, where $J = N/2$. How are the
gilmore-lie-groups_F01386	141	■ 1.000	$\theta(t)$	$\theta(t)$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_F01387	141	■ 1.000	$\langle a(t) \rangle$	$\langle a(t) \rangle$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_F01388	141	■ 1.000	$\langle a(t) \rangle^*$	$\langle a(t) \rangle^*$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_F01389	141	■ 1.000	$\langle a \rangle$	$\langle a \rangle$	field expectation values $\langle a \rangle$ and $\langle a \rangle^*$. Conversely, the semiclassical hamiltonian
gilmore-lie-groups_F01390	141	■ 1.000	$\langle a \rangle^*$	$\langle a \rangle^*$	field expectation values $\langle a \rangle$ and $\langle a \rangle^*$. Conversely, the semiclassical hamiltonian
gilmore-lie-groups_F01391	141	■ 1.000	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$. Construct a self-consistent model by requiring that both sets of
gilmore-lie-groups_F01392	141	■ 1.000	$T(\beta = 1/k_B T)$	$T(\beta = 1/k_B T)$	equilibrium at temperature T ($\beta = 1/k_B T$).
gilmore-lie-groups_F01393	141	■ 1.000	$\langle \sigma_+^{(i)} \rangle_T$	$\langle \sigma_+^{(i)} \rangle_T$	a. Assume $\langle \sigma_+^{(i)} \rangle_T$ has some fixed unknown value, and similarly for the other atomic
gilmore-lie-groups_F01394	141	■ 1.000	$a^\dagger, a, a^\dagger a$	$a^\dagger, a, a^\dagger a$	expectation values for the operators $a^\dagger, a, a^\dagger a$.
gilmore-lie-groups_F01395	142	■ 1.000	$\sigma_z, \sigma_+, \sigma_-$	$\sigma_z, \sigma_+, \sigma_-$	operators $\sigma_z, \sigma_+, \sigma_-$.
gilmore-lie-groups_F01396	142	■ 1.000	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$ is there a nontrivial solution (e.g., $\langle J_+ \rangle_T \neq 0$)?
gilmore-lie-groups_F01397	142	■ 1.000	$\langle J_+ \rangle_T \neq 0$	$\langle J_+ \rangle_T \neq 0$	$\epsilon, \hbar\omega, \lambda, N$ is there a nontrivial solution (e.g., $\langle J_+ \rangle_T \neq 0$)?
gilmore-lie-groups_F01398	142	■ 1.000	$\lambda^2 / \epsilon \hbar \omega$	$\lambda^2 / \epsilon \hbar \omega$	e. Show that a thermodynamic phase transition occurs as $\lambda^2 / \epsilon \hbar \omega$ increases through
gilmore-lie-groups_F01399	142	■ 1.000	$a(t), b(t)$	$a(t), b(t)$	17. The two complex parameters $a(t), b(t)$ in the evolution equation (7.58) can be ex-
gilmore-lie-groups_F01400	142	■ 1.000	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	pressed in terms of their real and imaginary parts. These obey $a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$
gilmore-lie-groups_F01401	142	■ 0.542	(a_r, a_i, b_r, b_i)	(a_r, a_i, b_r, b_i)	compute the length of the vector (a_r, a_i, b_r, b_i) and scale this length back to +1.

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gilmore-lie-groups_ F01402	142	■ 1.000	$\lambda_3 \rightarrow 0$	$\lambda_3 \rightarrow 0$	Simplify (7.67) by taking the limit $\lambda_3 \rightarrow 0$. Simplify (7.70) by taking the limit $\lambda_n \rightarrow 0$
gilmore-lie-groups_ F01403	142	■ 1.000	$\lambda_n \rightarrow 0$	$\lambda_n \rightarrow 0$	Simplify (7.67) by taking the limit $\lambda_3 \rightarrow 0$. Simplify (7.70) by taking the limit $\lambda_n \rightarrow 0$
gilmore-lie-groups_ F01404	142	■ 1.000	$d=\delta=0$	$d = \delta = 0$	and setting $d = \delta = 0$.
gilmore-lie-groups_ F01405	142	■ 1.000	$\langle J_+ J_- + J_- J_+ \rangle$	$\langle J_+ J_- + J_- J_+ \rangle$	<i>metric</i> operator products such as $\langle J_+ J_- + J_- J_+ \rangle$ or $\langle a a^\dagger + a^\dagger a \rangle$ than it is to construct
gilmore-lie-groups_ F01406	142	■ 1.000	$\langle a a^\dagger + a^\dagger a \rangle$	$\langle a a^\dagger + a^\dagger a \rangle$	<i>metric</i> operator products such as $\langle J_+ J_- + J_- J_+ \rangle$ or $\langle a a^\dagger + a^\dagger a \rangle$ than it is to construct
gilmore-lie-groups_ F01407	142	■ 1.000	$\langle J_+ J_- \rangle$	$\langle J_+ J_- \rangle$	generating functions for <i>ordered</i> products of operators such as $\langle J_+ J_- \rangle$ or $\langle a^\dagger a \rangle$. Show
gilmore-lie-groups_ F01408	142	■ 1.000	$\langle a^\dagger a \rangle$	$\langle a^\dagger a \rangle$	generating functions for <i>ordered</i> products of operators such as $\langle J_+ J_- \rangle$ or $\langle a^\dagger a \rangle$. Show
gilmore-lie-groups_ F01409	143	■ 0.749	$X_j = A_j^s Y_s$	$X_j = A_j^s Y_s$	Under a change of basis $X_j = A_j^s Y_s$ this $n \times n$ matrix undergoes a similarity trans-
gilmore-lie-groups_ F01410	144	■ 1.000	$N=p \ q$	$N = pq$	This is an $n \times n$ ($n = p + q$) matrix algebra which is $N = pq$ dimensional. The
gilmore-lie-groups_ F01411	144	■ 1.000	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	3. $\mathfrak{so}(n)$ Upper triangular In this case nonzero elements occur on and above the
gilmore-lie-groups_ F01412	145	■ 1.000	$\hat{n} =$	$\hat{n} =$	Example We consider the algebra consisting of the six photon operators $\hat{n} =$
gilmore-lie-groups_ F01413	145	■ 0.993	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^{\dagger 2}, a^2, a^\dagger, a, I = [a, a^\dagger]$	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^{\dagger 2}, a^2, a^\dagger, a, I = [a, a^\dagger]$	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^{\dagger 2}, a^2, a^\dagger, a, I = [a, a^\dagger]$. This is isomorphic to the algebra
gilmore-lie-groups_ F01414	145	■ 1.000	$X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$	$X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$	Example The Lie algebra $\mathfrak{so}(4)$ has six generators $X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$ and
gilmore-lie-groups_ F01415	146	■ 1.000	$\mathfrak{so}(n), n > 4$	$\mathfrak{so}(n), n > 4$	other orthogonal Lie algebras, $\mathfrak{so}(n), n > 4$. We will have to wait until Chapter 10
gilmore-lie-groups_ F01416	146	■ 0.999	$\mathfrak{su}(n) (n \geq 2), \mathfrak{so}(n) (n > 4)$	$\mathfrak{su}(n) (n \geq 2), \mathfrak{so}(n) (n > 4)$	Example The Lie algebras $\mathfrak{su}(n) (n \geq 2), \mathfrak{so}(n) (n > 4)$, and $\mathfrak{sp}(n) (n \geq 1)$ are
gilmore-lie-groups_ F01417	146	■ 0.999	$\mathfrak{sp}(n) (n \geq 1)$	$\mathfrak{sp}(n) (n \geq 1)$	Example The Lie algebras $\mathfrak{su}(n) (n \geq 2), \mathfrak{so}(n) (n > 4)$, and $\mathfrak{sp}(n) (n \geq 1)$ are
gilmore-lie-groups_ F01418	147	■ 0.999	$[Z, X_i]$	$[Z, X_i]$	This means that $[Z, X_i]$ can be expressed as a linear combination of basis vectors
gilmore-lie-groups_ F01419	147	■ 1.000	X_j	X_j	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01420	147	■ 1.000	$j \geq i$	$j \geq i$	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01421	147	■ 1.000	$X_i, X_{i+1}, \dots, X_n$	$X_i, X_{i+1}, \dots, X_n$	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01422	147	■ 0.593	$i=1,2, \dots, n$	$i = 1, 2, \dots, n$	subalgebra for each value of $i = 1, 2, \dots, n$. Since this subalgebra is mapped onto
gilmore-lie-groups_ F01423	147	■ 1.000	V_i	V_i	Each V_i is an invariant subalgebra in $V_j, i > j$. The original algebra is V_1 .
gilmore-lie-groups_ F01424	147	■ 1.000	$V_j, i > j$	$V_j, i > j$	Each V_i is an invariant subalgebra in $V_j, i > j$. The original algebra is V_1 .
gilmore-lie-groups_ F01425	148	■ 1.000	$[V_2, V_2] \subseteq V_2, V_2$	$[V_2, V_2] \subseteq V_2, V_2$	Since in particular $[V_2, V_2] \subseteq V_2, V_2$ is a subalgebra in the original algebra. Further,
gilmore-lie-groups_ F01426	148	■ 1.000	V_2, V_2	V_2, V_2	since the commutator of anything in the original algebra with V_2 is in V_2, V_2 is an
gilmore-lie-groups_ F01427	149	■ 1.000	$\mathfrak{g}' = \mathfrak{g} - V_0$	$\mathfrak{g}' = \mathfrak{g} - V_0$	(iv) Form the difference $\mathfrak{g}' = \mathfrak{g} - V_0$. This is a Lie algebra (under the “mod” operation: set
gilmore-lie-groups_ F01428	149	■ 1.000	$\mathfrak{g}'$	$\mathfrak{g}'$	Cartan–Killing inner product for $\mathfrak{g}'$.

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gilmore-lie-groups_ F01429	150	■ 1.000	$V_{\{0\}^{\prime}}=0$	$V_0' = 0$	In this case $V_0' = 0$ and the two photon algebra has the decomposition
gilmore-lie-groups_ F01430	150	■ 1.000	$a^{\dagger 2}$	$a^{\dagger 2}$	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01431	150	■ 1.000	$a^{\{2\}}$	a^2	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01432	150	■ 1.000	$a^{\dagger 2}+a^{\{2\}}$	$a^{\dagger 2} + a^2$	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01433	150	■ 1.000	$a^{\dagger} a+\frac{1}{2}$	$a^{\dagger} a + \frac{1}{2}$	the Cartan–Killing form is negative-definite on it. The other two operators, $a^{\dagger} a + \frac{1}{2}$
gilmore-lie-groups_ F01434	150	■ 1.000	$a^{\dagger 2}-a^{\{2\}}$	$a^{\dagger 2} - a^2$	and $a^{\dagger 2} - a^2$, are noncompact.
gilmore-lie-groups_ F01435	151	■ 1.000	$\hat{n}, a^{\dagger}, a, I$	$\hat{n}, a^{\dagger}, a, I$	a. The photon algebra $\hat{n}, a^{\dagger}, a, I$ (Eq. (8.5)).
gilmore-lie-groups_ F01436	151	■ 0.786	$\left[b_i, b_j^{\dagger}\right]=I \delta_{i j}, 1 \leq i, j \leq n$	$\left[b_i, b_j^{\dagger}\right]=I \delta_{i j}, 1 \leq i, j \leq n$	$[b_i, b_j^{\dagger}]=I \delta_{i j}, 1 \leq i, j \leq n$. Commutators involving bilinear (trilinear, . . .) products
gilmore-lie-groups_ F01437	151	■ 1.000	$b_{\{i\}}, b_{\{j\}}^{\dagger}, I$	$b_i, b_j^{\dagger}, I$	a. $b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01438	151	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{j\}}, b_{\{i\}}, b_{\{j\}}^{\dagger}, I$	$b_i^{\dagger} b_j, b_i, b_j^{\dagger}, I$	c. $b_i^{\dagger} b_j, b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01439	151	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{j\}}^{\dagger}, b_{\{i\}}^{\dagger} b_{\{j\}}+\frac{1}{2} \delta_{i j}, b_i b_j$	$b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{i j}, b_i b_j$	d. $b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{i j}, b_i b_j$.
gilmore-lie-groups_ F01440	151	■ 0.998	$b_{\{i\}}^{\dagger} b_{\{j\}}^{\dagger}, b_{\{i\}}^{\dagger} b_{\{j\}}+\frac{1}{2} \delta_{i j}, b_i b_j, b_i, b_j^{\dagger}, I$	$b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{i j}, b_i b_j, b_i, b_j^{\dagger}, I$	e. $b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{i j}, b_i b_j, b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01441	151	■ 1.000	$b, b^{\dagger}, b, b^{\dagger} b$	$b, b^{\dagger} b, b^{\dagger} b^{\dagger} b$	f. $b, b^{\dagger} b, b^{\dagger} b^{\dagger} b$.
gilmore-lie-groups_ F01442	151	■ 0.788	$b^{\dagger}$	$b^{\dagger}$	g. $b^{\dagger}, b^{\dagger} b, b^{\dagger} b b$.
gilmore-lie-groups_ F01443	151	■ 0.788	$b^{\dagger} b, b^{\dagger} b b$	$b^{\dagger} b, b^{\dagger} b b$	g. $b^{\dagger}, b^{\dagger} b, b^{\dagger} b b$.
gilmore-lie-groups_ F01444	151	■ 1.000	$\left\{f_i, f_j^{\dagger}\right\}=\delta_{i j}$	$\left\{f_i, f_j^{\dagger}\right\}=\delta_{i j}$	$\{f_i, f_j^{\dagger}\}=\delta_{i j}$, but their bilinear combinations close under the same commutation re-
gilmore-lie-groups_ F01445	151	■ 0.985	$f_i^{\dagger} f_j^{\dagger}$	$f_i^{\dagger} f_j^{\dagger}$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j + \frac{1}{2} \delta_{i j}, f_i f_j$.
gilmore-lie-groups_ F01446	151	■ 0.985	$f_i^{\dagger} f_j+\frac{1}{2} \delta_{i j}$	$f_i^{\dagger} f_j + \frac{1}{2} \delta_{i j}$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j + \frac{1}{2} \delta_{i j}, f_i f_j$.
gilmore-lie-groups_ F01447	151	■ 0.985	$f_i f_j$	$f_i f_j$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j + \frac{1}{2} \delta_{i j}, f_i f_j$.
gilmore-lie-groups_ F01448	151	■ 1.000	$\left(x_i\right)$	$\left(x_i\right)$	of the position $\left(x_i\right)$ and momentum $\left(\partial_j\right)$ operators for n independent directions:
gilmore-lie-groups_ F01449	151	■ 1.000	$\left(\partial_j\right)$	$\left(\partial_j\right)$	of the position $\left(x_i\right)$ and momentum $\left(\partial_j\right)$ operators for n independent directions:
gilmore-lie-groups_ F01450	151	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}, I$	x_i, ∂_j, I	a. x_i, ∂_j, I .
gilmore-lie-groups_ F01451	151	■ 1.000	$x_{\{i\}} \partial_{\{j\}}$	$x_i \partial_j$	b. $x_i \partial_j$.
gilmore-lie-groups_ F01452	151	■ 1.000	$x_{\{i\}} \partial_{\{j\}}, x_{\{i\}}, \partial_{\{j\}}, I$	$x_i \partial_j, x_i, \partial_j, I$	c. $x_i \partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01453	151	■ 0.991	$x_{\{i\}} x_{\{j\}}, x_{\{i\}} \partial_{\{j\}}+\frac{1}{2} I \delta_{i j}, \partial_i \partial_j$	$x_i x_j, x_i \partial_j + \frac{1}{2} I \delta_{i j}, \partial_i \partial_j$	d. $x_i x_j, x_i \partial_j + \frac{1}{2} I \delta_{i j}, \partial_i \partial_j$.

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gilmore-lie-groups-F01454	151	■ 1.000	$x_{\{i\}} x_{\{j\}}, x_{\{i\}} \partial_{\{j\}}, \partial_{\{i\}} \partial_{\{j\}}, \partial_{\{i\}} \partial_{\{j\}}, x_{\{i\}}, \partial_{\{j\}}, I$	$x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$	e. $x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$.
gilmore-lie-groups-F01455	151	■ 1.000	$\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$	$\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$	f. $\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$.
gilmore-lie-groups-F01456	151	■ 1.000	$x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$	$x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$	g. $x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$.
gilmore-lie-groups-F01457	151	■ 0.799	2+1	2 + 1	6. The Lorentz, Poincaré, and Galilei groups in 2 + 1 dimensions (x , y and t) have Lie
gilmore-lie-groups-F01458	151	■ 0.799	(x , y	$(x, y$	6. The Lorentz, Poincaré, and Galilei groups in 2 + 1 dimensions (x , y and t) have Lie
gilmore-lie-groups-F01459	151	■ 0.799	t)	$t)$	6. The Lorentz, Poincaré, and Galilei groups in 2 + 1 dimensions (x , y and t) have Lie
gilmore-lie-groups-F01463	152	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}$	x_i, ∂_j	d. For each Lie algebra, express the generators in terms of the operators x_i, ∂_j .
gilmore-lie-groups-F01464	152	■ 0.781	$b_{\{i\}}^{\dagger}, b_{\{j\}}$	$b_i^{\dagger}, b_j$	e. For each Lie algebra, express the generators in terms of the boson operators $b_i^{\dagger}, b_j$,
gilmore-lie-groups-F01465	152	■ 1.000	$g_{\{i\} \{j\}} = C_{\{i\} \{r\}}^{\{s\}} C_{\{j\} \{s\}}^{\{r\}}$	$g_{ij} = C_{ir}^s C_{js}^r$	7. In a semisimple Lie algebra the Cartan–Killing metric $g_{ij} = C_{ir}^s C_{js}^r$ is nonsingular and
gilmore-lie-groups-F01466	152	■ 1.000	$g^{\{i\} \{j\}}$	g^{ij}	therefore the contravariant metric g^{ij} is well defined. Show that the bilinear operator
gilmore-lie-groups-F01467	152	■ 0.940	$\mathcal{C}^2 = g^{\{i\} \{j\}} X_{\{i\}} X_{\{j\}}$	$\mathcal{C}^2 = g^{ij} X_i X_j$	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $[\mathcal{C}^2, X_k] = 0$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups-F01468	152	■ 0.940	$\left[\mathcal{C}^2, X_k \right] = 0$	$[\mathcal{C}^2, X_k] = 0$	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $[\mathcal{C}^2, X_k] = 0$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups-F01469	152	■ 1.000	$\mathcal{C}^2$	$\mathcal{C}^2$	therefore be proportional to $\mathcal{C}^2$.
gilmore-lie-groups-F01470	152	■ 1.000	$C_{\{i\} \{j\} \{k\}} = C_{\{i\} \{j\}}^{\{r\}} g_{\{r\} \{k\}}$	$C_{ijk} = C_{ij}^r g_{rk}$	8. Show that $C_{ijk} = C_{ij}^r g_{rk}$ is a third order antisymmetric tensor: $C_{ijk} = C_{jki} = C_{kij} =$
gilmore-lie-groups-F01471	152	■ 1.000	$C_{\{i\} \{j\} \{k\}} = C_{\{j\} \{k\} \{i\}} = C_{\{k\} \{i\} \{j\}} =$	$C_{ijk} = C_{jki} = C_{kij} =$	8. Show that $C_{ijk} = C_{ij}^r g_{rk}$ is a third order antisymmetric tensor: $C_{ijk} = C_{jki} = C_{kij} =$
gilmore-lie-groups-F01472	152	■ 1.000	$-C_{\{k\} \{j\} \{i\}} = -C_{\{j\} \{i\} \{k\}} = -C_{\{i\} \{k\} \{j\}}$	$-C_{kji} = -C_{jik} = -C_{ikj}$	$-C_{kji} = -C_{jik} = -C_{ikj}$. (Hint: use the Jacobi identity.)
gilmore-lie-groups-F01473	154	■ 0.989	$\mathfrak{su}(1, 1)$	$\mathfrak{su}(1, 1)$	$\mathfrak{su}(1, 1)$) have the same complex extension and have their different commutation
gilmore-lie-groups-F01474	154	■ 1.000	Z, X	Z, X	It would be very useful to find vectors Z, X in the Lie algebra that obeyed the
gilmore-lie-groups-F01475	154	■ 1.000	$X = \sum_{i=1}^N a^{\{i\}} X_{\{i\}}$	$X = \sum_{i=1}^N a^i X_i$	we write $X = \sum_{i=1}^N a^i X_i$, where X_i form a basis set. Then
gilmore-lie-groups-F01476	154	■ 1.000	$a^{\{i\}}$	a^i	This equation has a nonzero solution for the coefficients a^i provided the secular
gilmore-lie-groups-F01477	154	■ 1.000	$\phi_{\{j\}}(Z)$	$\phi_j(Z)$	coefficients $\phi_j(Z)$ are homogeneous polynomials of degree j in the coefficients
gilmore-lie-groups-F01478	154	■ 0.973	$z^{\{i\}} \left(Z = \sum z^{\{i\}} X_{\{i\}} \right)$	$z^i \left(Z = \sum z^i X_i \right)$	$z^i (Z = \sum z^i X_i)$ that describe Z :
gilmore-lie-groups-F01479	155	■ 1.000	$a^{\dagger}, a, I = \left[a, a^{\dagger} \right]$	$a^{\dagger}, a, I = [a, a^{\dagger}]$	$a^{\dagger}, a, I = [a, a^{\dagger}]$ is
gilmore-lie-groups-F01480	155	■ 1.000	$(-\lambda)^N = 0$	$(-\lambda)^N = 0$	The converse is true. If the secular equation of an $N \times N$ matrix is $(-\lambda)^N = 0$, then
gilmore-lie-groups-F01481	155	■ 0.852	$X = \sum a_{\{i\}} X_{\{i\}} \in \mathfrak{su}(2)$	$X = \sum a_i X_i \in \mathfrak{su}(2)$	Example For $X = \sum a_i X_i \in \mathfrak{su}(2)$ the defining 2×2 matrix representation
gilmore-lie-groups-F01482	155	■ 0.953	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(X)$ are

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gilmore-lie-groups_ F01483	155	■ 0.953	$\mathfrak{R e g}(X)$	$\Re g(X)$	$\mathfrak{d e f}(X)$ and the regular 3×3 matrix representation $\Re g(X)$ are
gilmore-lie-groups_ F01484	155	■ 1.000	$\phi_2(\mathbf{a}) \geq 0$	$\phi_2(\mathbf{a}) \geq 0$	Since $\phi_2(\mathbf{a}) \geq 0$, this secular equation cannot be solved over the field of real num-
gilmore-lie-groups_ F01485	155	■ 0.760	$\mathfrak{s u}(2))$	$\mathfrak{s u}(2))$	ization to find the three (three is the dimension of $\mathfrak{s u}(2))$ roots: $\lambda = 0, \lambda = \pm i a$,
gilmore-lie-groups_ F01486	155	■ 0.760	$\lambda=0, \lambda=\pm i a$	$\lambda = 0, \lambda = \pm i a$	ization to find the three (three is the dimension of $\mathfrak{s u}(2))$ roots: $\lambda=0, \lambda=\pm i a$,
gilmore-lie-groups_ F01487	155	■ 0.999	$a^2=+a_1^2+a_2^2+a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$.
gilmore-lie-groups_ F01488	156	■ 0.861	$Y=\sum b_i Y_i \in \mathfrak{s u}(1,1)$	$Y = \sum b_i Y_i \in \mathfrak{s u}(1,1)$	Example For $Y = \sum b_i Y_i \in \mathfrak{s u}(1,1)$ the defining 2×2 matrix representation
gilmore-lie-groups_ F01489	156	■ 0.976	$\mathfrak{d e f}(Y)$	$\mathfrak{d e f}(Y)$	$\mathfrak{d e f}(Y)$ and the regular 3×3 matrix representation $\Re g(Y)$ are
gilmore-lie-groups_ F01490	156	■ 0.976	$\mathfrak{R e g}(Y)$	$\Re g(Y)$	$\mathfrak{d e f}(Y)$ and the regular 3×3 matrix representation $\Re g(Y)$ are
gilmore-lie-groups_ F01491	156	■ 0.999	$\left(a_1, a_2, a_3\right) \rightarrow\left(i b_1, i b_2, b_3\right)$	$\left(a_1, a_2, a_3\right) \rightarrow\left(i b_1, i b_2, b_3\right)$	$\left(a_1, a_2, a_3\right) \rightarrow\left(i b_1, i b_2, b_3\right)$, the coefficient $\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$ of the secular
gilmore-lie-groups_ F01492	156	■ 0.999	$\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$	$\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$	$\left(a_1, a_2, a_3\right) \rightarrow\left(i b_1, i b_2, b_3\right)$, the coefficient $\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$ of the secular
gilmore-lie-groups_ F01493	156	■ 0.953	$\phi_2(\mathbf{b})=-b_1^2-b_2^2+b_3^2$	$\phi_2(\mathbf{b})=-b_1^2-b_2^2+b_3^2$	equation for $\mathfrak{s u}(2)$ maps to $\phi_2(\mathbf{b})=-b_1^2-b_2^2+b_3^2$ for $\mathfrak{s u}(1,1)$. This same rotation
gilmore-lie-groups_ F01494	156	■ 1.000	ϕ_2	ϕ_2	For each algebra the functional forms of the nonzero coefficient ϕ_2 in the secular
gilmore-lie-groups_ F01495	157	■ 0.515	$\Re e g$	$\Re g$	equation of its regular representation, $\Re g$. Since the number of independent roots
gilmore-lie-groups_ F01496	157	■ 1.000	$\phi_j\left(z^i\right)$	$\phi_j\left(z^i\right)$	of the secular equation is equal to the number of independent coefficients $\phi_j\left(z^i\right)$,
gilmore-lie-groups_ F01497	157	■ 1.000	$X=Z$	$X=Z$	least one zero root (put $X=Z$ in (9.1)). For simple Lie algebras of dimension N ,
gilmore-lie-groups_ F01498	157	■ 1.000	$l^2 \sim N$	$l^2 \sim N$	$l^2 \sim N$, so describing commutation relations in terms of rank rather than dimension
gilmore-lie-groups_ F01499	157	■ 1.000	$\mathfrak{m}$	$\mathfrak{m}$	If $\mathfrak{m}$ is some matrix Lie algebra of $n \times n$ matrices, then any operator in $\mathfrak{m}$ can be
gilmore-lie-groups_ F01503	158	■ 0.955	ϕ_j	ϕ_j	the coordinates a_s^r by the operators X_s^r in the coefficients ϕ_j of the secular equation.
gilmore-lie-groups_ F01505	158	■ 0.751	$R^3,=\epsilon_{i j k}=+1$	$R^3,=\epsilon_{i j k}=+1$	(e.g., in $R^3,=\epsilon_{i j k}=+1$ for $(i j k)$ a cyclic permutation of (123), -1 for a cyclic
gilmore-lie-groups_ F01506	158	■ 0.751	$(i j k)$	$(i j k)$	(e.g., in $R^3,=\epsilon_{i j k}=+1$ for $(i j k)$ a cyclic permutation of (123), -1 for a cyclic
gilmore-lie-groups_ F01507	159	■ 1.000	$\phi_j\left(X^r{}_s\right)$	$\phi_j\left(X^r{}_s\right)$	these invariant operators $\phi_j\left(X^r{}_s\right)$ remain invariant when the matrices are replaced
gilmore-lie-groups_ F01508	159	■ 1.000	$O(n)$	$O(n)$	Example The orthogonal groups $O(n)$ and their subgroups $SO(n)$ have Lie alge-
gilmore-lie-groups_ F01509	159	■ 0.651	$d \times d$	$d \times d$	the $d \times d$ (the dimension of $\mathfrak{s o}(n)$ is $d=n(n-1) / 2$) regular matrix representation
gilmore-lie-groups_ F01510	159	■ 0.651	$\mathfrak{s o}(n)$	$\mathfrak{s o}(n)$	the $d \times d$ (the dimension of $\mathfrak{s o}(n)$ is $d=n(n-1) / 2$) regular matrix representation
gilmore-lie-groups_ F01511	159	■ 0.651	$d=n(n-1) / 2$	$d=n(n-1) / 2$	the $d \times d$ (the dimension of $\mathfrak{s o}(n)$ is $d=n(n-1) / 2$) regular matrix representation
gilmore-lie-groups_ F01512	159	■ 1.000	$\mathfrak{d e f}(X)^t=-\mathfrak{d e f}(X)$	$\mathfrak{d e f}(X)^t=-\mathfrak{d e f}(X)$	Lie algebra consists of antisymmetric matrices, $\mathfrak{d e f}(X)^t=-\mathfrak{d e f}(X)$, and we find

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gilmore-lie-groups_F01513	159	■ 0.986	[n / 2]	$[n/2]$	even coefficients. Therefore the algebra $\mathfrak{so}(n)$ has rank $[n/2]$.
gilmore-lie-groups_F01514	159	■ 0.996	j=2	$j = 2$	setting $j = 2$ in Fig. 9.2 for the generators X_{ij} of $SO(n)$. Since $X_{ij} = -X_{ji}$, it is
gilmore-lie-groups_F01515	159	■ 0.996	$X_{\{i\ j\}} = -X_{\{j\ i\}}$	$X_{ij} = -X_{ji}$	setting $j = 2$ in Fig. 9.2 for the generators X_{ij} of $SO(n)$. Since $X_{ij} = -X_{ji}$, it is
gilmore-lie-groups_F01516	159	■ 1.000	$\mathfrak{so}(5)$	$\mathfrak{so}(5)$	higher order Casimir invariant operators. For example, for $\mathfrak{so}(5)$ the fourth order
gilmore-lie-groups_F01518	160	■ 0.940	$v^m = \epsilon^{ijklm} X_{\{i\ j\}} X_{\{k\ l\}}$	$v^m = \epsilon^{ijklm} X_{ij} X_{kl}$	where the components of the five-vector $\mathbf{v}$ are $v^m = \epsilon^{ijklm} X_{ij} X_{kl}$, for example
gilmore-lie-groups_F01519	160	■ 1.000	n / 2	$n/2$	square. Its square root, of order $n/2$, should be taken as an appropriate functionally
gilmore-lie-groups_F01520	160	■ 1.000	Z=0	$Z = 0$	the opposite extreme, the choice $Z = 0$ is not clever since all X obey the same
gilmore-lie-groups_F01521	160	■ 1.000	$[Z, X] = 0$	$[Z, X] = 0X$	eigenvalue equation $[Z, X] = 0X$.)
gilmore-lie-groups_F01522	160	■ 1.000	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$	dimensional algebra of two photon operators spanned by $\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$,
gilmore-lie-groups_F01523	160	■ 1.000	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$. A useful choice for Z is
gilmore-lie-groups_F01524	161	■ 1.000	$\pm 2z_1, \pm z_1, 0, 0$	$\pm 2z_1, \pm z_1, 0, 0$	representation are $\pm 2z_1, \pm z_1, 0, 0$ and the commutation relations can be summa-
gilmore-lie-groups_F01525	161	■ 0.999	$X_{\{2,0\}} = a^{\dagger 2}, X_{\{1,0\}} = a^\dagger, X_{\{0,0\}} = \hat{n} + \frac{1}{2} I, X_{\{-1,0\}} = a, X_{\{-2,0\}} = a^2$	$X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$	where $X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$. We
gilmore-lie-groups_F01526	161	■ 1.000	$k, l \in \{-2, -1, 0, +1, +2\}$	$k, l \in \{-2, -1, 0, +1, +2\}$	also see that if $k, l \in \{-2, -1, 0, +1, +2\}$
gilmore-lie-groups_F01527	161	■ 1.000	$k+l$	$k + l$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01528	161	■ 1.000	$\{-2, -1, 0, +1, +2\}$	$\{-2, -1, 0, +1, +2\}$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01529	161	■ 1.000	$k+l=0$	$k + l = 0$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01530	161	■ 0.968	$(0,0): \hat{n} + \frac{1}{2}$	$(0,0): \hat{n} + \frac{1}{2}$	$(0,0): \hat{n} + \frac{1}{2}$ and I .
gilmore-lie-groups_F01531	162	■ 1.000	$\alpha_1, \alpha_2, \dots, \alpha_l$	$\alpha_1, \alpha_2, \dots, \alpha_l$	(ii) the number of independent roots $\alpha_1, \alpha_2, \dots, \alpha_l$ of the secular equation; these l inde-
gilmore-lie-groups_F01532	162	■ 0.570	$(\alpha_1, \alpha_2, \dots, \alpha_l)$	$(\alpha_1, \alpha_2, \dots, \alpha_l)$	$\alpha_1, \alpha_2, \dots, \alpha_l$ in a root space ;
gilmore-lie-groups_F01533	162	■ 1.000	V_β	V_β	(i) If α and β are roots with subspaces V_α and V_β in the Lie algebra, then
gilmore-lie-groups_F01534	162	■ 1.000	$V_{\alpha+\beta}$	$V_{\alpha+\beta}$	That is, the commutator of any vector in V_α with any vector in V_β is a vector in $V_{\alpha+\beta}$.
gilmore-lie-groups_F01535	162	■ 1.000	$\alpha + \beta$	$\alpha + \beta$	If $\alpha + \beta$ is not a root, the commutator vanishes.
gilmore-lie-groups_F01536	162	■ 1.000	$H_1, H_2, \dots, H_l$	$H_1, H_2, \dots, H_l$	(ii) The l basis vectors $H_1, H_2, \dots, H_l$ in the l -dimensional subspace V_0 commute:
gilmore-lie-groups_F01537	162	■ 1.000	$V_\alpha (\alpha \neq 0)$	$V_\alpha (\alpha \neq 0)$	(iii) Each subspace $V_\alpha (\alpha \neq 0)$ is one dimensional. Therefore each subspace V_α is spanned
gilmore-lie-groups_F01538	162	■ 0.995	E_α	E_α	by an operator E_α that can be labeled by the root α . As a result (i.e., $[V_0, V_\alpha] \subset V_\alpha$),
gilmore-lie-groups_F01539	162	■ 0.995	$[V_0, V_\alpha] \subset V_\alpha$	$[V_0, V_\alpha] \subset V_\alpha$	by an operator E_α that can be labeled by the root α . As a result (i.e., $[V_0, V_\alpha] \subset V_\alpha$),

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gilmore-lie-groups_F01540	162	■ 1.000	$-\alpha$	$-\alpha$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01541	162	■ 1.000	$c \alpha$	$c\alpha$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01542	162	■ 1.000	$ c =1$	$ c = 1$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01543	162	■ 1.000	$E_{-\alpha}$	$E_{-\alpha}$	(v) The commutator of E_α and $E_{-\alpha}$ is in V_0 , so can be expanded as a linear superposition
gilmore-lie-groups_F01544	162	■ 1.000	α^i	α^i	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij}\alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01545	162	■ 1.000	$\alpha_{\{j\}}$	α_j	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij}\alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01546	162	■ 1.000	$\alpha^i = h^{ij} \alpha_j$	$\alpha^i = h^{ij}\alpha_j$	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij}\alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01547	163	■ 1.000	$h^{ij} = \delta^{ij}$	$h^{ij} = \delta^{ij}$	then $h^{ij} = \delta^{ij}$ and we can identify α^i with α_i : $\alpha^i = \alpha_i$.
gilmore-lie-groups_F01548	163	■ 1.000	$\alpha_i : \alpha^i = \alpha_i$	$\alpha_i : \alpha^i = \alpha_i$	then $h^{ij} = \delta^{ij}$ and we can identify α^i with α_i : $\alpha^i = \alpha_i$.
gilmore-lie-groups_F01549	163	■ 1.000	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$ when $\alpha + \beta$ is a nonzero root.
gilmore-lie-groups_F01550	163	■ 1.000	$E_\alpha, E_\beta, E_\gamma$	$E_\alpha, E_\beta, E_\gamma$	$E_\alpha, E_\beta, E_\gamma$ of three nonzero roots that sum to zero
gilmore-lie-groups_F01551	163	■ 1.000	$\alpha, \beta + k\alpha$	$\alpha, \beta + k\alpha$	are not roots. By applying the Jacobi identity to roots $\alpha, \beta + k\alpha$, and $-\alpha$ we obtain
gilmore-lie-groups_F01553	164	■ 1.000	$N_{\alpha, \beta + n\alpha} = 0$	$N_{\alpha, \beta + n\alpha} = 0$	The initial condition $N_{\alpha, \beta + n\alpha} = 0$ leads to
gilmore-lie-groups_F01554	164	■ 1.000	$N_{-\alpha, \beta - m\alpha}^2 = N_{\alpha, \beta - (m+1)\alpha}^2 = 0$	$N_{-\alpha, \beta - m\alpha}^2 = N_{\alpha, \beta - (m+1)\alpha}^2 = 0$	The other boundary condition $N_{-\alpha, \beta - m\alpha}^2 = N_{\alpha, \beta - (m+1)\alpha}^2 = 0$ leads to
gilmore-lie-groups_F01555	164	■ 1.000	$N_{\alpha, \beta + k\alpha}^2 = (n - k)(m + k + 1)(\alpha \cdot \alpha)/2 \geq 0$	$N_{\alpha, \beta + k\alpha}^2 = (n - k)(m + k + 1)(\alpha \cdot \alpha)/2 \geq 0$	(i) $N_{\alpha, \beta + k\alpha}^2 = (n - k)(m + k + 1)(\alpha \cdot \alpha)/2 \geq 0$. We use this expression because it shows
gilmore-lie-groups_F01556	164	■ 1.000	$\alpha \cdot \beta > 0$	$\alpha \cdot \beta > 0$	clearly how the boundary conditions are imposed. We note that $\alpha \cdot \beta > 0$ when $m -$
gilmore-lie-groups_F01557	164	■ 1.000	$m -$	$m -$	clearly how the boundary conditions are imposed. We note that $\alpha \cdot \beta > 0$ when $m -$
gilmore-lie-groups_F01558	164	■ 1.000	$n > 0$	$n > 0$	$n > 0$ and $\alpha \cdot \beta < 0$ when $m - n < 0$.
gilmore-lie-groups_F01559	164	■ 1.000	$\alpha \cdot \beta < 0$	$\alpha \cdot \beta < 0$	$n > 0$ and $\alpha \cdot \beta < 0$ when $m - n < 0$.
gilmore-lie-groups_F01560	164	■ 1.000	$m - n < 0$	$m - n < 0$	$n > 0$ and $\alpha \cdot \beta < 0$ when $m - n < 0$.
gilmore-lie-groups_F01563	165	■ 0.456	$l(l$	$l(l$	constants of a semisimple Lie algebra. The basis vectors are the l (l is the rank)
gilmore-lie-groups_F01564	165	■ 0.977	$\mathbf{H} = (H_1, H_2, \dots, H_l)$	$\mathbf{H} = (H_1, H_2, \dots, H_l)$	operators $\mathbf{H} = (H_1, H_2, \dots, H_l)$ and the “shift” operators E_α , one corresponding
gilmore-lie-groups_F01565	165	■ 0.994	$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$	$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$	to each nonzero root. The root vector $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$ has l components. The
gilmore-lie-groups_F01566	166	■ 1.000	C_2	C_2	Figure 9.7. Root space C_2 . The roots are expressed in terms of a Cartesian
gilmore-lie-groups_F01569	166	■ 1.000	$\mathbf{e}_1$	$\mathbf{e}_1$	If we choose orthogonal vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ in a two-dimensional Euclidean space,
gilmore-lie-groups_F01570	166	■ 1.000	$\mathbf{e}_2$	$\mathbf{e}_2$	If we choose orthogonal vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ in a two-dimensional Euclidean space,
gilmore-lie-groups_F01571	166	■ 1.000	$\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm \mathbf{e}_1 \pm \mathbf{e}_2$	$\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm \mathbf{e}_1 \pm \mathbf{e}_2$	the nonzero roots for C_2 are $\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm \mathbf{e}_1 \pm \mathbf{e}_2$. The 10 basis vectors in the Lie

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gilmore-lie-groups_ F01572	166	■ 0.986	$H_{\{i\}}, i=1,2$	$H_i, i = 1, 2$	algebra are $H_i, i = 1, 2$, and E_α , with α the eight nonzero roots. We normalize
gilmore-lie-groups_ F01573	166	■ 1.000	$\sum \alpha \cdot \alpha = 2$	$\sum \alpha \cdot \alpha = 2$	these roots by $\sum \alpha \cdot \alpha = 2$ so that
gilmore-lie-groups_ F01609	168	■ 1.000	$\mathfrak{sl}(2; \mathbb{C}))$	$\mathfrak{sl}(2; \mathbb{C}))$	extension $\mathfrak{sl}(2; \mathbb{C}))$. Root spaces classify commutation relations of these complex
gilmore-lie-groups_ F01610	168	■ 1.000	$\alpha_{\{i\}}$	α_i	real subalgebras of these complex algebras – some roots α_i and structure constants
gilmore-lie-groups_ F01611	168	■ 1.000	$\frac{1}{2}\{a^\dagger, a\}$	$\frac{1}{2}\{a^\dagger, a\}$	1. Construct the regular representation for the two-photon operator algebra: $\frac{1}{2}\{a^\dagger, a\}$,
gilmore-lie-groups_ F01612	168	■ 1.000	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$. Determine the secular equation for this matrix. Determine the rank
gilmore-lie-groups_ F01613	168	■ 1.000	$X_{\{i\ j\}}=x^{\{i\}} \partial_{\{j\}}-x^{\{j\}} \partial_{\{i\}}(1 \leq i < j \leq 4)$	$X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$	ical basis $X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$ these operators are $\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$ and
gilmore-lie-groups_ F01614	168	■ 1.000	$\mathcal{C}_2=\sum_{i < j} X_{\{i\ j\}}^2$	$\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$	ical basis $X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$ these operators are $\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$ and
gilmore-lie-groups_ F01615	168	■ 0.980	$\mathcal{C}_2^{\prime}=\mathcal{C}_2^{\prime}(\text{prime})=X_{\{12\}} X_{\{34\}}-X_{\{13\}} X_{\{24\}}+X_{\{14\}} X_{\{23\}}$	$\mathcal{C}_2' = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	$\mathcal{C}_2' = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$.
gilmore-lie-groups_ F01616	168	■ 0.994	$\mathfrak{su}(4)$	$\mathfrak{su}(4)$	3. The Lie algebra $\mathfrak{su}(4)$ has a 4×4 defining matrix representation and a 15×15 regular
gilmore-lie-groups_ F01617	168	■ 1.000	$\mathfrak{so}(2n+1)$	$\mathfrak{so}(2n+1)$	4. For $\mathfrak{so}(2n+1)$ the invariant operators (Casimir operators) are of degree 2, 4, ..., $2n$.
gilmore-lie-groups_ F01618	168	■ 1.000	$2, 4, \ldots, 2n$	$2, 4, \ldots, 2n$	4. For $\mathfrak{so}(2n+1)$ the invariant operators (Casimir operators) are of degree <u>$2, 4, \ldots, 2n$</u> .
gilmore-lie-groups_ F01619	168	■ 1.000	$\mathfrak{so}(2n)$	$\mathfrak{so}(2n)$	This is true also for $\mathfrak{so}(2n)$, with one difference: the invariant operator of degree $2n$
gilmore-lie-groups_ F01620	168	■ 0.600	$\mathcal{C}_n^{\prime}=\epsilon^{i_1 i_2 \cdots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \cdots X_{i_{2n-1} i_{2n}}$	$\mathcal{C}_n' = \epsilon^{i_1 i_2 \cdots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \cdots X_{i_{2n-1} i_{2n}}$	$\mathcal{C}_n' = \epsilon^{i_1 i_2 \cdots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \cdots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}_2'$ for $\mathfrak{so}(4)$ and $\mathcal{C}_3'$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01621	168	■ 0.600	$\mathcal{C}_2^{\prime}$	$\mathcal{C}_2'$	$\mathcal{C}_n' = \epsilon^{i_1 i_2 \cdots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \cdots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}_2'$ for $\mathfrak{so}(4)$ and $\mathcal{C}_3'$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01622	168	■ 0.600	$\mathcal{C}_3^{\prime}$	$\mathcal{C}_3'$	$\mathcal{C}_n' = \epsilon^{i_1 i_2 \cdots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \cdots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}_2'$ for $\mathfrak{so}(4)$ and $\mathcal{C}_3'$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01623	168	■ 0.600	$\mathfrak{so}(6)$	$\mathfrak{so}(6)$	$\mathcal{C}_n' = \epsilon^{i_1 i_2 \cdots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \cdots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}_2'$ for $\mathfrak{so}(4)$ and $\mathcal{C}_3'$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01624	168	■ 1.000	$\mathfrak{su}(4)=\mathfrak{so}(6)$	$\mathfrak{su}(4) = \mathfrak{so}(6)$	5. In Chapter 11 we will show that <u>$\mathfrak{su}(4) = \mathfrak{so}(6)$</u> . Both Lie algebras have invariant
gilmore-lie-groups_ F01625	168	■ 1.000	$a_{\{i\}}^{\dagger} a_{\{j\}}+\frac{1}{2} \delta_{\{i\} \{j\}}$	$a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$	two modes. This algebra is ten dimensional. It contains the four operators $a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$
gilmore-lie-groups_ F01626	169	■ 0.999	$(1 \leq i, j \leq 2)$	$(1 \leq i, j \leq 2)$	<u>$1 \leq i, j \leq 2$</u> and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_ F01627	169	■ 0.999	$a_{\{i\}}^{\dagger} a_{\{j\}}^{\dagger}$	$a_i^\dagger a_j^\dagger$	$(1 \leq i, j \leq 2)$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_ F01628	169	■ 0.999	$a_{\{i\}} a_{\{j\}} \left(a_{\{i\}} a_{\{j\}}=a_{\{j\}} a_{\{i\}} \right)$	$a_i a_j (a_i a_j = a_j a_i)$	$(1 \leq i, j \leq 2)$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_ F01629	169	■ 1.000	$a_{\{i\}}^{\dagger} a_{\{i\}}+\frac{1}{2} \rightarrow H_{\{i\}}, a_{\{i\}}^{\dagger} a_{\{j\}}^{\dagger} \rightarrow E_{+\mathbf{e}_i+\mathbf{e}_j}(i \neq j), a_{\{i\}}^{\dagger} a_{\{j\}} \rightarrow E_{+\mathbf{e}_i-\mathbf{e}_j}(i \neq j), a_{\{i\}}^{\dagger} a_{\{i\}} \rightarrow E_{+\mathbf{e}_i}, a_{\{j\}}^{\dagger} a_{\{j\}} \rightarrow E_{+\mathbf{e}_j}, a_{\{i\}}^{\dagger} a_{\{j\}} \rightarrow E_{+\mathbf{e}_i+\mathbf{e}_j}(i \neq j), a_{\{i\}}^{\dagger} a_{\{j\}} \rightarrow E_{+\mathbf{e}_i-\mathbf{e}_j}(i \neq j)$	$a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+\mathbf{e}_i+\mathbf{e}_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+\mathbf{e}_i-\mathbf{e}_j} (i \neq j)$	identification is: <u>$a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+\mathbf{e}_i+\mathbf{e}_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+\mathbf{e}_i-\mathbf{e}_j} (i \neq j)$</u> ,

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gilmore-lie-groups_ F01630	169	■ 1.000	$a_{\{i\}} a_{\{j\}} \rightarrow E_{-\mathbf{e}_{\{i\}} - \mathbf{e}_{\{j\}}} (i \neq j), a_{\{i\}}^{\dagger} a_{\{i\}}^{\dagger} \rightarrow E_{+2\mathbf{e}_{\{i\}}}, a_{\{i\}} \rightarrow E_{-2\mathbf{e}_{\{i\}}}$	$a_i a_j \leftrightarrow E_{-\mathbf{e}_i - \mathbf{e}_j} (i \neq j), a_i^{\dagger} a_i^{\dagger} \leftrightarrow E_{+2\mathbf{e}_i}, a_i a_i \leftrightarrow E_{-2\mathbf{e}_i}$	$a_i a_j \leftrightarrow E_{-\mathbf{e}_i - \mathbf{e}_j} (i \neq j), a_i^{\dagger} a_i^{\dagger} \leftrightarrow E_{+2\mathbf{e}_i}, a_i a_i \leftrightarrow E_{-2\mathbf{e}_i}.$
gilmore-lie-groups_ F01631	169	■ 1.000	$D_{\{2\}}$	D_2	is six dimensional. Show that the resulting root space diagram is D_2 (Fig. 9.6). Why
gilmore-lie-groups_ F01632	169	■ 0.995	$f_{\{i\}}^{\dagger} f_{\{i\}}^{\dagger} = 0$	$f_i^{\dagger} f_i^{\dagger} = 0$	the difference? (Hint: $f_i^{\dagger} f_i^{\dagger} = 0$.)
gilmore-lie-groups_ F01633	169	■ 1.000	$X, Y \in \mathfrak{su}(2)$	$X, Y \in \mathfrak{su}(2)$	ular representation for the former. Choose $X, Y \in \mathfrak{su}(2)$ and compute $(X, Y) =$
gilmore-lie-groups_ F01634	169	■ 0.577	$\text{tr}[\mathfrak{d} \mathfrak{e} \mathfrak{f}(X) \mathfrak{d} \mathfrak{e} \mathfrak{f}(Y)]$	$\text{tr}[\mathfrak{d} \mathfrak{e} \mathfrak{f}(X) \mathfrak{d} \mathfrak{e} \mathfrak{f}(Y)]$	$\text{tr} [\mathfrak{d} \mathfrak{e} \mathfrak{f}(X) \mathfrak{d} \mathfrak{e} \mathfrak{f}(Y)]$ by taking the trace of the 2×2 matrices in $\mathfrak{su}(2)$ that represent
gilmore-lie-groups_ F01635	169	■ 0.569	$\mathfrak{su}(2) : (X, Y) = \text{tr}[\mathfrak{R} \mathfrak{e} \mathfrak{g}(X) \mathfrak{R} \mathfrak{e} \mathfrak{g}(Y)]$	$\mathfrak{su}(2) : (X, Y) = \text{tr}[\mathfrak{R} \mathfrak{e} \mathfrak{g}(X) \mathfrak{R} \mathfrak{e} \mathfrak{g}(Y)]$	the regular matrix representation of $\mathfrak{su}(2)$: $(X, Y) = \text{tr} [\mathfrak{R} \mathfrak{e} \mathfrak{g}(X) \mathfrak{R} \mathfrak{e} \mathfrak{g}(Y)]$. Show that
gilmore-lie-groups_ F01636	169	■ 0.995	$(n^2 - 1) \times (n^2 - 1)$	$(n^2 - 1) \times (n^2 - 1)$	the $n \times n$ defining matrix representation and in the $(n^2 - 1) \times (n^2 - 1)$ regular matrix
gilmore-lie-groups_ F01637	169	■ 1.000	$Y = X$	$Y = X$	constant? (Hint: set $Y = X$ and choose a special X , for example $X = H_1$.)
gilmore-lie-groups_ F01638	169	■ 1.000	$X = H_{\{1\}}$	$X = H_1$	constant? (Hint: set $Y = X$ and choose a special X , for example $X = H_1$.)
gilmore-lie-groups_ F01639	169	■ 0.999	$1 \leq i, j \leq N$	$1 \leq i, j \leq N$	form (b boson operators; f fermion operators; $1 \leq i, j \leq N$):
gilmore-lie-groups_ F01640	169	■ 1.000	$\mathbf{R} = \frac{1}{2} \sum_{\alpha > 0} \alpha$	$\mathbf{R} = \frac{1}{2} \sum_{\alpha > 0} \alpha$	11. Compute $\mathbf{R} = \frac{1}{2} \sum_{\alpha > 0} \alpha$, half the sum over all positive roots, in each of the simple Lie
gilmore-lie-groups_ F01641	169	■ 1.000	$B_{\{n\}}, R_{\{i\}} = \frac{1}{2} (2n + 1) - i$	$B_n, R_i = \frac{1}{2} (2n + 1) - i$	algebras. For example, for $B_n, R_i = \frac{1}{2} (2n + 1) - i$ and the spectrum is
gilmore-lie-groups_ F01642	169	■ 0.996	$\mathbf{M}$	$\mathbf{M}$	where $\mathbf{M}$ is the highest weight in the representation. For the $(2j + 1)$ dimensional
gilmore-lie-groups_ F01643	169	■ 1.000	$\mathfrak{so}(3), \mathbf{M} = j, \mathbf{R} = R_1 = \frac{1}{2}$	$\mathfrak{so}(3), \mathbf{M} = j, \mathbf{R} = R_1 = \frac{1}{2}$	representation of $\mathfrak{so}(3)$, $\mathbf{M} = j$, $\mathbf{R} = R_1 = \frac{1}{2}$ and $\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$
gilmore-lie-groups_ F01644	169	■ 1.000	$\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$	$\mathcal{C}^2(j) = \left(j + \frac{1}{2}\right)^2 - \left(0 + \frac{1}{2}\right)^2 =$	representation of $\mathfrak{so}(3)$, $\mathbf{M} = j$, $\mathbf{R} = R_1 = \frac{1}{2}$ and $\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$
gilmore-lie-groups_ F01645	169	■ 1.000	$j(j + 1)$	$j(j + 1)$	$j(j + 1).$
gilmore-lie-groups_ F01646	169	■ 1.000	$A_{\{n-1\}}$	A_{n-1}	a. Show that the Weyl group for A_{n-1} is of order $n!$, the Weyl group for D_n is of
gilmore-lie-groups_ F01647	169	■ 1.000	$D_{\{n\}}$	D_n	a. Show that the Weyl group for A_{n-1} is of order $n!$, the Weyl group for D_n is of
gilmore-lie-groups_ F01648	169	■ 0.919	$2^{n-1} n!$	$2^{n-1} n!$	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_ F01649	169	■ 0.919	$B_{\{n\}}$	B_n	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_ F01650	169	■ 0.919	$C_{\{n\}}$	C_n	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_ F01651	169	■ 0.919	$2^n n!$	$2^n n!$	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_ F01652	170	■ 1.000	$f_{\{r\}}(\Gamma)$	$f_r(\Gamma)$	the index $f_r(\Gamma)$.
gilmore-lie-groups_ F01653	170	■ 1.000	$\mathcal{C}^r$	$\mathcal{C}^r$	In this expression $\mathcal{C}^r$ is the value of the r th Casimir invariant in the representation
gilmore-lie-groups_ F01654	170	■ 1.000	$\mathfrak{so}(2n), \mathfrak{so}(2n + 1), \mathfrak{sp}(2n)$	$\mathfrak{so}(2n), \mathfrak{so}(2n + 1), \mathfrak{sp}(2n)$	15. The matrix Lie algebras $\mathfrak{so}(2n), \mathfrak{so}(2n + 1), \mathfrak{sp}(2n)$ have the form $\sum_{ij} a^{ij} M_{ij}$, where

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gilmore-lie-groups_ F01655	170	■ 1.000	$\sum_{ij} a^{ij} M_{ij}$	$\sum_{ij} a^{ij} M_{ij}$	15. The matrix Lie algebras $\mathfrak{so}(2n)$, $\mathfrak{so}(2n+1)$, $\mathfrak{sp}(2n)$ have the form $\sum_{ij} a^{ij} M_{ij}$, where
gilmore-lie-groups_ F01656	170	■ 1.000	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	M is $2n \times 2n$ for $\mathfrak{so}(2n), \mathfrak{sp}(2n)$ and $(2n+1) \times (2n+1)$ for $\mathfrak{so}(2n+1)$, and suitable
gilmore-lie-groups_ F01657	170	■ 1.000	$(2n+1) \times (2n+1)$	$(2n+1) \times (2n+1)$	M is $2n \times 2n$ for $\mathfrak{so}(2n), \mathfrak{sp}(2n)$ and $(2n+1) \times (2n+1)$ for $\mathfrak{so}(2n+1)$, and suitable
gilmore-lie-groups_ F01658	170	■ 1.000	a^{ij}	a^{ij}	reality restrictions are imposed on the coefficients a^{ij} .
gilmore-lie-groups_ F01659	170	■ 1.000	$\phi_r(a^{ij})$	$\phi_r(a^{ij})$	b. Write down the coefficients $\phi_r(a^{ij})$ that occur in the secular equation for each of
gilmore-lie-groups_ F01660	170	■ 0.999	$\epsilon_{i_1 i_2 \dots i_l}$	$\epsilon_{i_1 i_2 \dots i_l}$	d. Express the even coefficients in terms of the Levi-Civita skew tensors $\epsilon_{i_1 i_2 \dots i_l}$
gilmore-lie-groups_ F01661	170	■ 1.000	$(l=2n, 2n+1, 2n)$	$(l=2n, 2n+1, 2n)$	$(l=2n, 2n+1, 2n)$.
gilmore-lie-groups_ F01662	170	■ 1.000	$\phi_{2n}(a^{ij})$	$\phi_{2n}(a^{ij})$	g. Show that $\phi_{2n}(a^{ij})$ is a perfect square for $\mathfrak{so}(2n)$; write down its square root; show
gilmore-lie-groups_ F01663	171	■ 0.999	$\text{tr } M^2 = -2\theta \cdot \theta$	$\text{tr } M^2 = -2\theta \cdot \theta$	a. Show $\text{tr } M^2 = -2\theta \cdot \theta$.
gilmore-lie-groups_ F01664	171	■ 0.987	$\text{tr } \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$	$\text{tr } \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$	b. Show $\text{tr } \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$.
gilmore-lie-groups_ F01665	171	■ 1.000	$[\mathbf{J}, \text{tr } \mathcal{M}^2] = 0$	$[\mathbf{J}, \text{tr } \mathcal{M}^2] = 0$	c. Show $[\mathbf{J}, \text{tr } \mathcal{M}^2] = 0$.
gilmore-lie-groups_ F01666	171	■ 0.999	$\text{tr } \mathcal{M}^{2n+1} = 0$	$\text{tr } \mathcal{M}^{2n+1} = 0$	d. Show $\text{tr } \mathcal{M}^{2n+1} = 0$ and $\text{tr } \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$.
gilmore-lie-groups_ F01667	171	■ 0.999	$\text{tr } \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$	$\text{tr } \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$	d. Show $\text{tr } \mathcal{M}^{2n+1} = 0$ and $\text{tr } \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$.
gilmore-lie-groups_ F01668	171	■ 1.000	$[X_i, X_j] = C_{ij}^k X_k$	$[X_i, X_j] = C_{ij}^k X_k$	commutation relations $[X_i, X_j] = C_{ij}^k X_k$. There are two linear vector spaces, $V^{(1)}$
gilmore-lie-groups_ F01669	171	■ 1.000	$V^{(1)}$	$V^{(1)}$	commutation relations $[X_i, X_j] = C_{ij}^k X_k$. There are two linear vector spaces, $V^{(1)}$
gilmore-lie-groups_ F01670	171	■ 1.000	$V^{(2)}$	$V^{(2)}$	and $V^{(2)}$, that carry irreducible representations of this Lie algebra: $X_i \rightarrow \Gamma^{(1)}(X_i) =$
gilmore-lie-groups_ F01671	171	■ 1.000	$X_i \rightarrow \Gamma^{(1)}(X_i) =$	$X_i \rightarrow \Gamma^{(1)}(X_i) =$	and $V^{(2)}$, that carry irreducible representations of this Lie algebra: $X_i \rightarrow \Gamma^{(1)}(X_i) =$
gilmore-lie-groups_ F01672	171	■ 1.000	Y_i	Y_i	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_ F01673	171	■ 1.000	$X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$	$X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_ F01674	171	■ 1.000	$g^{ij} Y_i Z_j$	$g^{ij} Y_i Z_j$	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_ F01675	171	■ 0.857	$(Y+Z)_k$	$(Y+Z)_k$	$(Y+Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$).
gilmore-lie-groups_ F01676	171	■ 0.857	$\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$	$\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$	$(Y+Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$).
gilmore-lie-groups_ F01677	171	■ 1.000	$X^0 = I_n$	$X^0 = I_n$	of X , starting at $X^0 = I_n$:
gilmore-lie-groups_ F01678	171	■ 1.000	$\lambda_i(X)$	$\lambda_i(X)$	eigenvalues $\lambda_i(X)$ or the independent coefficients $\phi_i(X)$ of the secular equation
gilmore-lie-groups_ F01679	171	■ 1.000	$\phi_i(X)$	$\phi_i(X)$	eigenvalues $\lambda_i(X)$ or the independent coefficients $\phi_i(X)$ of the secular equation
gilmore-lie-groups_ F01680	171	■ 0.967	$e^{i\phi J_z}$	$e^{i\phi J_z}$	d. Compute $e^{i\phi J_z}$ for the $(2j+1)$ dimensional matrix representations of $SU(2)$ by

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gilmore-lie-groups_ F01681	171	■ 0.999	$j=\frac{1}{2}, 1, \frac{3}{2}, 2$	$j = \frac{1}{2}, 1, \frac{3}{2}, 2$	computing the vanderMonde matrices. Show that for $j = \frac{1}{2}, 1, \frac{3}{2}, 2$ the resulting
gilmore-lie-groups_ F01682	172	■ 1.000	$f_{\{j\}}(\phi)$	$f_j(\phi)$	e. Invert each of these van der Monde matrices and determine the functions $f_j(\phi)$ in
gilmore-lie-groups_ F01683	172	■ 1.000	$e^{\{X\}}$	e^X	the expansions of e^X for $X \in \mathfrak{su}(2)$. In particular, show
gilmore-lie-groups_ F01684	172	■ 1.000	$X \in \mathfrak{su}(\frac{s}{u}(2))$	$X \in \mathfrak{su}(2)$	the expansions of e^X for $X \in \mathfrak{su}(2)$. In particular, show
gilmore-lie-groups_ F01686	257	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	the charge on the proton) and spin $\frac{1}{2}$. The Dirac equation $H_D \psi = E \psi$ is a matrix
gilmore-lie-groups_ F01705	172	■ 1.000	$j=\frac{1}{2}$	$j = \frac{1}{2}$	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:
gilmore-lie-groups_ F01706	172	■ 1.000	$l=1$	$l = 1$	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:
gilmore-lie-groups_ F01707	173	■ 1.000	$l+1$	$l + 1$	grams to construct rank $l + 1$ root space diagrams. However, it is not easy
gilmore-lie-groups_ F01708	174	■ 1.000	m, n	m, n	The coefficients $N_{\alpha, \beta}$ are defined in terms of the nonnegative integers m, n by
gilmore-lie-groups_ F01709	174	■ 1.000	$\beta+k \alpha$	$\beta + k\alpha$	where $\beta + k\alpha$ is a root only for $k = -m, \dots, +n$. The roots are normalized by
gilmore-lie-groups_ F01710	174	■ 1.000	$k=-m, \ldots, +n$	$k = -m, \dots, +n$	where $\beta + k\alpha$ is a root only for $k = -m, \dots, +n$. The roots are normalized by
gilmore-lie-groups_ F01711	174	■ 0.999	$\beta^{\wedge\prime}$	β'	The root β' is obtained by reflecting β in the hyperplane orthogonal to α . These two
gilmore-lie-groups_ F01712	174	■ 1.000	$2(\alpha \cdot \beta) / (\alpha \cdot \alpha)=n$	$2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$	If we write $2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$ and $2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$, where n and n' are
gilmore-lie-groups_ F01713	174	■ 1.000	$2(\alpha \cdot \beta) / (\beta \cdot \beta)=n^{\wedge\prime}$	$2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$	If we write $2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$ and $2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$, where n and n' are
gilmore-lie-groups_ F01714	174	■ 0.992	$\pm \mathbf{e}_1$	$\pm \mathbf{e}_1$	vectors $\pm \mathbf{e}_1$. To construct rank-two root spaces, add a noncollinear vector to this
gilmore-lie-groups_ F01715	174	■ 1.000	$A_2, B_2=C_2, D_2$	$A_2, B_2 = C_2, D_2$	are $A_2, B_2 \equiv C_2, D_2$ and G_2 , as shown in Fig. 9.6.
gilmore-lie-groups_ F01734	175	■ 0.998	$(l-1)$	$(l - 1)$	this way from rank- $(l - 1)$ root space diagrams. We find by this building-up process
gilmore-lie-groups_ F01735	175	■ 0.998	D_l	D_l	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_ F01736	175	■ 0.998	$\pm \left(\mathbf{e}_i + \mathbf{e}_j \right)$	$\pm (\mathbf{e}_i + \mathbf{e}_j)$	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_ F01737	175	■ 0.998	A_{l-1}	A_{l-1}	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_ F01738	175	■ 0.998	B_l, C_l	B_l, C_l	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_ F01739	175	■ 1.000	$\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$	$\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$	adding roots $\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$ to D_l . The root spaces A_{l-1}, D_l, B_l, C_l are all inequivalent
gilmore-lie-groups_ F01740	175	■ 1.000	A_{l-1}, D_l, B_l, C_l	A_{l-1}, D_l, B_l, C_l	adding roots $\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$ to D_l . The root spaces A_{l-1}, D_l, B_l, C_l are all inequivalent
gilmore-lie-groups_ F01743	178	■ 1.000	A_1	A_1	describe the rank-one algebra A_1 .
gilmore-lie-groups_ F01744	178	■ 1.000	$S L(l ; C)$	$SL(l; C)$	space has the same complex extension, for example, $SL(l; C)$ for A_{l-1} .
gilmore-lie-groups_ F01745	178	■ 0.997	$\pm \left(\mathbf{e}_1 - \mathbf{e}_2 \right)$	$\pm (\mathbf{e}_1 - \mathbf{e}_2)$	Remark 2. The root space D_2 consists of two orthogonal sets of roots $\pm(\mathbf{e}_1 - \mathbf{e}_2)$
gilmore-lie-groups_ F01746	178	■ 1.000	$\pm \left(\mathbf{e}_1 + \mathbf{e}_2 \right)$	$\pm (\mathbf{e}_1 + \mathbf{e}_2)$	and $\pm(\mathbf{e}_1 + \mathbf{e}_2)$. The decomposition is shown in Fig. 10.3. Orthogonal root spaces

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gilmore-lie-groups_ F01747	178	■ 0.964	B_2	B_2	Remark 3. The root spaces B_2 and C_2 are equivalent, as is easily seen by ro-
gilmore-lie-groups_ F01748	178	■ 0.513	$S_0(5)$	$SO(5)$	tation. The root space B_2 describes $SO(5)$ while C_2 describes $Sp(2) = U(2; \mathbb{Q})$,
gilmore-lie-groups_ F01749	178	■ 0.513	$Sp(2) = U(2; \mathbb{Q})$	$Sp(2) = U(2; \mathbb{Q})$	tation. The root space B_2 describes $SO(5)$ while C_2 describes $Sp(2) = U(2; \mathbb{Q})$,
gilmore-lie-groups_ F01750	178	■ 1.000	$U(2; \mathbb{Q})$	$U(2; \mathbb{Q})$	have a four-dimensional “spinor” representation based on $U(2; \mathbb{Q})$ in the same way
gilmore-lie-groups_ F01751	178	■ 0.975	$SO(3)(B_1)$	$SO(3)(B_1)$	that $SO(3)(B_1)$ has a two-dimensional spinor representation based on $U(1; \mathbb{Q})$ or
gilmore-lie-groups_ F01752	178	■ 0.975	$U(1; \mathbb{Q})$	$U(1; \mathbb{Q})$	that $SO(3)(B_1)$ has a two-dimensional spinor representation based on $U(1; \mathbb{Q})$ or
gilmore-lie-groups_ F01753	178	■ 0.995	$SU(2)(A_1)$	$SU(2)(A_1)$	$SU(2)(A_1)$.
gilmore-lie-groups_ F01771	184	■ 1.000	C_l	C_l	root spaces C_l and D_l :
gilmore-lie-groups_ F01773	180	■ 1.000	n_{ij}	n_{ij}	where n_{ij} is 0, 1, 2, or 3. Each fundamental root is represented by a dot. Dots i
gilmore-lie-groups_ F01774	180	■ 1.000	$0, 1, 2$	$0, 1, 2$	where n_{ij} is 0, 1, 2, or 3. Each fundamental root is represented by a dot. Dots i
gilmore-lie-groups_ F01775	180	■ 1.000	$G_2 + B_3$	$G_2 + B_3$	represented by orthogonal root spaces $G_2 + B_3$ is shown in Fig. 10.5.
gilmore-lie-groups_ F01776	180	■ 0.665	$\mathbf{v}_i$	$\mathbf{v}_i$	Observation 2 If $\mathbf{v}_i$ is an orthonormal system of vectors in the root space and $\mathbf{u}$ is a unit
gilmore-lie-groups_ F01777	180	■ 0.665	$\mathbf{u}$	$\mathbf{u}$	Observation 2 If $\mathbf{v}_i$ is an orthonormal system of vectors in the root space and $\mathbf{u}$ is a unit
gilmore-lie-groups_ F01778	180	■ 1.000	$\mathbf{u} \cdot \mathbf{v}_i$	$\mathbf{u} \cdot \mathbf{v}_i$	vector, then the direction cosines $\mathbf{u} \cdot \mathbf{v}_i$ obey
gilmore-lie-groups_ F01779	181	■ 1.000	$\mathbf{u}_i = \alpha_i / \alpha_i $	$\mathbf{u}_i = \alpha_i / \alpha_i $	vertices. With $\mathbf{u}_i = \alpha_i / \alpha_i $ the inner product
gilmore-lie-groups_ F01780	181	■ 1.000	$2 \mathbf{u}_i \cdot \mathbf{u}_j \leq -1$	$2 \mathbf{u}_i \cdot \mathbf{u}_j \leq -1$	cannot be positive since $2 \mathbf{u}_i \cdot \mathbf{u}_j \leq -1$ if α_i and α_j are connected.
gilmore-lie-groups_ F01781	181	■ 0.979	(B, C, F)	(B, C, F)	For the diagrams (B, C, F) with a single double link, the Schwarz inequality
gilmore-lie-groups_ F01782	182	■ 1.000	$\mathbf{u}_i, \mathbf{v}_j$	$\mathbf{u}_i, \mathbf{v}_j$	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_j = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups_ F01783	182	■ 1.000	$\mathbf{v}_i = \alpha_j / \alpha_j $	$\mathbf{v}_i = \alpha_j / \alpha_j $	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_j = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups_ F01784	182	■ 0.985	$p \geq q$	$p \geq q$	This has the following solutions with $p \geq q$
gilmore-lie-groups_ F01785	182	■ 0.657	(D, E)	(D, E)	For the diagrams (D, E) Observation 2 applied to the vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ defined
gilmore-lie-groups_ F01786	182	■ 0.997	$p \geq q \geq r \geq 2$	$p \geq q \geq r \geq 2$	This has the following solutions with $p \geq q \geq r \geq 2$
gilmore-lie-groups_ F01788	327	■ 0.991	E_6	E_6	E_6 , 162, 168
gilmore-lie-groups_ F01789	327	■ 0.546	E_7	E_7	E_7 , 162, 168
gilmore-lie-groups_ F01790	327	■ 0.731	E_8	E_8	E_8 , 162, 168
gilmore-lie-groups_ F01791	182	■ 1.000	$l - 1$	$l - 1$	inductively from each root space diagram of rank $l - 1$. The completeness of this
gilmore-lie-groups_ F01793	184	■ 1.000	$U(l)$	$U(l)$	group $U(l)$:

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gilmore-lie-groups_F01814	185	■ 0.852	D, E	D, E	Dynkin diagrams of type (D, E) to obtain the inequality of Eq. (10.19).
gilmore-lie-groups_F01815	185	■ 1.000	$1 \leq i, j \leq n$	$1 \leq i, j \leq n$	8. A Lie algebra is spanned by n^2 operators of the form $a_i^\dagger a_j$, with $1 \leq i, j \leq n$. Show
gilmore-lie-groups_F01816	185	■ 1.000	$n=3$	$n = 3$	For $n = 3$ the subspaces transform like an angular momentum vector and a quadrupole
gilmore-lie-groups_F01817	185	■ 1.000	$\hat{\mathbf{Q}}$	$\hat{\mathbf{Q}}$	traceless tensor $\hat{\mathbf{Q}}$ and a scalar N :
gilmore-lie-groups_F01818	185	■ 1.000	$\hat{N}$	$\hat{N}$	The operator $\hat{N}$ commutes with all operators $a_i^\dagger a_j$. Interpret these commutation re-
gilmore-lie-groups_F01819	185	■ 0.998	$\mathbf{L} + \hat{\mathbf{Q}}$	$\mathbf{L} + \hat{\mathbf{Q}}$	simple Lie algebra $\mathbf{L} + \hat{\mathbf{Q}}$).
gilmore-lie-groups_F01820	185	■ 0.623	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	the discussion of Problem 8 are the dimensions of the spaces $\mathbf{L}(3 \rightarrow n(n-1)/2)$, $\mathbf{Q}$
gilmore-lie-groups_F01821	185	■ 0.623	$\mathbf{Q}$	$\mathbf{Q}$	the discussion of Problem 8 are the dimensions of the spaces $\mathbf{L}(3 \rightarrow n(n-1)/2)$, $\mathbf{Q}$
gilmore-lie-groups_F01822	185	■ 1.000	$(6 \rightarrow n(n+1)/2)$	$(6 \rightarrow n(n+1)/2)$	$(6 \rightarrow n(n+1)/2)$, and the definition of $\hat{N}(3 \rightarrow n)$.
gilmore-lie-groups_F01823	185	■ 1.000	$\hat{N}(3 \rightarrow n)$	$\hat{N}(3 \rightarrow n)$	$(6 \rightarrow n(n+1)/2)$, and the definition of $\hat{N}(3 \rightarrow n)$.
gilmore-lie-groups_F01824	186	■ 1.000	$Z = r^i X_i$	$Z = r^i X_i$	Lie algebra with elements $Z = r^i X_i$ (r^i are real numbers, X_i the generators of the
gilmore-lie-groups_F01825	186	■ 1.000	r^i	r^i	Lie algebra with elements $Z = r^i X_i$ (r^i are real numbers, X_i the generators of the
gilmore-lie-groups_F01826	186	■ 1.000	$\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$	$\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$	of the form $\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$. This equation cannot be solved in
gilmore-lie-groups_F01827	186	■ 1.000	h^i	h^i	where h^i and e^α are complex numbers and the “diagonal” and “shift” operators
gilmore-lie-groups_F01828	186	■ 1.000	e^α	e^α	where h^i and e^α are complex numbers and the “diagonal” and “shift” operators
gilmore-lie-groups_F01829	187	■ 1.000	A_{n-1}, D_n, B_n, C_n	A_{n-1}, D_n, B_n, C_n	classical simple Lie algebras A_{n-1}, D_n, B_n, C_n . These decompositions are: block
gilmore-lie-groups_F01830	187	■ 1.000	a_1, a_2, a_3	a_1, a_2, a_3	for certain choices of the real parameters a_1, a_2, a_3 while in the compact case this
gilmore-lie-groups_F01831	187	■ 1.000	(a_1, a_2, a_3)	(a_1, a_2, a_3)	is not possible. If the real parameters (a_1, a_2, a_3) and (b_1, b_2, b_3) are allowed to
gilmore-lie-groups_F01832	187	■ 1.000	(b_1, b_2, b_3)	(b_1, b_2, b_3)	is not possible. If the real parameters (a_1, a_2, a_3) and (b_1, b_2, b_3) are allowed to
gilmore-lie-groups_F01833	188	■ 0.998	H_i, E_α	H_i, E_α	The Cartan–Killing inner product for the basis vectors H_i, E_α is
gilmore-lie-groups_F01834	188	■ 1.000	$\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$	$\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$	of basis vectors $\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$:
gilmore-lie-groups_F01835	189	■ 0.826	$H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	$H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_F01836	189	■ 0.826	$\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$	$\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_F01837	189	■ 0.826	$\alpha \neq 0$	$\alpha \neq 0$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_F01838	189	■ 1.000	$\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	$\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	generators H_i and $\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$ span a noncompact subspace.

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gilmore-lie-groups_ F01839	189	■ 0.999	$A_{\{1\}}, \mathfrak{sl}(2; \mathbb{R})$	$A_1, \mathfrak{sl}(2; \mathbb{R})$	Two real forms of A_1 , $\mathfrak{sl}(2; \mathbb{R})$ and $\mathfrak{su}(1, 1)$, are obtained as follows
gilmore-lie-groups_ F01840	189	■ 1.000	$h_{\{r\}}, h_{\{i\}}; a_{\{r\}}, a_{\{i\}}; b_{\{r\}}, b_{\{i\}}$	$h_r, h_i; a_r, a_i; b_r, b_i$	Here the six parameters $h_r, h_i; a_r, a_i; b_r, b_i$ are real.
gilmore-lie-groups_ F01841	190	■ 0.850	$n_{\{+\}}, n_{\{-\}}$	n_+, n_-	It is useful to specify how compact a real form is by specifying its index (n_+, n_-) ,
gilmore-lie-groups_ F01842	190	■ 1.000	$n_{\{+\}}$	n_+	where n_+ is the dimension of the subspace on which the nonsingular Cartan–
gilmore-lie-groups_ F01843	190	■ 1.000	$n_{\{-\}}$	n_-	Killing inner product is positive-definite and n_- is the dimension of the subspace
gilmore-lie-groups_ F01844	190	■ 1.000	$\chi = n_{\{+\}} - n_{\{-\}}$	$\chi = n_+ - n_-$	be abbreviated to a single integer, the character $\chi = n_+ - n_-$, to describe a real
gilmore-lie-groups_ F01845	190	■ 1.000	$i H_{\{i\}}, i \frac{1}{\sqrt{2}} \left(E_{\{\alpha\}} + E_{\{-\alpha\}} \right)$	$i H_i, i \frac{1}{\sqrt{2}} (E_\alpha + E_{-\alpha})$	the compact real form spanned by real linear combinations of $i H_i, i \frac{1}{\sqrt{2}} (E_\alpha + E_{-\alpha})$,
gilmore-lie-groups_ F01846	190	■ 0.917	$g_{\{\mu, \nu\}} = \operatorname{diag}(1, 1, 1, -1)$	$g_{\mu, \nu} = \operatorname{diag}(1, 1, 1, -1)$	mation of space-time (x, y, z, ct) with indefinite metric $g_{\mu, \nu} = \operatorname{diag}(1, 1, 1, -1)$ to
gilmore-lie-groups_ F01847	190	■ 0.870	$x, y, z, i c t$	$x, y, z, i c t$	space-time with imaginary time $(x, y, z, i c t)$ and positive-definite metric $g_{\mu, \nu} =$
gilmore-lie-groups_ F01848	190	■ 0.870	$g_{\{\mu, \nu\}} =$	$g_{\mu, \nu} =$	space-time with imaginary time $(x, y, z, i c t)$ and positive-definite metric $g_{\mu, \nu} =$
gilmore-lie-groups_ F01849	190	■ 0.672	$\operatorname{diag}(1, 1, 1, 1)$	$\operatorname{diag}(1, 1, 1, 1)$	<u>$\operatorname{diag}(1, 1, 1, 1)$</u> .
gilmore-lie-groups_ F01850	190	■ 1.000	$\mathfrak{p} \rightarrow \mathfrak{p}' = i \mathfrak{p}$	$\mathfrak{p} \rightarrow \mathfrak{p}' = i \mathfrak{p}$	“Minkowski’s trick”: $\mathfrak{p} \rightarrow \mathfrak{p}' = i \mathfrak{p}$. The mapping, commutation relations, and
gilmore-lie-groups_ F01851	191	■ 1.000	$\mathfrak{g}'^{\prime}, \mathfrak{h}$	$\mathfrak{g}', \mathfrak{h}$	In $\mathfrak{g}'$, $\mathfrak{h}$ is the maximal compact subalgebra and $\mathfrak{p}'$ consists of all the noncompact
gilmore-lie-groups_ F01852	191	■ 1.000	$\mathfrak{p}'^{\prime}$	$\mathfrak{p}'$	In $\mathfrak{g}'$, $\mathfrak{h}$ is the maximal compact subalgebra and $\mathfrak{p}'$ consists of all the noncompact
gilmore-lie-groups_ F01853	191	■ 1.000	$T^2 = I$	$T^2 = I$	compact real form of that Lie algebra to itself that obey $T^2 = I$.
gilmore-lie-groups_ F01854	192	■ 1.000	$\mathfrak{u}(n, \mathbb{F})$	$\mathfrak{u}(n, \mathbb{F})$	In a block matrix decomposition the compact Lie algebras $\mathfrak{u}(n, \mathbb{F})$ have the form
gilmore-lie-groups_ F01855	192	■ 0.949	$A_{\{p\}} = -A_{\{p\}}^{\dagger}, A_{\{q\}} = -A_{\{q\}}^{\dagger}$	$A_p = -A_p^{\dagger}, A_q = -A_q^{\dagger}$	where $A_p = -A_p^{\dagger}, A_q = -A_q^{\dagger}$, and B is an arbitrary $p \times q$ matrix. Under the
gilmore-lie-groups_ F01856	192	■ 0.999	$I_{\{p+q\}}$	I_{p+q}	i . This is equivalent to changing the metric I_{p+q} that is preserved by $\mathfrak{u}(n; \mathbb{F})$ to the
gilmore-lie-groups_ F01857	192	■ 0.999	$\mathfrak{u}(n; \mathbb{F})$	$\mathfrak{u}(n; \mathbb{F})$	i . This is equivalent to changing the metric I_{p+q} that is preserved by $\mathfrak{u}(n; \mathbb{F})$ to the
gilmore-lie-groups_ F01858	192	■ 1.000	$I_{\{p, q\}}$	$I_{p, q}$	metric $I_{p, q}$ that is preserved by $\mathfrak{u}(p, q; \mathbb{F})$, where $p + q = n$. The factor i can be
gilmore-lie-groups_ F01859	192	■ 1.000	$\mathfrak{u}(p, q; \mathbb{F})$	$\mathfrak{u}(p, q; \mathbb{F})$	metric $I_{p, q}$ that is preserved by $\mathfrak{u}(p, q; \mathbb{F})$, where $p + q = n$. The factor i can be
gilmore-lie-groups_ F01860	192	■ 0.995	$\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	$\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	For the fields $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$ related to the root spaces $(D, B), A, C$ the real forms
gilmore-lie-groups_ F01861	192	■ 0.995	$(D, B), A, C$	$(D, B), A, C$	For the fields $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$ related to the root spaces $(D, B), A, C$ the real forms
gilmore-lie-groups_ F01862	192	■ 0.998	$\mathfrak{sp}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$.
gilmore-lie-groups_ F01863	193	■ 1.000	$\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$	$\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$	The transformation from <u>$\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$</u> to $\mathfrak{sp}(2n; \mathbb{R})$ is somewhat less familiar.
gilmore-lie-groups_ F01864	193	■ 1.000	α, β, γ	α, β, γ	where α, β, γ and δ are real. With these replacements the Lie algebra of $n \times n$

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gilmore-lie-groups_ F01865	193	■ 0.999	$\mathfrak{o}(2n)$	$\mathfrak{ou}(2n)$	these matrices $\mathfrak{ou}(2n)$, since they form an <u>orthogonal</u> representation of the <u>unitary</u>
gilmore-lie-groups_ F01866	193	■ 1.000	$\underline{2n} \times 2n$	$\underline{2n} \times 2n$	algebra in terms of $\underline{2n} \times 2n$ matrices. Similarly, the Lie algebra of $n \times n$ quaternion
gilmore-lie-groups_ F01867	194	■ 1.000	R^{2n}	R^{2n}	metric metric in R^{2n} . It is possible to choose coordinates $p_1, q_1, p_2, q_2, \dots, p_n, q_n$
gilmore-lie-groups_ F01868	194	■ 1.000	$p_{\{1\}}, q_{\{1\}}, p_{\{2\}}, q_{\{2\}}, \ldots, p_{\{n\}}, q_{\{n\}}$	$p_1, q_1, p_2, q_2, \dots, p_n, q_n$	metric metric in R^{2n} . It is possible to choose coordinates $p_1, q_1, p_2, q_2, \dots, p_n, q_n$
gilmore-lie-groups_ F01869	194	■ 1.000	$v_{\{i\}}^{\prime} G_{ij} v_j$	$v_i' G_{ij} v_j$	in this space so that the inner product between two vectors $v_i' G_{ij} v_j$ is
gilmore-lie-groups_ F01870	194	■ 0.967	$M \in \operatorname{Sp}(2n; \mathbb{R})$	$M \in \operatorname{Sp}(2n; \mathbb{R})$	Then symplectic transformations $M \in \operatorname{Sp}(2n; \mathbb{R})$ leave this metric matrix G invari-
gilmore-lie-groups_ F01871	194	■ 1.000	$M^t G M = G$	$M^t G M = G$	ant: $M^t G M = G$. Symplectic transformations leave invariant the canonical form
gilmore-lie-groups_ F01872	194	■ 1.000	$\mathfrak{p} \rightarrow i\mathfrak{p}$	$\mathfrak{p} \rightarrow i\mathfrak{p}$	Application of the map $\mathfrak{p} \rightarrow i\mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.
gilmore-lie-groups_ F01873	194	■ 1.000	$\mathfrak{so}^*(2n)$	$\mathfrak{so}^*(2n)$	Application of the map $\mathfrak{p} \rightarrow i\mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.
gilmore-lie-groups_ F01874	194	■ 1.000	$\mathfrak{su}^*(2n)$	$\mathfrak{su}^*(2n)$	Application of the map $\mathfrak{p} \rightarrow i\mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.
gilmore-lie-groups_ F01875	194	■ 0.991	A_{2n-1}	A_{2n-1}	Remark The real forms $\mathfrak{so}^*(2n)$ of D_n and $\mathfrak{su}^*(2n)$ of A_{2n-1} do not occur explicitly
gilmore-lie-groups_ F01876	195	■ 1.000	$\mathfrak{su}(2), B_1(\mathfrak{so}(3))$	$\mathfrak{su}(2), B_1(\mathfrak{so}(3))$	the compact real form is $\mathfrak{su}(2), B_1(\mathfrak{so}(3))$, and $C_1(\mathfrak{sp}(1))$ are equivalent, as are B_2
gilmore-lie-groups_ F01877	195	■ 1.000	$C_1(\mathfrak{sp}(1))$	$C_1(\mathfrak{sp}(1))$	the compact real form is $\mathfrak{su}(2), B_1(\mathfrak{so}(3))$, and $C_1(\mathfrak{sp}(1))$ are equivalent, as are B_2
gilmore-lie-groups_ F01878	195	■ 1.000	$(\mathfrak{so}(5))$	$(\mathfrak{so}(5))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_ F01879	195	■ 1.000	$C_2(\mathfrak{sp}(2))$	$C_2(\mathfrak{sp}(2))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_ F01880	195	■ 1.000	$A_3(\mathfrak{su}(4))$	$A_3(\mathfrak{su}(4))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_ F01881	195	■ 1.000	$D_3(\mathfrak{so}(6))$	$D_3(\mathfrak{so}(6))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_ F01903	234	■ 0.679	D_3	D_3	7. One real form of D_3 is the conformal group $SO(4, 2)$.
gilmore-lie-groups_ F01918	327	■ 0.999	F_4	F_4	$F_4, 162, 168$
gilmore-lie-groups_ F01944	196	■ 1.000	$T(\mathfrak{p}) = -\mathfrak{p}$	$T(\mathfrak{p}) = -\mathfrak{p}$	that obeys $T(\mathfrak{p}) = -\mathfrak{p}$ is analytically continued by $\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$; the subspace $\mathfrak{h}$
gilmore-lie-groups_ F01945	196	■ 1.000	$T(\mathfrak{h}) = +\mathfrak{h}$	$T(\mathfrak{h}) = +\mathfrak{h}$	of $\mathfrak{g}$ that obeys $T(\mathfrak{h}) = +\mathfrak{h}$ is the maximal compact subalgebra of the noncompact
gilmore-lie-groups_ F01946	196	■ 1.000	$\mathfrak{g}' : \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}$	$\mathfrak{g}' : \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}$	real form $\mathfrak{g}' : \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}$. For the simple classical (matrix) Lie alge-
gilmore-lie-groups_ F01947	197	■ 1.000	$1 \leq i$	$1 \leq i$	1. Four operators $a_i^\dagger a_j$ can be constructed from boson operators for two modes $\mathfrak{a}_1, \mathfrak{a}_2$,
gilmore-lie-groups_ F01948	197	■ 1.000	$j \leq 2$	$j \leq 2$	$j \leq 2$. These operators close under commutation.
gilmore-lie-groups_ F01949	197	■ 1.000	$a_1^\dagger a_1 + a_2^\dagger a_2$	$a_1^\dagger a_1 + a_2^\dagger a_2$	Conclude that $a_1^\dagger a_1 + a_2^\dagger a_2$ spans the maximum commutative subalgebra, $a_1^\dagger a_2 -$
gilmore-lie-groups_ F01950	197	■ 1.000	$a_1^\dagger a_2 -$	$a_1^\dagger a_2 -$	Conclude that $a_1^\dagger a_1 + a_2^\dagger a_2$ spans the maximum commutative subalgebra, $a_1^\dagger a_2 -$

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gilmore-lie-groups_ F01951	197	■ 1.000	$a_1^\dagger a_1 - a_2^\dagger a_2$	$a_1^\dagger a_1 - a_2^\dagger a_2$	$a_2^\dagger a_1$ spans the maximal compact subalgebra, and the two generators $a_1^\dagger a_1 - a_2^\dagger a_2$
gilmore-lie-groups_ F01952	197	■ 1.000	$a_1^\dagger a_2 + a_2^\dagger a_1$	$a_1^\dagger a_2 + a_2^\dagger a_1$	and $a_1^\dagger a_2 + a_2^\dagger a_1$ are noncompact.
gilmore-lie-groups_ F01953	197	■ 1.000	$i \sigma_j$	$i \sigma_j$	Show that the three matrices that result are exactly $i \sigma_j$, where σ_j are the Pauli spin
gilmore-lie-groups_ F01954	197	■ 1.000	σ_j	σ_j	Show that the three matrices that result are exactly $i \sigma_j$, where σ_j are the Pauli spin
gilmore-lie-groups_ F01955	197	■ 0.982	$k \rightarrow 1$	$k \rightarrow 1$	2. The classical matrix groups $SO(n)$ are not simply connected, so they are $k \rightarrow 1$
gilmore-lie-groups_ F01956	197	■ 0.844	$\overline{SO(n)} = Spin(n)$	$\overline{SO(n)} = Spin(n)$	images of their universal covering groups $\overline{SO(n)} = Spin(n)$, for some integer k .
gilmore-lie-groups_ F01957	197	■ 1.000	$Spin(n)$	$Spin(n)$	Show that the covering groups $Spin(n)$ are classical matrix groups for $n = 3, 4, 5, 6$,
gilmore-lie-groups_ F01958	197	■ 1.000	$n=3,4,5,6$	$n = 3, 4, 5, 6$	Show that the covering groups $Spin(n)$ are classical matrix groups for $n = 3, 4, 5, 6$,
gilmore-lie-groups_ F01959	197	■ 1.000	$n>6$	$n > 6$	Show that for $n > 6$ the groups $Spin(n)$ are not equal to any classical matrix Lie
gilmore-lie-groups_ F01960	198	■ 0.968	$H_i: n_1, n_2, \dots, n_r\rangle$	$H_i: n_1, n_2, \dots, n_r\rangle$	eigenvalues under the operators $H_i: n_1, n_2, \dots, n_r\rangle$,
gilmore-lie-groups_ F01961	198	■ 1.000	$H_j \rightarrow$	$H_j \rightarrow$	c. For the orthogonal groups $SO(n)$, impose suitable reality conditions (i.e., $H_j \rightarrow$
gilmore-lie-groups_ F01962	198	■ 1.000	$i H_j$	$i H_j$	$i H_j$, etc.), choose a Hilbert space containing the state $ l, 0, \dots, 0\rangle$ and show that
gilmore-lie-groups_ F01963	198	■ 1.000	$ l, 0, \dots, 0\rangle$	$ l, 0, \dots, 0\rangle$	$i H_j$, etc.), choose a Hilbert space containing the state $ l, 0, \dots, 0\rangle$ and show that
gilmore-lie-groups_ F01964	198	■ 0.989	$-m^2$	$-m^2$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01965	198	■ 0.989	$e^{im\phi}$	$e^{im\phi}$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01966	198	■ 0.989	$-l(l+1)$	$-l(l+1)$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01967	198	■ 0.989	$Y_m^l(\theta, \phi)$	$Y_m^l(\theta, \phi)$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01968	198	■ 1.000	$J_i, i=1,2,3$	$J_i, i=1,2,3$	matrix elements of a <i>hermitian</i> representation of $J_i, i=1,2,3$. Exponentials of the
gilmore-lie-groups_ F01969	198	■ 0.926	$EXP(ir^k J_k)$	$EXP(ir^k J_k)$	form $EXP(ir^k J_k)$, with r^k real, provide <i>unitary</i> representations of the compact Lie
gilmore-lie-groups_ F01970	198	■ 0.926	r^k	r^k	form $EXP(ir^k J_k)$, with r^k real, provide <i>unitary</i> representations of the compact Lie
gilmore-lie-groups_ F01971	198	■ 1.000	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$.
gilmore-lie-groups_ F01972	199	■ 0.826	K_3	K_3	$SU(2)$ with J_3 diagonal, or in any matrix representation of $SU(1,1)$ with K_3
gilmore-lie-groups_ F01973	199	■ 1.000	$e^{im\phi} \delta_{m'm}$	$e^{im\phi} \delta_{m'm}$	diagonal, the matrix is diagonal with matrix elements $e^{im\phi} \delta_{m'm}$. Show that the
gilmore-lie-groups_ F01974	199	■ 1.000	$\phi \rightarrow \phi + 4\pi$	$\phi \rightarrow \phi + 4\pi$	single-valuedness condition under $\phi \rightarrow \phi + 4\pi$ requires that $m = \frac{1}{2}(n_1 - n_2)$ is
gilmore-lie-groups_ F01975	199	■ 1.000	$m = \frac{1}{2}(n_1 - n_2)$	$m = \frac{1}{2}(n_1 - n_2)$	single-valuedness condition under $\phi \rightarrow \phi + 4\pi$ requires that $m = \frac{1}{2}(n_1 - n_2)$ is
gilmore-lie-groups_ F01976	199	■ 1.000	$J_\pm$	$J_\pm$	integer or half-integer. Show that the shift operators $J_\pm$ and $K_\pm$ require that all m
gilmore-lie-groups_ F01977	199	■ 1.000	$K_\pm$	$K_\pm$	integer or half-integer. Show that the shift operators $J_\pm$ and $K_\pm$ require that all m

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gilmore-lie-groups_F01978	199	■ 1.000	$n_{\{1\}}, n_{\{2\}}$	n_1, n_2	c. Relax the assumption that all indices n_1, n_2 in the basis states $ n_1, n_2\rangle = n_1\rangle \otimes$
gilmore-lie-groups_F01979	199	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}\right\rangle=\left n_{\{1\}}\right\rangle\otimes$ $\otimes$	$ n_1, n_2\rangle = n_1\rangle \otimes$	c. Relax the assumption that all indices n_1, n_2 in the basis states $ n_1, n_2\rangle = n_1\rangle \otimes$
gilmore-lie-groups_F01980	199	■ 1.000	$\left n_{\{2\}}\right\rangle$	$ n_2\rangle$	$ n_2\rangle$ must be nonnegative integers. Construct the matrix elements of the operators
gilmore-lie-groups_F01981	199	■ 0.999	$J_{\{3\}}, J_{\{ \pm \}}, K_{\{3\}}, K_{\{ \pm \}}$	$J_3, J_{\pm}, K_3, K_{\pm}$	$J_3, J_{\pm}, K_3, K_{\pm}$ under this relaxed assumption. Show that all commutation relations
gilmore-lie-groups_F01982	199	■ 1.000	$J_{\{x\}}$	J_x	d. Show that the matrices for J_x and J_y are also hermitian provided that
gilmore-lie-groups_F01983	199	■ 1.000	$J_{\{y\}}$	J_y	d. Show that the matrices for J_x and J_y are also hermitian provided that
gilmore-lie-groups_F01984	199	■ 0.749	$(0,0)$	$(0,0)$	Show that the lattice sites in quadrants I and III of Fig. 11.2, with vertices at $(0,0)$
gilmore-lie-groups_F01985	199	■ 0.975	$\left \{ \}_m^{\{j\}}\right\rangle=\left n_{\{1\}}, n_{\{2\}}\right\rangle$ $\rangle, j=\frac{1}{2}\left(n_{\{1\}}+n_{\{2\}}\right), m=\frac{1}{2}\left(n_{\{1\}}-n_{\{2\}}\right)$	$ _m^j\rangle = n_1, n_2\rangle, j = \frac{1}{2}(n_1 + n_2), m = \frac{1}{2}(n_1 - n_2)$	e. With the identification $ _m^j\rangle = n_1, n_2\rangle, j = \frac{1}{2}(n_1 + n_2), m = \frac{1}{2}(n_1 - n_2)$, show
gilmore-lie-groups_F01986	199	■ 1.000	$n_{\{1\}}$	n_1	successive application of these operators to remain in QI or QIII, n_1 and n_2 must
gilmore-lie-groups_F01987	199	■ 1.000	$n_{\{2\}}$	n_2	successive application of these operators to remain in QI or QIII, n_1 and n_2 must
gilmore-lie-groups_F01996	200	■ 0.998	$\left n_{\{1\}}+k, n_{\{2\}}-k\right\rangle$	$ n_1 + k, n_2 - k\rangle$	i. In order to ensure that a set of states $ n_1 + k, n_2 - k\rangle$ (k integer) mapped into each
gilmore-lie-groups_F01997	200	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}=-1\right\rangle$	$ n_1, n_2 = -1\rangle$	(lowest m) state must be $ n_1, n_2 = -1\rangle$ for $n_1 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_F01998	200	■ 1.000	$n_{\{1\}}=0,1,2, \ldots$	$n_1 = 0, 1, 2, \dots$	(lowest m) state must be $ n_1, n_2 = -1\rangle$ for $ n_1 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_F01999	200	■ 0.995	$\left \{ \}_m^{\{j\}}\right\rangle=\left n_{\{1\}}, n_{\{2\}}\right\rangle$ $\rangle$	$ _m^j\rangle = n_1, n_2\rangle$	this bounded discrete series of representations are $ _m^j\rangle = n_1, n_2\rangle$ with $j + \frac{1}{2} =$
gilmore-lie-groups_F02000	200	■ 0.995	$j+\frac{1}{2}=$	$j + \frac{1}{2} =$	this bounded discrete series of representations are $ _m^j\rangle = n_1, n_2\rangle$ with $j + \frac{1}{2} =$
gilmore-lie-groups_F02001	200	■ 0.993	$0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the
gilmore-lie-groups_F02002	200	■ 0.993	$2j+1=0,1,2,3, \ldots$	$2j + 1 = 0, 1, 2, 3, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the
gilmore-lie-groups_F02003	200	■ 0.993	$m=j+1, j+2, \ldots$	$m = j + 1, j + 2, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the
gilmore-lie-groups_F02004	200	■ 1.000	$\mathcal{D}_{\{+\}^{\{j\}}}$	$\mathcal{D}_+^j$	discrete series of representations that is bounded below: $\mathcal{D}_+^j$.
gilmore-lie-groups_F02005	200	■ 0.831	$\mathbf{j}$	j	j. In order to ensure that a set of states $ n_1 + k, n_2 - k\rangle$ (k integer) mapped into
gilmore-lie-groups_F02006	200	■ 1.000	$\left n_{\{1\}}=-1, n_{\{2\}}\right\rangle$	$ n_1 = -1, n_2\rangle$	(highest m) state must be $ n_1 = -1, n_2\rangle$ for $n_2 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_F02007	200	■ 1.000	$n_{\{2\}}=0,1,2, \ldots$	$n_2 = 0, 1, 2, \dots$	(highest m) state must be $ n_1 = -1, n_2\rangle$ for $ n_2 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_F02008	200	■ 1.000	$m=-j-1,-j-2, \ldots$	$m = -j - 1, -j - 2, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = -j - 1, -j - 2, \dots$. This is
gilmore-lie-groups_F02009	200	■ 0.999	$\mathcal{D}_{\{-\}^{\{j\}}}$	$\mathcal{D}_-^j$	the discrete series of representations that is bounded above: $\mathcal{D}_-^j$.
gilmore-lie-groups_F02010	201	■ 0.998	$\left(\mathcal{D}_{\{+\}^{\{j\}}}\right)$	$(\mathcal{D}_+^j)$	$(\mathcal{D}_+^j)$ or from QII $(\mathcal{D}_-^j)$.
gilmore-lie-groups_F02011	201	■ 0.998	$\left(\mathcal{D}_{\{-\}^{\{j\}}}\right)$	$(\mathcal{D}_-^j)$	$(\mathcal{D}_-^j)$ or from QII $(\mathcal{D}_-^j)$.

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gilmore-lie-groups_F02012	201	■ 0.999	$\mid n_{\{1\}+k, n_{\{2\}}-$	$\mid n_1 + k, n_2-$	l. Now relax the condition that n_1 and n_2 are integers. The set of states $\mid n_1 + k, n_2-$
gilmore-lie-groups_F02013	201	■ 1.000	$k\langle k=\ldots,-2,-1,0,+1,+2,\ldots\rangle$	$k\rangle(k=\ldots,-2,-1,0,+1,+2,\ldots)$	$k\rangle(k=\ldots,-2,-1,0,+1,+2,\ldots)$ connected by $K_{\pm}$ carries a hermitian repre-
gilmore-lie-groups_F02014	201	■ 1.000	$\mid p, q\rangle$	$\mid p, q\rangle$	vertices of the four quadrants. If this state is $\mid p, q\rangle$, with $-1 \leq p, q \leq 0$ then the
gilmore-lie-groups_F02015	201	■ 1.000	$-1 \leq p, q \leq 0$	$-1 \leq p, q \leq 0$	vertices of the four quadrants. If this state is $\mid p, q\rangle$, with $-1 \leq p, q \leq 0$ then the
gilmore-lie-groups_F02016	201	■ 1.000	$\frac{1}{2}(p-q)=$	$\frac{1}{2}(p-q) =$	single-valuedness condition requires $\frac{1}{2}(p-q) =$ integer or half-integer. In the lat-
gilmore-lie-groups_F02017	201	■ 0.891	$p-q=0$	$p-q=0$	values of the $U(1)$ index m . Therefore $p-q=0$ and $-1 \leq p=q \leq 0$. The states
gilmore-lie-groups_F02018	201	■ 0.891	$-1 \leq p=q \leq 0$	$-1 \leq p=q \leq 0$	values of the $U(1)$ index m . Therefore $p-q=0$ and $-1 \leq p=q \leq 0$. The states
gilmore-lie-groups_F02019	201	■ 0.679	$\mid m^j\rangle$	$\mid m^j\rangle$	$\mid m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_F02020	201	■ 0.679	$-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$	$-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$	$\mid m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_F02021	201	■ 0.679	$\mathcal{D}^p$	$\mathcal{D}^p$	$\mid m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_F02022	201	■ 1.000	$j + \frac{1}{2} = i\beta$	$j + \frac{1}{2} = i\beta$	m. By setting $j + \frac{1}{2} = i\beta$ (β real) the matrix elements in Eq. (11.30) become
gilmore-lie-groups_F02023	201	■ 1.000	$j' < -\frac{1}{2}$	$j' < -\frac{1}{2}$	Show that states with $j' < -\frac{1}{2}$ obtained by reflection through the diagonal contain-
gilmore-lie-groups_F02024	201	■ 0.693	$(n_1, n_2) = (-\frac{1}{2}, -\frac{1}{2})$	$(n_1, n_2) = (-\frac{1}{2}, -\frac{1}{2})$	ing the central point in the shaded square with coordinates $(n_1, n_2) = (-\frac{1}{2}, -\frac{1}{2})$
gilmore-lie-groups_F02025	201	■ 1.000	$j > -\frac{1}{2}$	$j > -\frac{1}{2}$	support representations equivalent to those with index $j > -\frac{1}{2}$. The relation among
gilmore-lie-groups_F02026	201	■ 0.311	$j + \frac{1}{2} = -(j' + \frac{1}{2})$	$j + \frac{1}{2} = -(j' + \frac{1}{2})$	indices is $j + \frac{1}{2} = -(j' + \frac{1}{2})$ and the relation among states is $\mid m'^j\rangle \simeq \mid_{m=m'}^j$.
gilmore-lie-groups_F02027	201	■ 0.311	$\mid k_{m'}^{j'}\rangle \simeq \mid_{m=m'}^j$	$\mid k_{m'}^{j'}\rangle \simeq \mid_{m=m'}^j$	indices is $j + \frac{1}{2} = -(j' + \frac{1}{2})$ and the relation among states is $\mid m'^j\rangle \simeq \mid_{m=m'}^j$.
gilmore-lie-groups_F02028	201	■ 0.996	$X = h^i H_i + e^\alpha E_\alpha$	$X = h^i H_i + e^\alpha E_\alpha$	$X = h^i H_i + e^\alpha E_\alpha$, with h^i, e^α real. Show that the real subalgebra spanned by the
gilmore-lie-groups_F02029	201	■ 0.996	h^i, e^α	h^i, e^α	$X = h^i H_i + e^\alpha E_\alpha$, with h^i, e^α real. Show that the real subalgebra spanned by the
gilmore-lie-groups_F02030	201	■ 1.000	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$ linear combinations of the form $(E_\alpha - E_{-\alpha})/\sqrt{2}$ is the maximal compact
gilmore-lie-groups_F02031	201	■ 1.000	$(E_\alpha - E_{-\alpha})/\sqrt{2}$	$(E_\alpha - E_{-\alpha})/\sqrt{2}$	$\frac{1}{2}(n-l)$ linear combinations of the form $(E_\alpha - E_{-\alpha})/\sqrt{2}$ is the maximal compact
gilmore-lie-groups_F02032	202	■ 1.000	$\mathfrak{sp}(p, q)$	$\mathfrak{sp}(p, q)$	6. The noncompact real form $\mathfrak{sp}(p, q)$ of the symplectic algebra was constructed from
gilmore-lie-groups_F02033	202	■ 1.000	$\mathfrak{sp}(p+q)$	$\mathfrak{sp}(p+q)$	the compact real form $\mathfrak{sp}(p+q)$ by “Minkowski’s trick,” or analytic continuation.
gilmore-lie-groups_F02034	203	■ 0.999	$\mathfrak{p}, i\mathfrak{p}$	$\mathfrak{p}, i\mathfrak{p}$	countered subspaces $\mathfrak{p}, i\mathfrak{p}$ on which the Cartan–Killing inner product was
gilmore-lie-groups_F02035	203	■ 1.000	$i\mathfrak{p}$	$i\mathfrak{p}$	negative-definite (on $\mathfrak{p}$) or positive-definite (on $i\mathfrak{p}$). In both cases these
gilmore-lie-groups_F02036	203	■ 1.000	g_{ij}	g_{ij}	metric g_{ij} is definite, either negative or positive. Manifolds with a definite
gilmore-lie-groups_F02037	203	■ 1.000	$\operatorname{EXP}(\mathfrak{p})$	$\operatorname{EXP}(\mathfrak{p})$	each point in the space $\operatorname{EXP}(\mathfrak{p})$ or $\operatorname{EXP}(i\mathfrak{p})$ is the image of the origin

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gilmore-lie-groups_F02038	203	■ 1.000	$\operatorname{EXP}(i\,\mathfrak{p})$	$\operatorname{EXP}(ip)$	each point in the space $\operatorname{EXP}(\mathfrak{p})$ or $\operatorname{EXP}(ip)$ is the image of the origin
gilmore-lie-groups_F02039	203	■ 0.997	$S^2\sim S\,U(2)\,/\,U(1)$	$S^2\sim SU(2)/U(1)$	These were $S^2\sim SU(2)/U(1)$, which is compact, and its dual $H_{2+}^2=SL(2;\mathbb{R})/$
gilmore-lie-groups_F02040	203	■ 0.997	$H_{2+}^2=S\,L(2\,;\,\mathbb{R})\,/\,$	$H_{2+}^2=SL(2;\mathbb{R})/$	These were $S^2\sim SU(2)/U(1)$, which is compact, and its dual $H_{2+}^2=SL(2;\mathbb{R})/$
gilmore-lie-groups_F02041	203	■ 1.000	$S\,O(2)=S\,U(1,1)\,/\,U(1)$	$SO(2)=SU(1,1)/U(1)$	$SO(2)=SU(1,1)/U(1)$, which is the upper sheet of the two-sheeted hyperboloid.
gilmore-lie-groups_F02042	203	■ 1.000	$H_1^2=S\,L(2\,;\,\mathbb{R})\,/\,S\,O(1,1)$	$H_1^2=SL(2;\mathbb{R})/SO(1,1)$	“Between” these two spaces occurs $H_1^2=SL(2;\mathbb{R})/SO(1,1)$, which is the single-
gilmore-lie-groups_F02043	203	■ 0.943	$\mathfrak{s}\,u(2)-\mathfrak{u}(1)$	$\mathfrak{su}(2)-\mathfrak{u}(1)$	The Cartan–Killing inner product in the linear vector subspace $\mathfrak{su}(2)-\mathfrak{u}(1)$ is
gilmore-lie-groups_F02044	203	■ 0.999	$S\,U(2)\,/\,U(1)\,\sim S^2$	$SU(2)/U(1)\sim S^2$	Killing metric on the space $SU(2)/U(1)\sim S^2$, the sphere. On S^2 the Cartan–Killing
gilmore-lie-groups_F02045	203	■ 0.991	$\mathfrak{s}\,u(1,1)-\mathfrak{u}(1)\,\simeq\,\mathfrak{s}\,\mathfrak{l}(2\,;\,\mathbb{R})-\mathfrak{s}\,\mathfrak{o}(2)$	$\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$	The Cartan–Killing inner product on $\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$ is
gilmore-lie-groups_F02046	203	■ 1.000	$H_{2+}^2=S\,U(1,1)\,/\,S\,O(2)$	$H_{2+}^2=SU(1,1)/SO(2)$	positive-definite. It maps to a positive-definite metric on $H_{2+}^2=SU(1,1)/SO(2)$.
gilmore-lie-groups_F02049	204	■ 0.998	$\mathfrak{s}\,u(1,1)-\mathfrak{s}\,\mathfrak{o}(1,1)$	$\mathfrak{su}(1,1)-\mathfrak{so}(1,1)$	obtained by exponentiating $\mathfrak{su}(1,1)-\mathfrak{so}(1,1)$. The Cartan–Killing inner product
gilmore-lie-groups_F02050	204	■ 0.966	$\operatorname{EXP}[\mathfrak{s}\,u(1,1)-\mathfrak{s}\,\mathfrak{o}(1,1)]=S\,U(1,1)\,/\,S\,O(1,1)$	$\operatorname{EXP}[\mathfrak{su}(1,1)-\mathfrak{so}(1,1)]=SU(1,1)/SO(1,1)$	the topological space $\operatorname{EXP}[\mathfrak{su}(1,1)-\mathfrak{so}(1,1)]=SU(1,1)/SO(1,1)$ is indefinite.
gilmore-lie-groups_F02051	204	■ 0.952	$\mathfrak{s}\,\mathfrak{o}(n),\,\mathfrak{s}\,u(n),\,\mathfrak{s}\,\mathfrak{p}(n)$	$\mathfrak{so}(n),\mathfrak{su}(n),\mathfrak{sp}(n)$	algebra $\mathfrak{g}$ (i.e., $\mathfrak{so}(n),\mathfrak{su}(n),\mathfrak{sp}(n)$) the Cartan decompositions have the form (11.10)
gilmore-lie-groups_F02052	205	■ 1.000	$\mathfrak{p}(i\,\mathfrak{p})$	$\mathfrak{p}(ip)$	On the linear vector space $\mathfrak{p}$ (ip) the Cartan–Killing inner product is negative
gilmore-lie-groups_F02053	205	■ 1.000	$P=G\,/\,H=\operatorname{EXP}(\mathfrak{p})$	$P=G/H=\operatorname{EXP}(\mathfrak{p})$	The space $P=G/H=\operatorname{EXP}(\mathfrak{p})$ (e.g., S^2) is compact. The exponential of a
gilmore-lie-groups_F02054	205	■ 1.000	P	P	identity. The space P is not topologically equivalent to any Euclidean space, in
gilmore-lie-groups_F02055	205	■ 0.999	$P^{\prime}=\operatorname{EXP}(i\,\mathfrak{p})$	$P'=G'/H=\operatorname{EXP}(ip)$	The space $P'=G'/H=\operatorname{EXP}(ip)$ (i.e., H_{2+}^2) is noncompact. The exponential of
gilmore-lie-groups_F02056	205	■ 1.000	$P^{\prime}=\operatorname{EXP}(i\,\mathfrak{p})$	$P'=\operatorname{EXP}(ip)$	simply goes away from this point without ever returning. The space $P'=\operatorname{EXP}(ip)$ is
gilmore-lie-groups_F02057	205	■ 1.000	$n=\operatorname{dim}\,i\,\mathfrak{p}$	$n=\dim\,ip$	topologically equivalent to a Euclidean space R^n , where $n=\dim\,ip$. Geometrically
gilmore-lie-groups_F02058	205	■ 1.000	$P=\operatorname{EXP}(\mathfrak{p})$	$P=\operatorname{EXP}(\mathfrak{p})$	The Riemannian spaces $P=\operatorname{EXP}(\mathfrak{p})$ and $P'=\operatorname{EXP}(ip)$ are symmetric but not
gilmore-lie-groups_F02059	205	■ 1.000	$\mathfrak{g}=\mathfrak{k}+\mathfrak{p}$	$\mathfrak{g}=\mathfrak{k}+\mathfrak{p}$	If $\mathfrak{g}$ is simple with a Cartan decomposition of the form $\mathfrak{g}=\mathfrak{k}+\mathfrak{p}$, with standard
gilmore-lie-groups_F02060	205	■ 1.000	$[\mathfrak{k},\mathfrak{k}]\,\subseteq\,\mathfrak{k},[\mathfrak{k},\mathfrak{p}]\,\subseteq\,\mathfrak{p}$	$[\mathfrak{k},\mathfrak{k}]\subseteq\mathfrak{k},[\mathfrak{k},\mathfrak{p}]\subseteq\mathfrak{p}$	commutation relations $[\mathfrak{k},\mathfrak{k}]\subseteq\mathfrak{k},[\mathfrak{k},\mathfrak{p}]\subseteq\mathfrak{p}$, and $[\mathfrak{p},\mathfrak{p}]\subseteq\mathfrak{k}$, the quotient coset $P=$
gilmore-lie-groups_F02061	205	■ 1.000	$[\mathfrak{p},\mathfrak{p}]\,\subseteq\,\mathfrak{k}$	$[\mathfrak{p},\mathfrak{p}]\subseteq\mathfrak{k}$	commutation relations $[\mathfrak{k},\mathfrak{k}]\subseteq\mathfrak{k},[\mathfrak{k},\mathfrak{p}]\subseteq\mathfrak{p}$, and $[\mathfrak{p},\mathfrak{p}]\subseteq\mathfrak{k}$, the quotient coset $P=$
gilmore-lie-groups_F02062	205	■ 1.000	$P=$	$P=$	commutation relations $[\mathfrak{k},\mathfrak{k}]\subseteq\mathfrak{k},[\mathfrak{k},\mathfrak{p}]\subseteq\mathfrak{p}$, and $[\mathfrak{p},\mathfrak{p}]\subseteq\mathfrak{k}$, the quotient coset $P=$
gilmore-lie-groups_F02063	205	■ 1.000	$G\,/\,K$	G/K	G/K is a globally symmetric space as every point “looks like” every other point.
gilmore-lie-groups_F02064	205	■ 1.000	$P=G\,/\,H$	$P=G/H$	of points (coset representatives $P=G/H$ or $P'=G'/H$) in the Lie group.
gilmore-lie-groups_F02065	205	■ 1.000	$P^{\prime}=\operatorname{EXP}(i\,\mathfrak{p})$	$P'=G'/H$	of points (coset representatives $P=G/H$ or $P'=G'/H$) in the Lie group.
gilmore-lie-groups_F02066	205	■ 0.993	$S\,O(p,\,q)\,/\,S\,O(p)\,\times\,S\,O(q)$	$SO(p,q)/SO(p)\times SO(q)$	resentation. The secular equations for the spaces $SO(p,q)/SO(p)\times SO(q)$,

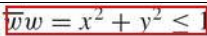
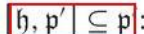
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gilmore-lie-groups_ F02067	206	■ 1.000	$B^{\dagger} B$	$B^{\dagger} B$	It is easy to check that the function ϕ_j depends on the $q \times q$ matrix $B^{\dagger} B$ or the
gilmore-lie-groups_ F02068	206	■ 1.000	$B B^{\dagger}$	$B B^{\dagger}$	$p \times p$ matrix $B B^{\dagger}$, whichever is smaller. The rank of these spaces is $\min(p, q)$.
gilmore-lie-groups_ F02069	206	■ 1.000	$\min(p, q)$	$\min(p, q)$	$p \times p$ matrix $B B^{\dagger}$, whichever is smaller. The rank of these spaces is $\min(p, q)$.
gilmore-lie-groups_ F02070	206	■ 1.000	$P(\mathcal{P})$	$P(P')$	(iv) the number of invariant (Laplace–Beltrami) operators defined over the space $P(P')$;
gilmore-lie-groups_ F02072	206	■ 0.999	$G^{\prime}$	G'	form G'/H , where G' is simple and noncompact and H is the maximal compact
gilmore-lie-groups_ F02078	313	■ 0.994	$n-1$	$n-1$	($n-1$)st order equation.
gilmore-lie-groups_ F02082	207	■ 1.000	$SO(p, q)/SO(p) \otimes SO(q)$	$SO(p, q)/SO(p) \otimes SO(q)$	example, $SO(p, q)/SO(p) \otimes SO(q)$ and $SO(p+q)/SO(p) \otimes SO(q)$ are dual.
gilmore-lie-groups_ F02106	207	■ 1.000	$G^{\prime}/H \rightarrow G/H$	$G'/H \rightarrow G/H$	subgroup in G' . To each there is a compact real form under $G'/H \rightarrow G/H$. For
gilmore-lie-groups_ F02107	207	■ 1.000	$SO(p+q)/SO(p) \otimes SO(q)$	$SO(p+q)/SO(p) \otimes SO(q)$	example, $SO(p, q)/SO(p) \otimes SO(q)$ and $SO(p+q)/SO(p) \otimes SO(q)$ are dual.
gilmore-lie-groups_ F02108	207	■ 1.000	$P, P^{\prime}$	P, P'	The metric tensor on the spaces P, P' is computed by defining a metric at the
gilmore-lie-groups_ F02109	207	■ 0.987	$dx(\mathrm{Id})$	$dx(\mathrm{Id})$	If $dx(\mathrm{Id})$ are infinitesimal displacements at the identity that are translated to
gilmore-lie-groups_ F02110	207	■ 1.000	$dx(p)$	$dx(p)$	infinitesimal displacements $dx(p)$ at point p , then these two sets of infinitesimals
gilmore-lie-groups_ F02111	208	■ 1.000	$M^i_{\mu}(p)$	$M^i_{\mu}(p)$	The matrix $M^i_{\mu}(p)$ is not easy to compute in general. For the rank-one spaces
gilmore-lie-groups_ F02112	208	■ 0.998	$SO(n, 1)/SO(n), SU(n, 1)/U(n), Sp(n, 1)/Sp(n) \times Sp(1)$	$SO(n, 1)/SO(n), SU(n, 1)/U(n), Sp(n, 1)/Sp(n) \times Sp(1)$	$SO(n, 1)/SO(n), SU(n, 1)/U(n), Sp(n, 1)/Sp(n) \times Sp(1)$ defined by
gilmore-lie-groups_ F02113	208	■ 1.000	$M^i_{\mu}(X)$	$M^i_{\mu}(X)$	the matrix $M^i_{\mu}(X)$ is determined from
gilmore-lie-groups_ F02114	208	■ 1.000	W^{-1}	W^{-1}	The matrix $M^i_{\mu}(X)$ is given by W^{-1} . Since the Cartan–Killing inner product is I_n
gilmore-lie-groups_ F02115	208	■ 1.000	$H_2^2=SO(2, 1)/SO(2)$	$H_2^2=SO(2, 1)/SO(2)$	the noncompact hyperboloid $H_2^2=SO(2, 1)/SO(2)$ and the compact sphere
gilmore-lie-groups_ F02116	209	■ 0.999	$S^2=SO(3)/SO(2)$	$S^2=SO(3)/SO(2)$	$S^2=SO(3)/SO(2)$. The metric tensors in the two cases are the 2×2 matrices
gilmore-lie-groups_ F02117	209	■ 1.000	$SO(2+1)/SO(2)$	$SO(2+1)/SO(2)$	ric space $SO(2+1)/SO(2)$ is parameterized by the interior of the unit circle
gilmore-lie-groups_ F02118	209	■ 0.999	$Y^2=1-\left(x^2+y^2\right) \geq 0$	$Y^2=1-\left(x^2+y^2\right) \geq 0$	$Y^2=1-\left(x^2+y^2\right) \geq 0$.
gilmore-lie-groups_ F02119	209	■ 1.000	x, y	x, y	for the compact case, the sphere. Instead of using the pair x, y as coordinates,
gilmore-lie-groups_ F02120	209	■ 1.000	$x^i, i=1, 2, \ldots, N$	$x^i, i=1, 2, \ldots, N$	we use indexed coordinates $x^i, i=1, 2, \ldots, N$, and set $N=2$ at the end of this
gilmore-lie-groups_ F02121	209	■ 1.000	$N=2$	$N=2$	we use indexed coordinates $x^i, i=1, 2, \ldots, N$, and set $N=2$ at the end of this
gilmore-lie-groups_ F02122	210	■ 1.000	$g^{ij} \simeq \delta^{ij}-x^i x^j$	$g^{ij} \simeq \delta^{ij}-x^i x^j$	The inverse (contravariant metric) is $g^{IJ} \simeq \delta^{IJ}-x^I x^J$, but we will not need this
gilmore-lie-groups_ F02123	210	■ 0.900	$g^{ij} \rightarrow \delta^{ij}$	$g^{ij} \rightarrow \delta^{ij}$	result. In the neighborhood of the identity ($g^{ij} \rightarrow \delta^{ij}$)
gilmore-lie-groups_ F02124	210	■ 1.000	$R_{\mu \sigma, \alpha \beta} =$	$R_{\mu \sigma, \alpha \beta} =$	delta function at the origin, and the resulting fully covariant metric tensor $R_{\mu \sigma, \alpha \beta} =$

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gilmore-lie-groups_ F02125	210	■ 1.000	$\backslash\delta_{\alpha\mu}\backslash\alpha\ \backslash\mu\ \backslash\delta_{\beta\sigma}\backslash\sigma\backslash\delta_{\alpha\sigma}\backslash\alpha\ \backslash\sigma\ \backslash\delta_{\beta\mu}\backslash\mu$	$\delta_{\alpha\mu}\delta_{\beta\sigma} - \delta_{\alpha\sigma}\delta_{\beta\mu}$	$\delta_{\alpha\mu}\delta_{\beta\sigma} - \delta_{\alpha\sigma}\delta_{\beta\mu}$ exhibits the full spectrum of expected symmetries.
gilmore-lie-groups_ F02126	210	■ 0.874	R=2	$R = 2$	For $N = 2$ (sphere S^2), $R = 2$.
gilmore-lie-groups_ F02127	210	■ 1.000	$H_{\{2\}}^{\{2\}}$	H_2^2	The computation can be carried out just as easily for the noncompact space H_2^2 .
gilmore-lie-groups_ F02128	210	■ 0.848	$\backslash\Gamma_{\mu\nu}\backslash\nu\backslash\sigma\ \backslash\rightarrow\backslash\delta_{\mu\nu}\backslash\nu\ \backslash\sigma\backslash\sigma$	$\Gamma_{\mu\nu}^\sigma \rightarrow -\delta_{\mu\nu}x^\sigma$	for example $\Gamma_{\mu\nu}^\sigma \rightarrow -\delta_{\mu\nu}x^\sigma$. The end result for H_2^2 is that $R = -2$.
gilmore-lie-groups_ F02129	210	■ 0.848	R=-2	$R = -2$	for example $\Gamma_{\mu\nu}^\sigma \rightarrow -\delta_{\mu\nu}x^\sigma$. The end result for H_2^2 is that $R = -2$.
gilmore-lie-groups_ F02130	211	■ 0.998	$d\ s^{\{2\}}=g_{\{\mu\}\ \{\nu\}}\ d\ x^{\{\mu\}}\ d\ x^{\{\nu\}}>0$	$ds^2 = g_{\mu\nu}dx^\mu dx^\nu > 0$	positive-definite ($ds^2 = g_{\mu\nu} \ dx^\mu dx^\nu > 0$, equality $\Rightarrow dx = 0$) nor negative-
gilmore-lie-groups_ F02131	211	■ 0.998	$\backslash\rightarrow\ d\ x=0$	$\Rightarrow dx = 0$	positive-definite ($ds^2 = g_{\mu\nu} \ dx^\mu dx^\nu > 0$, equality $\Rightarrow dx = 0$) nor negative-
gilmore-lie-groups_ F02132	211	■ 0.958	$\backslash\left(d\ s^{\{2\}}<0\backslash\right)$	$(ds^2 < 0)$	definite ($ds^2 < 0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-
gilmore-lie-groups_ F02133	211	■ 0.958	$\backslash g \ \backslash\neq\ 0$	$\ g\ \neq 0$	definite ($ds^2 < 0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-
gilmore-lie-groups_ F02134	211	■ 1.000	$\backslash\mathfrak{g}^{\backslash\prime\ \backslash\prime}$	$\mathfrak{g}''$	To make these statements more explicit, assume a Lie algebra $\mathfrak{g}''$ (noncompact)
gilmore-lie-groups_ F02135	211	■ 1.000	$\backslash\mathfrak{h}^{\backslash\prime\ \backslash\prime}$	$\mathfrak{h}''$	Then $\mathfrak{h}''$ and $\mathfrak{p}''$ are orthogonal subspaces in $\mathfrak{g}''$ under the Cartan–Killing inner
gilmore-lie-groups_ F02136	211	■ 1.000	$\backslash\mathfrak{p}^{\backslash\prime\ \backslash\prime}$	$\mathfrak{p}''$	Then $\mathfrak{h}''$ and $\mathfrak{p}''$ are orthogonal subspaces in $\mathfrak{g}''$ under the Cartan–Killing inner
gilmore-lie-groups_ F02137	211	■ 1.000	$H^{\backslash\prime\ \backslash\prime}=$	$H'' =$	indefinite. The space is curved and typically multiply connected. The space $H'' =$
gilmore-lie-groups_ F02138	211	■ 1.000	$\backslash\operatorname{EXP}\backslash\left(\backslash\mathfrak{h}^{\backslash\prime\ \backslash\prime}\backslash\right)$	$\operatorname{EXP}(\mathfrak{h}'')$	$\operatorname{EXP}(\mathfrak{h}'')$ is also an interesting pseudo-Riemannian symmetric space.
gilmore-lie-groups_ F02139	211	■ 1.000	$T_{\{1\}},\ T_{\{2\}}$	T_1, T_2	spaces. Begin with a compact simple Lie algebra $\mathfrak{g}$ and suppose T_1, T_2 are two
gilmore-lie-groups_ F02140	211	■ 0.819	$T_{\{1\}}^{\{2\}}=I$	$T_1^2 = I$	metric-preserving mappings of the Lie algebra onto itself that obey $T_1^2 = I, T_2^2 = I$
gilmore-lie-groups_ F02141	211	■ 0.819	$T_{\{2\}}^{\{2\}}=I$	$T_2^2 = I$	metric-preserving mappings of the Lie algebra onto itself that obey $T_1^2 = I, T_2^2 = I$
gilmore-lie-groups_ F02142	211	■ 1.000	$T_{\{1\}}\ \backslash\neq\ T_{\{2\}}$	$T_1 \neq T_2$	(cf. Section 11.3) and $T_1 \neq T_2$. Define the eigenspaces of $\mathfrak{g}$ under T_1, T_2 as $\mathfrak{g}_{\pm,\pm}$:
gilmore-lie-groups_ F02143	211	■ 1.000	$\backslash\mathfrak{g}_{\backslash\pm,\ \backslash\pm}$	$\mathfrak{g}_{\pm,\pm}$	(cf. Section 11.3) and $T_1 \neq T_2$. Define the eigenspaces of $\mathfrak{g}$ under T_1, T_2 as $\mathfrak{g}_{\pm,\pm}$:
gilmore-lie-groups_ F02144	212	■ 1.000	$\backslash\mathfrak{h}^{\backslash\prime\ \backslash\prime},\ \backslash\mathfrak{p}^{\backslash\prime\ \backslash\prime}$	$\mathfrak{h}'', \mathfrak{p}''$	The subspaces $\mathfrak{h}'', \mathfrak{p}''$ obey commutation relations (12.20). The Cartan–Killing inner
gilmore-lie-groups_ F02145	212	■ 0.996	$\backslash\mathfrak{p}^{\backslash\prime\ \backslash\prime}$	$\mathfrak{p}''$	product is indefinite on both $\mathfrak{h}''$ and $\mathfrak{p}''$ as long as $T_1 \neq T_2$.
gilmore-lie-groups_ F02146	212	■ 0.999	$T_{\{1\}}=$	$T_1 =$	For $\mathfrak{su}(2)$ the only two mappings are $T_1 =$ block diagonal decomposition and
gilmore-lie-groups_ F02147	212	■ 1.000	$T_{\{2\}}=$	$T_2 =$	$T_2 =$ complex conjugation. The eigenspace decomposition is
gilmore-lie-groups_ F02151	213	■ 1.000	$T_{\{3\}}=T_{\{1\}}\ T_{\{2\}}$	$T_3 = T_1 T_2$	generate a third, $T_3 = T_1 T_2$ and that $T_1 T_2 = T_2 T_1$. Show that $T_3 \neq I$ if $T_1 \neq T_2$. Show
gilmore-lie-groups_ F02152	212	■ 1.000	$\backslash\mathfrak{g}_{\backslash+,+}=0,\ \backslash\mathfrak{g}_{\backslash+,-}=i\ \backslash\sigma_3,\ \backslash\mathfrak{g}_{\backslash-,+}=i\ \backslash\sigma_2,\ \backslash\mathfrak{g}_{\backslash-,-}=i\ \backslash\sigma_1$	$\mathfrak{g}_{+,+} = 0, \mathfrak{g}_{+,-} = i\sigma_3, \mathfrak{g}_{-,+} = i\sigma_2, \mathfrak{g}_{-,-} = i\sigma_1$	This gives $\mathfrak{g}_{+,+} = 0, \mathfrak{g}_{+,-} = i\sigma_3, \mathfrak{g}_{-,+} = i\sigma_2, \mathfrak{g}_{-,-} = i\sigma_1$. Note that each mapping
gilmore-lie-groups_ F02153	212	■ 1.000	$T_{\{i\}}$	T_i	T_i has one positive and two negative eigenvalues, and chooses a different generator

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gilmore-lie-groups_ F02154	212	■ 0.999	$\mathfrak{h}^{\prime}$	$\mathfrak{h}'$	for the maximal compact subalgebra $\mathfrak{h}'$ of the noncompact real form $\mathfrak{g}'$.
gilmore-lie-groups_ F02155	212	■ 0.984	$P=G / K$	$P = G/K$	Globally symmetric spaces have the form $P = G/K$, where $\mathfrak{g}$ is a real form of
gilmore-lie-groups_ F02156	212	■ 1.000	$\phi_j(B, B^{\dag})$	$\phi_j(B, B^\dagger)$	1. Show that the invariant polynomials $\phi_j(B, B^\dagger)$ in (12.4) actually depend on the invari-
gilmore-lie-groups_ F02157	212	■ 0.585	$(p-q)$	$(p - q)$	The remaining $(p - q)$ or $(q - p)$ (whichever is positive) eigenvalues of the larger
gilmore-lie-groups_ F02158	212	■ 0.585	$(q-p)$	$(q - p)$	The remaining $(p - q)$ or $(q - p)$ (whichever is positive) eigenvalues of the larger
gilmore-lie-groups_ F02159	213	■ 1.000	Δ^2	Δ^2	2. The second order Laplace–Beltrami operator Δ^2 is constructed from the second order
gilmore-lie-groups_ F02160	213	■ 1.000	$G / H=P$	$G/H = P$	$G/H = P$.
gilmore-lie-groups_ F02161	213	■ 1.000	$\Delta^2=g^{ij}\left(\partial_i\partial_j-\Gamma_{ij}^k\partial_k\right)$	$\Delta^2 = g^{ij}(\partial_i\partial_j - \Gamma_{ij}^k\partial_k)$	tensor on P as $\Delta^2 = g^{ij}(\partial_i\partial_j - \Gamma_{ij}^k\partial_k)$.
gilmore-lie-groups_ F02162	213	■ 1.000	$x^2+y^2\leq 1$	$x^2 + y^2 \leq 1$	unit disk $x^2 + y^2 \leq 1$.
gilmore-lie-groups_ F02163	213	■ 1.000	S^n	S^n	e. Extend these results to the sphere S^n and its dual, $H^n, n > 2$.
gilmore-lie-groups_ F02164	213	■ 1.000	$H^n, n>2$	$H^n, n > 2$	e. Extend these results to the sphere S^n and its dual, $H^n, n > 2$.
gilmore-lie-groups_ F02165	213	■ 1.000	$T_1^2=T_2^2=I$	$T_1^2 = T_2^2 = I$	3. Show that the two metric-preserving mappings T_1 and T_2 that satisfy $T_1^2 = T_2^2 = I$
gilmore-lie-groups_ F02166	213	■ 1.000	$T_1T_2=T_2T_1$	$T_1T_2 = T_2T_1$	generate a third, $T_3 = T_1T_2$ and that $T_1T_2 = T_2T_1$. Show that $T_3 \neq I$ if $T_1 \neq T_2$. Show
gilmore-lie-groups_ F02167	213	■ 1.000	$T_3\neq I$	$T_3 \neq I$	generate a third, $T_3 = T_1T_2$ and that $T_1T_2 = T_2T_1$. Show that $T_3 \neq I$ if $T_1 \neq T_2$. Show
gilmore-lie-groups_ F02168	213	■ 1.000	$\mathfrak{g}=\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{-,-}+\mathfrak{g}_{-,-}+\mathfrak{g}_{-,-}$	$\mathfrak{g} = \mathfrak{g}_{+,+} + \mathfrak{g}_{+,-} + \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-}$	compact Lie algebra $\mathfrak{g} = \mathfrak{g}_{+,+} + \mathfrak{g}_{+,-} + \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-}$ that is available by choosing
gilmore-lie-groups_ F02169	213	■ 1.000	$3!/1!=6$	$3!/1! = 6$	first, one of these three involutions, and then a second (there are $3!/1!=6$ choices).
gilmore-lie-groups_ F02170	213	■ 1.000	$b_3=0$	$b_3 = 0$	from (11.2) by setting $b_3 = 0$:
gilmore-lie-groups_ F02171	213	■ 1.000	$\phi_j(\mathfrak{p})$	$\phi_j(\mathfrak{p})$	5. Show that the coefficients $\phi_j(\mathfrak{p})$ in the secular equation for a symmetric space are
gilmore-lie-groups_ F02172	213	■ 0.992	$\phi_j(\mathfrak{h},\mathfrak{p})$	$\phi_j(\mathfrak{h}, \mathfrak{p})$	obtained from the coefficients $\phi_j(\mathfrak{h}, \mathfrak{p})$ in the secular equation for the parent Lie
gilmore-lie-groups_ F02173	213	■ 0.998	$\mathfrak{h}=0$	$\mathfrak{h} = 0$	algebra (Eq. (12.3)) by setting $\mathfrak{h} = 0$.
gilmore-lie-groups_ F02174	213	■ 0.891	$x_0^2-x_1^2-x_2^2=1$	$x_0^2 - x_1^2 - x_2^2 = 1$	by the condition $x_0^2 - x_1^2 - x_2^2 = 1$ in the Lorentz 3-space with signature (1, 2).
gilmore-lie-groups_ F02175	213	■ 1.000	$-ds^2=dx_0^2-dx_1^2-$	$-ds^2 = dx_0^2 - dx_1^2 -$	b. Show that the invariant metric is induced from the metric $-ds^2 = dx_0^2 - dx_1^2 -$
gilmore-lie-groups_ F02176	213	■ 0.973	dx_2^2	dx_2^2	dx_2^2 in this Lorentz 3-space.
gilmore-lie-groups_ F02177	214	■ 1.000	x_1, x_2	x_1, x_2	c. Use coordinates x_1, x_2 to parameterize the points in H_2^2 , and show
gilmore-lie-groups_ F02178	214	■ 1.000	$(r,\theta), x_1=r\cos(\theta), x_2=r\sin(\theta)$	$(r, \theta), x_1 = r\cos(\theta), x_2 = r\sin(\theta)$	e. Introduce polar coordinates $(r, \theta), x_1 = r\cos(\theta), x_2 = r\sin(\theta)$. Show that
gilmore-lie-groups_ F02179	214	■ 1.000	$S O(1,2)$	$SO(1, 2)$	f. Determine the action of a group operation in $SO(1, 2)$ on the point $(x_1, x_2) \in H_2^2$.
gilmore-lie-groups_ F02180	214	■ 1.000	$\left(x_1, x_2\right) \in H_2^2$	$(x_1, x_2) \in H_2^2$	f. Determine the action of a group operation in $SO(1, 2)$ on the point $(x_1, x_2) \in H_2^2$.

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gilmore-lie-groups_F02181	214	■ 1.000	$g_{\{i\ j\}}(x)$	$g_{ij}(x)$	7. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.
gilmore-lie-groups_F02182	214	■ 1.000	$X_{\{r\ s\}}=g_{\{r\ t\}}x^{\{t\}}\partial_{\{s\}}-$	$X_{rs} = g_{rt}x^t\partial_s -$	a. Show that the generators of infinitesimal rotations at a point are $X_{rs} = g_{rt}x^t\partial_s -$
gilmore-lie-groups_F02183	214	■ 1.000	$g_{\{s\ t\}}x^{\{t\}}\partial_{\{r\}}$	$g_{st}x^t\partial_r$	$g_{st}x^t\partial_r$.
gilmore-lie-groups_F02184	214	■ 1.000	$\left[X_{\{a\ b\}}, \Delta\right]=0$	$[X_{ab}, \Delta] = 0$	b. Show $[X_{ab}, \Delta] = 0$, where $\Delta = G^{ab;rs}X_{ab}X_{rs}$ is the Laplace–Beltrami operator
gilmore-lie-groups_F02185	214	■ 1.000	$\Delta=G^{\{a\ b\ ;\ r\ s\}}X_{\{a\ b\}}X_{\{r\ s\}}$	$\Delta = G^{ab;rs}X_{ab}X_{rs}$	b. Show $[X_{ab}, \Delta] = 0$, where $\Delta = G^{ab;rs}X_{ab}X_{rs}$ is the Laplace–Beltrami operator
gilmore-lie-groups_F02186	214	■ 0.546	$G_{\{a\ b\ ;\ r\ s\}}=\operatorname{tr}\left\{\frac{\mathrm{d}\mathfrak{e}\mathfrak{f}}{\mathrm{d}\mathfrak{e}\mathfrak{f}}\left(X_{\{a\ b\}}\right)\frac{\mathrm{d}\mathfrak{e}\mathfrak{f}}{\mathrm{d}\mathfrak{e}\mathfrak{f}}\left(X_{\{r\ s\}}\right)\right\}$	$G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{ab}) \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{rs}) \}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{ab}) \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups_F02187	214	■ 0.546	$G^{\{a\ b\ ;\ r\ s\}}$	$G^{ab;rs}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{ab}) \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups_F02188	214	■ 0.546	$G_{\{a\ b\ ;\ r\ s\}}$	$G_{ab;rs}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{ab}) \mathfrak{d}\mathfrak{e}\mathfrak{f}(X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups_F02189	214	■ 0.989	Δ	Δ	c. Show that Δ consists of terms that are both quadratic and linear in the operators
gilmore-lie-groups_F02190	214	■ 1.000	$\partial_{\{r\}}$	∂_r	∂_r , and that
gilmore-lie-groups_F02191	214	■ 0.765	$\Gamma_{\{r\ s\}}^{\{t\}}$	Γ_{rs}^t	The function Γ_{rs}^t is <i>not</i> a tensor. The components of the Christoffel symbol are
gilmore-lie-groups_F02192	214	■ 1.000	$\left(r,\phi_2,\phi_3,\ldots,\phi_n\right)$	$(r, \phi_2, \phi_3, \dots, \phi_n)$	8. Use radial coordinates $(r, \phi_2, \phi_3, \dots, \phi_n)$ on the sphere $S^n \subset R^{n+1}$.
gilmore-lie-groups_F02193	214	■ 1.000	$S^{\{n\}} \subset R^{\{n+1\}}$	$S^n \subset R^{n+1}$	8. Use radial coordinates $(r, \phi_2, \phi_3, \dots, \phi_n)$ on the sphere $S^n \subset R^{n+1}$.
gilmore-lie-groups_F02194	215	■ 0.995	$S\ O(n+1)$	$SO(n+1)$	c. Compare this with the second order Casimir operator for $SO(n+1)$:
gilmore-lie-groups_F02195	215	■ 1.000	$f_{\{1\}}(\phi)$	$f_1(\phi)$	Compute $f_1(\phi)$ and $f_2(\phi)$.
gilmore-lie-groups_F02196	215	■ 1.000	$f_{\{2\}}(\phi)$	$f_2(\phi)$	Compute $f_1(\phi)$ and $f_2(\phi)$.
gilmore-lie-groups_F02197	215	■ 0.994	$a_{\{i\}}^{\dagger}a_{\{j\}}\left(\mathcal{H}=h_{ij}(t)a_i^{\dagger}a_j,1\leq i,j\leq\right)$	$a_i^{\dagger}a_j\left(\mathcal{H}=h_{ij}(t)a_i^{\dagger}a_j,1\leq i,j\leq\right)$	is a linear superposition of the bilinear products $a_i^{\dagger}a_j$ ($\mathcal{H}=h_{ij}(t)a_i^{\dagger}a_j$, $1\leq i,j\leq$
gilmore-lie-groups_F02198	215	■ 0.985	$i\ \mathcal{H}$	$i\mathcal{H}$	n), so that $i\mathcal{H}$ is a time-dependent element in the Lie algebra $\mathfrak{u}(n)$. Assume the
gilmore-lie-groups_F02199	215	■ 1.000	$S\ U(n)\ /\ U(n-1)$	$SU(n)/U(n-1)$	trajectory exists in the rank one symmetric space $SU(n)/U(n-1)$. Write down the
gilmore-lie-groups_F02200	215	■ 1.000	$V^{\{n\}}$	$V^{(n)}$	10. Conformal group The inner product on an n -dimensional linear vector space $V^{(n)}$
gilmore-lie-groups_F02201	215	■ 0.999	$(x,\ x)_{\{m\}}=m_{\{i\ j\}}x^{\{i\}}x^{\{j\}}$	$(x,x)_m = m_{ij}x^ix^j$	is defined by $(x,x)_m = m_{ij}x^ix^j$. Define coordinates y in an $n+2$ dimensional linear
gilmore-lie-groups_F02202	215	■ 0.999	$n+2$	$n+2$	is defined by $(x,x)_m = m_{ij}x^ix^j$. Define coordinates y in an $n+2$ dimensional linear
gilmore-lie-groups_F02203	215	■ 1.000	$W^{\{n+2\}}$	$W^{(n+2)}$	vector space $W^{(n+2)}$ as follows
gilmore-lie-groups_F02204	215	■ 1.000	$(y,\ y)_{\{M\}}=M_{\{\mu\nu\}}y^{\{\mu\}}y^{\{\nu\}}=(s\ x,\ s\ x)_{\{m\}}-\frac{2}{2}s\left[s(x,\ x)_{\{m\}}\right]=0$	$(y,y)_M = M_{\mu\nu}y^{\mu}y^{\nu} = (sx, sx)_m - \frac{2}{2}s[s(x, x)_m] = 0$	a. Show $(y, y)_M = M_{\mu\nu}y^{\mu}y^{\nu} = (sx, sx)_m - \frac{2}{2}s[s(x, x)_m] = 0$.
gilmore-lie-groups_F02205	215	■ 1.000	$G=O(n)$	$G = O(n)$	$G = O(n)$.
gilmore-lie-groups_F02206	215	■ 1.000	$O(n+1,1)$	$O(n+1,1)$	c. Show that the Lie group H that preserves inner products in $W^{(n+2)}$ is $O(n+1,1)$.

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gilmore-lie-groups_F02287	215	■ 1.000	$n_{\{1\}}, n_{\{2\}} \backslash \text{left}(n_{\{1\}}+n_{\{2\}}=n \backslash \text{right})$	$n_1, n_2 \left(n_1 + n_2 = n \right)$	d. If the metric m has signature $n_1, n_2 \left(n_1 + n_2 = n \right)$, show that $G = O(n_1, n_2)$ and
gilmore-lie-groups_F02288	215	■ 1.000	$G=O \backslash \text{left}(n_{\{1\}}, n_{\{2\}} \backslash \text{right})$	$G = O \left(n_1, n_2 \right)$	d. If the metric m has signature $n_1, n_2 \left(n_1 + n_2 = n \right)$, show that $G = O(n_1, n_2)$ and
gilmore-lie-groups_F02289	215	■ 1.000	$H=O \backslash \text{left}(n_{\{1\}}+1, n_{\{2\}}+1 \backslash \text{right})$	$H = O \left(n_1 + 1, n_2 + 1 \right)$	$H = O(n_1 + 1, n_2 + 1)$.
gilmore-lie-groups_F02210	215	■ 1.000	$S \ 0 \backslash \text{left}(n_{\{1\}}+1, n_{\{2\}}+1 \backslash \text{right}) \ / \ S \ 0 \backslash \text{left}(n_{\{1\}}, n_{\{2\}} \backslash \text{right})$	$SO \left(n_1 + 1, n_2 + 1 \right) / SO \left(n_1, n_2 \right)$	f. Construct the quotient space $SO(n_1 + 1, n_2 + 1) / SO(n_1, n_2)$.
gilmore-lie-groups_F02211	215	■ 0.996	$y \rightarrow y^{\prime}$	$y \rightarrow y'$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x^i = y^i / y'^{m+1}$.
gilmore-lie-groups_F02212	215	■ 0.996	$x \rightarrow x^{\prime}$	$x \rightarrow x'$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x^i = y^i / y'^{m+1}$.
gilmore-lie-groups_F02213	215	■ 0.996	$x^{\prime \ i}=y^{\prime \ i} / y^{\prime \ m+1}$	$x^i = y^i / y'^{m+1}$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x^i = y^i / y'^{m+1}$.
gilmore-lie-groups_F02214	215	■ 0.997	$(+1,-1,-1,-1)$	$(+1, -1, -1, -1)$	h. The Lorentz metric $(+1, -1, -1, -1)$ leaves the four-momentum invariant:
gilmore-lie-groups_F02215	216	■ 0.988	$L_{\mu \nu}$	$L_{\mu \nu}$	The operators $L_{\mu \nu}$ are the infinitesimal generators of the Lorentz group $SO(3, 1)$
gilmore-lie-groups_F02216	216	■ 1.000	P_{μ}	P_{μ}	and P_{μ} generate translations. Taken together $L_{\mu \nu}$ and P_{μ} generate the Poincaré
gilmore-lie-groups_F02217	216	■ 1.000	K_{μ}	K_{μ}	group. The operator S generates dilations and the four operators K_{μ} generate
gilmore-lie-groups_F02218	216	■ 1.000	$x_{\mu}=g_{\mu \nu} \ x^{\nu}$	$x_{\mu} = g_{\mu \nu} x^{\nu}$	conformal transformations. Above $x_{\mu} = g_{\mu \nu} x^{\nu}$.
gilmore-lie-groups_F02219	216	■ 0.989	$e^{c^{\mu} K_{\mu}} \left(x^{\nu} \right) = x'^{\nu} = \frac{x^{\nu} + c^{\nu} \left(x, x \right)}{1 + 2 \left(c, x \right) + \left(c, c \right) \left(x, x \right)}$	$e^{c^{\mu} K_{\mu}} \left(x^{\nu} \right) = x'^{\nu} = \frac{x^{\nu} + c^{\nu} \left(x, x \right)}{1 + 2 \left(c, x \right) + \left(c, c \right) \left(x, x \right)}$	k. Show that $e^{c^{\mu} K_{\mu}} \left(x^{\nu} \right) = x'^{\nu} = \frac{x^{\nu} + c^{\nu} \left(x, x \right)}{1 + 2 \left(c, x \right) + \left(c, c \right) \left(x, x \right)}$.
gilmore-lie-groups_F02220	216	■ 0.999	$P_{\mu} \rightarrow P_{\mu}^{\prime}=x_{\mu}$	$P_{\mu} \rightarrow P_{\mu}' = x_{\mu}$	m. Discuss the duality created by $P_{\mu} \rightarrow P_{\mu}' = x_{\mu}$ and $K_{\mu} \rightarrow K_{\mu}' = 2(x, \partial) \partial_{\mu} -$
gilmore-lie-groups_F02221	216	■ 0.999	$K_{\mu} \rightarrow K_{\mu}^{\prime}=2 \left(x, \partial \right) \partial_{\mu} - \partial_{\mu}$	$K_{\mu} \rightarrow K_{\mu}' = 2(x, \partial) \partial_{\mu} -$	m. Discuss the duality created by $P_{\mu} \rightarrow P_{\mu}' = x_{\mu}$ and $K_{\mu} \rightarrow K_{\mu}' = 2(x, \partial) \partial_{\mu} -$
gilmore-lie-groups_F02222	216	■ 0.719	$\left(\partial, \partial \right) x_{\mu}$	$(\partial, \partial) x_{\mu}$	$(\partial, \partial) x_{\mu}$.
gilmore-lie-groups_F02223	216	■ 1.000	$z=x+i \ y$	$z = x + i y$	11. The upper half of the complex plane has coordinates $z = x + i y$. This upper half-
gilmore-lie-groups_F02224	216	■ 1.000	$P \ S \ L(2, \mathbb{R})$	$PSL(2, \mathbb{R})$	and denoted $PSL(2, \mathbb{R})$.
gilmore-lie-groups_F02225	216	■ 1.000	$M, -M \in S \ L(2 \ ; \ \mathbb{R})$	$M, -M \in SL(2; \mathbb{R})$	a. Show that $M, -M \in SL(2; \mathbb{R})$ generate identical transformations. The group
gilmore-lie-groups_F02226	216	■ 1.000	$y^{\prime}>0$	$y' > 0$	In particular, show that $y' > 0$ if $y > 0$ and $y' = 0$ if $y = 0$. The transformation
gilmore-lie-groups_F02227	216	■ 1.000	$y>0$	$y > 0$	In particular, show that $y' > 0$ if $y > 0$ and $y' = 0$ if $y = 0$. The transformation
gilmore-lie-groups_F02228	216	■ 1.000	$y^{\prime}=0$	$y' = 0$	In particular, show that $y' > 0$ if $y > 0$ and $y' = 0$ if $y = 0$. The transformation
gilmore-lie-groups_F02229	216	■ 0.997	$(y=0)$	$(y = 0)$	$(y = 0)$, onto itself.
gilmore-lie-groups_F02230	217	■ 1.000	$d \ z^{\prime}=d \ z \ / c \ z+d ^2$	$dz' = dz / cz + d ^2$	d. Show $dz' = dz / cz + d ^2$
gilmore-lie-groups_F02231	217	■ 1.000	$d \ \mu=d \ x \ d y \ / \ y^2$	$d\mu = dx dy / y^2$	e. Show that the invariant measure is $d\mu = dx dy / y^2$
gilmore-lie-groups_F02232	217	■ 1.000	$z_{\{2\}}$	z_2	f. Show that the distance between two points z_1 and z_2 is
gilmore-lie-groups_F02233	217	■ 1.000	$w=x+i \ y$	$w = x + i y$	12. The unit disk in the complex plane $w = x + i y$ consists of those points that satisfy

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gilmore-lie-groups_F02234	217	■ 1.000	$\bar{w}w = x^2 + y^2 \leq 1$	$\bar{w}w = x^2 + y^2 \leq 1$	 . The unit disk, with a suitable metric, provides a second represen-
gilmore-lie-groups_F02235	217	■ 1.000	$M, -M \in SU(1, 1)$	$M, -M \in SU(1, 1)$	a. Show that $M, -M \in SU(1, 1)$ generate identical mappings of the unit disk into
gilmore-lie-groups_F02236	217	■ 1.000	$w=e^{i \phi} \rightarrow w^{\prime}=e^{i \psi}$	$w = e^{i\phi} \rightarrow w' = e^{i\psi}$	b. Show that  . Compute $\psi(\phi)$.
gilmore-lie-groups_F02237	217	■ 1.000	$\psi(\phi)$	$\psi(\phi)$	b. Show that $w = e^{i\phi} \rightarrow w' = e^{i\psi}$. Compute $\psi(\phi)$.
gilmore-lie-groups_F02238	217	■ 1.000	$w_{\{1\}}$	w_1	e. Show that the distance between two points w_1 and w_2 in this unit disk is
gilmore-lie-groups_F02239	217	■ 1.000	$w_{\{2\}}$	w_2	e. Show that the distance between two points w_1 and w_2 in this unit disk is
gilmore-lie-groups_F02240	217	■ 1.000	w	w	13. Show that the mapping from z in the upper half-plane to w in the unit disk given by
gilmore-lie-groups_F02241	217	■ 1.000	$z_{\{0\}}$	z_0	is conformal, that is, it preserves angles. Here z_0 is any point in the upper half-plane.
gilmore-lie-groups_F02242	218	■ 1.000	$z_{\{0\}}=i$	$z_0 = i$	b. Choose $z_0 = i$ and $e^{i\phi} = i$ to give the canonical map
gilmore-lie-groups_F02243	218	■ 1.000	$e^{i \phi}=i$	$e^{i\phi} = i$	b. Choose $z_0 = i$ and $e^{i\phi} = i$ to give the canonical map
gilmore-lie-groups_F02244	219	■ 1.000	$Y_{\{r\}}$	Y_r	called contraction . In this process a new set of basis vectors Y_r is related to the
gilmore-lie-groups_F02245	219	■ 1.000	$Y_{\{r\}}=$	$Y_r =$	initial set of basis vectors X_i through a parameter-dependent change of basis: $Y_r =$
gilmore-lie-groups_F02246	219	■ 1.000	$M_{\{r\}}^{\{i\}}(\epsilon) X_{\{i\}}$	$M_r^{\ i}(\epsilon)X_i$	$M_r^{\ i}(\epsilon)X_i$. The structure constants have the transformation properties of a tensor:
gilmore-lie-groups_F02247	219	■ 0.986	$C_{\{r \ s\}}^{\{t\}}(\epsilon)=M_{\{r\}}^{\{i\}}(\epsilon) M_{\{s\}}^{\{j\}}(\epsilon) C_{\{i \ j\}}^{\{k\}}\left(M(\epsilon)^{-1}\right)_{\{k\}}^{\{t\}}$	$C_{rs}^{\ t}(\epsilon) = M_r^{\ i}(\epsilon)M_s^{\ j}(\epsilon)C_{ij}^{\ k}(M(\epsilon)^{-1})_k^{\ t}$	$C_{rs}^{\ t}(\epsilon) = M_r^{\ i}(\epsilon)M_s^{\ j}(\epsilon)C_{ij}^{\ k}(M(\epsilon)^{-1})_k^{\ t}$ (cf. Eq. (4.22)). As long as the change of
gilmore-lie-groups_F02248	219	■ 1.000	$C_{\{r \ s\}}^{\{t\}}(\epsilon)$	$C_{rs}^{\ t}(\epsilon)$	If the transformation becomes singular, the structure constants $C_{rs}^{\ t}(\epsilon)$ may
gilmore-lie-groups_F02249	220	■ 0.998	$\mathfrak{g} \rightarrow \mathfrak{g}^{\prime}$	$\mathfrak{g} \rightarrow \mathfrak{g}'$	then the Inönü–Wigner contraction of $\mathfrak{g} \rightarrow \mathfrak{g}'$ involves the following change of
gilmore-lie-groups_F02250	220	■ 1.000	$\operatorname{dim}(\mathfrak{h})$	$\dim(\mathfrak{h})$	where $\dim(\mathfrak{h})$ is the dimension of the subalgebra $\mathfrak{h}$. The commutation relations of
gilmore-lie-groups_F02251	220	■ 1.000	$\epsilon \rightarrow 0$	$\epsilon \rightarrow 0$	$\mathfrak{g}'$ are well defined for all values of ϵ , including the singular limit $\epsilon \rightarrow 0$:
gilmore-lie-groups_F02252	220	■ 1.000	$\left[\mathfrak{h}, \mathfrak{p}^{\prime}\right] \subseteq \mathfrak{p}^{\prime}$	$[\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}'$	subalgebra $\mathfrak{h}$ and the subalgebra $\mathfrak{p}'$, where $\mathfrak{p}'$ is commutative and  :
gilmore-lie-groups_F02254	220	■ 0.998	$L_{\{1\}}=X_{\{23\}}=$	$L_1 = X_{23} =$	The infinitesimal generators of the Lie group $SO(3)$ may be chosen as $L_1 = X_{23} =$
gilmore-lie-groups_F02255	220	■ 1.000	$x_{\{2\}} \partial_{\{3\}}-x_{\{3\}} \partial_{\{2\}}=\epsilon_{1 \ j \ k} x_{\{j\}} \partial_{\{k\}}$	$x_2\partial_3 - x_3\partial_2 = \epsilon_{1jk}x_j\partial_k$	$x_2\partial_3 - x_3\partial_2 = \epsilon_{1jk}x_j\partial_k$, with L_2 and L_3 defined by cyclic permutation. The com-
gilmore-lie-groups_F02256	220	■ 1.000	$L_{\{2\}}$	L_2	$x_2\partial_3 - x_3\partial_2 = \epsilon_{1jk}x_j\partial_k$, with L_2 and L_3 defined by cyclic permutation. The com-
gilmore-lie-groups_F02257	220	■ 1.000	$L_{\{3\}}$	L_3	$x_2\partial_3 - x_3\partial_2 = \epsilon_{1jk}x_j\partial_k$, with L_2 and L_3 defined by cyclic permutation. The com-
gilmore-lie-groups_F02258	221	■ 1.000	$L_{\{1\}}$	L_1	finitesimal generator L_3) the operators L_1 and L_2 go to
gilmore-lie-groups_F02259	221	■ 1.000	$I \leq O(2)=E(2)$	$ISO(2) = E(2)$	The commutation relations of the contracted algebra, $ISO(2) = E(2)$, are
gilmore-lie-groups_F02260	221	■ 1.000	$L_{\{3\}}, P_{\{1\}}, P_{\{2\}}$	L_3, P_1, P_2	The three operators L_3, P_1, P_2 generate the group of Euclidean motions of the plane,

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gilmore-lie-groups_ F02261	221	■ 0.983	$R^2, \text{ISO}(2)$	$R^2, ISO(2)$	$E(2)$, or inhomogeneous orthogonal transformations in the plane $R^2, ISO(2)$. This
gilmore-lie-groups_ F02262	221	■ 1.000	$P_1=\partial_1$	$P_1 = \partial_1$	the origin in the x - and y -directions, generated by $P_1 = \partial_1$ and $P_2 = \partial_2$.
gilmore-lie-groups_ F02263	221	■ 1.000	$P_2=\partial_2$	$P_2 = \partial_2$	the origin in the x - and y -directions, generated by $P_1 = \partial_1$ and $P_2 = \partial_2$.
gilmore-lie-groups_ F02264	221	■ 0.698	$x^2+y^2+z^2=R^2$	$x^2 + y^2 + z^2 = R^2$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown in
gilmore-lie-groups_ F02265	221	■ 0.698	$0, 0, R$	$0, 0, R$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown in
gilmore-lie-groups_ F02266	221	■ 1.000	$R \rightarrow \infty$	$R \rightarrow \infty$	In the limit $R \rightarrow \infty$ we find
gilmore-lie-groups_ F02267	221	■ 0.992	$-P_2, +P_1$	$-P_2, +P_1$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_ F02268	221	■ 0.992	$-y$	$-y$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_ F02269	221	■ 0.992	$+x$	$+x$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_ F02270	221	■ 1.000	θ_1, θ_2	θ_1, θ_2	directions. In addition, the parameters θ_1, θ_2 and d_1, d_2 are related by
gilmore-lie-groups_ F02271	221	■ 1.000	d_1, d_2	d_1, d_2	directions. In addition, the parameters θ_1, θ_2 and d_1, d_2 are related by
gilmore-lie-groups_ F02272	221	■ 1.000	$R \theta_i (i=1,2)$	$R\theta_i (i = 1, 2)$	with the product $R\theta_i (i = 1, 2)$ approaching a well-defined limit. This corresponds
gilmore-lie-groups_ F02273	221	■ 1.000	$\theta_2=d_1 / R$	$\theta_2 = d_1/R$	to a rotation through an angle $\theta_2 = d_1/R$ about the y -axis producing a displacement
gilmore-lie-groups_ F02274	221	■ 1.000	d_1	d_1	of d_1 in the x -direction, and a rotation through an angle $\theta_1 = d_2/R$ about the x -axis
gilmore-lie-groups_ F02275	221	■ 1.000	$\theta_1=d_2 / R$	$\theta_1 = d_2/R$	of d_1 in the x -direction, and a rotation through an angle $\theta_1 = d_2/R$ about the x -axis
gilmore-lie-groups_ F02276	221	■ 1.000	$-d_2$	$-d_2$	producing a displacement of $-d_2$ in the y -direction.
gilmore-lie-groups_ F02278	223	■ 0.418	P_i	P_i	The operators P_i describe displacements in the x -, y -, and z -directions ($i = 1, 2, 3$).
gilmore-lie-groups_ F02279	223	■ 0.418	$i=1,2,3$	$i = 1, 2, 3$	The operators P_i describe displacements in the x -, y -, and z -directions ($i = 1, 2, 3$).
gilmore-lie-groups_ F02280	223	■ 1.000	$\text{ISO}(3)$	$ISO(3)$	The contracted group is $ISO(3)$, the Euclidean, or inhomogeneous orthogonal
gilmore-lie-groups_ F02281	223	■ 1.000	$\text{SO}(3) \rightarrow \text{ISO}(2)$	$SO(3) \rightarrow ISO(2)$	As in the case $SO(3) \rightarrow ISO(2)$, we can contract the second order Casimir
gilmore-lie-groups_ F02282	223	■ 1.000	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	The two invariant operators $\mathbf{P} \cdot \mathbf{P} = \nabla^2$ and $\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$ form a complete set of
gilmore-lie-groups_ F02283	223	■ 1.000	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	The two invariant operators $\mathbf{P} \cdot \mathbf{P} = \nabla^2$ and $\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$ form a complete set of
gilmore-lie-groups_ F02285	224	■ 0.592	$[\text{SO}(3,1)]$	$[SO(3,1)]$	The group $ISO(3, 1)$ consists of proper Lorentz transformations $[SO(3, 1)]$ that
gilmore-lie-groups_ F02286	224	■ 0.997	$\text{SO}(3,2)$	$SO(3, 2)$	This group can be contracted from either $SO(4, 1)$ or $SO(3, 2)$.
gilmore-lie-groups_ F02287	224	■ 0.998	$\epsilon^{\alpha\beta\gamma\mu\nu}$	$\epsilon^{\alpha\beta\gamma\mu\nu}$	where $\epsilon^{\alpha\beta\gamma\mu\nu}$ is the Levi–Civita symbol (antisymmetric tensor) on five symbols,
gilmore-lie-groups_ F02288	224	■ 1.000	W_{α}	W_{α}	and W_{α} is similarly defined. The contracted limit of W^{α} is nonzero only if one of
gilmore-lie-groups_ F02289	224	■ 1.000	W^{α}	W^{α}	and W_{α} is similarly defined. The contracted limit of W^{α} is nonzero only if one of

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gilmore-lie-groups_F02290	224	■ 0.615	v	v	the four remaining symbols (e.g., v) is 5:
gilmore-lie-groups_F02291	225	■ 1.000	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu)$	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu)$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu)$ is fairly complicated. Since $W^\alpha W_\alpha$ is invariant,
gilmore-lie-groups_F02292	225	■ 1.000	$W^\alpha W_\alpha$	$W^\alpha W_\alpha$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu)$ is fairly complicated. Since $W^\alpha W_\alpha$ is invariant,
gilmore-lie-groups_F02293	225	■ 1.000	$\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$	$\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$	with $\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$.
gilmore-lie-groups_F02294	225	■ 0.862	$\mathfrak{u}(2)$	$\mathfrak{u}(2)$	The Lie algebra $\mathfrak{u}(2)$ is spanned by infinitesimal generators $J_3, J_\pm, J_0$ with com-
gilmore-lie-groups_F02295	225	■ 0.862	$J_3, J_\pm, J_0$	$J_3, J_\pm, J_0$	The Lie algebra $\mathfrak{u}(2)$ is spanned by infinitesimal generators $J_3, J_\pm, J_0$ with com-
gilmore-lie-groups_F02296	225	■ 0.996	$h_3, h_\pm, h_0$	$h_3, h_\pm, h_0$	The operators $h_3, h_\pm, h_0$ are related to $J_3, J_\pm, J_0$ by the following change of basis
gilmore-lie-groups_F02297	226	■ 1.000	$c \rightarrow 0$	$c \rightarrow 0$	In the limit $c \rightarrow 0$ the change of basis transformation becomes singular but the
gilmore-lie-groups_F02298	226	■ 1.000	$J_0 \rightarrow h_0$	$J_0 \rightarrow h_0$	Under contraction $J_0 \rightarrow h_0$ but the second Casimir operator has a more interesting
gilmore-lie-groups_F02299	226	■ 1.000	$(h_0/2c)^2$	$(h_0/2c)^2$	The operator $(h_0/2c)^2$ is proportional to the square of the first Casimir operator. It
gilmore-lie-groups_F02300	226	■ 1.000	$c \rightarrow 0, (ch_3)^2 \rightarrow 0$	$c \rightarrow 0, (ch_3)^2 \rightarrow 0$	in the Lie algebra. In the limit $c \rightarrow 0$, $(ch_3)^2 \rightarrow 0$ and the remaining operators go
gilmore-lie-groups_F02301	227	■ 1.000	$\hat{n} + \frac{1}{2}I, a^\dagger, a$	$\hat{n} + \frac{1}{2}I, a^\dagger, a$	This is a quadratic operator in the generators $\hat{n} + \frac{1}{2}I, a^\dagger, a$, and I of H_4 . The value
gilmore-lie-groups_F02302	227	■ 1.000	$\mathfrak{h}_4$	$\mathfrak{h}_4$	An arbitrary element in the Lie algebra $\mathfrak{u}(2)$ and its counterpart in the algebra $\mathfrak{h}_4$
gilmore-lie-groups_F02303	227	■ 1.000	θ_0	θ_0	exist. In addition, θ_0 should diverge so that $\theta_0 - \theta_3/2c^2$ remains well defined.
gilmore-lie-groups_F02304	227	■ 1.000	$\theta_0 - \theta_3/2c^2$	$\theta_0 - \theta_3/2c^2$	exist. In addition, θ_0 should diverge so that $\theta_0 - \theta_3/2c^2$ remains well defined.
gilmore-lie-groups_F02305	227	■ 0.999	h_3	h_3	The action of the operators h_3 on the angular momentum state $ J, M\rangle$ is
gilmore-lie-groups_F02306	227	■ 0.999	$ J, M\rangle$	$ J, M\rangle$	The action of the operators h_3 on the angular momentum state $ J, M\rangle$ is
gilmore-lie-groups_F02307	227	■ 1.000	$ J, -J\rangle$	$ J, -J\rangle$	It is useful to measure states from the “lowest” state $ J, -J\rangle$ in the angular mo-
gilmore-lie-groups_F02308	227	■ 1.000	$M = -J$	$M = -J$	$M = -J$, and the n th state when
gilmore-lie-groups_F02309	227	■ 0.985	J	J	representations J of dimension $(2J + 1)$ as c becomes smaller and smaller. Specif-
gilmore-lie-groups_F02310	227	■ 0.985	$(2J + 1)$	$(2J + 1)$	representations J of dimension $(2J + 1)$ as c becomes smaller and smaller. Specif-
gilmore-lie-groups_F02311	228	■ 1.000	$n = J + M$	$n = J + M$	applying the angular momentum shift up operator $n = J + M$ times to the ground
gilmore-lie-groups_F02312	229	■ 1.000	$\lim_{c \rightarrow 0} \zeta/c \rightarrow \alpha$	$\lim_{c \rightarrow 0} \zeta/c \rightarrow \alpha$	contracts under $\lim_{c \rightarrow 0} \zeta/c \rightarrow \alpha$ to the BCH formula for H_4
gilmore-lie-groups_F02313	229	■ 1.000	$P_m^l(\cos \theta)$	$P_m^l(\cos \theta)$	$P_m^l(\cos \theta)$ are related by (Arecchi <i>et al.</i> , 1972; Gilmore, 1974b)
gilmore-lie-groups_F02314	229	■ 1.000	$u \rightarrow$	$u \rightarrow$	These polynomials are contracted to harmonic oscillator wavefunctions under $u \rightarrow$
gilmore-lie-groups_F02315	229	■ 1.000	$x/\sqrt{l}$	$x/\sqrt{l}$	$x/\sqrt{l}$ and $l + m = n$:

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gilmore-lie-groups_ F02316	229	■ 1.000	l+m=n	$l + m = n$	$x/\sqrt{l}$ and $l + m = n$:
gilmore-lie-groups_ F02317	230	■ 1.000	c \rightarrow 0, l \rightarrow \infty, l+m=n, 2 l c^2=1	$c \rightarrow 0, l \rightarrow \infty, l + m = n, 2lc^2 = 1$	The limit is taken as $c \rightarrow 0, l \rightarrow \infty, l + m = n, 2lc^2 = 1$. The limit inside the
gilmore-lie-groups_ F02318	230	■ 1.000	1 / \sqrt{\pi}	$1/\sqrt{\pi}$	first square root is $1/\sqrt{\pi}$, that within the second is $(2^n n!)^{-1}$. The result of this
gilmore-lie-groups_ F02319	230	■ 1.000	\left(2^{\{n\} n!}\right)^{-1}	$(2^n n!)^{-1}$	first square root is $1/\sqrt{\pi}$, that within the second is $(2^n n!)^{-1}$. The result of this
gilmore-lie-groups_ F02320	230	■ 1.000	H_{\{n\}}(x)	$H_n(x)$	and $H_n(x)$ is the n th Hermite polynomial.
gilmore-lie-groups_ F02321	231	■ 1.000	I S 0(2)	$ISO(2)$	representations of $ISO(2)$. Since $ISO(2)$ is a noncompact group it has no faith-
gilmore-lie-groups_ F02322	231	■ 1.000	P_{\{ \pm \}}	$P_{\pm}$	a. Compute the matrix elements of the operators $P_{\pm}$ in the algebra $\mathfrak{iso}(2)$ and show
gilmore-lie-groups_ F02323	231	■ 1.000	\mathfrak{iso}(2)	$\mathfrak{iso}(2)$	a. Compute the matrix elements of the operators $P_{\pm}$ in the algebra $\mathfrak{iso}(2)$ and show
gilmore-lie-groups_ F02324	231	■ 1.000	J_{\{k\}}(x)	$J_k(x)$	where $J_k(x)$ is the k th Bessel function (Arecchi <i>et al.</i> , 1972; Gilmore, 1974b).
gilmore-lie-groups_ F02325	231	■ 0.992	A_{\{k\}}^l	A_k^l	compute the coefficients A_k^l and the arguments θ', ϕ' explicitly. Contract these
gilmore-lie-groups_ F02326	231	■ 0.992	\theta^{\{\prime\}}, \phi^{\{\prime\}}	θ', ϕ'	compute the coefficients A_k^l and the arguments θ', ϕ' explicitly. Contract these
gilmore-lie-groups_ F02327	231	■ 0.998	\mathbf{L} \cdot \mathbf{L}	$\mathbf{L} \cdot \mathbf{L}$	f. Show that the operator $\mathbf{L} \cdot \mathbf{L}$ contracts to ∇^2 in the plane.
gilmore-lie-groups_ F02328	231	■ 0.998	\nabla^2	∇^2	f. Show that the operator $\mathbf{L} \cdot \mathbf{L}$ contracts to ∇^2 in the plane.
gilmore-lie-groups_ F02329	231	■ 1.000	\mathfrak{u}(2) \rightarrow \mathfrak{h}_4	$\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$	2. Under the contraction $\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$ the representations of the unitary group $U(2)$ con-
gilmore-lie-groups_ F02330	232	■ 1.000	N_{\{n\}}	N_n	$H_n(x)$ is the n th Hermite polynomial, and N_n is the usual normalization coefficient,
gilmore-lie-groups_ F02331	232	■ 0.993	N_{\{n\}}=1 / \sqrt{2^{\{n\} n!}\sqrt{\pi}}	$N_n = 1/\sqrt{2^n n! \sqrt{\pi}}$	$N_n = 1/\sqrt{2^n n! \sqrt{\pi}}$.
gilmore-lie-groups_ F02332	232	■ 0.984	J_{\{3\}}, J_{\{ \pm \}}\left(\left[J_{\{3\}}, J_{\{ \pm \}}\right]= \pm J_{\{ \pm \}}, \left[J_{\{ \pm \}}, J_{\{+\}}\right]=2 J_{\{3\}}\right)	$J_3, J_{\pm} ([J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3)$	3. Contract the Lie algebra $\mathfrak{su}(2)$ spanned by $J_3, J_{\pm} ([J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3)$
gilmore-lie-groups_ F02333	232	■ 0.993	\left(P^{\{\prime\}}, T^{\{\prime\}}, V^{\{\prime\}}\right)	(P', T', V')	operators on the right (P', T', V') satisfy an isomorphic set of commutation relations.
gilmore-lie-groups_ F02334	232	■ 0.991	\partial_x, \partial_t, t \partial_x	$\partial_x, \partial_t, t\partial_x$	The operators $\partial_x, \partial_t, t\partial_x$ generate the Galilean group in one dimension. Conclude
gilmore-lie-groups_ F02335	232	■ 1.000	\mathfrak{a}_1	$\mathfrak{a}_1$	that if the Lie algebra $\mathfrak{a}_1$ is contracted with respect to one of its shift operators the
gilmore-lie-groups_ F02336	232	■ 0.985	\mathfrak{gal}(1)	$\mathfrak{gal}(1)$	Galilean algebra $\mathfrak{gal}(1)$ results.
gilmore-lie-groups_ F02337	232	■ 1.000	S^{\{n\}}=S 0(n+1) / S 0(n)	$S^n = SO(n+1)/SO(n)$	metric and measure on the sphere $S^n = SO(n+1)/SO(n)$ reduce to the familiar
gilmore-lie-groups_ F02338	232	■ 1.000	R^{\{n\}}=I S 0(n) / S 0(n)	$R^n = ISO(n)/SO(n)$	metric and measure on $R^n = ISO(n)/SO(n)$.
gilmore-lie-groups_ F02339	233	■ 1.000	\tau=(\zeta / \zeta) \tan (\zeta)	$\tau = (\zeta/ \zeta) \tan(\zeta)$	and show $\tau = (\zeta/ \zeta) \tan(\zeta)$.
gilmore-lie-groups_ F02340	233	■ 1.000	\alpha=\lim _{c \rightarrow 0} \zeta / c	$\alpha = \lim_{c \rightarrow 0} \zeta/c$	$\alpha = \lim_{c \rightarrow 0} \zeta/c$.
gilmore-lie-groups_ F02341	233	■ 1.000	\langle X\rangle=\operatorname{tr} X e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}	$\langle X \rangle = \operatorname{tr} X e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}$	$\langle X \rangle = \operatorname{tr} X e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}$, and a generating function for expectation values is $\langle e^{\alpha X} \rangle =$
gilmore-lie-groups_ F02342	233	■ 1.000	\left\langle e^{\alpha X}\right\rangle=	$\langle e^{\alpha X} \rangle =$	$\langle X \rangle = \operatorname{tr} X e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}$, and a generating function for expectation values is $\langle e^{\alpha X} \rangle =$

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gilmore-lie-groups_ F02343	233	■ 1.000	$\operatorname{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}$	$\operatorname{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}$	$\operatorname{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \operatorname{tr} e^{-\beta \mathcal{H}}$. When the operators X and $\mathcal{H}$ are elements in a finite dimensional
gilmore-lie-groups_ F02344	233	■ 0.998	$\mathcal{H} = \epsilon J_3$	$\mathcal{H} = \epsilon J_3$	a. Assume $\mathcal{H} = \epsilon J_3$ and X is in the Lie algebra $\mathfrak{su}(2)$. Show that in the 2×2 defining
gilmore-lie-groups_ F02345	233	■ 1.000	$2j+1=2$	$2j+1=2$	c. Show that in the 2×2 matrix representation with $j = \frac{1}{2}$ and $2j+1=2$,
gilmore-lie-groups_ F02346	234	■ 0.974	$\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$	$\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$	are analytic continuations of $\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4 =$
gilmore-lie-groups_ F02347	234	■ 0.974	$\mathcal{C}^4 =$	$\mathcal{C}^4 =$	are analytic continuations of $\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4 =$
gilmore-lie-groups_ F02348	234	■ 0.812	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $SO(6)$.
gilmore-lie-groups_ F02349	234	■ 0.812	$Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$	$Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $SO(6)$.
gilmore-lie-groups_ F02350	234	■ 0.812	$SO(6)$	$SO(6)$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $SO(6)$.
gilmore-lie-groups_ F02351	234	■ 0.995	$SO(4) \otimes SO(2)$	$SO(4) \otimes SO(2)$	b. Contract $SO(4, 2)$ with respect to the subgroup $SO(4) \otimes SO(2)$.
gilmore-lie-groups_ F02352	234	■ 0.933	$A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$	$A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$	of part a. If we define $A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$ and $B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$, then show that
gilmore-lie-groups_ F02353	234	■ 0.933	$B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$	$B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$	of part a. If we define $A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$ and $B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$, then show that
gilmore-lie-groups_ F02354	234	■ 1.000	$SO(4, 2) / [SO(4) \otimes SO(2)]$	$SO(4, 2) / [SO(4) \otimes SO(2)]$	$SO(4, 2) / [SO(4) \otimes SO(2)]$ and $I[SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$.
gilmore-lie-groups_ F02355	234	■ 1.000	$I[SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$	$I[SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$	$SO(4, 2) / [SO(4) \otimes SO(2)]$ and $I[SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$.
gilmore-lie-groups_ F02356	234	■ 1.000	X_{α}	X_{α}	Operators X_i span $\mathfrak{h}$ and X_{α} span $\mathfrak{p}$.
gilmore-lie-groups_ F02357	234	■ 0.987	$Y_{\alpha} = \lim_{\epsilon \rightarrow 0} \epsilon X_{\alpha}$	$Y_{\alpha} = \lim_{\epsilon \rightarrow 0} \epsilon X_{\alpha}$	b. Show that in the contracted limit $Y_{\alpha} = \lim_{\epsilon \rightarrow 0} \epsilon X_{\alpha}$ a metric tensor on $\mathfrak{p}$ is well
gilmore-lie-groups_ F02358	236	■ 1.000	$ \psi\rangle$	$ \psi\rangle$	state $ \psi\rangle$ in their respective coordinate systems. They describe the state by mathe-
gilmore-lie-groups_ F02359	236	■ 1.000	$\langle S \psi \rangle$	$\langle S \psi \rangle$	tical functions $\langle S \psi \rangle$ and $\langle S' \psi \rangle$. The two observers know the relation between
gilmore-lie-groups_ F02360	236	■ 1.000	$\langle S' \psi \rangle$	$\langle S' \psi \rangle$	tical functions $\langle S \psi \rangle$ and $\langle S' \psi \rangle$. The two observers know the relation between
gilmore-lie-groups_ F02361	236	■ 0.998	$\langle S' S \rangle$	$\langle S' S \rangle$	tions from one coordinate system to the other is $\langle S' S \rangle$. The set of transformations
gilmore-lie-groups_ F02362	236	■ 1.000	$\langle S S' \rangle$	$\langle S S' \rangle$	S has seen, he applies the appropriate transformation, $\langle S S' \rangle$, to his mathematical
gilmore-lie-groups_ F02363	237	■ 1.000	$ \psi'\rangle$	$ \psi'\rangle$	by S' to describe a <i>new</i> physical state $ \psi'\rangle$
gilmore-lie-groups_ F02364	237	■ 0.997	$SO(3)$	$SO(3)$	example, a rotation in $SO(3)$) is expressed by $\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$,
gilmore-lie-groups_ F02365	237	■ 0.997	$\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$	$\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$	example, a rotation in $SO(3)$) is expressed by $\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$,
gilmore-lie-groups_ F02366	237	■ 1.000	$\langle S' H S' \rangle =$	$\langle S' H S' \rangle =$	the invariance under the transformation $\langle S' S \rangle$ is represented by $\langle S' H S' \rangle =$
gilmore-lie-groups_ F02367	237	■ 1.000	$\langle S H S \rangle$	$\langle S H S \rangle$	$\langle S H S \rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups_ F02368	237	■ 1.000	$2p_z$	$2p_z$	$\langle S H S \rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry

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gilmore-lie-groups_F02369	237	■ 1.000	$2\ p_{\mathrm{x}}$	$2p_x$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups_F02370	237	■ 1.000	$2\ p_{\mathrm{y}}$	$2p_y$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups_F02371	237	■ 0.996	$p^{\mu} p_{\mu} = g_{\mu\nu} \ p^{\mu} \ p^{\nu} = (m\ c)^2$	$p^{\mu} p_{\mu} = g_{\mu\nu} p^{\mu} p^{\nu} = (mc)^2$	the relativistic dispersion relation for the free particle: $p^{\mu} p_{\mu} = g_{\mu\nu} p^{\mu} p^{\nu} = (mc)^2$.
gilmore-lie-groups_F02372	237	■ 1.000	$\mathbf{p}$	$\mathbf{p}$	In terms of the energy E and the three-momentum $\mathbf{p}$ this is
gilmore-lie-groups_F02373	237	■ 1.000	$p_{\mu} \rightarrow \pi_{\mu} = p_{\mu} - (q/c) A_{\mu}$	$p_{\mu} \rightarrow \pi_{\mu} = p_{\mu} - (q/c) A_{\mu}$	principle of minimal electromagnetic coupling: $p_{\mu} \rightarrow \pi_{\mu} = p_{\mu} - (q/c) A_{\mu}$, where
gilmore-lie-groups_F02374	237	■ 0.974	Φ	Φ	the four-vector potential A consists of the scalar potential Φ and the vector potential
gilmore-lie-groups_F02375	237	■ 0.997	$\mathbf{B} = \nabla \times \mathbf{A}$	$\mathbf{B} = \nabla \times \mathbf{A}$	$\mathbf{A}$. These obey $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla\Phi - (1/c)(\partial\mathbf{A}/\partial t)$. For an electron $q =$
gilmore-lie-groups_F02376	237	■ 0.997	$\mathbf{E} = -\nabla\Phi - (1/c)(\partial\mathbf{A}/\partial t)$	$\mathbf{E} = -\nabla\Phi - (1/c)(\partial\mathbf{A}/\partial t)$	$\mathbf{A}$. These obey $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla\Phi - (1/c)(\partial\mathbf{A}/\partial t)$. For an electron $q =$
gilmore-lie-groups_F02377	237	■ 0.997	$q =$	$q =$	$\mathbf{A}$. These obey $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla\Phi - (1/c)(\partial\mathbf{A}/\partial t)$. For an electron $q =$
gilmore-lie-groups_F02378	237	■ 0.928	$-e$	$-e$	$-e$, where e is the charge on the proton, positive by convention. In the Coulomb
gilmore-lie-groups_F02379	237	■ 1.000	$\Phi = e/r$	$\Phi = e/r$	field established by a proton, $\Phi = e/r$ and $\mathbf{A} = \mathbf{0}$, so that $E \rightarrow E + e^2/r$. Here
gilmore-lie-groups_F02380	237	■ 1.000	$\mathbf{A} = \mathbf{0}$	$\mathbf{A} = \mathbf{0}$	field established by a proton, $\Phi = e/r$ and $\mathbf{A} = \mathbf{0}$, so that $E \rightarrow E + e^2/r$. Here
gilmore-lie-groups_F02381	237	■ 1.000	$E \rightarrow E + e^2/r$	$E \rightarrow E + e^2/r$	field established by a proton, $\Phi = e/r$ and $\mathbf{A} = \mathbf{0}$, so that $E \rightarrow E + e^2/r$. Here
gilmore-lie-groups_F02382	237	■ 1.000	$\mathbf{p} \rightarrow (\hbar/i)\nabla$	$\mathbf{p} \rightarrow (\hbar/i)\nabla$	dispersion relation to a wave equation is to replace $\mathbf{p} \rightarrow (\hbar/i)\nabla$ and allow the
gilmore-lie-groups_F02383	237	■ 1.000	$\psi(\mathbf{x})$	$\psi(\mathbf{x})$	resulting equation to act on a spacial function $\psi(\mathbf{x})$. This prescription results in the
gilmore-lie-groups_F02384	238	■ 0.999	$\langle S' H S' \rangle = \langle S H S \rangle$	$\langle S' H S' \rangle = \langle S H S \rangle$	variant) in form under rotations: $\langle S' H S' \rangle = \langle S H S \rangle$, where $\langle S' S \rangle \in SO(3)$.
gilmore-lie-groups_F02385	238	■ 0.999	$\langle S' S \rangle \in SO(3)$	$\langle S' S \rangle \in SO(3)$	variant) in form under rotations: $\langle S' H S' \rangle = \langle S H S \rangle$, where $\langle S' S \rangle \in SO(3)$.
gilmore-lie-groups_F02386	238	■ 0.623	mc^2	mc^2	Since the binding energy is about 13.6 eV and the electron rest energy mc^2 is about
gilmore-lie-groups_F02387	238	■ 0.906	$E = mc^2 + W$	$E = mc^2 + W$	510 000 eV, it makes sense to write $E = mc^2 + W$, where the principal part of the
gilmore-lie-groups_F02388	238	■ 1.000	W	W	relativistic energy E is the electron rest energy and the nonrelativistic energy W is
gilmore-lie-groups_F02389	238	■ 0.884	$(\simeq 0.0025\ \%)$	$(\simeq 0.0025\%)$	a small perturbation of either $\simeq 0.0025\%$. Under this substitution, and neglecting
gilmore-lie-groups_F02390	238	■ 1.000	$(W + e^2/r)^2 / mc^2$	$(W + e^2/r)^2 / mc^2$	terms of order $(W + e^2/r)^2 / mc^2$, we obtain the nonrelativistic form of Eq. (14.4),
gilmore-lie-groups_F02391	238	■ 0.727	$[A, B] / i \hbar = \{A, B\}$	$[A, B] / i\hbar = \{A, B\}$	tor brackets $([A, B] / i\hbar = \{A, B\})$ is an attempt to fit quantum mechanics into the
gilmore-lie-groups_F02392	238	■ 0.860	(r, θ, ϕ)	(r, θ, ϕ)	useful to introduce spherical coordinates (r, θ, ϕ) . In this coordinate system the
gilmore-lie-groups_F02393	239	■ 1.000	$\mathcal{L}^2(S^2)$	$\mathcal{L}^2(S^2)$	The second order differential operator $\mathcal{L}^2(S^2)$ is the Laplacian on the sphere S^2 . Its
gilmore-lie-groups_F02394	239	■ 0.980	$\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$	$\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$	$\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $l = 0, 1, 2, \dots$
gilmore-lie-groups_F02395	239	■ 0.980	(l, m)	(l, m)	is $\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $l = 0, 1, 2, \dots$

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gilmore-lie-groups_ F02396	239	■ 0.980	$l=0,1,2, \ldots$	$l = 0, 1, 2, \dots$	is $\mathcal{L}^2(S^2)Y_m^l(\theta, \phi) = -l(l+1)Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $l = 0, 1, 2, \dots$
gilmore-lie-groups_ F02397	239	■ 0.999	$-l \leq m \leq +l$	$-l \leq m \leq +l$	and $-l \leq m \leq +l$. The negative sign and discrete spectrum characteristically indi-
gilmore-lie-groups_ F02398	239	■ 1.000	A, B, C	A, B, C	The values of the coefficients A, B, C that are obtained for the Klein–Gordon
gilmore-lie-groups_ F02404	239	■ 1.000	$n=$	$n =$	relativistic problems are expressed in terms of the radial quantum number $n =$
gilmore-lie-groups_ F02405	239	■ 1.000	$0,1,2, \ldots$	$0, 1, 2, \dots$	$0, 1, 2, \dots$ and the angular momentum quantum number $l = 0, 1, 2, \dots$, mass m
gilmore-lie-groups_ F02406	239	■ 1.000	$m_{\text{red}}^{-1}=m_e^{-1}+M_p^{-1}$	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$, and the fine structure constant (Gabrielse <i>et al.</i> , 2006)
gilmore-lie-groups_ F02409	271	■ 1.000	$\mathcal{L}^2$	$\mathcal{L}^2$	The angular part of the Laplacian operator, $\mathcal{L}^2$, acts on spherical harmonics on S^{D-1} ,
gilmore-lie-groups_ F02412	240	■ 1.000	$f(r)$	$f(r)$	equation for $f(r)$
gilmore-lie-groups_ F02418	240	■ 1.000	$\hbar$	$\hbar$	three “different” areas of physics: e (electromagnetism), $\hbar$ (quantum mechanics),
gilmore-lie-groups_ F02420	241	■ 1.000	$\alpha \rightarrow Z\alpha$	$\alpha \rightarrow Z\alpha$	replacing $\alpha \rightarrow Z\alpha$ everywhere.
gilmore-lie-groups_ F02421	241	■ 1.000	$ \text{left} W_1 \text{right} =\frac{1}{2} m c^2 \alpha^2$	$ W_1 = \frac{1}{2}mc^2\alpha^2$	ground state energy $ W_1 = \frac{1}{2}mc^2\alpha^2$. These normalized energy levels decrease to
gilmore-lie-groups_ F02422	241	■ 0.999	$1 / N^2$	$1/N^2$	zero like $1/N^2$, where $N = n + l + 1$ is the principal quantum number. The ener-
gilmore-lie-groups_ F02423	241	■ 0.999	$N=n+l+1$	$N = n + l + 1$	zero like $1/N^2$, where $N = n + l + 1$ is the principal quantum number. The ener-
gilmore-lie-groups_ F02424	241	■ 1.000	$N^{\prime}$	N'	ergy mc^2 as N' increases. When this limit is removed these energies (also rescaled
gilmore-lie-groups_ F02425	241	■ 1.000	$\frac{1}{2} m c^2 \alpha^2$	$\frac{1}{2}mc^2\alpha^2$	by dividing by $\frac{1}{2}mc^2\alpha^2$) can be plotted on the same graph. At the resolution shown,
gilmore-lie-groups_ F02426	241	■ 1.000	$\frac{1}{2} m c^2 (Z \alpha)^2$	$\frac{1}{2}mc^2(Z\alpha)^2$	energy for the same ion: $\frac{1}{2}mc^2(Z\alpha)^2$. The energy difference between the $1s$ ground
gilmore-lie-groups_ F02427	241	■ 1.000	$1 s$	$1s$	energy for the same ion: $\frac{1}{2}mc^2(Z\alpha)^2$. The energy difference between the $1s$ ground
gilmore-lie-groups_ F02428	242	■ 1.000	$g_i \psi\rangle$	$g_i \psi\rangle$	eigenstate of H with eigenvalue E , then $g_i \psi\rangle$ is also an eigenstate of H with the
gilmore-lie-groups_ F02429	242	■ 1.000	$ \psi\rangle=\psi_{2 p_z}(\mathbf{x})$	$ \psi\rangle = \psi_{2p_z}(\mathbf{x})$	To illustrate this idea, assume that $ \psi\rangle = \psi_{2p_z}(\mathbf{x})$. A rotation by $\pi/2$ radians about
gilmore-lie-groups_ F02430	242	■ 1.000	$\pi / 2$	$\pi/2$	To illustrate this idea, assume that $ \psi\rangle = \psi_{2p_z}(\mathbf{x})$. A rotation by $\pi/2$ radians about
gilmore-lie-groups_ F02431	242	■ 1.000	$\psi_{2 p_x}(\mathbf{x})$	$\psi_{2p_x}(\mathbf{x})$	the y -axis maps this state to $\psi_{2p_x}(\mathbf{x})$ and a rotation by $\pi/2$ radians about the x -axis
gilmore-lie-groups_ F02432	242	■ 1.000	$-\psi_{2 p_y}(\mathbf{x})$	$-\psi_{2p_y}(\mathbf{x})$	maps this state to $-\psi_{2p_y}(\mathbf{x})$. By invariance (of the hamiltonian) under the rotation
gilmore-lie-groups_ F02433	242	■ 1.000	$H=\mathbf{p} \cdot \mathbf{p} / 2 m-e^2 / r$	$H = \mathbf{p} \cdot \mathbf{p}/2m - e^2/r$	$H = \mathbf{p} \cdot \mathbf{p}/2m - e^2/r$. The scalar $\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$ is invariant under rotations, as
gilmore-lie-groups_ F02434	242	■ 1.000	$\mathbf{p} \cdot \mathbf{p}=-\hbar^2 \nabla^2$	$\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$	$H = \mathbf{p} \cdot \mathbf{p}/2m - e^2/r$. The scalar $\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$ is invariant under rotations, as
gilmore-lie-groups_ F02435	242	■ 1.000	$-e^2 / r$	$-e^2/r$	is also the potential energy term $-e^2/r$. Rotation operators can be expressed in
gilmore-lie-groups_ F02436	242	■ 1.000	$i: \epsilon_{i j k} x_j \partial_k$	$i : \epsilon_{ijk} x_j \partial_k$	terms of the infinitesimal generators of rotations about axis i : $\epsilon_{ijk} x_j \partial_k$. These ge-

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gilmore-lie-groups-F02437	242	■ 1.000	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i) \epsilon_{ijk} x_j \partial_k$	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i) \epsilon_{ijk} x_j \partial_k$	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i) \epsilon_{ijk} x_j \partial_k$. Finite rotations can be expressed as exponentials
gilmore-lie-groups-F02438	242	■ 0.998	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	The angular momentum operators $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ share the same commutation relations
gilmore-lie-groups-F02439	242	■ 0.992	$\mathbf{r} \times \nabla$	$\mathbf{r} \times \nabla$	as the infinitesimal generators of rotations $\mathbf{r} \times \nabla$, up to the proportionality factor
gilmore-lie-groups-F02440	242	■ 1.000	$\hbar/i$	$\hbar/i$	$\hbar/i$. The commutation relations are
gilmore-lie-groups-F02441	242	■ 1.000	$\left(L_+\right)$	(L_+)	raising (L_+) and lowering (L_-) operators by $L_{\pm} = L_x \pm iL_y$. The commutation
gilmore-lie-groups-F02442	242	■ 1.000	$\left(L_-\right)$	(L_-)	raising (L_+) and lowering (L_-) operators by $L_{\pm} = L_x \pm iL_y$. The commutation
gilmore-lie-groups-F02443	242	■ 1.000	$L_{\pm} = L_x \pm iL_y$	$L_{\pm} = L_x \pm iL_y$	raising (L_+) and lowering (L_-) operators by $L_{\pm} = L_x \pm iL_y$. The commutation
gilmore-lie-groups-F02444	242	■ 1.000	$L_z = \hbar \frac{1}{2} (a_1^\dagger a_1 - a_2^\dagger a_2)$, $L_+ = \hbar a_1^\dagger a_2$, $L_- = \hbar a_2^\dagger a_1$	$L_z = \hbar \frac{1}{2} (a_1^\dagger a_1 - a_2^\dagger a_2)$, $L_+ = \hbar a_1^\dagger a_2$, $L_- = \hbar a_2^\dagger a_1$	follows: $L_z = \hbar \frac{1}{2} (a_1^\dagger a_1 - a_2^\dagger a_2)$, $L_+ = \hbar a_1^\dagger a_2$, $L_- = \hbar a_2^\dagger a_1$. As a result, the angu-
gilmore-lie-groups-F02445	242	■ 1.000	$ n_1 n_2\rangle$	$ n_1 n_2\rangle$	lar momentum operators have matrix representations with basis vectors $ n_1 n_2\rangle$
gilmore-lie-groups-F02446	242	■ 0.838	$ ^j_m\rangle$	$ ^j_m\rangle$	$= ^j_m\rangle$, with $n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$,
gilmore-lie-groups-F02447	242	■ 0.838	$n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$	$n_1 = 0, 1, 2, \dots, n_2 = 0, 1, 2, \dots, n_1 + n_2 = 2j, n_1 - n_2 = 2m$	$= ^j_m\rangle$, with $n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$.
gilmore-lie-groups-F02448	243	■ 1.000	$-j \leq m \leq +j$	$-j \leq m \leq +j$	$-j \leq m \leq +j$. These basis vectors describe the finite-dimensional irreducible
gilmore-lie-groups-F02449	243	■ 0.994	$((x, y, z) \rightarrow (r, \theta, \phi))$	$((x, y, z) \rightarrow (r, \theta, \phi))$	operators. In spherical coordinates $((x, y, z) \rightarrow (r, \theta, \phi))$ with $x = r \sin \theta \cos \phi$
gilmore-lie-groups-F02450	243	■ 0.994	$x = r \sin \theta \cos \phi$	$x = r \sin \theta \cos \phi$	operators. In spherical coordinates $((x, y, z) \rightarrow (r, \theta, \phi))$ with $x = r \sin \theta \cos \phi$
gilmore-lie-groups-F02451	243	■ 1.000	$Y_{-l}^l(\theta, \phi)$	$Y_{-l}^l(\theta, \phi)$	construct the extremal function $Y_{-l}^l(\theta, \phi)$ by solving the equation $L_- Y_{-l}^l(\theta, \phi) = 0$
gilmore-lie-groups-F02452	243	■ 1.000	$L_- Y_{-l}^l(\theta, \phi) = 0$	$L_- Y_{-l}^l(\theta, \phi) = 0$	construct the extremal function $Y_{-l}^l(\theta, \phi)$ by solving the equation $L_- Y_{-l}^l(\theta, \phi) = 0$
gilmore-lie-groups-F02453	266	■ 1.000	$l = 0$	$l = 0$	the relativistic solution is clearly incorrect? (Hint: set $l = 0$.)
gilmore-lie-groups-F02466	244	■ 1.000	L_+	L_+	of the shift up operator L_+
gilmore-lie-groups-F02467	244	■ 0.815	$(l = 0, 1, 2, 3)$	$(l = 0, 1, 2, 3)$	The lowest spherical harmonics $(l = 0, 1, 2, 3)$ are collected in Table 14.2.
gilmore-lie-groups-F02468	244	■ 0.662	θ, ϕ	θ, ϕ	generators acting on the sphere parameterized by coordinates (θ, ϕ) , is $-l(l + 1)$,
gilmore-lie-groups-F02469	244	■ 1.000	$\hbar^2 l(l + 1)$	$\hbar^2 l(l + 1)$	of the angular momentum operator, $\mathbf{L} \cdot \mathbf{L}$, is $\hbar^2 l(l + 1)$. This means physically that
gilmore-lie-groups-F02470	244	■ 1.000	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	number $N = n + l + 1$ are degenerate, with energy $E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$. There are
gilmore-lie-groups-F02471	244	■ 1.000	$\sum_{l=0}^{l=N-1} (2l + 1) = N^2$	$\sum_{l=0}^{l=N-1} (2l + 1) = N^2$	$\sum_{l=0}^{l=N-1} (2l + 1) = N^2$ states with this energy. This N^2 -fold degeneracy is larger
gilmore-lie-groups-F02472	244	■ 1.000	$1/r^2$	$1/r^2$	related to a constant of motion that is peculiar to $1/r^2$ force laws. This constant of
gilmore-lie-groups-F02473	244	■ 1.000	$d\mathbf{p}/dt = -K\mathbf{r}/r^3$	$d\mathbf{p}/dt = -K\mathbf{r}/r^3$	planetary motion, for which the force law has the form $d\mathbf{p}/dt = -K\mathbf{r}/r^3$, where

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gilmore-lie-groups_ F02474	245	■ 1.000	K=G M m, G	$K = GMm, G$	$K = GMm, G$ is the universal gravitational constant, M and m are the two attracting
gilmore-lie-groups_ F02475	245	■ 1.000	$\mathbf{r}=x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}}$	$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$	masses, and $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ is the vector from one mass to the other. The time
gilmore-lie-groups_ F02476	245	■ 1.000	$\mathbf{p} \times \mathbf{L}$	$\mathbf{p} \times \mathbf{L}$	derivative of the vector $\mathbf{p} \times \mathbf{L}$ is
gilmore-lie-groups_ F02477	245	■ 0.999	$\mathbf{L}$	$\mathbf{L}$	In going from the first line in Eq. (14.22) to the second, we use the fact that $\mathbf{L}$ is
gilmore-lie-groups_ F02478	245	■ 1.000	$1 / r$	$1/r$	law for a $1/r$ potential. In going from the second line to the third, we express the
gilmore-lie-groups_ F02479	245	■ 0.967	$\mathbf{r} \times \mathbf{L}$	$\mathbf{r} \times \mathbf{L}$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\dot{\mathbf{r}}$. We also
gilmore-lie-groups_ F02480	245	■ 0.967	$\dot{\mathbf{r}}$	$\dot{\mathbf{r}}$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\dot{\mathbf{r}}$. We also
gilmore-lie-groups_ F02481	245	■ 1.000	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$	use the identity $(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$. The result is that the Laplace–
gilmore-lie-groups_ F02482	245	■ 0.985	$d\mathbf{M}/dt = 0$	$d\mathbf{M}/dt = 0$	Runge–Lenz vector $\mathbf{M}$ is a constant of motion: $d\mathbf{M}/dt = 0$, where
gilmore-lie-groups_ F02483	245	■ 1.000	$[H, \mathbf{M}] = 0$	$[H, \mathbf{M}] = 0$	with the nonrelativistic hamiltonian: $[H, \mathbf{M}] = 0$. The six operators L_i, M_j obey
gilmore-lie-groups_ F02484	245	■ 1.000	L_i, M_j	L_i, M_j	with the nonrelativistic hamiltonian: $[H, \mathbf{M}] = 0$. The six operators L_i, M_j obey
gilmore-lie-groups_ F02485	245	■ 0.993	$(E < 0)$	$(E < 0)$	bound states ($E < 0$) or $SO(3, 1)$ for excited states ($E > 0$). The operators $\mathbf{L}$ and
gilmore-lie-groups_ F02486	245	■ 0.993	$(E > 0)$	$(E > 0)$	bound states ($E < 0$) or $SO(3, 1)$ for excited states ($E > 0$). The operators $\mathbf{L}$ and
gilmore-lie-groups_ F02487	246	■ 1.000	$\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$	$\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$	Laplace–Runge–Lenz vector by a scale factor as follows: $\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$.
gilmore-lie-groups_ F02488	246	■ 1.000	$E > 0$	$E > 0$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02489	246	■ 1.000	$- \rightarrow +$	$- \rightarrow +$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02490	246	■ 1.000	$SO(4) \rightarrow SO(3, 1)$	$SO(4) \rightarrow SO(3, 1)$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02491	246	■ 0.834	$\mathbf{A}$	$\mathbf{A}$	Figs. 10.3, 10.8(b)). It is useful to introduce two vector operators $\mathbf{A}$ and $\mathbf{B}$ as
gilmore-lie-groups_ F02492	246	■ 0.834	$\mathbf{B}$	$\mathbf{B}$	Figs. 10.3, 10.8(b)). It is useful to introduce two vector operators $\mathbf{A}$ and $\mathbf{B}$ as
gilmore-lie-groups_ F02493	246	■ 0.996	$A_3 = \frac{1}{2}(a_1^\dagger a_1 -$	$A_3 = \frac{1}{2}(a_1^\dagger a_1 -$	momentum operators $\mathbf{A}$ in terms of two independent boson modes: $A_3 = \frac{1}{2}(a_1^\dagger a_1 -$
gilmore-lie-groups_ F02494	246	■ 0.910	$a_2^\dagger a_2)$	$a_2^\dagger a_2)$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2$, $A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_ F02495	246	■ 0.910	$A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1)$	$A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1)$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_ F02496	246	■ 0.910	$\hbar \rightarrow 1$	$\hbar \rightarrow 1$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_ F02497	246	■ 1.000	b_1, b_2	b_1, b_2	b_1, b_2 and their creation operators is also introduced.
gilmore-lie-groups_ F02498	246	■ 1.000	$ p_1, p_2\rangle$	$ p_1, p_2\rangle$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_ F02499	246	■ 1.000	$p_1 + p_2 = 2j_a$	$p_1 + p_2 = 2j_a$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis

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gilmore-lie-groups_F02500	246	■ 1.000	$p_{\{1\}}-p_{\{2\}}=m_{\{a\}}$	$p_1 - p_2 = m_a$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02501	246	■ 1.000	$2 j_{\{a\}}+1$	$2j_a + 1$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02502	246	■ 0.968	$p_{\{1\}}=2 j_{\{a\}}, p_{\{2\}}=0 ; p_{\{1\}}=2 j_{\{a\}}-1, p_{\{2\}}=1$	$p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$	states correspond to $p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$; etc. For $\mathbf{B}$ the basis
gilmore-lie-groups_F02503	246	■ 1.000	$\left q_{\{1\}}, q_{\{2\}}\right\rangle$	$ q_1, q_2\rangle$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02504	246	■ 1.000	$q_{\{1\}}+q_{\{2\}}=2 j_{\{b\}}$	$q_1 + q_2 = 2j_b$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02505	246	■ 1.000	$q_{\{1\}}-q_{\{2\}}=m_{\{b\}}$	$q_1 - q_2 = m_b$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02506	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A}=j_{\{a\}}\left(j_{\{a\}}+1\right)$	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02507	246	■ 1.000	$\mathbf{B} \cdot \mathbf{B}=j_{\{b\}}\left(j_{\{b\}}+1\right)$	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02508	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A}=\mathbf{B} \cdot \mathbf{B}$	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02509	246	■ 1.000	$j_{\{a\}}=j_{\{b\}}$	$j_a = j_b$	Problem 14.15), $j_a = j_b$ and the set of states related by the shift operators is $(2j +$
gilmore-lie-groups_F02510	246	■ 1.000	$(2 j+$	$(2j+$	Problem 14.15), $j_a = j_b$ and the set of states related by the shift operators is $(2j +$
gilmore-lie-groups_F02511	246	■ 1.000	$2 j+1=N=n+l+1$	$2j + 1 = N = n + l + 1$	$1)^2$ fold degenerate, where $2J + 1 = N = n + l + 1$.
gilmore-lie-groups_F02512	246	■ 0.996	$\operatorname{good} l$	good l	States with good l and m quantum numbers can be constructed from these states
gilmore-lie-groups_F02513	247	■ 1.000	$M_{+}^{\prime}=$	$M'_+ =$	spherical harmonics, is determined in a straightforward way. For example, $M'_+ =$
gilmore-lie-groups_F02514	247	■ 1.000	$A_{+}-B_{+}=a_{1}^{\dagger} a_{2}-b_{1}^{\dagger} b_{2}$	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$, so that
gilmore-lie-groups_F02515	247	■ 0.998	$\mathbf{M}^{\prime}$	$\mathbf{M}'$	The operators $\mathbf{L}$ and $\mathbf{M}'$ are infinitesimal generators for the orthogonal group $SO(4)$.
gilmore-lie-groups_F02516	247	■ 0.608	$\mathbf{W}=\mathbf{L} \times \mathbf{M}$	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$	the vectors $\mathbf{L}$ and $\mathbf{M}$. Their cross product $\mathbf{W} = \mathbf{L} \times \mathbf{M}$ is orthogonal to both and
gilmore-lie-groups_F02517	247	■ 1.000	$\mathbf{r} \cdot \mathbf{L}=0$	$\mathbf{r} \cdot \mathbf{L} = 0$	since $\mathbf{r} \cdot \mathbf{L} = 0$. The momentum vector moves in the same plane, since $\mathbf{p} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_F02518	247	■ 1.000	$\mathbf{p} \cdot \mathbf{L}=0$	$\mathbf{p} \cdot \mathbf{L} = 0$	since $\mathbf{r} \cdot \mathbf{L} = 0$. The momentum vector moves in the same plane, since $\mathbf{p} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_F02519	248	■ 0.986	$\mathbf{W}$	$\mathbf{W}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02520	248	■ 0.986	$p_{\{z\}}=0, p_{\{x\}}=\mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}} \quad \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	$p_z = 0, p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02521	248	■ 0.986	$p_{\{y\}}=$	$p_y =$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02522	248	■ 1.000	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$. The two nonzero components of the momentum vector are not
gilmore-lie-groups_F02523	248	■ 1.000	$m K / L$	mK/L	momentum moves in the same plane on a circular orbit (radius mK/L) with the
gilmore-lie-groups_F02524	248	■ 1.000	$m M / L$	mM/L	center displaced from the origin by mM/L .
gilmore-lie-groups_F02525	248	■ 1.000	$p_{\{0\}}=\sqrt{-2 E / m}$	$p_0 = \sqrt{-2E/m}$	With $p_0 = \sqrt{-2E/m}$, define the unit vector $\hat{\mathbf{u}} \in S^3 \subset R^4$ by the projective trans-

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gilmore-lie-groups_ F02526	248	■ 1.000	$\hat{\mathbf{u}} \in S^3 \subset R^4$	$\hat{\mathbf{u}} \in S^3 \subset R^4$	With $p_0 = \sqrt{-2E/m}$, define the unit vector $\hat{\mathbf{u}} \in S^3 \subset R^4$ by the projective trans-
gilmore-lie-groups_ F02527	248	■ 1.000	$\hat{\mathbf{w}}$	$\hat{\mathbf{w}}$	Here $\hat{\mathbf{w}}$ is a unit vector in R^4 that is orthogonal to all vectors in the physical space R^3 .
gilmore-lie-groups_ F02528	248	■ 0.995	$\hat{\mathbf{u}}$	$\hat{\mathbf{u}}$	preserves angles (conformal). It is a simple matter to check that $\hat{\mathbf{u}}$ is a unit vector.
gilmore-lie-groups_ F02529	248	■ 1.000	S^3	S^3	The circular trajectory in R^3 (Eq. (14.33)) lifts to a circle in S^3 . Reversibly, circles in
gilmore-lie-groups_ F02530	248	■ 0.803	$S O(4) / S O(3)$	$SO(4)/SO(3)$	$SO(4)/SO(3)$ act to change the lengths of both $\mathbf{L}$ and $\mathbf{M}$. All group operations in
gilmore-lie-groups_ F02531	248	■ 1.000	$p_{\{0\}}$	p_0	$SO(4)$ keep p_0 fixed. In this way the group $SO(4)$ maps states with principal quan-
gilmore-lie-groups_ F02532	249	■ 1.000	$N^2=(n+l+1)^2$	$N^2 = (n + l + 1)^2$	irreducible representations of dimension $N^2 = (n + l + 1)^2$.
gilmore-lie-groups_ F02533	249	■ 1.000	$-N^3$	$-N^3$	plied each energy eigenvalue by $-N^3$, where N is the principal quantum number.
gilmore-lie-groups_ F02534	249	■ 1.000	$\mathbf{x}^{\prime} = M \mathbf{x}$	$\mathbf{x}' = M\mathbf{x}$	linear vector space: $\mathbf{x}' = M\mathbf{x}$, with M a transformation in the group and the inner
gilmore-lie-groups_ F02535	249	■ 1.000	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	product defined by $(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$. As always, the metric-preserving
gilmore-lie-groups_ F02536	249	■ 0.749	$\mathbf{y}$	$\mathbf{y}$	It is useful to define a new N -vector $\mathbf{y}$ as a scaled version of the original vector:
gilmore-lie-groups_ F02537	249	■ 1.000	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_ F02538	249	■ 1.000	$z_1 = \lambda$	$z_1 = \lambda$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_ F02539	249	■ 1.000	$z_2 =$	$z_2 =$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_ F02540	249	■ 1.000	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$. With these definitions we find the conformal condition
gilmore-lie-groups_ F02541	249	■ 1.000	$N + 2$	$N + 2$	The conformal condition defines an inner product in the $N + 2$ dimensional linear
gilmore-lie-groups_ F02542	249	■ 0.998	$\mathbf{y}, z_1, z_2$	$\mathbf{y}, z_1, z_2$	vector space that is nondiagonal in the coordinates $\mathbf{y}, z_1, z_2$ but diagonal in the
gilmore-lie-groups_ F02543	249	■ 0.986	$\mathbf{y}, y_{N+1}, y_{N+2}$	$\mathbf{y}, y_{N+1}, y_{N+2}$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_ F02544	249	■ 0.986	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_ F02545	249	■ 0.986	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_ F02546	249	■ 1.000	R^N	R^N	sional space. If the group that preserves the metric G in R^N is $SO(p, q)$, the group
gilmore-lie-groups_ F02547	250	■ 0.999	R^{N+2}	R^{N+2}	that preserves the metric in R^{N+2} is $SO(p + 1, q + 1)$. We seek to construct a
gilmore-lie-groups_ F02548	250	■ 0.999	$SO(p + 1, q + 1)$	$SO(p + 1, q + 1)$	that preserves the metric in R^{N+2} is $SO(p + 1, q + 1)$. We seek to construct a
gilmore-lie-groups_ F02549	250	■ 1.000	$S^3 \subset R^4 (E < 0)$	$S^3 \subset R^4 (E < 0)$	lifted to the coordinates on a sphere $S^3 \subset R^4 (E < 0)$ or a two-sheeted hyperboloid
gilmore-lie-groups_ F02550	250	■ 1.000	$H^3 \subset R^4 (E > 0)$	$H^3 \subset R^4 (E > 0)$	$H^3 \subset R^4 (E > 0)$ by the following projective transformations:
gilmore-lie-groups_ F02551	250	■ 1.000	$\mathbf{u}^t G \mathbf{u} = 1$	$\mathbf{u}^t G \mathbf{u} = 1$	condition $\mathbf{u}^t G \mathbf{u} = 1$.

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gilmore-lie-groups_F02552	251	■ 0.999	$\operatorname{diag}\left(1, \pm I_3, -1, +1\right)$	$\operatorname{diag}\left(1, \pm I_3, -1, +1\right)$	lows. The group that leaves invariant the conformal metric $\operatorname{diag}(1, \pm I_3, -1, +1)$ is
gilmore-lie-groups_F02553	251	■ 0.984	$S O(5, 1)$	$SO(5, 1)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02554	251	■ 0.984	$E<0$	$E < 0$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02555	251	■ 0.984	$S O(2, 4)$	$SO(2, 4)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02556	251	■ 1.000	$z_{\{1\}}=z_{\{2\}}, \ y_{\{4\}}=\lambda$	$z_1 = z_2, y_4 = \lambda$	the condition $\mathbf{u}'G\mathbf{u} = 1$ is satisfied, so that $z_1 = z_2$, $y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_F02557	251	■ 1.000	$y_{\{5\}}=0$	$y_5 = 0$	the condition $\mathbf{u}'G\mathbf{u} = 1$ is satisfied, so that $z_1 = z_2$, $y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_F02558	251	■ 0.934	$\left(y_{\{0\}}, \ \mathbf{y}=\lambda\mathbf{u}, y_{\{4\}}=\frac{1}{2}\left(z_{\{1\}}+z_{\{2\}}\right), \ y_{\{5\}}=\frac{1}{2}\left(z_{\{1\}}-z_{\{2\}}\right)\right)$	$\left(y_0, \mathbf{y} = \lambda\mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2)\right)$	coordinates are labeled $(y_0, \mathbf{y} = \lambda\mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2))$ Transform-
gilmore-lie-groups_F02559	251	■ 0.908	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array}\right]$	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array}\right]$	$SO(5, 1)$ or $SO(2, 4)$ of the form $\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array}\right]$, with M a 5×5 matrix that preserves the
gilmore-lie-groups_F02560	251	■ 0.997	$\operatorname{diag}\left(1, \pm I_3, -1\right)$	$\operatorname{diag}\left(1, \pm I_3, -1\right)$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02561	251	■ 0.997	$R^{\{5\}}$	R^5	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02562	251	■ 0.997	$S O(1, 4)$	$SO(1, 4)$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02563	251	■ 0.761	$\mathbf{u}^{\prime}$	$\mathbf{u}'$	The inner product of $\mathbf{u}'$ with itself satisfies
gilmore-lie-groups_F02564	251	■ 1.000	$A^tGA-C^tC=G$	$A^tGA - C^tC = G$	condition (e.g., $A^tGA - C^tC = G$, etc.) it is a simple matter to show that this
gilmore-lie-groups_F02565	251	■ 1.000	u	u	follows (Bander and Itzykson, 1966a, 1966b). For $E < 0$ introduce a four-vector u
gilmore-lie-groups_F02566	251	■ 1.000	$\left(u_{\{0\}} \rightarrow u_{\{4\}}\right)$	$(u_0 \rightarrow u_4)$	as usual $(u_0 \rightarrow u_4)$
gilmore-lie-groups_F02567	252	■ 0.967	$1/\sqrt{2m E }$	$1/\sqrt{2m E }$	vector $\mathbf{L}$ and the scaled (by $1/\sqrt{2m E }$) Runge–Lenz vector $\mathbf{M}'$ as follows:
gilmore-lie-groups_F02568	252	■ 1.000	$L_{\{i\}}, \ M_{\{i\}}^{\prime}$	L_i, M'_i	The operators L_i, M'_i , and B_μ are the infinitesimal generators of $SO(4, 1)$ as follows,
gilmore-lie-groups_F02569	252	■ 1.000	$B_{\{\mu\}}$	B_μ	The operators L_i, M'_i , and B_μ are the infinitesimal generators of $SO(4, 1)$ as follows,
gilmore-lie-groups_F02570	252	■ 0.993	$a_{\{i\}}, \ a_{\{j\}}^{\dagger}$	$a_i, a_j^\dagger$	That is, for the operators $\mathbf{A}$ we introduce annihilation and creation operators $a_i, a_j^\dagger$
gilmore-lie-groups_F02571	252	■ 1.000	$b_{\{i\}}, \ b_{\{j\}}^{\dagger}$	$b_i, b_j^\dagger$	for two independent modes, and similarly we introduce operators $b_i, b_j^\dagger$ to describe
gilmore-lie-groups_F02572	252	■ 1.000	$\left m_{\{1\}}, \ m_{\{2\}}; \ n_{\{1\}}, \ n_{\{2\}}\right\rangle$	$ m_1, m_2; n_1, n_2\rangle$	B. Basis states on which these operators act have the form $m_1, m_2; n_1, n_2\rangle$ where,
gilmore-lie-groups_F02573	252	■ 1.000	$j_{\{a\}}=\frac{1}{2}\left(m_{\{1\}}+m_{\{2\}}\right)$	$j_a = \frac{1}{2}(m_1 + m_2)$	directly to the condition $j_a = j_b$, where $j_a = \frac{1}{2}(m_1 + m_2)$ and $j_b = \frac{1}{2}(n_1 + n_2)$.
gilmore-lie-groups_F02574	252	■ 1.000	$j_{\{b\}}=\frac{1}{2}\left(n_{\{1\}}+n_{\{2\}}\right)$	$j_b = \frac{1}{2}(n_1 + n_2)$	directly to the condition $j_a = j_b$, where $j_a = \frac{1}{2}(m_1 + m_2)$ and $j_b = \frac{1}{2}(n_1 + n_2)$.
gilmore-lie-groups_F02575	253	■ 1.000	$N=2 \ j_{\{a\}}+1=2 \ j_{\{b\}}+1=\left(j_{\{a\}}+j_{\{b\}}\right)+1$	$N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$	principal quantum number $N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$ must change
gilmore-lie-groups_F02576	253	■ 1.000	$j_{\{a\}}$	j_a	$j_a = j_b$. Operators that change j_a have the form $a_i^\dagger$ or $a_i^\dagger a_j^\dagger$, but they do not simul-

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gilmore-lie-groups_F02577	253	■ 1.000	$j_{\{b\}}$	j_b	taneously change j_b . Only operators that simultaneously add or subtract one exci-
gilmore-lie-groups_F02578	253	■ 0.991	$\left(H_{\{1\}}, H_{\{2\}}, H_{\{3\}}, H_{\{4\}}\right)=\left(a_{\{1\}}^{\dagger} a_{\{1\}}, a_{\{2\}}^{\dagger} a_{\{2\}}, b_{\{1\}}^{\dagger} b_{\{1\}}\right)$	$(H_1, H_2, H_3, H_4) = \left(a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1\right.$	number operators for the four boson modes: $(H_1, H_2, H_3, H_4) = (a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1,$
gilmore-lie-groups_F02579	253	■ 0.829	$b_{\{2\}}^{\dagger} b_{\{2\}}$	$b_2^\dagger b_2$	$b_2^\dagger b_2)$. The remaining twelve operators have eigenoperator commutation relations
gilmore-lie-groups_F02580	253	■ 1.000	$\pm \frac{1}{2}$	$\pm \frac{1}{2}$	All these roots have equal length, and inner products among these roots are all $\pm \frac{1}{2}$
gilmore-lie-groups_F02581	253	■ 1.000	$\pi / 4$	$\pi / 4$	or make angles of $\pi / 4$ or $3 \pi / 4$ radians with each other. The algebra is uniquely a
gilmore-lie-groups_F02582	253	■ 1.000	$3 \pi / 4$	$3 \pi / 4$	or make angles of $\pi / 4$ or $3 \pi / 4$ radians with each other. The algebra is uniquely a
gilmore-lie-groups_F02583	253	■ 1.000	$A_{\{3\}}=D_{\{3\}}$	$A_3 = D_3$	real form of $A_3 = D_3$.
gilmore-lie-groups_F02584	254	■ 1.000	$a_{\{1\}}^{\dagger} a_{\{1\}}+a_{\{2\}}^{\dagger} a_{\{2\}}+b_{\{1\}}^{\dagger} b_{\{1\}}+b_{\{2\}}^{\dagger} b_{\{2\}}$	$a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups_F02585	254	■ 1.000	$\mathfrak{so}(4)+\mathfrak{so}(2)$	$\mathfrak{so}(4) + \mathfrak{so}(2)$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups_F02586	254	■ 0.890	$\mathfrak{so}(4,2)=\mathfrak{su}(2,2)$	$\mathfrak{so}(4,2) = \mathfrak{su}(2,2)$	$\mathfrak{so}(4,2) = \mathfrak{su}(2,2)$. This is the conformal algebra.
gilmore-lie-groups_F02587	254	■ 1.000	$(q_{\{1\}}, q_{\{2\}}, q_{\{3\}}, q_{\{4\}})$	(q_1, q_2, q_3, q_4)	Coordinates (q_1, q_2, q_3, q_4) in the fictitious space R^4 are related to coordinates
gilmore-lie-groups_F02588	254	■ 0.559	$(Q_{\{1\}}, Q_{\{2\}}, Q_{\{3\}})$	(Q_1, Q_2, Q_3)	(Q_1, Q_2, Q_3) in the real space by the 4×4 transformation
gilmore-lie-groups_F02589	254	■ 1.000	$Q_{\{4\}}$	Q_4	The transformation is constructed so that the “fourth” real coordinate Q_4 is identi-
gilmore-lie-groups_F02590	254	■ 1.000	$q_{\{1\}}^2+q_{\{2\}}^2+q_{\{3\}}^2+q_{\{4\}}^2 \neq 0$	$q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$	cally zero. This transformation is invertible provided $q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$. The
gilmore-lie-groups_F02591	254	■ 1.000	$R=\sqrt{Q_{\{1\}}^2+Q_{\{2\}}^2+Q_{\{3\}}^2}$	$R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$	distance $R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$ in R^3 and the distance $q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$
gilmore-lie-groups_F02592	254	■ 1.000	$q=\sqrt{q_{\{1\}}^2+q_{\{2\}}^2+q_{\{3\}}^2+q_{\{4\}}^2}$	$q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$	distance $R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$ in R^3 and the distance $q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$
gilmore-lie-groups_F02593	254	■ 1.000	$R=q^2$	$R = q^2$	in R^4 are related by $R = q^2$.
gilmore-lie-groups_F02594	254	■ 1.000	$P_{\{4\}}=0$	$P_4 = 0$	A constraint condition must be applied to force $P_4 = 0$. This condition is
gilmore-lie-groups_F02595	254	■ 1.000	$P^2=P_{\{1\}}^2+P_{\{2\}}^2+P_{\{3\}}^2=\left(1 / 4 R p^2\right)-\left(\zeta^2 / 4 R^2\right) \rightarrow$	$P^2 = P_1^2 + P_2^2 + P_3^2 = \left(1 / 4 R p^2\right) - \left(\zeta^2 / 4 R^2\right) \rightarrow$	With this constraint we find $P^2 = P_1^2 + P_2^2 + P_3^2 = \left(1 / 4 R p^2\right) - \left(\zeta^2 / 4 R^2\right) \rightarrow$
gilmore-lie-groups_F02596	254	■ 0.980	$\left(1 / 4 R\right)\left(p_{\{1\}}^2+p_{\{2\}}^2+p_{\{3\}}^2+p_{\{4\}}^2\right)$	$(1 / 4 R)\left(p_1^2+p_2^2+p_3^2+p_4^2\right)$	$(1 / 4 R)\left(p_1^2+p_2^2+p_3^2+p_4^2\right)$. With these transformations the hamiltonian in the
gilmore-lie-groups_F02597	255	■ 1.000	$\left(q_{\{1\}}, q_{\{2\}}, q_{\{3\}}, q_{\{4\}} ; p_{\{1\}}, p_{\{2\}}, p_{\{3\}}, p_{\{4\}}\right)$	$\left(q_1, q_2, q_3, q_4 ; p_1, p_2, p_3, p_4\right)$	$\left(q_1, q_2, q_3, q_4 ; p_1, p_2, p_3, p_4\right)$ on the left and its transpose on the right (Sadovskii
gilmore-lie-groups_F02598	255	■ 1.000	$\zeta=0$	$\zeta = 0$	(iii) preserves $\zeta = 0$.
gilmore-lie-groups_F02599	255	■ 1.000	$M^t G_{\{1\}} M=G_{\{1\}}$	$M^t G_1 M = G_1$	symplectic metric-preserving condition: $M^t G_1 M = G_1$. Preserving the condition
gilmore-lie-groups_F02600	255	■ 1.000	$M^t G_{\{2\}} M=G_{\{2\}}$	$M^t G_2 M = G_2$	$M^t G_2 M = G_2$.
gilmore-lie-groups_F02601	256	■ 0.988	$\operatorname{Sp}(8 ; \mathbb{R})$	$\operatorname{Sp}(8 ; \mathbb{R})$	$(+4,-4)$. The group that preserves the antisymmetric metric is $\operatorname{Sp}(8 ; \mathbb{R})$ and the

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gilmore-lie-groups-F02602	256	■ 1.000	$S_0(4,4)$	$SO(4,4)$	group that preserves the symmetric metric is $SO(4,4)$. The group that satisfies both
gilmore-lie-groups-F02603	256	■ 0.998	(q, p)	(q, p)	coordinates (q, p) to coordinates (s, r) :
gilmore-lie-groups-F02604	256	■ 0.998	(s, r)	(s, r)	coordinates (q, p) to coordinates (s, r) :
gilmore-lie-groups-F02605	256	■ 1.000	$U(2,2)$	$U(2,2)$	The noncompact group $U(2,2)$ preserves the constraint Eq. (14.49).
gilmore-lie-groups-F02606	256	■ 0.825	$\left(V(r)=-e^2/r\right)$	$(V(r) = -e^2/r)$	charged particle interacting with an external Coulomb field $V(r) = -e^2/r$ can be
gilmore-lie-groups-F02607	257	■ 1.000	$\mathfrak{so}(4,1)$	$\mathfrak{so}(4,1)$	close under commutation to span a Lie algebra that is isomorphic with $\mathfrak{so}(4,1)$.
gilmore-lie-groups-F02608	257	■ 1.000	$\mathfrak{so}(4,2)$	$\mathfrak{so}(4,2)$	Five additional operators can be introduced that extend the algebra to $\mathfrak{so}(4,2)$.
gilmore-lie-groups-F02609	257	■ 1.000	$J_i, M_i, A_i, \Gamma_i (i=1,2,3)$	$J_i, M_i, A_i, \Gamma_i (i = 1, 2, 3)$	The four triplets $J_i, M_i, A_i, \Gamma_i (i = 1, 2, 3)$ have transformation properties of three-
gilmore-lie-groups-F02610	257	■ 0.996	A_4, Γ_4, Γ_5	A_4, Γ_4, Γ_5	vectors under rotations. The three additional operators A_4, Γ_4, Γ_5 close under com-
gilmore-lie-groups-F02611	257	■ 1.000	$\Phi(r)=e/r$	$\Phi(r) = e/r$	in the Coulomb field $\Phi(r) = e/r$ of a proton can be expressed in terms of operators
gilmore-lie-groups-F02612	257	■ 1.000	A_4, Γ_4	A_4, Γ_4	of type A_4, Γ_4 , and Γ_5 . These operators are displayed in Table 14.3, along with the
gilmore-lie-groups-F02613	257	■ 1.000	Γ_5	Γ_5	of type A_4, Γ_4 , and Γ_5 . These operators are displayed in Table 14.3, along with the
gilmore-lie-groups-F02614	257	■ 0.860	$(\mathbf{E}, \mathbf{B})$	$(\mathbf{E}, \mathbf{B})$	scribed by the Dirac equation. The electromagnetic field $(\mathbf{E}, \mathbf{B})$ is described by the
gilmore-lie-groups-F02615	257	■ 1.000	$A_\mu=(\phi, \mathbf{A})$	$A_\mu = (\phi, \mathbf{A})$	four-vector potential $A_\mu = (\phi, \mathbf{A})$. The electron has charge $q = -e$ (where e is
gilmore-lie-groups-F02616	257	■ 1.000	$q=-e$	$q = -e$	four-vector potential $A_\mu = (\phi, \mathbf{A})$. The electron has charge $q = -e$ (where e is
gilmore-lie-groups-F02617	257	■ 1.000	$H_D\psi=E\psi$	$H_D\psi = E\psi$	the charge on the proton) and spin $\frac{1}{2}$. The Dirac equation $H_D\psi = E\psi$ is a matrix
gilmore-lie-groups-F02621	260	■ 1.000	Γ_4	Γ_4	Γ_4 or the compact generator Γ_5 , depending on the sign of the parameter C .
gilmore-lie-groups-F02634	258	■ 1.000	$\pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$	$\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$	replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under
gilmore-lie-groups-F02635	258	■ 0.971	γ_i	γ_i	The 4×4 matrices β and γ_i can be chosen as
gilmore-lie-groups-F02636	258	■ 0.982	σ_i	σ_i	Here σ_i are the standard Pauli 2×2 spin matrices (cf., Eq. (3.39), Problem 3.1).
gilmore-lie-groups-F02637	259	■ 1.000	$\mathbf{p} \rightarrow \pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$	$\mathbf{p} \rightarrow \pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$	As before, the substitution $\mathbf{p} \rightarrow \pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$ is in order in the event there is a
gilmore-lie-groups-F02638	259	■ 1.000	J_i, M_i, A_i, Γ_i	J_i, M_i, A_i, Γ_i	three-vectors J_i, M_i, A_i, Γ_i defined in Table 14.3. The Dirac hamiltonian is ex-
gilmore-lie-groups-F02639	259	■ 0.990	Γ_4, Γ_5	Γ_4, Γ_5	Gordon operator (see Table 14.3). In short, the operators Γ_4, Γ_5 are modified but
gilmore-lie-groups-F02640	259	■ 1.000	$D=d/dr$	$D = d/dr$	with $D = d/dr$. The operators r and D behave under commutation like the boson
gilmore-lie-groups-F02641	259	■ 1.000	rD^2+r	$rD^2 + r$	The linear combinations $rD^2 + r$ and $rD^2 - r$ are compact and noncompact, re-
gilmore-lie-groups-F02642	259	■ 1.000	rD^2-r	$rD^2 - r$	The linear combinations $rD^2 + r$ and $rD^2 - r$ are compact and noncompact, re-

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gilmore-lie-groups_ F02643	260	■ 1.000	$C^2=\Gamma_5^2-\Gamma_4^2-M_4^2=-a$	$C^2 = \Gamma_5^2 - \Gamma_4^2 - M_4^2 = -a$	The Casimir operator for this algebra is $C^2 = \Gamma_5^2 - \Gamma_4^2 - M_4^2 = -a$. The represen-
gilmore-lie-groups_ F02644	260	■ 0.978	$a \rightarrow A$	$a \rightarrow A$	follows ($a \rightarrow A$)
gilmore-lie-groups_ F02645	260	■ 1.000	$C<0$	$C < 0$	If $C < 0$ we can choose $e^{-\theta} + C e^{\theta} = 0$, so that the resulting equation becomes
gilmore-lie-groups_ F02646	260	■ 1.000	$e^{-\theta}+C e^{\theta}=0$	$e^{-\theta} + C e^{\theta} = 0$	If $C < 0$ we can choose $e^{-\theta} + C e^{\theta} = 0$, so that the resulting equation becomes
gilmore-lie-groups_ F02647	260	■ 1.000	$u(r)=e^{\theta M_4} R(r)$	$u(r) = e^{\theta M_4} R(r)$	where $u(r) = e^{\theta M_4} R(r)$. If A is the Casimir invariant of this representation
gilmore-lie-groups_ F02648	260	■ 0.798	$N=-\frac{1}{2}+$	$N = -\frac{1}{2}+$	of $\mathfrak{su}(1, 1)$, the discrete spectrum of the compact operator Γ_5 is $N = -\frac{1}{2} +$
gilmore-lie-groups_ F02649	260	■ 1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A}+1+n, \ n=0,1,2, \ldots$	$\sqrt{\left(\frac{1}{2}\right)^2 - A + 1 + n}, n = 0, 1, 2, \dots$	$\sqrt{(\frac{1}{2})^2 - A + 1 + n}, n = 0, 1, 2, \dots$. This result leads directly to the eigenvalue
gilmore-lie-groups_ F02650	261	■ 1.000	$W=-\frac{1}{2} m c^2 \alpha^2 \left(1 / N^2\right)$	$W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$	relativistic case) $W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$, where $N = l + 1 + k, k = 0, 1, 2, \dots$.
gilmore-lie-groups_ F02651	261	■ 1.000	$N=l+1+k, \ k=0,1,2, \ldots$	$N = l + 1 + k, k = 0, 1, 2, \dots$	relativistic case) $W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$, where $N = l + 1 + k, k = 0, 1, 2, \dots$.
gilmore-lie-groups_ F02652	261	■ 1.000	$m_{\{l\}}$	m_l	tum operators, the quantum number m_l (eigenvalue of L_z) is also invariant under
gilmore-lie-groups_ F02653	261	■ 0.449	$ N, \ln \rangle \rightarrow N \pm 1, l m\rangle$	$ N, \ln\rangle \leftrightarrow N \pm 1, lm\rangle$	algebra are $ N, lm\rangle \leftrightarrow N \pm 1, lm\rangle$. The states on which these operators act are
gilmore-lie-groups_ F02660	261	■ 1.000	$L_{-} Y_{m=-l}^l(\theta, \phi)=0$	$L_{-} Y_{m=-l}^l(\theta, \phi) = 0$	known, this relation can be used, as was the relation $L_{-} Y_{m=-l}^l(\theta, \phi) = 0$, to deter-
gilmore-lie-groups_ F02662	261	■ 1.000	$C>0$	$C > 0$	If $C > 0$ we can choose θ so that $e^{-\theta} - C e^{\theta} = 0$. Equation (14.66) reduces to
gilmore-lie-groups_ F02663	261	■ 1.000	$e^{-\theta}-C e^{\theta}=0$	$e^{-\theta} - C e^{\theta} = 0$	If $C > 0$ we can choose θ so that $e^{-\theta} - C e^{\theta} = 0$. Equation (14.66) reduces to
gilmore-lie-groups_ F02664	261	■ 1.000	$k \simeq e^{-\theta}$	$k \simeq e^{-\theta}$	$k \simeq e^{-\theta}$ with $E = \hbar^2 k^2 / 2m$. The asymptotic form of the wave function is (Gilmore
gilmore-lie-groups_ F02665	261	■ 1.000	$E=\hbar^2 k^2 / 2 m$	$E = \hbar^2 k^2 / 2m$	$k \simeq e^{-\theta}$ with $E = \hbar^2 k^2 / 2m$. The asymptotic form of the wave function is (Gilmore
gilmore-lie-groups_ F02666	262	■ 1.000	$\delta(j)=\arg [\Gamma(j+1-i(\alpha / k))]$	$\delta(j) = \arg[\Gamma(j + 1 - i(\alpha/k))]$	where $\delta(j) = \arg[\Gamma(j + 1 - i(\alpha/k))]$ is part of the scattering phase shift, and the
gilmore-lie-groups_ F02667	262	■ 1.000	$j=-\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}$	$j = -\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A}$	expression for j is given by $j = -\frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}$.
gilmore-lie-groups_ F02668	262	■ 1.000	$-1 / r$	$-1/r$	attracted than a simple $-1/r$ potential suggests. It is possible to represent this
gilmore-lie-groups_ F02669	262	■ 1.000	$-1 / r^2$	$-1/r^2$	extra attraction by adding a term of the form $-1/r^2$ to the potential to represent
gilmore-lie-groups_ F02670	262	■ 1.000	$V(r)=-e^2 / r \rightarrow-e^2 / r-\mu_l\left(\hbar^2 / 2 m\right) / r^2$	$V(r) = -e^2/r \rightarrow -e^2/r - \mu_l (\hbar^2/2m) / r^2$	and Klein–Gordon equations is $V(r) = -e^2/r \rightarrow -e^2/r - \mu_l (\hbar^2/2m) / r^2$. This
gilmore-lie-groups_ F02671	262	■ 1.000	$j \rightarrow j^{\prime}=j+\Delta j$	$j \rightarrow j' = j + \Delta j$	This change produces a change in the value of $j \rightarrow j' = j + \Delta j$, where $\Delta j =$
gilmore-lie-groups_ F02672	262	■ 1.000	$\Delta j=$	$\Delta j =$	This change produces a change in the value of $j \rightarrow j' = j + \Delta j$, where $\Delta j =$
gilmore-lie-groups_ F02673	262	■ 1.000	$-\mu_l /(2 l+1)$	$-\mu_l / (2l + 1)$	$-\mu_l / (2l + 1)$ in the nonrelativistic case. This change produces a change in the
gilmore-lie-groups_ F02674	262	■ 1.000	Δj	Δj	The quantum defect Δj causes the Rydberg states to be bound more strongly than in

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gilmore-lie-groups_F02675	263	■ 1.000	$G \supset S O(3)$	$G \supset SO(3)$	group, we are prodded to search for a larger group $G \supset SO(3)$ that explains the
gilmore-lie-groups_F02676	264	■ 0.987	$3^{\text{\texttt{\texttt{deg}}}}$	3^{deg}	of relativity. Show that the existence of the 3^{deg} microwave background radiation is
gilmore-lie-groups_F02677	264	■ 1.000	$\mathbf{F}=m\mathbf{g}$	$\mathbf{F} = m\mathbf{g}$	tional force $\mathbf{F} = m\mathbf{g}$ in the “down” direction. Assume that the other elevator is in
gilmore-lie-groups_F02678	264	■ 1.000	$\mathbf{A}=\frac{1}{2}\mathbf{B}\times\mathbf{r}$	$\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$	taken as $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$, so that $\mathbf{B} = \nabla \times \mathbf{A}$. Derive the Klein–Gordon equation for an
gilmore-lie-groups_F02679	264	■ 0.996	$(r,\theta,\phi)=\left(\theta_3,\theta_2,\theta_1\right)$	$(r,\theta,\phi) = (\theta_3,\theta_2,\theta_1)$	4. Introduce spherical coordinates as follows: $(r,\theta,\phi) = (\theta_3,\theta_2,\theta_1)$ and
gilmore-lie-groups_F02680	264	■ 1.000	$\mathcal{L}^2\left(S^1\right)=\partial^2/\partial\theta_1^2$	$\mathcal{L}^2\left(S^1\right) = \partial^2/\partial\theta_1^2$	Show that $\mathcal{L}^2(S^1) = \partial^2/\partial\theta$. Show that
gilmore-lie-groups_F02681	264	■ 1.000	$\mathcal{L}^2\left(S^3\right)$	$\mathcal{L}^2\left(S^3\right)$	Generalize this result to $\mathcal{L}^2(S^3)$ recursively using $\mathcal{L}^2(S^2)$ and $(\partial/\partial\cos\theta_3)^2$. Do this
gilmore-lie-groups_F02682	264	■ 1.000	$\left(\partial/\partial\cos\theta_3\right)^2$	$(\partial/\partial\cos\theta_3)^2$	Generalize this result to $\mathcal{L}^2(S^3)$ recursively using $\mathcal{L}^2(S^2)$ and $(\partial/\partial\cos\theta_3)^2$. Do this
gilmore-lie-groups_F02683	264	■ 1.000	$\mathcal{L}^2\left(S^n\right)$	$\mathcal{L}^2\left(S^n\right)$	more generally for $\mathcal{L}^2(S^n)$.
gilmore-lie-groups_F02684	265	■ 1.000	$r\rightarrow\infty$	$r \rightarrow \infty$	a. Show that the singular points of Eq. (14.9) occur at $r = 0$ and $r \rightarrow \infty$.
gilmore-lie-groups_F02685	265	■ 1.000	$\gamma=\frac{1}{2}\pm\sqrt{\left(\frac{1}{2}\right)^2-A}$	$\gamma = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - A}$	Show that $\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$ and $\lambda = \pm\sqrt{-C}$.
gilmore-lie-groups_F02686	265	■ 1.000	$\lambda=\pm\sqrt{-C}$	$\lambda = \pm\sqrt{-C}$	Show that $\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$ and $\lambda = \pm\sqrt{-C}$.
gilmore-lie-groups_F02687	265	■ 1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A}$	$\sqrt{\left(\frac{1}{2}\right)^2 - A}$	c. Show that if $\sqrt{(\frac{1}{2})^2 - A}$ is real, the solution with the positive sign is always square
gilmore-lie-groups_F02688	265	■ 1.000	$\pm\sqrt{-C}$	$\pm\sqrt{-C}$	with the negative sign square integrable? Show that if $C < 0$ the solution $\pm\sqrt{-C}$
gilmore-lie-groups_F02689	265	■ 1.000	$R(r)=r^{\gamma}e^{\lambda r}f(r)$	$R(r) = r^{\gamma}e^{\lambda r}f(r)$	d. Show that a solution of the form $R(r) = r^{\gamma}e^{\lambda r}f(r)$ can be found where the function
gilmore-lie-groups_F02690	265	■ 1.000	$f(r)=\sum_{j=0}^{\infty}f_jr^j$	$f(r) = \sum_{j=0}^{\infty}f_jr^j$	f. Represent the function $f(r)$ as an ascending power series: $f(r) = \sum_{j=0}^{\infty}f_jr^j$.
gilmore-lie-groups_F02691	265	■ 1.000	$j\rightarrow\infty$	$j \rightarrow \infty$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_F02692	265	■ 1.000	$f(r)\rightarrow e^{-2\lambda r}$	$f(r) \rightarrow e^{-2\lambda r}$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_F02693	265	■ 1.000	$\lambda<0$	$\lambda < 0$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_F02694	265	■ 0.963	r^n	r^n	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_F02695	265	■ 0.963	$f_n\neq 0$	$f_n \neq 0$	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_F02696	265	■ 0.963	$f_{n+1}=0\left(\Rightarrow f_{n+2}=\right)$	$f_{n+1} = 0 (\Rightarrow f_{n+2} =$	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_F02697	265	■ 1.000	$f_{n+3}=\cdots=0$	$f_{n+3} = \cdots = 0$	$f_{n+3} = \cdots = 0$), show that the quantization conditon $2\lambda(n+\gamma)+B=0$ must be
gilmore-lie-groups_F02698	265	■ 1.000	$2\lambda(n+\gamma)+B=0$	$2\lambda(n+\gamma)+B=0$	$f_{n+3} = \cdots = 0$), show that the quantization conditon $2\lambda(n+\gamma)+B=0$ must be
gilmore-lie-groups_F02699	266	■ 1.000	$\frac{1}{r}r^{\gamma}f(r)e^{\lambda r}$	$\frac{1}{r}r^{\gamma}f(r)e^{\lambda r}$	The radial part of the wavefunction $\frac{1}{r}r^{\gamma}f(r)e^{\lambda r}$ has exactly n nodes in the open
gilmore-lie-groups_F02700	266	■ 0.907	$(0,\infty)$	$(0,\infty)$	interval $[0,\infty)$.

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gilmore-lie-groups_F02781	266	■ 1.000	$Z e / r$	Ze/r	show that the potential is Ze/r and the solutions of the relativistic and nonrelativistic
gilmore-lie-groups_F02782	266	■ 1.000	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$	$N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$	$N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$ and $N = n + l + 1$
gilmore-lie-groups_F02783	266	■ 1.000	$e^{\lambda r}$	$e^{\lambda r}$	8. The radial part of the wavefunction dies off like $e^{\lambda r}$ for large r , where $\lambda < 0$ for bound
gilmore-lie-groups_F02784	266	■ 1.000	λ^{-1}	λ^{-1}	states. The parameter λ^{-1} has the dimensions of length, and $a \simeq 1/ \lambda $ characterizes
gilmore-lie-groups_F02785	266	■ 1.000	$a \simeq 1/ \lambda $	$a \simeq 1/ \lambda $	states. The parameter λ^{-1} has the dimensions of length, and $a \simeq 1/ \lambda $ characterizes
gilmore-lie-groups_F02786	266	■ 0.910	(n, l)	(n, l)	the size of a bound state orbit. Show that bound states with quantum numbers (n, l)
gilmore-lie-groups_F02717	267	■ 1.000	$a_{\mathrm{B}}=\hbar^2 / m e^2=0.529 \times 10^{-8} \mathrm{~cm}$	$a_B = \hbar^2/m e^2 = 0.529 \times 10^{-8} \text{ cm}$	In these expressions $a_B = \hbar^2/m e^2 = 0.529 \times 10^{-8} \text{ cm}$ is the Bohr radius: the char-
gilmore-lie-groups_F02718	267	■ 0.906	n, l	n, l	sizes of the atoms in the (n, l) states differ between the relativistic and nonrelativistic
gilmore-lie-groups_F02719	267	■ 1.000	$W=-\frac{1}{2} m c^2 \alpha^2 / N^2$	$W = -\frac{1}{2} m c^2 \alpha^2 / N^2$	binding energy $W = -\frac{1}{2} m c^2 \alpha^2 / N^2$ is the reduced mass: $1/m = 1/m_1 + 1/m_2$ of
gilmore-lie-groups_F02720	267	■ 1.000	$1 / m=1 / m_1+1 / m_2$	$1/m = 1/m_1 + 1/m_2$	binding energy $W = -\frac{1}{2} m c^2 \alpha^2 / N^2$ is the reduced mass: $1/m = 1/m_1 + 1/m_2$ of
gilmore-lie-groups_F02721	267	■ 1.000	$(E=0)$	$(E = 0)$	$(E > 0)$; and parabolic at the separatrix $(E = 0)$. If the radial force field includes a
gilmore-lie-groups_F02722	267	■ 1.000	$1 / r^3$	$1/r^3$	radial $1/r^3$ perturbation
gilmore-lie-groups_F02723	267	■ 1.000	$\alpha=\sqrt{1-\eta}, \eta=C / K a$	$\alpha = \sqrt{1 - \eta}, \eta = C/K a$	where $\alpha = \sqrt{1 - \eta}, \eta = C/K a$. This can be treated as an ellipse that is slowly
gilmore-lie-groups_F02724	267	■ 1.000	$\alpha \simeq 1$	$\alpha \simeq 1$	rotating, $\alpha \simeq 1$. In this case the parameters a and ϵ have their usual meanings
gilmore-lie-groups_F02725	267	■ 1.000	η	η	η is a measure of the strength of the perturbation to the strength of the Coulomb
gilmore-lie-groups_F02726	268	■ 0.651	$E=\sqrt{\left(m c^2\right)^2+(\mathbf{p} c)^2}-K / r$	$E = \sqrt{(m c^2)^2 + (\mathbf{p} c)^2} - K/r$	a. Expand the relativistic energy $E = \sqrt{(m c^2)^2 + (\mathbf{p} c)^2} - K/r$ to fourth order in $\mathbf{p}$
gilmore-lie-groups_F02727	268	■ 1.000	$E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$	$E = (m c^2) + (p^2/2m) - (p^2/2m)^2/(2m c^2) - K/r = m c^2 + W$	and show $E = (m c^2) + (p^2/2m) - (p^2/2m)^2/(2m c^2) - K/r = m c^2 + W$.
gilmore-lie-groups_F02728	268	■ 1.000	$-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$	$-(p^2/2m)^2/(2m c^2)$	b. Replace the quartic term $-(p^2/2m)^2/(2m c^2)$ by $-(W + K/r)^2/(2m c^2)$ and ex-
gilmore-lie-groups_F02729	268	■ 1.000	$-(W+K / r)^2 /\left(2 m c^2\right)$	$-(W + K/r)^2/(2m c^2)$	b. Replace the quartic term $-(p^2/2m)^2/(2m c^2)$ by $-(W + K/r)^2/(2m c^2)$ and ex-
gilmore-lie-groups_F02730	268	■ 1.000	$K^{\prime}$	K'	Evaluate K' and C' and show $K' = K(1 + W/m c^2)$ and $C' = -K^2/(2m c^2)$.
gilmore-lie-groups_F02731	268	■ 1.000	$C^{\prime}$	C'	Evaluate K' and C' and show $K' = K(1 + W/m c^2)$ and $C' = -K^2/(2m c^2)$.
gilmore-lie-groups_F02732	268	■ 1.000	$K^{\prime}=K\left(1+W / m c^2\right)$	$K' = K(1 + W/m c^2)$	Evaluate K' and C' and show $K' = K(1 + W/m c^2)$ and $C' = -K^2/(2m c^2)$.
gilmore-lie-groups_F02733	268	■ 1.000	$C^{\prime}=-K^2 /\left(2 m c^2\right)$	$C' = -K^2/(2m c^2)$	Evaluate K' and C' and show $K' = K(1 + W/m c^2)$ and $C' = -K^2/(2m c^2)$.
gilmore-lie-groups_F02734	268	■ 1.000	$K \rightarrow K^{\prime}$	$K \rightarrow K'$	c. Argue that the classical motion involves a renormalized coupling $K \rightarrow K'$ as well
gilmore-lie-groups_F02735	268	■ 1.000	$C=2 C^{\prime}$	$C = 2C'$	as a $1/r^3$ component to the force, with $C = 2C'$.
gilmore-lie-groups_F02736	268	■ 1.000	$\delta \theta \simeq \eta / 2$	$\delta \theta \simeq \eta/2$	d. Show that the advance in the perihelion of the orbit is $\delta \theta \simeq \eta/2$ per period.

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gilmore-lie-groups_F02737	268	■ 1.000	\epsilon=0.206	$\epsilon = 0.206$	e. Evaluate η for the planet Mercury, for which $\epsilon = 0.206$ and the period is $T = 0.24$
gilmore-lie-groups_F02738	268	■ 1.000	T=0.24	$T = 0.24$	e. Evaluate η for the planet Mercury, for which $\epsilon = 0.206$ and the period is $T = 0.24$
gilmore-lie-groups_F02739	268	■ 1.000	m=	$m =$	correct and the mass of the particle varies with its state of motion according to $m =$
gilmore-lie-groups_F02740	268	■ 1.000	$m_{\{0\}} / \sqrt{1 - (v / c)^{2}}$	$m_0 / \sqrt{1 - (v/c)^2}$	$m_0 / \sqrt{1 - (v/c)^2}$. (2) The mass of a particle is a constant of nature and Newton's
gilmore-lie-groups_F02741	268	■ 1.000	$\mathbf{F}(r)=\left(-K / r^2+p(r)\right) \hat{\mathbf{r}}$	$\mathbf{F}(r) = (-K/r^2 + p(r)) \hat{\mathbf{r}}$	force has the form $\mathbf{F}(r) = (-K/r^2 + p(r))\hat{\mathbf{r}}$, where $p(r)$ is a small perturbation. The
gilmore-lie-groups_F02742	268	■ 1.000	$p(r)$	$p(r)$	force has the form $\mathbf{F}(r) = (-K/r^2 + p(r))\hat{\mathbf{r}}$, where $p(r)$ is a small perturbation. The
gilmore-lie-groups_F02743	268	■ 1.000	$1 / r=\left(m K / L^2\right)\left(1+(M / m K) \cos \theta\right)$	$1/r = (mK/L^2)(1 + (M/mK) \cos \theta)$	with $1/r = (mK/L^2)(1 + (M/mK) \cos \theta)$. Here L is the particle's orbital angular
gilmore-lie-groups_F02744	268	■ 1.000	L	L	with $1/r = (mK/L^2)(1 + (M/mK) \cos \theta)$. Here L is the particle's orbital angular
gilmore-lie-groups_F02745	268	■ 1.000	C / r^3	C/r^3	momentum and T is its period. If the perturbing term is of the form C/r^3 the integral
gilmore-lie-groups_F02746	268	■ 1.000	$C \times 2 \pi \frac{m K}{L^2}$	$C \times 2\pi \frac{mK}{L^2}$	is $C \times 2\pi \frac{mK}{L^2}$. The perturbations due to special and General Relativity are
gilmore-lie-groups_F02747	268	■ 1.000	$K=G M m$	$K = GMm$	For planetary motion $K = GMm$. When $M \gg m$, ω is (almost) independent of m .
gilmore-lie-groups_F02748	268	■ 1.000	$M \gg m, \omega$	$M \gg m, \omega$	For planetary motion $K = GMm$. When $M \gg m$, ω is (almost) independent of m .
gilmore-lie-groups_F02749	269	■ 1.000	$\omega=42^{\prime \prime}$	$\omega = 42''$	planetary distance from the Sun. What is the precession for the Earth? Use $\omega = 42''$
gilmore-lie-groups_F02750	269	■ 1.000	L_{-}	L_-	12. The action of the angular momentum shift down operator L_- on the lowest m -value
gilmore-lie-groups_F02751	269	■ 1.000	$Y_{m}^l(\theta, \phi)=P_{-l}^l(\theta) e^{-i l \phi}$	$Y_m^l(\theta, \phi) = P_{-l}^l(\theta)e^{-il\phi}$	a. Write $Y_m^l(\theta, \phi) = P_{-l}^l(\theta)e^{-il\phi}$ and show
gilmore-lie-groups_F02752	269	■ 1.000	$P_{-l}^l=(\sin \theta)^l$	$P_{-l}^l = (\sin \theta)^l$	b. Show $P_{-l}^l = (\sin \theta)$ satisfies this equation.
gilmore-lie-groups_F02753	269	■ 1.000	$N_{\{l\}}$	N_l	a normalization coefficient N_l and enforcing the condition
gilmore-lie-groups_F02754	269	■ 1.000	$Y_{\pm l}^l(\theta, \phi)$	$Y_{\pm l}^l(\theta, \phi)$	$Y_{\pm l}^l(\theta, \phi)$:
gilmore-lie-groups_F02755	269	■ 0.999	$N_{\{0\}}=\sqrt{1 / 4 \pi}$	$N_0 = \sqrt{1/4\pi}$	Compare these results with Table 14.2 using initial condition $N_0 = \sqrt{1/4\pi}$. Com-
gilmore-lie-groups_F02756	269	■ 1.000	$N_{\{3\}}$	N_3	pute N_3 .
gilmore-lie-groups_F02757	269	■ 0.999	$\left \left\langle L_{+}\right L_{-}^l\right\rangle =\sqrt{(l+m)(l-m)}$	$\langle_{m'}^l L_{+} _m^l\rangle = \sqrt{(l+m')(l-m)}$	f. Use the numerical value of the matrix elements $\langle_{m'}^l L_{+} _m^l\rangle = \sqrt{(l+m')(l-m)}$
gilmore-lie-groups_F02758	269	■ 1.000	$\delta_{m^{\prime}, m+1}$	$\delta_{m', m+1}$	$\delta_{m', m+1}$ and the coordinate representation of the shift up operator L_+ to construct
gilmore-lie-groups_F02759	269	■ 1.000	$l=N-1$	$l = N - 1$	numbers: $l = N - 1$, where in general the principal quantum number $N = n + l + 1$.
gilmore-lie-groups_F02760	269	■ 0.980	$n=0$	$n = 0$	These functions have no nodes in the interval $(0, \infty)$ (since $n = 0$).
gilmore-lie-groups_F02761	269	■ 0.848	$\mathbf{r}$	$\mathbf{r}$	15. $\mathbf{r}$ is the position vector from the sun to a planet, or from the proton to the electron
gilmore-lie-groups_F02762	270	■ 1.000	$\mathbf{M} \cdot \mathbf{L}=\mathbf{0}$	$\mathbf{M} \cdot \mathbf{L} = \mathbf{0}$	a. $\mathbf{M} \cdot \mathbf{L} = \mathbf{0}$.

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gilmore-lie-groups_ F02763	270	■ 1.000	$\mathbf{M} \cdot \mathbf{M} = (2 \mathbf{L} \cdot \mathbf{L} / m) (\mathbf{p} \cdot \mathbf{p} / 2m - K / r) + K^2$	$\mathbf{M} \cdot \mathbf{M} = (2 \mathbf{L} \cdot \mathbf{L} / m) (\mathbf{p} \cdot \mathbf{p} / 2m - K / r) + K^2$	b. $\mathbf{M} \cdot \mathbf{M} = (2 \mathbf{L} \cdot \mathbf{L} / m) (\mathbf{p} \cdot \mathbf{p} / 2m - K / r) + K^2$.
gilmore-lie-groups_ F02764	270	■ 1.000	$\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L} / m - K r$	$\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L} / m - K r$	c. $\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L} / m - K r$.
gilmore-lie-groups_ F02765	270	■ 0.999	$\mathbf{M} \cdot \mathbf{r} = M r \cos \theta$	$\mathbf{M} \cdot \mathbf{r} = M r \cos \theta$	d. $\mathbf{M} \cdot \mathbf{r} = M r \cos \theta$.
gilmore-lie-groups_ F02766	270	■ 1.000	$r = (\mathbf{L} \cdot \mathbf{L} / m K / 1 + (M / K) \cos \theta)$	$r = (\mathbf{L} \cdot \mathbf{L} / m K / 1 + (M / K) \cos \theta)$	e. $r = (\mathbf{L} \cdot \mathbf{L} / m K / 1 + (M / K) \cos \theta)$.
gilmore-lie-groups_ F02767	270	■ 1.000	$L^2 / m K$	$L^2 / m K$	in a $1/r$ potential to conclude that $L^2 / m K$ is the semimajor axis of the elliptical
gilmore-lie-groups_ F02768	270	■ 1.000	$\epsilon = M / K$	$\epsilon = M / K$	orbit and $\epsilon = M / K$ is the eccentricity of the orbit.
gilmore-lie-groups_ F02769	270	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = \left(-1 / 4 \hbar^2 \right) (\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$	$\mathbf{A} \cdot \mathbf{A} = \left(-1 / 4 \hbar^2 \right) (\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$	16. Show that $\mathbf{A} \cdot \mathbf{A} = \left(-1 / 4 \hbar^2 \right) (\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$. Show that $\mathbf{B} \cdot \mathbf{B}$
gilmore-lie-groups_ F02770	270	■ 1.000	$\mathbf{B} \cdot \mathbf{B}$	$\mathbf{B} \cdot \mathbf{B}$	16. Show that $\mathbf{A} \cdot \mathbf{A} = \left(-1 / 4 \hbar^2 \right) (\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$. Show that $\mathbf{B} \cdot \mathbf{B}$
gilmore-lie-groups_ F02771	270	■ 0.880	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = 0$	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = 0$	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_ F02772	270	■ 1.000	$p_x = \mathbf{p} \cdot \mathbf{M} / M$	$p_x = \mathbf{p} \cdot \mathbf{M} / M$	18. Compute $p_x = \mathbf{p} \cdot \mathbf{M} / M$ and $p_y = \mathbf{p} \cdot \mathbf{W} / W$. Show $p_x^2 + (p_y - a)^2 = r^2$. Explicitly
gilmore-lie-groups_ F02773	270	■ 1.000	$p_y = \mathbf{p} \cdot \mathbf{W} / W$	$p_y = \mathbf{p} \cdot \mathbf{W} / W$	18. Compute $p_x = \mathbf{p} \cdot \mathbf{M} / M$ and $p_y = \mathbf{p} \cdot \mathbf{W} / W$. Show $p_x^2 + (p_y - a)^2 = r^2$. Explicitly
gilmore-lie-groups_ F02774	270	■ 1.000	$p_x^2 + (p_y - a)^2 = r^2$	$p_x^2 + (p_y - a)^2 = r^2$	18. Compute $p_x = \mathbf{p} \cdot \mathbf{M} / M$ and $p_y = \mathbf{p} \cdot \mathbf{W} / W$. Show $p_x^2 + (p_y - a)^2 = r^2$. Explicitly
gilmore-lie-groups_ F02775	270	■ 0.900	$0, a$	$0, a$	compute the displacement vector a (i.e., $(0, a)$) and the radius r of circular motion.
gilmore-lie-groups_ F02776	270	■ 1.000	$x^a y^b z^c$	$x^a y^b z^c$	19. Show that the number of independent monomials of the form $x^a y^b z^c$, with a, b, c
gilmore-lie-groups_ F02777	270	■ 0.996	$a + b + c = l$	$a + b + c = l$	nonnegative integers and $a + b + c = l$ is $N(l, 3) = (l + 3 - 1) / (3 - 1)!$. In N -
gilmore-lie-groups_ F02778	270	■ 0.996	$N(l, 3) = (l + 3 - 1) / (3 - 1)!$	$N(l, 3) = (l + 3 - 1) / (3 - 1)!$	nonnegative integers and $a + b + c = l$ is $N(l, 3) = (l + 3 - 1) / (3 - 1)!$. In N -
gilmore-lie-groups_ F02779	270	■ 0.997	$x_1, x_2, \dots, x_N$	$x_1, x_2, \dots, x_N$	l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_ F02780	270	■ 0.997	$3 \rightarrow N$	$3 \rightarrow N$	l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_ F02781	270	■ 1.000	$r^l Y_m^l(\theta, \phi)$	$r^l Y_m^l(\theta, \phi)$	a. Show that the functions $r^l Y_m^l(\theta, \phi)$ are homogeneous polynomials in x, y, z of
gilmore-lie-groups_ F02782	270	■ 1.000	x, y, z	x, y, z	a. Show that the functions $r^l Y_m^l(\theta, \phi)$ are homogeneous polynomials in x, y, z of
gilmore-lie-groups_ F02783	270	■ 1.000	$l - 2$	$l - 2$	ference between the number of homogeneous polynomials of degree l and $l - 2$
gilmore-lie-groups_ F02784	270	■ 1.000	$\dim \{ Y_m^l \} = N(l, 3) - N(l - 2, 3) = 2l + 1$	$\dim \{ Y_m^l \} = N(l, 3) - N(l - 2, 3) = 2l + 1$	on three variables $\dim \{ Y_m^l \} = N(l, 3) - N(l - 2, 3) = 2l + 1$.
gilmore-lie-groups_ F02785	270	■ 0.999	$\dim \{ \mathcal{Y}_{lm}^n \} = N(n, 4) - N(n - 2, 4) = (n + 1)^2$	$\dim \{ \mathcal{Y}_{lm}^n \} = N(n, 4) - N(n - 2, 4) = (n + 1)^2$	degree n is $\dim \{ \mathcal{Y}_{lm}^n \} = N(n, 4) - N(n - 2, 4) = (n + 1)^2$.
gilmore-lie-groups_ F02786	270	■ 0.995	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$ for $n = 0, 1, 2; l = 0, \dots, n - 1$; and $-l \leq m \leq +l$.
gilmore-lie-groups_ F02787	270	■ 0.995	$n = 0, 1, 2; l = 0, \dots, n - 1$	$n = 0, 1, 2; l = 0, \dots, n - 1$	$\psi(\mathbf{x})_{nlm}$ for $n = 0, 1, 2; l = 0, \dots, n - 1$; and $-l \leq m \leq +l$.
gilmore-lie-groups_ F02788	271	■ 1.000	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$	where a and b are half odd integers: $\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$.

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gilmore-lie-groups_ F02789	271	■ 1.000	$N(l, d) =$	$N(l, d) =$	f. Show homogeneous polynomials satisfy the recursion relation: $N(l, d) =$
gilmore-lie-groups_ F02790	271	■ 1.000	$N(l, d-1) + N(l-1, d)$	$N(l, d-1) + N(l-1, d)$	$N(l, d-1) + N(l-1, d)$.
gilmore-lie-groups_ F02791	271	■ 1.000	S^{n-1}	S^{n-1}	spaces of spherical harmonics on spheres S^n and S^{n-1} ;
gilmore-lie-groups_ F02792	271	■ 1.000	$(l+1)^2 = l^2 + (2l+1)$	$(l+1)^2 = l^2 + (2l+1)$	For the case $n = 3$ this gives $l^2 = l^2 + (2l+1)$. The initialization for all n
gilmore-lie-groups_ F02793	271	■ 1.000	$\mathcal{Y}^0(S^n) = 1 = \dim \mathcal{Y}^0(S^n)$	$\mathcal{Y}^0(S^n) = 1 = \dim \mathcal{Y}^0(S^n)$	is $\mathcal{Y}^0(S^n) = 1 = \dim \mathcal{Y}^0(S^n)$.
gilmore-lie-groups_ F02794	271	■ 1.000	$\dim \mathcal{Y}^l(S^n) = \frac{(l+n-2)!}{l!(n-1)!} (2l+n-1)$	$\dim \mathcal{Y}^l(S^n) = \frac{(l+n-2)!}{l!(n-1)!} (2l+n-1)$	b. $\dim \mathcal{Y}^l(S^n) = \frac{(l+n-2)!}{l!(n-1)!} (2l+n-1)$.
gilmore-lie-groups_ F02795	271	■ 1.000	S^{D-1}	S^{D-1}	The angular part of the Laplacian operator, $\mathcal{L}^2$, acts on spherical harmonics on S^{D-1} ,
gilmore-lie-groups_ F02796	271	■ 1.000	$\mathcal{Y}^l(S^{D-1})$	$\mathcal{Y}^l(S^{D-1})$	$\mathcal{Y}^l(S^{D-1})$. These spherical harmonics are eigenfunctions of this (Laplace–Beltrami)
gilmore-lie-groups_ F02797	271	■ 1.000	$-\left[(l+\alpha)^2 - \alpha^2\right]$	$-\left[(l+\alpha)^2 - \alpha^2\right]$	operator with eigenvalue $-\left[(l+\alpha)^2 - \alpha^2\right]$, and α is a quantity that depends on the
gilmore-lie-groups_ F02798	271	■ 0.709	$SO(D)$	$SO(D)$	Lie algebra of $SO(D)$: it is half the sum over all positive roots of the algebra. For
gilmore-lie-groups_ F02799	271	■ 1.000	$\alpha = D-2$	$\alpha = D-2$	$\alpha = D-2$.
gilmore-lie-groups_ F02800	271	■ 1.000	$\psi(\mathbf{x}) = \left(1/r^{(D-1)/2}\right) \mathcal{Y}^l$	$\psi(\mathbf{x}) = \left(1/r^{(D-1)/2}\right) \mathcal{Y}^l$	a. Show that $\psi(\mathbf{x}) = (1/r^{(D-1)/2}) \mathcal{Y}^l(\text{angles})$ is a clever ansatz that reduces the
gilmore-lie-groups_ F02801	272	■ 1.000	$a - 1/r^2$	$a - 1/r^2$	a $-1/r^2$ perturbation. Show that the bound state energy and scattering phase shifts
gilmore-lie-groups_ F02802	272	■ 0.999	$l(l+1) \rightarrow l(l+1) - \mu_l$	$l(l+1) \rightarrow l(l+1) - \mu_l$	are given by the substitution $l(l+1) \rightarrow l(l+1) - \mu_l$. Argue that electrons in the s
gilmore-lie-groups_ F02803	272	■ 0.999	s	s	are given by the substitution $l(l+1) \rightarrow l(l+1) - \mu_l$. Argue that electrons in the s
gilmore-lie-groups_ F02804	272	■ 0.953	$\mu_0 \gg \mu_1 > \dots$	$\mu_0 \gg \mu_1 > \dots$	d -state electrons, . . .) so that $\mu_0 \gg \mu_1 > \dots$.
gilmore-lie-groups_ F02805	272	■ 1.000	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i = -L_{ji}$	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i = -L_{ji}$	angular momentum operators $L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i = -L_{ji}$.
gilmore-lie-groups_ F02806	272	■ 1.000	$Q_{ij} =$	$Q_{ij} =$	angular momentum operators together with the quadrupole tensor operators $Q_{ij} =$
gilmore-lie-groups_ F02807	272	■ 1.000	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$.
gilmore-lie-groups_ F02808	272	■ 1.000	$[a_i, a_j^\dagger] = 1$	$[a_i, a_j^\dagger] = 1$	as the single boson operators $a_i^\dagger$ and a_j , as well as their commutator $[a_i, a_j^\dagger] = 1$.
gilmore-lie-groups_ F02809	272	■ 1.000	$a_i a_j$	$a_i a_j$	$a_i a_j$. Show that this algebra is simple and describe its structure. Show that this
gilmore-lie-groups_ F02810	273	■ 0.999	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	The electromagnetic field $\mathbf{E}(\mathbf{x}, t)$, $\mathbf{B}(\mathbf{x}, t)$ is determined by Maxwell's
gilmore-lie-groups_ F02811	273	■ 0.935	$\mathbf{E}(\mathbf{k}), \mathbf{B}(\mathbf{k})$	$\mathbf{E}(\mathbf{k}), \mathbf{B}(\mathbf{k})$	four linear constraints on the six amplitudes $\mathbf{E}(\mathbf{k}), \mathbf{B}(\mathbf{k})$. Why? At a more
gilmore-lie-groups_ F02812	274	■ 1.000	$\mathbf{E}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t)$	and magnetic fields, $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, or their components after Fourier transfor-
gilmore-lie-groups_ F02813	274	■ 1.000	$\mathbf{B}(\mathbf{x}, t)$	$\mathbf{B}(\mathbf{x}, t)$	and magnetic fields, $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, or their components after Fourier transfor-
gilmore-lie-groups_ F02814	274	■ 1.000	$\mathbf{E}(\mathbf{k})$	$\mathbf{E}(\mathbf{k})$	mation, $\mathbf{E}(\mathbf{k})$ and $\mathbf{B}(\mathbf{k})$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_ F02815	274	■ 1.000	$\mathbf{B}(\mathbf{k})$	$\mathbf{B}(\mathbf{k})$	mation, $\mathbf{E}(\mathbf{k})$ and $\mathbf{B}(\mathbf{k})$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.

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gilmore-lie-groups_ F02816	274	■ 1.000	$\mathbf{k} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	$k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	tion, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $\mathbf{k} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_ F02817	274	■ 0.798	k_4	k_4	Here $\mathbf{k}$ is essentially a three-momentum vector and k_4 is essentially an energy.
gilmore-lie-groups_ F02818	274	■ 1.000	$j=1$	$j = 1$	straints so derived reduce, for $j=1$, to Maxwell's equations. This derivation is
gilmore-lie-groups_ F02819	274	■ 1.000	$\mathbf{k} \cdot \mathbf{k} = 0$	$k \cdot k = 0$	ton four-vector k no longer obeys $k \cdot k = 0$. In this case the manifestly covariant
gilmore-lie-groups_ F02820	275	■ 0.999	$(x, y, z, i c t)$	$(x, y, z, i c t)$	This requires that the coordinates $(x, y, z, i c t)$ and $(x, y, z, i c t)'$ for observers S
gilmore-lie-groups_ F02821	275	■ 0.999	$(x, y, z, i c t)^{\prime}$	$(x, y, z, i c t)'$	This requires that the coordinates $(x, y, z, i c t)$ and $(x, y, z, i c t)'$ for observers S
gilmore-lie-groups_ F02822	275	■ 0.865	Λ	Λ	The 4×4 matrix transformations Λ belong to the Lie group $O(3, 1)$. The infinites-
gilmore-lie-groups_ F02823	275	■ 1.000	$\mathbf{k} \cdot \mathbf{a} = \Lambda \mathbf{k} \cdot \mathbf{a}$	$k \cdot a = \Lambda k \cdot a$	Homogeneous Lorentz transformations leave invariant inner products: $\mathbf{k} \cdot \mathbf{a} = \Lambda \mathbf{k} \cdot \mathbf{a}$.
gilmore-lie-groups_ F02824	275	■ 1.000	Λa	Λa	Λa , where k and a are four vectors and $\Lambda \in O(3, 1)$. The infinitesimal generators
gilmore-lie-groups_ F02825	275	■ 1.000	$\Lambda \in O(3, 1)$	$\Lambda \in O(3, 1)$	Λa , where k and a are four vectors and $\Lambda \in O(3, 1)$. The infinitesimal generators
gilmore-lie-groups_ F02826	275	■ 0.936	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$ satisfy the following commutation relations:
gilmore-lie-groups_ F02827	276	■ 0.988	$\mathbf{a} = (x, y, z, c t)$	$a = (x, y, z, c t)$	tion can be written as a 5×5 matrix, in terms of the 4-vector $a = (x, y, z, c t)$:
gilmore-lie-groups_ F02828	276	■ 0.995	$(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial (c t))$	$(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial (c t))$	subgroup are $(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial (c t))$.
gilmore-lie-groups_ F02829	276	■ 1.000	$\{\Lambda, 0\}$	$\{\Lambda, 0\}$	groups. These are the subgroup of homogeneous Lorentz transformations $\{\Lambda, 0\}$
gilmore-lie-groups_ F02830	276	■ 1.000	$\{I, a\}$	$\{I, a\}$	and the invariant subgroup of translations $\{I, a\}$. Both their representations play a
gilmore-lie-groups_ F02831	276	■ 1.000	Γ^k	Γ^k	We may define a basis state for the one-dimensional representation Γ^k of $\{I, a\}$ as
gilmore-lie-groups_ F02832	276	■ 1.000	$ k\rangle$	$ k\rangle$	$ k\rangle$:
gilmore-lie-groups_ F02833	277	■ 1.000	$D_2 = A_1 + A_1$	$D_2 = A_1 + A_1$	The Lie algebra $D_2 = A_1 + A_1$ is semisimple: it is the direct sum of two simple
gilmore-lie-groups_ F02834	277	■ 1.000	$\mathbf{J}^{(1)}$	$\mathbf{J}^{(1)}$	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$
gilmore-lie-groups_ F02835	277	■ 1.000	D^j	D^j	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$
gilmore-lie-groups_ F02836	277	■ 1.000	$\mathbf{J}^{(2)}$	$\mathbf{J}^{(2)}$	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$
gilmore-lie-groups_ F02837	277	■ 0.970	$2j' + 1$	$2j' + 1$	has $2j' + 1$ dimensional irreducible representations $D^{j'}$. Any element in $SO(3, 1)$
gilmore-lie-groups_ F02838	277	■ 0.970	$D^{j'}$	$D^{j'}$	has $2j' + 1$ dimensional irreducible representations $D^{j'}$. Any element in $SO(3, 1)$
gilmore-lie-groups_ F02839	277	■ 1.000	$(2j + 1)(2j' + 1)$	$(2j + 1)(2j' + 1)$	can be expressed in a $(2j + 1)(2j' + 1)$ dimensional representation $D^{jj'}$ as follows
gilmore-lie-groups_ F02840	277	■ 1.000	$D^{jj'}$	$D^{jj'}$	can be expressed in a $(2j + 1)(2j' + 1)$ dimensional representation $D^{jj'}$ as follows
gilmore-lie-groups_ F02841	278	■ 1.000	$\mathbf{J}^{(j)}$	$\mathbf{J}^{(j)}$	where $\mathbf{J}^{(j)}$ are the three $(2j + 1) \times (2j + 1)$ angular momentum matrices. The
gilmore-lie-groups_ F02842	278	■ 0.990	$D^{jj'}(\Lambda)$	$D^{jj'}(\Lambda)$	Basis states for the action of Λ through the representation $D^{jj'}(\Lambda)$ can be com-

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gilmore-lie-groups_ F02843	278	■ 1.000	$S_0(3) \subset S_0(3,1)$	$SO(3) \subset SO(3,1)$	Under restriction to the subgroup $SO(3) \subset SO(3,1)$ this representation is reducible
gilmore-lie-groups_ F02844	278	■ 1.000	$j^{\prime}=0$	$j' = 0$	This representation remains irreducible only if $j' = 0$ or $j = 0$.
gilmore-lie-groups_ F02845	278	■ 1.000	$j=0$	$j = 0$	This representation remains irreducible only if $j' = 0$ or $j = 0$.
gilmore-lie-groups_ F02846	278	■ 1.000	$T_{\mu\nu}(x)$	$T_{\mu\nu}(x)$	A field $T_{\mu\nu}(x)$ is said to be manifestly covariant (obviously covariant) under
gilmore-lie-groups_ F02847	278	■ 1.000	$\Lambda \in S_0(3,1)$	$\Lambda \in SO(3,1)$	transformations of the homogeneous Lorentz group $\Lambda \in SO(3,1)$ if
gilmore-lie-groups_ F02848	278	■ 1.000	$x^{\prime}=x \Lambda^{-1}$	$x' = x\Lambda^{-1}$	nate system changes, the coordinates of the fixed point are changed by $x' = x\Lambda^{-1}$.
gilmore-lie-groups_ F02849	279	■ 0.774	$\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array} \right\rangle$	$\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array} \right\rangle$	$ _{\mu}^{j j'}\rangle$ separately.
gilmore-lie-groups_ F02855	280	■ 1.000	$\{\Lambda, a\}$	$\{\Lambda, a\}$	Suppose we have a representation of $\{\Lambda, a\}$ that is unitary and irreducible. Un-
gilmore-lie-groups_ F02856	280	■ 1.000	$\Gamma^k(\{I, a\})$	$\Gamma^k(\{I, a\})$	$\Gamma^k(\{I, a\})$ of $\{I, a\}$. The basis states are $ k; \xi\rangle$, where k is defined by the action of
gilmore-lie-groups_ F02857	280	■ 1.000	$ k ; \xi\rangle$	$ k; \xi\rangle$	$\Gamma^k(\{I, a\})$ of $\{I, a\}$. The basis states are $ k; \xi\rangle$, where k is defined by the action of
gilmore-lie-groups_ F02858	280	■ 1.000	ξ	ξ	and ξ is a helicity index that distinguishes different states with the same four-
gilmore-lie-groups_ F02859	280	■ 0.999	$k^{\prime}=\Lambda k$	$k' = \Lambda k$	subspace of states parameterized by $k' = \Lambda k$
gilmore-lie-groups_ F02860	280	■ 1.000	$M_{\xi^{\prime} \xi}(\Lambda)$	$M_{\xi'\xi}(\Lambda)$	where $M_{\xi'\xi}(\Lambda)$ is a matrix that remains to be determined.
gilmore-lie-groups_ F02861	280	■ 1.000	$k^{\prime}$	k'	k' with
gilmore-lie-groups_ F02862	281	■ 1.000	$M(\Lambda)$	$M(\Lambda)$	are present also. To construct the matrix $M(\Lambda)$, we choose one particular four-vector
gilmore-lie-groups_ F02863	281	■ 1.000	$k^{\{0\}}$	k^0	k^0 for each of the possible cases
gilmore-lie-groups_ F02864	281	■ 1.000	$\left k^{\{0\}} ; \xi\right\rangle$	$ k^0; \xi\rangle$	The effect of a homogeneous Lorentz transformation on the state $ k^0; \xi\rangle$ is de-
gilmore-lie-groups_ F02865	281	■ 0.822	$H_{k^{\{0\}}}$	H_{k^0}	That is, H_{k^0} is the stability subgroup of the little vector k^0 and C_k is a coset repre-
gilmore-lie-groups_ F02866	281	■ 0.822	$C_{\{k\}}$	C_k	That is, H_{k^0} is the stability subgroup of the little vector k^0 and C_k is a coset repre-
gilmore-lie-groups_ F02867	282	■ 1.000	$\theta_1=\theta_2=b_3=0, \theta_3, b_1, b_2$	$\theta_1 = \theta_2 = b_3 = 0, \theta_3, b_1, b_2$	The subalgebra leaving k^0 fixed is defined by $\theta_1 = \theta_2 = b_3 = 0, \theta_3, b_1, b_2$ arbi-
gilmore-lie-groups_ F02868	282	■ 1.000	$(0,0,1,-i)=T(0,0,1,+i)$	$(0,0,1,-i) = T(0,0,1,+i)$	$(0,0,1,-i) = T(0,0,1,+i)$ the infinitesimal generators are $Y_1 = J_1 - K_2, Y_2 =$
gilmore-lie-groups_ F02869	282	■ 1.000	$Y_1=J_1-K_2, Y_2=$	$Y_1 = J_1 - K_2, Y_2 =$	$(0,0,1,-i) = T(0,0,1,+i)$ the infinitesimal generators are $Y_1 = J_1 - K_2, Y_2 =$
gilmore-lie-groups_ F02870	282	■ 1.000	$J_2+K_1, Y_3=J_3$	$J_2 + K_1, Y_3 = J_3$	$J_2 + K_1, Y_3 = J_3$.
gilmore-lie-groups_ F02871	282	■ 1.000	$\mathbf{b}=0$	$\mathbf{b} = 0$	The subalgebra defined by $\mathbf{b} = 0$ is spanned by the angular momentum operators
gilmore-lie-groups_ F02872	283	■ 1.000	$D_{\xi^{\prime} \xi}\left(H_{k^{\{0\}}}\right)$	$D_{\xi'\xi}(H_{k^0})$	irreducible if and only if the representation $D_{\xi'\xi}(H_{k^0})$ of the little group is unitary
gilmore-lie-groups_ F02873	283	■ 0.999	$k \cdot k>0$	$k \cdot k > 0$	were described in Problem 5 of Chapter 11. Since $k \cdot k > 0$ describes negative mass

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gilmore-lie-groups_ F02874	283	■ 1.000	$j=0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$	$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	They are described by an integer or half-integer: $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. The angular
gilmore-lie-groups_ F02875	283	■ 1.000	$\kappa=\left(\kappa_1, \kappa_2\right)$	$\kappa = (\kappa_1, \kappa_2)$	can be labeled by a vector $\kappa = (\kappa_1, \kappa_2)$ in a two-dimensional Euclidean space,
gilmore-lie-groups_ F02876	283	■ 1.000	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$. If a state $ \kappa\rangle$ is in one such representation, so are all states
gilmore-lie-groups_ F02877	283	■ 1.000	$ \kappa\rangle$	$ \kappa\rangle$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$. If a state $ \kappa\rangle$ is in one such representation, so are all states
gilmore-lie-groups_ F02878	283	■ 0.999	$\left \kappa^{\prime}\right\rangle$	$ \kappa'\rangle$	$ \kappa'\rangle$ for which $\kappa' \cdot \kappa' = \kappa \cdot \kappa$. That is, $\kappa' = (\kappa'_1, \kappa'_2)$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_ F02879	283	■ 0.999	$\kappa^{\prime} \cdot \kappa^{\prime}=\kappa \cdot \kappa$	$\kappa' \cdot \kappa' = \kappa \cdot \kappa$	$ \kappa'\rangle$ for which $\kappa' \cdot \kappa' = \kappa \cdot \kappa$. That is, $\kappa' = (\kappa'_1, \kappa'_2)$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_ F02880	283	■ 0.999	$\kappa^{\prime}=\left(\kappa_1^{\prime}, \kappa_2^{\prime}\right)$	$\kappa' = (\kappa'_1, \kappa'_2)$	$ \kappa'\rangle$ for which $\kappa' \cdot \kappa' = \kappa \cdot \kappa$. That is, $\kappa' = (\kappa'_1, \kappa'_2)$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_ F02881	283	■ 1.000	$\kappa^{\prime}=R(\theta) \kappa$	$\kappa' = R(\theta)\kappa$	rotation: $\kappa' = R(\theta)\kappa$. The invariant length $\kappa \cdot \kappa$ parameterizes the representation.
gilmore-lie-groups_ F02882	283	■ 1.000	$\kappa \cdot \kappa$	$\kappa \cdot \kappa$	rotation: $\kappa' = R(\theta)\kappa$. The invariant length $\kappa \cdot \kappa$ parameterizes the representation.
gilmore-lie-groups_ F02883	283	■ 0.999	$\kappa \cdot \kappa>0 \quad \quad$	$\kappa \cdot \kappa > 0$	(i) $\kappa \cdot \kappa > 0$ little group = Identity
gilmore-lie-groups_ F02884	283	■ 0.996	$\kappa \cdot \kappa=0 \quad \quad$	$\kappa \cdot \kappa = 0$	(ii) $\kappa \cdot \kappa = 0$ little group = $ISO(2)$
gilmore-lie-groups_ F02885	283	■ 0.996	$=ISO(2)$	$= ISO(2)$	(ii) $\kappa \cdot \kappa = 0$ little group = $ISO(2)$
gilmore-lie-groups_ F02886	284	■ 1.000	κ^2	κ^2	The first case presents us with two problems. First, κ^2 is a continuous quantum
gilmore-lie-groups_ F02887	284	■ 1.000	$\kappa^2>0$	$\kappa^2 > 0$	$\kappa^2 > 0$ there must be an infinite number of states with this same continuous index,
gilmore-lie-groups_ F02888	284	■ 1.000	$\kappa=0$	$\kappa = 0$	for each four-momentum value. Therefore we require $\kappa = 0$. This leaves us with the
gilmore-lie-groups_ F02889	284	■ 0.961	$\left(Y_1 \rightarrow 0, Y_2 \rightarrow 0\right)$	$(Y_1 \rightarrow 0, Y_2 \rightarrow 0)$	following physically allowable representations of the little group $(Y_1 \rightarrow 0, Y_2 \rightarrow 0)$
gilmore-lie-groups_ F02890	284	■ 1.000	$\left(j, j^{\prime}\right)$	(j, j')	particle with transformation indices (j, j') contains all states of the form
gilmore-lie-groups_ F02891	285	■ 1.000	$\left\{H_{k^0}, 0\right\}$	$\{H_{k^0}, 0\}$	A. $\{H_{k^0}, 0\}$ on $ k^0; \xi\rangle$
gilmore-lie-groups_ F02892	285	■ 1.000	$H_{k^0}=\operatorname{EXP}\left(\Theta J_3+\theta_1 Y_1+\theta_2 Y_2\right)$	$H_{k^0} = \operatorname{EXP}(\Theta J_3 + \theta_1 Y_1 + \theta_2 Y_2)$	where $H_{k^0} = \operatorname{EXP}(\Theta J_3 + \theta_1 Y_1 + \theta_2 Y_2)$.
gilmore-lie-groups_ F02893	285	■ 0.494	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j j^{\prime}}\right\rangle\right _{\mu^{\prime}}^{\mu}$	$ k^0\rangle \Big _{\mu}^{j j'} \Big\rangle_{\mu'}$	B. $\{H_{k^0}, 0\}$ on $ k^0\rangle \Big _{\mu}^{j j'} \Big\rangle_{\mu'}$ The little group maps k^0 to k^0 but acts in a nontrivial way
gilmore-lie-groups_ F02894	285	■ 0.119	$\left.\left.\left.\left k^0\right\rangle\right _{j_0}^{j_0}\right\rangle\right _{j_0}^{j_0}$	$ k^0\rangle \Big _{j_0}^{j_0} \Big\rangle_{j_0}$	The state $ k^0\rangle \Big _{j_0}^{j_0} \Big\rangle_{j_0}$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_ F02895	285	■ 0.119	$\xi>0$	$\xi > 0$	The state $ k^0\rangle \Big _{j_0}^{j_0} \Big\rangle_{j_0}$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_ F02896	285	■ 0.119	$j=+\xi$	$j = +\xi$	The state $ k^0\rangle \Big _{j_0}^{j_0} \Big\rangle_{j_0}$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_ F02897	285	■ 0.983	$\left.\left.\left.\left k^0\right\rangle\right _{0}^{j_0}\right\rangle\right _{-j_0}^{j_0}$	$ k^0\rangle \Big _{0}^{j_0} \Big\rangle_{-j_0}$	The state $ k^0\rangle \Big _{0}^{j_0} \Big\rangle_{-j_0}$ transforms identically to $ k^0; \xi\rangle$ if $\xi < 0$ and $j' = -\xi$.
gilmore-lie-groups_ F02898	285	■ 0.983	$\xi<0$	$\xi < 0$	The state $ k^0\rangle \Big _{0}^{j_0} \Big\rangle_{-j_0}$ transforms identically to $ k^0; \xi\rangle$ if $\xi < 0$ and $j' = -\xi$.

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gilmore-lie-groups_F02899	285	■ 0.983	$j^{\{\prime\}}=-\xi$	$j' = -\xi$	The state $ k^0\rangle _0^{\dot{0} \dot{j}'}$ transforms identically to $ k^0;\xi\rangle$ if $\xi < 0$ and $j' = -\xi$.
gilmore-lie-groups_F02900	286	■ 0.560	$ k\rangle\left \begin{array}{cc}j & j^{\{\prime\}} \\ \mu & \mu^{\{\prime\}}\end{array}\right\rangle$	$ k\rangle\left \begin{array}{cc}j & j' \\ \mu & \mu'\end{array}\right\rangle$	basis states $ k;\xi\rangle$ or the direct product states $ k\rangle _{\mu}^{\dot{j} \dot{j}'}$:
gilmore-lie-groups_F02901	286	■ 1.000	$\langle k ; \xi \mid \psi \rangle$	$\langle k ; \xi \mid \psi \rangle$	The amplitudes of the projection of $ \psi\rangle$ onto the basis states are $\langle k ; \xi \psi \rangle$ in the first
gilmore-lie-groups_F02902	286	■ 0.741	$\langle k ; \mu^{\{\prime\}} \mid \psi \rangle$	$\langle k ; \mu^{\dot{j} \dot{j}'} \mid \psi \rangle$	case and $\langle k ; \mu^{\dot{j} \dot{j}'} \psi \rangle$ in the second. In both cases the sum extends over all k vectors
gilmore-lie-groups_F02903	286	■ 1.000	$\Lambda k \cdot \Lambda k = 0, k \neq 0$	$\Lambda k \cdot \Lambda k = 0, k \neq 0$	for which $\Lambda k \cdot \Lambda k = 0, k \neq 0$. In the first case the sum extends over the appropriate
gilmore-lie-groups_F02904	286	■ 1.000	$\xi = \pm 1$	$\xi = \pm 1$	helicity states ξ ($\xi = \pm 1$ for photons). In the second case the sum extends over the
gilmore-lie-groups_F02905	286	■ 0.712	$\mu, \mu^{\{\prime\}}: -j \leq \mu \leq +j, -j^{\{\prime\}} \leq \mu^{\{\prime\}} \leq +j^{\{\prime\}}$	$\mu, \mu': -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$	appropriate values of $\mu, \mu': -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$.
gilmore-lie-groups_F02906	286	■ 1.000	$\xi = j > 0$	$\xi = j > 0$	We discuss the positive helicity state $\xi = j > 0$ first. The amplitude $\langle k^0; j \psi \rangle$
gilmore-lie-groups_F02907	286	■ 1.000	$\langle k^0 ; j \mid \psi \rangle$	$\langle k^0 ; j \mid \psi \rangle$	We discuss the positive helicity state $\xi = j > 0$ first. The amplitude $\langle k^0; j \psi \rangle$
gilmore-lie-groups_F02908	286	■ 1.000	$ k^0 ; j \rangle$	$ k^0 ; j \rangle$	of the state $ k^0; j\rangle$ in any physical state $ \psi\rangle$ may be arbitrary. This is simply the
gilmore-lie-groups_F02909	286	■ 0.993	$\langle k^0 ; j_0^0 \mid \psi \rangle$	$\langle k^0 ; j_0^0 \mid \psi \rangle$	$\langle k^0 ; j_0^0 \psi \rangle$ in the same physical state $ \psi\rangle$ is the same. The amplitudes of the
gilmore-lie-groups_F02910	286	■ 0.999	$\langle k^0 ; m_0^0 \mid \psi \rangle, m \neq j$	$\langle k^0 ; m_0^0 \mid \psi \rangle, m \neq j$	states $\langle k^0 ; m_0^0 \psi \rangle, m \neq j$, must all vanish. These states are all superfluous –
gilmore-lie-groups_F02911	286	■ 0.719	$\{\cdot\}$	$\{\cdot\}$	The matrix within the bracket $\{\cdot\}$ is diagonal, with the coefficient $(j-j)k_3^0 = 0$
gilmore-lie-groups_F02912	286	■ 0.719	$(j-j) k_3^0=0$	$(j-j)k_3^0 = 0$	The matrix within the bracket $\{\cdot\}$ is diagonal, with the coefficient $(j-j)k_3^0 = 0$
gilmore-lie-groups_F02913	286	■ 0.999	$(m-j) k_3^0$	$(m-j)k_3^0$	multiplying the allowed amplitude $\langle k^0 ; j_0^0 \psi \rangle$ and nonzero coefficients $(m-j)k_3^0$
gilmore-lie-groups_F02914	286	■ 1.000	$\langle k^0 ; m_0^0 \mid \psi \rangle$	$\langle k^0 ; m_0^0 \mid \psi \rangle$	multiplying the amplitudes $\langle k^0 ; m_0^0 \psi \rangle$. Since $(m-j)k_3^0 \neq 0$, the amplitudes that
gilmore-lie-groups_F02915	286	■ 1.000	$(m-j) k_3^0 \neq 0$	$(m-j)k_3^0 \neq 0$	multiplying the amplitudes $\langle k^0 ; m_0^0 \psi \rangle$. Since $(m-j)k_3^0 \neq 0$, the amplitudes that
gilmore-lie-groups_F02916	286	■ 0.961	$m \neq j$	$m \neq j$	are absent in the description of a physical state $(m \neq j)$ must vanish.
gilmore-lie-groups_F02917	286	■ 1.000	$\xi = -j$	$\xi = -j$	For the negative helicity states $\xi = -j$ we have by a completely similar argument
gilmore-lie-groups_F02918	286	■ 0.556	$ k^0\rangle\left \begin{array}{cc}j & j^{\{\prime\}} \\ \mu & \mu^{\{\prime\}}\end{array}\right\rangle$	$ k^0\rangle\left \begin{array}{cc}j & j' \\ \mu & \mu'\end{array}\right\rangle$	and the subspace $ k^0\rangle _{\mu}^{\dot{j} \dot{j}'}$ into the subspace $ k\rangle _{\nu}^{\dot{j} \dot{j}'}$ through the following non-
gilmore-lie-groups_F02919	286	■ 0.556	$ k\rangle\left \begin{array}{cc}j & j^{\{\prime\}} \\ v & v^{\{\prime\}}\end{array}\right\rangle$	$ k\rangle\left \begin{array}{cc}j & j' \\ v & v'\end{array}\right\rangle$	and the subspace $ k^0\rangle _{\mu}^{\dot{j} \dot{j}'}$ into the subspace $ k\rangle _{\nu}^{\dot{j} \dot{j}'}$ through the following non-
gilmore-lie-groups_F02920	287	■ 0.847	$\langle k ; \mu^{\{\prime\}} \mid \psi \rangle$	$\langle k ; \mu^{\dot{j} \dot{j}'} \mid \psi \rangle$	The condition on the amplitude $\langle k ; \mu^{\dot{j} \dot{j}'} \psi \rangle$ in the subspace $ k\rangle$ is related to the
gilmore-lie-groups_F02921	287	■ 1.000	$ k^0\rangle$	$ k^0\rangle$	conditions (15.54) and (15.55) in the subspace $ k^0\rangle$ by a similarity transformation
gilmore-lie-groups_F02922	287	■ 1.000	$\xi = j$	$\xi = j$	For the positive helicity state $\xi = j$ the matrix $M^{jj'}(k^0) = M^{j0}(k^0)$ is given in

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gilmore-lie-groups_F02923	287	■ 1.000	$M^{\{j\}^{\prime}}\left(k^{\{0\}}\right)=M^{\{j\}^{\{0\}}}\left(k^{\{0\}}\right)$	$M^{jj'}\left(k^0\right)=M^{j0}\left(k^0\right)$	For the positive helicity state $\xi=j$ the matrix $M^{jj'}(k^0)=M^{j0}(k^0)$ is given in
gilmore-lie-groups_F02924	287	■ 0.943	$ k\rangle\left \begin{array}{ll}1&0\\ \mu&0\end{array}\right\rangle&\theta\ll\mu&\theta$	$ k\rangle\left \begin{array}{ll}1&0\\ \mu&0\end{array}\right\rangle$	as the linear constraint that must be satisfied in the subspace $ k\rangle\left \begin{array}{ll}1&0\\ \mu&0\end{array}\right\rangle$. The negative
gilmore-lie-groups_F02925	287	■ 1.000	$\langle x k\rangle\langle k;\{j\}_m^{\{0\}} \psi\rangle$	$\langle x k\rangle\langle k;_m^j \psi\rangle$	If we define complex fields $\langle x k\rangle\langle k;_m^j \psi\rangle$ by $\psi_{jm}(x)$, ($j=1, m=+1, 0, -1$
gilmore-lie-groups_F02926	287	■ 1.000	$\psi_{jm}(x), (j=1, m=+1, 0, -1$	$\psi_{jm}(x), (j=1, m=+1, 0, -1$	If we define complex fields $\langle x k\rangle\langle k;_m^j \psi\rangle$ by $\psi_{jm}(x)$, ($j=1, m=+1, 0, -1$
gilmore-lie-groups_F02927	288	■ 1.000	$D^{\{0\}}$	D^{01}	representation D^{01} and are
gilmore-lie-groups_F02928	289	■ 0.999	$k^{\{0\}}=(0,0,1,i)$	$k^0=(0,0,1,i)$	$k^0=(0,0,1,i)$ the only nonvanishing component of the field, $\langle k;_m^{j=1} \psi\rangle$, is the
gilmore-lie-groups_F02929	289	■ 0.999	$\langle k;_m^{j=1} \psi\rangle$	$\langle k;_m^{j=1} \psi\rangle$	$k^0=(0,0,1,i)$ the only nonvanishing component of the field, $\langle k;_m^{j=1} \psi\rangle$, is the
gilmore-lie-groups_F02930	289	■ 1.000	$m=+1$	$m=+1$	component with $m=+1$ (cf., Eq. (15.54)). The coordinates are $-(v_x+iv_y)$. The
gilmore-lie-groups_F02931	289	■ 1.000	$-(v_x+iv_y)$	$-(v_x+iv_y)$	component with $m=+1$ (cf., Eq. (15.54)). The coordinates are $-(v_x+iv_y)$. The
gilmore-lie-groups_F02932	289	■ 1.000	$\mathbf{v}=(v_x,v_y,0)$	$\mathbf{v}=(v_x,v_y,0)$	vector $\mathbf{v}=(v_x,v_y,0)$ represented by this coordinate is orthogonal to the spacial part
gilmore-lie-groups_F02933	289	■ 1.000	$\mathbf{k}^{\{0\}}=(0,0,1): \mathbf{k}^{\{0\}}\cdot\mathbf{v}=0$	$\mathbf{k}^0=(0,0,1): \mathbf{k}^0\cdot\mathbf{v}=0$	of the little vector $\mathbf{k}^0=(0,0,1): \mathbf{k}^0\cdot\mathbf{v}=0$. Under boosts B_z and rotations, the non-
gilmore-lie-groups_F02934	289	■ 1.000	B_z	B_z	of the little vector $\mathbf{k}^0=(0,0,1): \mathbf{k}^0\cdot\mathbf{v}=0$. Under boosts B_z and rotations, the non-
gilmore-lie-groups_F02935	289	■ 0.976	$\mathbf{k}$	$\mathbf{k}$	vanishing component of the boosted field is orthogonal to the spacial part of the $\mathbf{k}$
gilmore-lie-groups_F02936	289	■ 1.000	$\mathbf{k}\cdot\mathbf{v}(\mathbf{k})=0$	$\mathbf{k}\cdot\mathbf{v}(\mathbf{k})=0$	vector: $\mathbf{k}\cdot\mathbf{v}(\mathbf{k})=0$. Backtransforming from the Fourier to the spacial representation,
gilmore-lie-groups_F02937	290	■ 0.999	$\nabla\cdot\mathbf{E}=0$	$\nabla\cdot\mathbf{E}=0$	equations $\nabla\cdot\mathbf{E}=0$ and $\nabla\cdot\mathbf{B}=0$. Show this.
gilmore-lie-groups_F02938	290	■ 0.999	$\nabla\cdot\mathbf{B}=0$	$\nabla\cdot\mathbf{B}=0$	equations $\nabla\cdot\mathbf{E}=0$ and $\nabla\cdot\mathbf{B}=0$. Show this.
gilmore-lie-groups_F02939	290	■ 1.000	$\mathbf{x}(j)$	$\mathbf{x}(j)$	clearly expressed in the “manifestly covariant representation.” If particle j at $\mathbf{x}(j)$ has
gilmore-lie-groups_F02940	290	■ 1.000	e_j	e_j	electric charge e_j and magnetic charge m_j , the electric and magnetic charge densities
gilmore-lie-groups_F02941	290	■ 1.000	m_j	m_j	electric charge e_j and magnetic charge m_j , the electric and magnetic charge densities
gilmore-lie-groups_F02948	290	■ 0.997	T, P, TP	T, P, TP	T, P, TP . The result is unique up to scale factor:
gilmore-lie-groups_F02949	290	■ 1.000	$1/c$	$1/c$	The factor 4π is the surface area of the unit sphere in R^3 , and the factor $1/c$ on the
gilmore-lie-groups_F02950	291	■ 0.867	$\phi=\pi/2$	$\phi=\pi/2$	In particular, show that for $\phi=\pi/2$ this is the dual transformation $(\mathbf{B}, \mathbf{E})\rightarrow$
gilmore-lie-groups_F02951	291	■ 0.867	$(\mathbf{B}, \mathbf{E})\rightarrow$	$(\mathbf{B}, \mathbf{E})\rightarrow$	In particular, show that for $\phi=\pi/2$ this is the dual transformation $(\mathbf{B}, \mathbf{E})\rightarrow$
gilmore-lie-groups_F02952	291	■ 1.000	$(*)=0$	$(*)=0$	$(*)=0$, for $*$ = anyvector. Show
gilmore-lie-groups_F02953	291	■ 1.000	$*$	$*$	$(*)=0$, for $*$ = anyvector. Show
gilmore-lie-groups_F02954	291	■ 0.998	$D^{\{j\}^{\prime}+\{j\}^{\prime}}(\Lambda)$	$D^{jj'+j'j}(\Lambda)$	resentation $D^{jj'+j'j}(\Lambda)$, with $j-j'=\pm 2$. In the case with $(j, j')=(2, 0)$ a curl
gilmore-lie-groups_F02955	291	■ 0.998	$j-j^{\prime}=\pm 2$	$j-j'=\pm 2$	resentation $D^{jj'+j'j}(\Lambda)$, with $j-j'=\pm 2$. In the case with $(j, j')=(2, 0)$ a curl

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gilmore-lie-groups_ F02956	291	■ 0.998	$\left(j, j^{\prime}\right)=(2,0)$	$(j, j') = (2, 0)$	resentation $D^{jj'+j'j}(\Lambda)$, with $j - j' = \pm 2$. In the case with $\boxed{j, j') = (2, 0)}$ a curl
gilmore-lie-groups_ F02957	291	■ 0.990	$\mathbf{G}_{\mathbf{e}}$	$\mathbf{G}_{\mathbf{e}}$	The fields $\mathbf{G}_{\mathbf{e}}$ and $\mathbf{G}_{\mathbf{m}}$ are called the gravitoelectric and gravitomagnetic fields. These
gilmore-lie-groups_ F02958	291	■ 0.990	$\mathbf{G}_{\mathbf{m}}$	$\mathbf{G}_{\mathbf{m}}$	The fields $\mathbf{G}_{\mathbf{e}}$ and $\mathbf{G}_{\mathbf{m}}$ are called the gravitoelectric and gravitomagnetic fields. These
gilmore-lie-groups_ F02959	291	■ 0.866	$\mathbf{J} \cdot \nabla$	$\mathbf{J} \cdot \nabla$	latter case the curl operator is $\mathbf{J} \cdot \nabla$, where $\mathbf{J}$ is the 5×5 angular momentum operator:
gilmore-lie-groups_ F02960	292	■ 1.000	$F_{ij}=F_{ji}, \sum_i F_{ii}=0$	$F_{ij} = F_{ji}, \sum_i F_{ii} = 0$	The matrix components obey $\boxed{F_{ij} = F_{ji}, \sum_i F_{ii} = 0}$, and $\partial^i F_{ij} = 0$. The gravitoelec-
gilmore-lie-groups_ F02961	292	■ 1.000	$\partial^i F_{ij}=0$	$\partial^i F_{ij} = 0$	The matrix components obey $F_{ij} = F_{ji}, \sum_i F_{ii} = 0$, and $\partial^i F_{ij} = 0$. The gravitoelec-
gilmore-lie-groups_ F02962	292	■ 0.999	$U_{ij}=\sum_k m_k \left(\mathbf{x}_k(t) \mathbf{x}_k(t)\right)_{ij}$	$U_{ij} = \sum_k m_k (\mathbf{x}_k(t) \mathbf{x}_k(t))_{ij}$	dinates the source term for the gravitoelectric field is $U_{ij} = \sum_k m_k (\mathbf{x}_k(t) \mathbf{x}_k(t))_{ij}$,
gilmore-lie-groups_ F02963	292	■ 0.999	$\mathbf{G}_{\mathbf{m}}+i \mathbf{G}_{\mathbf{e}} \rightarrow e^{i \phi}\left(\mathbf{G}_{\mathbf{m}}+i \mathbf{G}_{\mathbf{e}}\right) \rightarrow e^{i \phi}\left(\mathbf{G}_{\mathbf{m}}+i \mathbf{G}_{\mathbf{e}}\right)$	$\mathbf{G}_{\mathbf{m}} + i \mathbf{G}_{\mathbf{e}} \rightarrow e^{i\phi} (\mathbf{G}_{\mathbf{m}} + i \mathbf{G}_{\mathbf{e}})$	$\boxed{\mathbf{G}_{\mathbf{m}} + i \mathbf{G}_{\mathbf{e}} \rightarrow e^{i\phi} (\mathbf{G}_{\mathbf{m}} + i \mathbf{G}_{\mathbf{e}})}$ and $\mathbf{J}_{\mathbf{m}} + i \mathbf{J}_{\mathbf{e}} \rightarrow e^{i\phi} (\mathbf{J}_{\mathbf{m}} + i \mathbf{J}_{\mathbf{e}})$. Show this.
gilmore-lie-groups_ F02964	292	■ 0.999	$\mathbf{J}_{\mathbf{m}}+i \mathbf{J}_{\mathbf{e}} \rightarrow e^{i \phi}\left(\mathbf{J}_{\mathbf{m}}+i \mathbf{J}_{\mathbf{e}}\right) \rightarrow e^{i \phi}\left(\mathbf{J}_{\mathbf{m}}+i \mathbf{J}_{\mathbf{e}}\right)$	$\mathbf{J}_{\mathbf{m}} + i \mathbf{J}_{\mathbf{e}} \rightarrow e^{i\phi} (\mathbf{J}_{\mathbf{m}} + i \mathbf{J}_{\mathbf{e}})$	$\mathbf{G}_{\mathbf{m}} + i \mathbf{G}_{\mathbf{e}} \rightarrow e^{i\phi} (\mathbf{G}_{\mathbf{m}} + i \mathbf{G}_{\mathbf{e}})$ and $\mathbf{J}_{\mathbf{m}} + i \mathbf{J}_{\mathbf{e}} \rightarrow e^{i\phi} (\mathbf{J}_{\mathbf{m}} + i \mathbf{J}_{\mathbf{e}})$. Show this.
gilmore-lie-groups_ F02965	292	■ 1.000	$D^{j j}(\Lambda)$	$D^{jj}(\Lambda)$	5. Construct the source-free field equations for gravitons for the $D^{jj}(\Lambda)$ representation,
gilmore-lie-groups_ F02966	292	■ 0.871	(J, M)	(J, M)	with $j = 1$. Show that there are seven constraints that correspond to (J, M) with
gilmore-lie-groups_ F02967	292	■ 1.000	$(J, M)=(0,0),(1,0),(1,-1),(2,0),(2 \pm 1)$	$(J, M) = (0, 0), (1, 0), (1, \pm 1), (2, 0), (2 \pm 1)$	$\boxed{(J, M) = (0, 0), (1, 0), (1, \pm 1), (2, 0), (2 \pm 1)}$. What are these equations in the stan-
gilmore-lie-groups_ F02968	293	■ 1.000	$N \lambda$	$N \lambda$	two (distance $N \lambda$), as time goes on and the distance increases the wavelength must
gilmore-lie-groups_ F02969	293	■ 1.000	$N \lambda^{\prime}$	$N \lambda'$	also increase to $N \lambda'$. This redshift source is sometimes confused with the Döppler
gilmore-lie-groups_ F02970	293	■ 1.000	$\rho(\mathbf{x}, t)=\rho(t)$	$\rho(\mathbf{x}, t) = \rho(t)$	a. Assume that the energy density in the universe has the form $\rho(\mathbf{x}, t) = \rho(t)$ (time
gilmore-lie-groups_ F02971	293	■ 1.000	$m_{\mathbf{e}}(t) / M_{\mathbf{p}}(t)$	$m_e(t)/M_p(t)$	b. Estimate the mass dependence of the electron–proton mass ratio $m_e(t)/M_p(t)$.
gilmore-lie-groups_ F02972	293	■ 0.981	$v=\frac{1}{2}\left(m c^2 / \hbar\right) \times\left \left(1 / n_1^2-1 / n_2^2\right)\right $	$v = \frac{1}{2} (mc^2/\hbar) \times (1/n_1^2 - 1/n_2^2) $	the hydrogen atom has frequency $\boxed{v = \frac{1}{2}(mc^2/\hbar) \times (1/n_1^2 - 1/n_2^2) }$ where n_1 and
gilmore-lie-groups_ F02973	293	■ 1.000	H_{α}	H_{α}	and m is the reduced mass of the electron–proton system. Show that H_{α} photons
gilmore-lie-groups_ F02974	294	■ 1.000	$\Phi(x)$	$\Phi(x)$	Here $\Phi(x)$ is the local Newtonian gravitational field. Find a locally linear coordinate
gilmore-lie-groups_ F02975	294	■ 1.000	$S^t g_{\text {grav }} S=g_{\text {flat }}$	$S^t g_{\text{grav}} S = g_{\text{flat}}$	transformation S that brings the curved metric to flat form: $S^t g_{\text{grav}} S = g_{\text{flat}}$. Interpret
gilmore-lie-groups_ F02976	294	■ 1.000	V	V	The integral on the left is over the surface bounding the volume V over which the
gilmore-lie-groups_ F02977	294	■ 0.867	$\mathbf{E}$	$\mathbf{E}$	integral on the right extends, $\mathbf{E}$ is the electric field and ρ is the charge density. For a
gilmore-lie-groups_ F02978	294	■ 0.523	$a, p(x)=q \delta(x)$	$a, p(x) = q\delta(x)$	charge q at the origin of a sphere of radius a , $\rho(x) = q\delta(x)$, The $\mathbf{E}$ field is spherically
gilmore-lie-groups_ F02979	294	■ 1.000	$ \mathbf{E}(a) =q / \mathbf{a} $	$ \mathbf{E}(a) = q/ \mathbf{a} $	By completely similar arguments Gauss’ Law in the plane R^2 gives $\boxed{ \mathbf{E}(a) = q/ \mathbf{a} }$.
gilmore-lie-groups_ F02980	294	■ 0.808	$\oint \mathbf{E} \cdot d \mathbf{S}=\int 2 \pi \rho d A$	$\left(\oint \mathbf{E} \cdot d \mathbf{S}=\int 2 \pi \rho d A\right)$	Assume a Gauss law ($\oint \mathbf{E} \cdot d \mathbf{S} = \int 2\pi \rho d A$) holds on the sphere S^2 . Place a charge
gilmore-lie-groups_ F02981	294	■ 1.000	$a=R \theta$	$a = R\theta$	sphere S^2 (c.f., Fig. 15.1). Show that its distance a from the north pole is $a = R\theta$

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gilmore-lie-groups_F02982	295	■ 1.000	$2 \pi R \sin \theta$	$2\pi R \sin \theta$	and the circumference of a circle of latitude through this point is $2\pi R \sin \theta$. Use
gilmore-lie-groups_F02983	295	■ 1.000	q/a	q/a	Conclude that the field is stronger than the q/a form it would have in a plane.
gilmore-lie-groups_F02984	295	■ 1.000	$a(a=R \theta)$	$a(a = R\theta)$	where a ($a = R\theta$) is the distance from the charge to the observation point.
gilmore-lie-groups_F02985	295	■ 1.000	$a=ct$	$a = ct$	charge depends on the distance from the observation point. In particular, if $a = ct$,
gilmore-lie-groups_F02986	295	■ 1.000	$H^n = SO(n, 1) / SO(n), R^n =$	$H^n = SO(n, 1)/SO(n), R^n =$	sure on the three Riemannian symmetric spaces $H^n \equiv SO(n, 1)/SO(n), R^n =$
gilmore-lie-groups_F02987	295	■ 0.999	$ISO(n) / SO(n)$	$ISO(n)/SO(n)$	$ISO(n)/SO(n)$, and $S^n = SO(n + 1)/SO(n)$ are
gilmore-lie-groups_F02988	295	■ 1.000	$k=(-1,0,+1)$	$k = (-1, 0, +1)$	where $k \equiv (-1, 0, +1)$ for H^n, R^n, S^n and radial coordinates are used:
gilmore-lie-groups_F02989	295	■ 1.000	H^n, R^n, S^n	H^n, R^n, S^n	where $k = (-1, 0, +1)$ for H^n, R^n, S^n and radial coordinates are used:
gilmore-lie-groups_F02990	295	■ 1.000	H^n, S^n	H^n, S^n	a. Derive the metric for H^n, S^n from Eq. (12.9) and the coordinate transformation
gilmore-lie-groups_F02991	295	■ 1.000	Ω	Ω	Compute Ω , the surface area of the unit sphere $S^{n-1} \subset H^n, R^n$, or S^n . (Hint: use
gilmore-lie-groups_F02992	295	■ 1.000	$S^{n-1} \subset H^n, R^n$	$S^{n-1} \subset H^n, R^n$	Compute Ω , the surface area of the unit sphere $S^{n-1} \subset H^n, R^n$, or S^n . (Hint: use
gilmore-lie-groups_F02993	295	■ 1.000	$\int e^{-x^2} dx = \sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$, carry the n -fold integral out in Cartesian and radial coordinates,
gilmore-lie-groups_F02994	295	■ 1.000	$\Omega=2 \pi^{n/2} / \Gamma(n/2)$	$\Omega = 2\pi^{n/2}/\Gamma(n/2)$	and show $\Omega = 2\pi^{n/2}/\Gamma(n/2)$.)
gilmore-lie-groups_F02995	296	■ 1.000	H^n	H^n	S^n , downward in H^n , with lookback time. Give a physical interpretation involving
gilmore-lie-groups_F02996	296	■ 1.000	R, c, t	R, c, t	on R, c, t ?
gilmore-lie-groups_F02997	296	■ 0.839	$(l, b)=\left(263^{\circ}, 48^{\circ}\right)$	$(l, b) = (263^{\circ}, 48^{\circ})$	the direction with galactic coordinates $(l, b) = (263^{\circ}, 48^{\circ})$.
gilmore-lie-groups_F02998	296	■ 1.000	$T(\theta, \phi ; t)=$	$T(\theta, \phi ; t) =$	b. Assume the temperature distribution of the microwave background is $T(\theta, \phi ; t) =$
gilmore-lie-groups_F02999	296	■ 1.000	$\sum_{l, m} A_m^l(t) Y_m^l(\theta, \phi)$	$\sum_{l, m} A_m^l(t) Y_m^l(\theta, \phi)$	$\sum_{l, m} A_m^l(t) Y_m^l(\theta, \phi)$. How do you use this information to determine a frame that
gilmore-lie-groups_F03000	296	■ 0.998	$\Delta U \Delta(1 / T) \geq k$	$\Delta U \Delta(1/T) \geq k$	the uncertainty relations of statistical mechanics: $\Delta U \Delta(1/T) \geq k$ in the entropy
gilmore-lie-groups_F03001	297	■ 1.000	$j=\frac{1}{2}, 1, 2$	$j = \frac{1}{2}, 1, 2$	why, or why not, the special frames defined by $j = \frac{1}{2}, 1, 2$ are the same. What
gilmore-lie-groups_F03002	297	■ 1.000	$v(t)$	$v(t)$	that it moves through the microwave background radiation with a velocity $v(t)$.
gilmore-lie-groups_F03003	297	■ 1.000	$v(t) \simeq v\left(t_{0}\right) e^{-\left(t-t_{0}\right) / \tau}$	$v(t) \simeq v\left(t_0\right) e^{-\left(t-t_0\right) / \tau}$	Show that its velocity decays to zero according to $v(t) \simeq v\left(t_0\right) e^{-\left(t-t_0\right) / \tau}$ because
gilmore-lie-groups_F03004	297	■ 1.000	τ	τ	it is moving through a viscous medium. Estimate τ and present your answer in
gilmore-lie-groups_F03005	297	■ 0.990	τ / T_p	τ / T_p	the form τ / T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03006	297	■ 0.990	T_p	T_p	the form τ / T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03007	297	■ 0.990	$T_p \simeq 13.7 \mathrm{BY}$	$T_p \simeq 13.7 \mathrm{BY}$	the form τ / T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03008	297	■ 1.000	T_{BH}	T_{BH}	in fact, assume it is a black hole with mass M , radius R at temperature T_{BH} .

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gilmore-lie-groups_ F03009	297	■ 1.000	$R=2\ G\ M\ /\ c^{\{2\}}$	$R = 2GM/c^2$	Use the standard relations for a neutral nonrotating black hole $R = 2GM/c^2$,
gilmore-lie-groups_ F03010	297	■ 1.000	$T_{\{B\ H\}}=\hbar\bar c^{\{3\}}\ /\ 8\ \pi\ k\ G\ M$	$T_{BH} = \hbar c^3/8\pi kGM$	$T_{BH} = \hbar c^3/8\pi kGM$. You can assume that the mass M is sufficiently large that
gilmore-lie-groups_ F03011	297	■ 1.000	$\gamma\pi\ R^{\{2\}}$	$\gamma\pi R^2$	(geometric) cross section for radiation on a black hole is $\gamma\pi R^2$, where $\gamma = 3^3/2^2$.
gilmore-lie-groups_ F03012	297	■ 1.000	$\gamma=3^{\{3\}}\ /\ 2^{\{2\}}$	$\gamma = 3^3/2^2$	(geometric) cross section for radiation on a black hole is $\gamma\pi R^2$, where $\gamma = 3^3/2^2$.
gilmore-lie-groups_ F03013	298	■ 1.000	$d\ y\ /\ d\ x=g(x)$	$dy/dx = g(x)$	constant that can be added to any indefinite integral of $dy/dx = g(x)$ is in
gilmore-lie-groups_ F03014	298	■ 1.000	$y: d\ y\ /\ d\ x=g(x)$	$y: dy/dx = g(x)$	dependent variable $y: dy/dx = g(x)$. This is treated in Section 16.1. In that section
gilmore-lie-groups_ F03015	298	■ 1.000	$d\ y\ /\ d\ x=p$	$dy/dx = p$	we set up the general formulation in terms of a constraint equation $dy/dx = p$ and
gilmore-lie-groups_ F03016	299	■ 1.000	$F(x,\ y,\ p)=0$	$F(x,y,p) = 0$	a surface equation $F(x,y,p) = 0$. The special forms of the surface and constraint
gilmore-lie-groups_ F03017	299	■ 1.000	$R,\ S,\ T$	R,S,T	metry is. Transformation to a set of canonical variables $R,\ S,\ T$ is algorithmic. The
gilmore-lie-groups_ F03018	299	■ 1.000	$R(x,\ y)$	$R(x,y)$	canonical variable $R(x,y)$ is the new independent variable (like x), $S(x,y)$ is the
gilmore-lie-groups_ F03019	299	■ 1.000	$S(x,\ y)$	$S(x,y)$	canonical variable $R(x,y)$ is the new independent variable (like x), $S(x,y)$ is the
gilmore-lie-groups_ F03020	299	■ 1.000	$T(x,\ y,\ p)$	$T(x,y,p)$	new dependent variable (like y), and $T(x,y,p)$ is the new constraint between S and
gilmore-lie-groups_ F03021	299	■ 1.000	$d\ y\ /\ d\ x$	dy/dx	R (like dy/dx). In this new coordinate system the surface and constraint equations
gilmore-lie-groups_ F03022	299	■ 0.999	$F(R,\ -,\ T)=0$	$F(R,-,T) = 0$	assume the desired forms $F(R,-,T) = 0$ and $dS/dR = f(R,-,T)$. The system
gilmore-lie-groups_ F03023	299	■ 0.999	$d\ S\ /\ d\ R=f(R,\ -,\ T)$	$dS/dR = f(R,-,T)$	assume the desired forms $F(R,-,T) = 0$ and $dS/dR = f(R,-,T)$. The system
gilmore-lie-groups_ F03024	299	■ 1.000	$y-G(x)=0$	$y - G(x) = 0$	If we write the solution in the form $y - G(x) = 0$, then the surface $y + c - G(x) =$
gilmore-lie-groups_ F03025	299	■ 1.000	$y+c-G(x)=$	$y + c - G(x) =$	If we write the solution in the form $y - G(x) = 0$, then the surface $y + c - G(x) =$
gilmore-lie-groups_ F03026	299	■ 1.000	$e^{\{c\ \partial/\ \partial y\}}$	$e^{c\partial/\partial y}$	represented by the Taylor series displacement operator $e^{c\partial/\partial y}$, for
gilmore-lie-groups_ F03027	300	■ 1.000	$F(x,\ y,\ p)=p-g(x)$	$F(x,y,p) = p - g(x)$	$F(x,y,p) = p - g(x)$ for the particular case at hand. There are two relations
gilmore-lie-groups_ F03028	300	■ 1.000	$x,\ y,\ p$	x,y,p	among the three variables x,y,p . They are given by the surface equation and
gilmore-lie-groups_ F03029	300	■ 1.000	$\partial/\partial x,\partial/\partial y,\partial/\partial p$	$\partial/\partial x,\partial/\partial y,\partial/\partial p$	It is useful to express the action of the three partial derivatives $\partial/\partial x,\partial/\partial y,\partial/\partial p$
gilmore-lie-groups_ F03030	300	■ 1.000	$F(x,\ y,\ p)$	$F(x,y,p)$	on the surface $F(x,y,p)$ defining the ordinary differential equation. It is also
gilmore-lie-groups_ F03031	300	■ 0.798	$\left(d^{\{n\}}\ y\ /\ d\ x^{\{n\}},\ n=1\right)$	$(d^ny/dx^n, n = 1)$	Estabrook, 1976). We do this for equations of first order $(d^ny/dx^n, n = 1)$ and
gilmore-lie-groups_ F03032	300	■ 1.000	$p^{\{m\}}=(d\ y\ /\ d\ x)^{\{m\}},\ m=1$	$p^m = (dy/dx)^m, m = 1$	first degree (depends on $p^m = (dy/dx)^m, m = 1$). The results are independent of
gilmore-lie-groups_ F03033	300	■ 1.000	$p-g(x)$	$p - g(x)$	$F(x,y,p) = 0$, is not of the form $p - g(x)$, so that $\frac{\partial}{\partial y}F(x,y,p) \neq 0$, then we
gilmore-lie-groups_ F03034	300	■ 1.000	$\frac{\partial}{\partial y}F(x,\ y,\ p)\ \neq\ 0$	$\frac{\partial}{\partial y}F(x,y,p) \neq 0$	$F(x,y,p) = 0$, is not of the form $p - g(x)$, so that $\frac{\partial}{\partial y}F(x,y,p) \neq 0$, then we
gilmore-lie-groups_ F03035	300	■ 0.999	$(R,\ S,\ T)$	(R,S,T)	(ii) A new “canonical” coordinate system (R,S,T) . In this coordinate system $R = R(x,y)$

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gilmore-lie-groups_ F03036	300	■ 0.999	$R=R(x, y)$	$R = R(x, y)$	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$
gilmore-lie-groups_ F03037	300	■ 1.000	$S=S(x, y)$	$S = S(x, y)$	is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03038	300	■ 1.000	$T=T(x, y, p)$	$T = T(x, y, p)$	is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03039	301	■ 1.000	$x \rightarrow x$	$x \rightarrow x$	In the simplest case Eq. (16.1), this one-parameter group is $x \rightarrow x$ and $y \rightarrow y + \epsilon$,
gilmore-lie-groups_ F03040	301	■ 1.000	$y \rightarrow y+\epsilon$	$y \rightarrow y + \epsilon$	In the simplest case Eq. (16.1), this one-parameter group is $x \rightarrow x$ and $y \rightarrow y + \epsilon$,
gilmore-lie-groups_ F03041	301	■ 1.000	$\xi=0, \eta=1$	$\xi = 0, \eta = 1$	so that $\xi = 0, \eta = 1$, and $\zeta = 0$.
gilmore-lie-groups_ F03042	301	■ 1.000	$\zeta(x, y, p)$	$\zeta(x, y, p)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03043	301	■ 1.000	$\xi(x, y)$	$\xi(x, y)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03044	301	■ 1.000	$\eta(x, y)$	$\eta(x, y)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03045	301	■ 1.000	$\eta_{\{x\}}=\partial \eta / \partial x$	$\eta_x = \partial \eta / \partial x$	to first order in ϵ , where $\eta_x = \partial \eta / \partial x$, etc. As a result
gilmore-lie-groups_ F03046	302	■ 1.000	$\xi(x, y), \eta(x, y)$	$\xi(x, y), \eta(x, y)$	Specifically, these equations are used to determine the functions $\xi(x, y), \eta(x, y)$,
gilmore-lie-groups_ F03047	302	■ 1.000	$y: p=p(x, y)$	$y : p = p(x, y)$	x and y : $p = p(x, y)$. This expression is substituted into the determining equation
gilmore-lie-groups_ F03048	302	■ 1.000	$X F(x, y, p(x, y))=0$	$X F(x, y, p(x, y)) = 0$	$X F(x, y, p(x, y)) = 0$, so that this equation depends only on two independent vari-
gilmore-lie-groups_ F03049	302	■ 1.000	$d_{\{\xi\}}, d_{\{\eta\}}$	d_ξ, d_η	and similarly for η . These representations are truncated at finite degrees d_ξ, d_η . The
gilmore-lie-groups_ F03050	302	■ 0.999	$X F=0$	$X F = 0$	determining equation $X F = 0$ is expanded into the form $\sum C_{ij} x^i y^j = 0$. Each
gilmore-lie-groups_ F03051	302	■ 0.999	$\sum C_{\{i j\}} x^{\{i\}} y^{\{j\}}=0$	$\sum C_{ij} x^i y^j = 0$	determining equation $X F = 0$ is expanded into the form $\sum C_{ij} x^i y^j = 0$. Each
gilmore-lie-groups_ F03052	302	■ 1.000	$C_{\{i j\}}$	C_{ij}	coefficient C_{ij} must vanish separately, by standard linear independence arguments.
gilmore-lie-groups_ F03053	302	■ 0.999	$\xi_{\{i j\}}, \eta_{\{i j\}}$	ξ_{ij}, η_{ij}	ξ_{ij}, η_{ij} . In general, there are more equations than unknowns. Since the equations
gilmore-lie-groups_ F03054	303	■ 1.000	$X(x, y, p) S(x, y)=1$	$X(x, y, p) S(x, y) = 1$	$X(x, y, p) S(x, y) = 1$. We require S to be independent of p , so the condition
gilmore-lie-groups_ F03055	303	■ 1.000	$X S=+1$	$X S = +1$	added to the solution. Further, it is not important that $X S = +1$: we could just as
gilmore-lie-groups_ F03056	303	■ 1.000	$X S=-1$	$X S = -1$	well choose a solution satisfying $X S = -1$ or, for that matter, $X S = k \neq 0$, where
gilmore-lie-groups_ F03057	303	■ 1.000	$X S=k \neq 0$	$X S = k \neq 0$	well choose a solution satisfying $X S = -1$ or, for that matter, $X S = k \neq 0$, where
gilmore-lie-groups_ F03058	303	■ 1.000	$X R=0$	$X R = 0$	transformation group. These functions obey $X R = 0$ and $X T = 0$, which are ex-
gilmore-lie-groups_ F03059	303	■ 1.000	$X T=0$	$X T = 0$	transformation group. These functions obey $X R = 0$ and $X T = 0$, which are ex-
gilmore-lie-groups_ F03060	304	■ 1.000	$R: T=T(R)$	$R : T = T(R)$	as a function of R : $T = T(R)$. This expression is used in the constraint equation,
gilmore-lie-groups_ F03061	304	■ 1.000	$x=x(R, S), y=y(R, S)$	$x = x(R, S), y = y(R, S)$	The inverse relation $x = x(R, S), y = y(R, S)$ is used to express the solution
gilmore-lie-groups_ F03065	305	■ 1.000	$y \rightarrow \alpha y$	$y \rightarrow \alpha y$	formation $y \rightarrow \alpha y$ and $x \rightarrow \beta x$. Under this transformation the equation trans-

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gilmore-lie-groups_ F03066	305	■ 1.000	$x \rightarrow \beta x$	$x \rightarrow \beta x$	formation $y \rightarrow \alpha y$ and $x \rightarrow \beta x$. Under this transformation the equation trans-
gilmore-lie-groups_ F03067	305	■ 1.000	$\alpha \left(xp + y - (\alpha \beta) xy^2 \right) = 0$	$\alpha (xp + y - (\alpha \beta) xy^2) = 0$	forms to $\alpha (xp + y - (\alpha \beta) xy^2) = 0$. The equation is invariant provided $\alpha \beta = 1$.
gilmore-lie-groups_ F03068	305	■ 1.000	$\alpha \beta = 1$	$\alpha \beta = 1$	forms to $\alpha (xp + y - (\alpha \beta) xy^2) = 0$. The equation is invariant provided $\alpha \beta = 1$.
gilmore-lie-groups_ F03069	305	■ 1.000	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	ant is $x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$. Since there is a one-parameter invariance
gilmore-lie-groups_ F03070	305	■ 1.000	$X(x, y, p)$	$X(x, y, p)$	it is possible to construct the infinitesimal generator $X(x, y, p)$ from this group
gilmore-lie-groups_ F03071	305	■ 1.000	$p = y^2 - y/x$	$p = y^2 - y/x$	The surface $p = y^2 - y/x$ is shown in Fig. 16.1. The value of p clearly depends
gilmore-lie-groups_ F03072	306	■ 1.000	$p: p(x, y) =$	$p: p(x, y) =$	we use the surface equation $F(x, y, p) = 0$ to find an expression for p : $p(x, y) =$
gilmore-lie-groups_ F03073	306	■ 1.000	$-y/x + y^2$	$-y/x + y^2$	$-y/x + y^2$. This is substituted into the determining equation $XF(x, y, p) = 0$ to
gilmore-lie-groups_ F03074	306	■ 1.000	$XF(x, y, p) = 0$	$XF(x, y, p) = 0$	$-y/x + y^2$. This is substituted into the determining equation $XF(x, y, p) = 0$ to
gilmore-lie-groups_ F03075	306	■ 1.000	$\eta: \xi = \xi_{00}, \eta = \eta_{00}$	$\eta: \xi = \xi_{00}, \eta = \eta_{00}$	We first attempt zeroth degree expressions for ξ and η : $\xi = \xi_{00}, \eta = \eta_{00}$. When these
gilmore-lie-groups_ F03076	306	■ 0.999	$y/x, 1$	$y/x, 1$	coefficients of the monomials $y/x, 1$, and xy depend on the unknown parameters
gilmore-lie-groups_ F03077	306	■ 0.999	xy	xy	coefficients of the monomials $y/x, 1$, and xy depend on the unknown parameters
gilmore-lie-groups_ F03078	306	■ 1.000	ξ_{00}, η_{00}	ξ_{00}, η_{00}	ξ_{00}, η_{00} as follows:
gilmore-lie-groups_ F03079	307	■ 1.000	$\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$	$\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$	four equations we determine $\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$, and from the coefficient
gilmore-lie-groups_ F03080	307	■ 1.000	xy^2	xy^2	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_ F03081	307	■ 1.000	$-\xi_{10} - \eta_{01} = 0$	$-\xi_{10} - \eta_{01} = 0$	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_ F03082	307	■ 1.000	$\xi(x, y) = x$	$\xi(x, y) = x$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution
gilmore-lie-groups_ F03083	307	■ 1.000	$\eta(x, y) = -y$	$\eta(x, y) = -y$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution
gilmore-lie-groups_ F03084	307	■ 1.000	$\zeta = -2p$	$\zeta = -2p$	(16.8) gives $\zeta = -2p$, so that the generator of infinitesimal displacements is
gilmore-lie-groups_ F03085	307	■ 1.000	$-y dS(y)/dy = 1$	$-y dS(y)/dy = 1$	y . Such a solution can be found if the equation $-y dS(y)/dy = 1$ can be solved.
gilmore-lie-groups_ F03086	307	■ 1.000	$-\ln(y)$	$-\ln(y)$	The solution, up to an additive constant, is $-\ln(y)$. We will adopt this solution,
gilmore-lie-groups_ F03087	307	■ 1.000	$S(x, y) = \ln(y)$	$S(x, y) = \ln(y)$	neglecting the negative sign: $S(x, y) = \ln(y)$.
gilmore-lie-groups_ F03088	307	■ 1.000	$y dx = -x dy$	$y dx = -x dy$	The first equation for the new independent variable simplifies to $y dx = -x dy$ or
gilmore-lie-groups_ F03089	307	■ 1.000	$d(xy) = 0$	$d(xy) = 0$	$d(xy) = 0$, from which we conclude that $R(x, y) = xy$ is an invariant coordinate
gilmore-lie-groups_ F03090	307	■ 1.000	$R(x, y) = xy$	$R(x, y) = xy$	$d(xy) = 0$, from which we conclude that $R(x, y) = xy$ is an invariant coordinate
gilmore-lie-groups_ F03091	307	■ 1.000	$-dp/2p$	$-dp/2p$	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to
gilmore-lie-groups_ F03092	307	■ 1.000	dx/x	dx/x	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to

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gilmore-lie-groups_ F03093	307	■ 1.000	$d x / x = -d p / 2 p$	$dx/x = -dp/2p$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2p) = 0$, so that $T(x, y, p) = x^2p$.
gilmore-lie-groups_ F03094	307	■ 1.000	$(1 / x) d \left(x^2 p \right) = 0$	$(1/x)d(x^2p) = 0$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2p) = 0$, so that $T(x, y, p) = x^2p$.
gilmore-lie-groups_ F03095	307	■ 1.000	$T(x, y, p) = x^2 p$	$T(x, y, p) = x^2p$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2p) = 0$, so that $T(x, y, p) = x^2p$.
gilmore-lie-groups_ F03101	308	■ 1.000	$T = T(R, S)$	$T = T(R, S)$	pendent of S . This surface $T = T(R, S)$ is plotted in Fig. 16.2. It has the desired
gilmore-lie-groups_ F03102	308	■ 1.000	$R: T(R) = R^2 - R$	$R: T(R) = R^2 - R$	The surface equation is solved for T as a function of R : $T(R) = R^2 - R$. This
gilmore-lie-groups_ F03103	309	■ 0.999	(R, S)	(R, S)	The inverse transformation, Eq. (16.30), from (R, S) to (x, y) is finally used to
gilmore-lie-groups_ F03104	309	■ 1.000	$x d / d x$	xd/dx	Remarks The operator $x d/dx$ is the infinitesimal generator for scaling transfor-
gilmore-lie-groups_ F03105	309	■ 1.000	$e^{\lambda x d / d x} x = e^{\lambda} x$	$e^{\lambda x d/dx} x = e^{\lambda} x$	mations, since $e^{\lambda x d/dx} x = e^{\lambda} x$. As a result, the infinitesimal generator X has the
gilmore-lie-groups_ F03106	309	■ 0.766	(x, y, p)	(x, y, p)	following effect on the coordinates (x, y, p) :
gilmore-lie-groups_ F03107	309	■ 1.000	$\ln (y)$	$\ln(y)$	From this scaling behavior, it is easy to see that $\ln(y)$ is linear in the Lie translation
gilmore-lie-groups_ F03108	309	■ 1.000	$\ln \left(e^{-\lambda} y \right) = \ln (y) - \lambda$	$\ln(e^{-\lambda}y) = \ln(y) - \lambda$	group parameter: $\ln(e^{-\lambda}y) = \ln(y) - \lambda$. The invariant operators come right out of
gilmore-lie-groups_ F03109	309	■ 1.000	$x^2 p$	x^2p	the scaling transformations: xy and x^2p are unchanged by the scaling transforma-
gilmore-lie-groups_ F03110	309	■ 1.000	$\ln \left(x y^2 \right)$	$\ln(xy^2)$	tion. None of these operators is unique. The operator $\ln(xy^2)$ is linear and x^3yp
gilmore-lie-groups_ F03111	309	■ 1.000	$x^3 y p$	x^3yp	tion. None of these operators is unique. The operator $\ln(xy^2)$ is linear and x^3yp
gilmore-lie-groups_ F03112	309	■ 1.000	$F \left(x y, x^2 p \right) = 0$	$F(xy, x^2p) = 0$	Eq. (16.36) has the form $F(R, -, T) = 0$ or more simply $F(xy, x^2p) = 0$. The
gilmore-lie-groups_ F03113	309	■ 1.000	$x^2 p = h(x y)$	$x^2p = h(xy)$	$x^2p = h(xy)$ or $dy/dx = x^{-2}h(xy)$. For the equation studied in Section 16.3,
gilmore-lie-groups_ F03114	309	■ 1.000	$d y / d x = x^{-2} h(x y)$	$dy/dx = x^{-2}h(xy)$	$x^2p = h(xy)$ or $dy/dx = x^{-2}h(xy)$. For the equation studied in Section 16.3,
gilmore-lie-groups_ F03115	309	■ 1.000	$h(z) = -z + z^2$	$h(z) = -z + z^2$	$h(z) = -z + z^2$. For the Riccati equation $dy/dx + y^2 - 2/x^2 = 0$, $h(z) = z^2 - 2$.
gilmore-lie-groups_ F03116	309	■ 1.000	$d y / d x + y^2 - 2 / x^2 = 0, h(z) = z^2 - 2$	$dy/dx + y^2 - 2/x^2 = 0, h(z) = z^2 - 2$	$h(z) = -z + z^2$. For the Riccati equation $dy/dx + y^2 - 2/x^2 = 0$, $h(z) = z^2 - 2$.
gilmore-lie-groups_ F03117	309	■ 1.000	$y^{\prime 2} + y^4 - x^{-4} = 0$	$y'^2 + y^4 - x^{-4} = 0$	For example, the first order, second degree equation $y'^2 + y^4 - x^{-4} = 0$ has
gilmore-lie-groups_ F03118	309	■ 1.000	$R^4 + T^2 = 1$	$R^4 + T^2 = 1$	canonical form $R^4 + T^2 = 1$. The original equation has two solution branches
gilmore-lie-groups_ F03119	310	■ 1.000	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$, corresponding to the two solution branches in the canonical
gilmore-lie-groups_ F03120	310	■ 1.000	$T = \pm \sqrt{1 - R^4}$	$T = \pm \sqrt{1 - R^4}$	coordinate system $T = \pm \sqrt{1 - R^4}$
gilmore-lie-groups_ F03130	310	■ 0.998	$y^{\{2\}}$	$y^{(2)}$	involves derivatives with respect to $y^{(2)}$ and is given by
gilmore-lie-groups_ F03234	312	■ 1.000	$F \left(x, y, y^{\prime}, y^{\prime \prime} \right) = 0$	$F(x, y, y', y'') = 0$	Suppose a second order equation $F(x, y, y', y'') = 0$ is invariant under the scal-
gilmore-lie-groups_ F03235	312	■ 1.000	$S = \ln y$	$S = \ln y$	ing group (16.36). Then the dependent coordinate is $S = \ln y$ and the surface equa-

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gilmore-lie-groups_ F03236	312	■ 1.000	$F(R, -, T, U) = 0$	$F(R, -, T, U) = 0$	tion can be expressed in terms of three invariant coordinates as $F(R, -, T, U) = 0$.
gilmore-lie-groups_ F03237	312	■ 1.000	$y, T = T(x, y, y')$	$y, T = T(x, y, y')$	Here as before R depends only on x and y , $T = T(x, y, y')$, and $U = U(x, y, y', y'')$
gilmore-lie-groups_ F03238	312	■ 1.000	$U = U(x, y, y', y'')$	$U = U(x, y, y', y'')$	Here as before R depends only on x and y , $T = T(x, y, y')$, and $U = U(x, y, y', y'')$
gilmore-lie-groups_ F03239	312	■ 1.000	dT/dR	dT/dR	nate? It is simple to see that the derivative dT/dR is invariant under the group. Not
gilmore-lie-groups_ F03240	312	■ 1.000	y''	y''	transformations Eq. (16.36) that is of first degree in y'' is
gilmore-lie-groups_ F03241	313	■ 1.000	$F(x, y, \dots, y^{(n)}) = 0$	$F(x, y, \dots, y^{(n)}) = 0$	equation $F(x, y, \dots, y^{(n)}) = 0$. As usual, we seek an infinitesimal generator
gilmore-lie-groups_ F03242	313	■ 1.000	$T(x, y, y^{(1)})$	$T(x, y, y^{(1)})$	In fact, only the first two invariant coordinates $R(x, y)$ and $T(x, y, y^{(1)})$ need be
gilmore-lie-groups_ F03243	313	■ 1.000	$dT^{(j)}/dR^{(j)}, j = 0$	$dT^{(j)}/dR^{(j)}, j = 0$	computed. The remaining invariant coordinates are $dT^{(j)}/dR^{(j)}, j = 0$ (for T) and
gilmore-lie-groups_ F03244	313	■ 1.000	$j = 1, 2, \dots, n - 1$	$j = 1, 2, \dots, n - 1$	$j = 1, 2, \dots, n - 1$. Each of these latter is of first degree in $y^{(j+1)}$. As a result,
gilmore-lie-groups_ F03245	313	■ 1.000	$y^{(j+1)}$	$y^{(j+1)}$	$j = 1, 2, \dots, n - 1$. Each of these latter is of first degree in $y^{(j+1)}$. As a result,
gilmore-lie-groups_ F03246	314	■ 1.000	$x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$	$x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$	scaling transformations $x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$. Under the scaling transformation
gilmore-lie-groups_ F03247	314	■ 1.000	$\delta(x) \rightarrow \delta(\lambda x) = \lambda^{-n} \delta(x)$	$\delta(x) \rightarrow \delta(\lambda x) = \lambda^{-n} \delta(x)$	$\delta(x) \rightarrow \delta(\lambda x) = \lambda^{-n} \delta(x)$ so that
gilmore-lie-groups_ F03248	314	■ 1.000	$\alpha = \lambda^{2-n}$	$\alpha = \lambda^{2-n}$	The equation is invariant provided $\alpha = \lambda^{2-n}$. The infinitesimal generators of sym-
gilmore-lie-groups_ F03249	314	■ 1.000	$R = R(x, u)$	$R = R(x, u)$	A new independent coordinate $R = R(x, u)$ satisfies $XR = 0$, where X is any linear
gilmore-lie-groups_ F03250	314	■ 1.000	$R \sim u x ^{n-2}$	$R \sim u x ^{n-2}$	combination of the generators in Eq. (16.52). A solution is $R \sim u x ^{n-2}$. As a result,
gilmore-lie-groups_ F03251	314	■ 1.000	$u \sim x ^{2-n} = k x ^{2-n}$	$u \sim x ^{2-n} = k x ^{2-n}$	$u \sim x ^{2-n} = k x ^{2-n}$. The constant of proportionality can be computed using the
gilmore-lie-groups_ F03252	314	■ 1.000	$V(S^{n-1}) = 2\pi^{n/2}/\Gamma(\frac{n}{2})$	$V(S^{n-1}) = 2\pi^{n/2}/\Gamma(\frac{n}{2})$	Here $V(S^{n-1}) = 2\pi^{n/2}/\Gamma(\frac{n}{2})$ is the surface area of a unit sphere in R^n . As a result,
gilmore-lie-groups_ F03253	314	■ 1.000	$R^n (n \neq 2)$	$R^n (n \neq 2)$	the solution of Laplace's equation in R^n ($n \neq 2$) with unit source term at the origin
gilmore-lie-groups_ F03254	314	■ 1.000	$u(x, t)$	$u(x, t)$	The heat equation on R^n for $u(x, t)$ with source term
gilmore-lie-groups_ F03255	314	■ 1.000	$u \rightarrow \alpha u, t \rightarrow \beta t$	$u \rightarrow \alpha u, t \rightarrow \beta t$	X_{ij} are Lie symmetries. Under the scaling transformation $u \rightarrow \alpha u, t \rightarrow \beta t$, and
gilmore-lie-groups_ F03256	314	■ 1.000	$x^i \rightarrow \lambda x^i$	$x^i \rightarrow \lambda x^i$	$x^i \rightarrow \lambda x^i$ the equation transforms as follows:
gilmore-lie-groups_ F03257	314	■ 1.000	$\alpha \lambda^n = 1$	$\alpha \lambda^n = 1$	the three scaling variables (since there is only one equation): $\alpha \lambda^n = 1$ and $\beta/\lambda^2 = 1$.
gilmore-lie-groups_ F03258	314	■ 1.000	$\beta/\lambda^2 = 1$	$\beta/\lambda^2 = 1$	the three scaling variables (since there is only one equation): $\alpha \lambda^n = 1$ and $\beta/\lambda^2 = 1$.
gilmore-lie-groups_ F03259	315	■ 1.000	$n + 1$	$n + 1$	From these relations it is possible to construct $n + 1$ additional Lie symmetries, so
gilmore-lie-groups_ F03260	315	■ 1.000	x^i, t	x^i, t	An invariant coordinate depending on the x^i, t and u is $R = ut^{n/2}e^{ x ^2/4t}$, from
gilmore-lie-groups_ F03261	315	■ 1.000	$R = ut^{n/2}e^{ x ^2/4t}$	$R = ut^{n/2}e^{ x ^2/4t}$	An invariant coordinate depending on the x^i, t and u is $R = ut^{n/2}e^{ x ^2/4t}$, from

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gilmore-lie-groups_ F03262	315	■ 1.000	$F(R, -, T) =$	$F(R, -, T) =$	3. Under what conditions is it possible to solve the canonical surface equation $F(R, -, T) =$
gilmore-lie-groups_ F03263	315	■ 0.980	$\int f(R, -, T(R)) dR$	$\int f(R, -, T(R)) dR$	$\int f(R, -, T(R)) dR$
gilmore-lie-groups_ F03265	316	■ 1.000	$\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)})$	$\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)})$	(ii) Construct the first prolongation $\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)})$ from the functions defin-
gilmore-lie-groups_ F03266	316	■ 1.000	$X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + \zeta \frac{\partial}{\partial p}$	$X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + \zeta \frac{\partial}{\partial p}$	(iii) Introduce the operator $X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + \zeta \frac{\partial}{\partial p}$. This describes a Taylor series
gilmore-lie-groups_ F03267	316	■ 1.000	$F = 0$	$F = 0$	(iv) Solve the determining equation $XF = 0$ when $F = 0$ for the functions $\xi(x, y)$ and
gilmore-lie-groups_ F03268	316	■ 1.000	$XR = 0, XS = 1, XT = 0$	$XR = 0, XS = 1, XT = 0$	(v) Solve the determining equations $XR = 0, XS = 1, XT = 0$ for the canonical coor-
gilmore-lie-groups_ F03269	316	■ 1.000	$F \rightarrow F(R, -, T) = 0$	$F \rightarrow F(R, -, T) = 0$	equation is independent of the new dependent variable: $F \rightarrow F(R, -, T) = 0$.
gilmore-lie-groups_ F03270	317	■ 1.000	$S: S = \int f(R, -, T(R)) + c$	$S: S = \int f(R, -, T(R)) + c$	(viii) Solve the constraint equation for $S: S = \int f(R, -, T(R)) + c$.
gilmore-lie-groups_ F03271	317	■ 1.000	$x = x(R, S)$	$x = x(R, S)$	(ix) Backsubstitute the original coordinates for the new coordinates, $x = x(R, S)$ and
gilmore-lie-groups_ F03272	317	■ 1.000	$y = y(R, S)$	$y = y(R, S)$	$y = y(R, S)$, to obtain the solution of the original equation.
gilmore-lie-groups_ F03273	317	■ 1.000	$n = 0, 1$	$n = 0, 1$	Show that the first two terms $n = 0, 1$ are exactly the determining equations
gilmore-lie-groups_ F03274	317	■ 1.000	$V(\mathbf{x})$	$V(\mathbf{x})$	mass m in the presence of a potential $V(\mathbf{x})$ is
gilmore-lie-groups_ F03275	317	■ 1.000	$m \rightarrow \alpha m, \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$m \rightarrow \alpha m, \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$m \rightarrow \alpha m, \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$. Assume also that the potential is homogeneous of degree
gilmore-lie-groups_ F03276	317	■ 1.000	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$ (Landau and Lifshitz, 1960). Under this scaling transformation
gilmore-lie-groups_ F03277	317	■ 1.000	$\alpha^1 \beta^{2-k}$	$\alpha^1 \beta^{2-k}$	a. Show that the scaled equation is identical to the original provided $\alpha^1 \beta^{2-k}$
gilmore-lie-groups_ F03278	317	■ 1.000	$\gamma^{-2} = 1$	$\gamma^{-2} = 1$	$\gamma^{-2} = 1$.
gilmore-lie-groups_ F03279	317	■ 1.000	$\alpha = 1$	$\alpha = 1$	b. Set $\alpha = 1$. Show that trajectories are invariant under the scaling transformation with
gilmore-lie-groups_ F03280	317	■ 1.000	$\gamma^2 = \beta^{2-k}$	$\gamma^2 = \beta^{2-k}$	$\gamma^2 = \beta^{2-k}$. Show that in the cases $k = -1, k = 0, k = +1, k = +2$ the following
gilmore-lie-groups_ F03281	317	■ 1.000	$k = -1, k = 0, k = +1, k = +2$	$k = -1, k = 0, k = +1, k = +2$	$\gamma^2 = \beta^{2-k}$. Show that in the cases $k = -1, k = 0, k = +1, k = +2$ the following
gilmore-lie-groups_ F03286	318	■ 0.785	γ^2	γ^2	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'
gilmore-lie-groups_ F03287	318	■ 0.785	β^3	β^3	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'
gilmore-lie-groups_ F03288	318	■ 0.785	R'	R'	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'
gilmore-lie-groups_ F03289	318	■ 1.000	T'	T'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and
gilmore-lie-groups_ F03290	318	■ 1.000	P'	P'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and
gilmore-lie-groups_ F03291	318	■ 1.000	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$. The second line is a statement of the integral of
gilmore-lie-groups_ F03292	318	■ 0.860	(β)	(β)	traveled (β) is proportional to the time elapsed (γ). The third line is a statement that
gilmore-lie-groups_ F03293	318	■ 0.860	(γ)	(γ)	traveled (β) is proportional to the time elapsed (γ). The third line is a statement that

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gilmore-lie-groups_F03294	318	■ 1.000	V=m g z	$V = mgz$	in a local gravitational potential of the form $V = mgz$, the distance fallen increases
gilmore-lie-groups_F03295	318	■ 1.000	\gamma=1	$\gamma = 1$	c. Fix $\gamma = 1$ and construct a table relating the mass and orbital scale under the four
gilmore-lie-groups_F03296	318	■ 1.000	\beta=1	$\beta = 1$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.
gilmore-lie-groups_F03297	318	■ 1.000	\sqrt{M}	$\sqrt{M}$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.
gilmore-lie-groups_F03298	318	■ 0.994	\alpha \beta^2 \gamma^{-2} = \beta^k	$\alpha \beta^2 \gamma^{-2} = \beta^k$	f. Show that the kinetic energy scales like $\alpha \beta^2 \gamma^{-2} = \beta^k$ (use a). Since the potential
gilmore-lie-groups_F03299	318	■ 1.000	\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$. Show that if a scaling transformation leaves the Lagrangian invari-
gilmore-lie-groups_F03300	318	■ 1.000	v_{i}	v_i	Show that the following six differential operators v_i are infinitesimal generators of
gilmore-lie-groups_F03301	318	■ 1.000	e^{\epsilon v_i} f(x, t)	$e^{\epsilon v_i} f(x, t)$	the invariance group of this equation. Show that $e^{\epsilon v_i} f(x, t)$ has the action shown for
gilmore-lie-groups_F03307	319	■ 1.000	u \partial_u	$u \partial_u$	all the rest, and is $u \partial_u$.
gilmore-lie-groups_F03317	319	■ 1.000	\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$	
gilmore-lie-groups_F03318	319	■ 1.000	\lambda^2 = 1/(1 + 4\epsilon t)	$\lambda^2 = 1/(1 + 4\epsilon t)$	where $\lambda^2 = 1/(1 + 4\epsilon t)$
gilmore-lie-groups_F03319	319	■ 1.000	D_2 = u \partial_u	$D_2 = u \partial_u$	Show that $D_2 = u \partial_u$ commutes with all remaining generators. Construct the commu-
gilmore-lie-groups_F03320	319	■ 1.000	SO(2+1, 1+1) = SO(3, 2)	$SO(2 + 1, 1 + 1) = SO(3, 2)$	$SO(2 + 1, 1 + 1) = SO(3, 2)$.
gilmore-lie-groups_F03321	319	■ 1.000	SO(3+1, 1+1) = SO(4, 2)	$SO(3 + 1, 1 + 1) = SO(4, 2)$	$SO(3 + 1, 1 + 1) = SO(4, 2)$ (Bateman, 1910). The extra generator commutes with
gilmore-lie-groups_F03322	319	■ 1.000	u_{xx} - u_t = 0	$u_{xx} - u_t = 0$	8. The heat equation in one dimension is $u_{xx} - u_t = 0$. The infinitesimal generator of
gilmore-lie-groups_F03323	319	■ 1.000	X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \dots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \dots	$X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \dots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \dots$	symmetries for this equation is $X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \dots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \dots$.
gilmore-lie-groups_F03324	320	■ 1.000	h(x, t)	$h(x, t)$	Here $h(x, t)$ is any function that satisfies the homogeneous heat equation. Construct
gilmore-lie-groups_F03325	320	■ 0.986	\left(t \rightarrow t' = T(t, x, \epsilon) = t + \epsilon \right)	$(t \rightarrow t' = T(t, x, \epsilon) = t +$	Under a one-parameter family of change of variables $(t \rightarrow t' = T(t, x, \epsilon) = t +$
gilmore-lie-groups_F03326	320	■ 1.000	\left. \epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x) \right\}	$\epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x)$	$\epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x)$ the action integral transforms to
gilmore-lie-groups_F03327	320	■ 1.000	dt'/dt = \partial T / \partial t + (\partial T / \partial x_i) dx_i / dt	$dt'/dt = \partial T / \partial t + (\partial T / \partial x_i) dx_i / dt$	where $dt'/dt = \partial T / \partial t + (\partial T / \partial x_i) dx_i / dt$. Show that if you differentiate the action
gilmore-lie-groups_F03328	320	■ 1.000	\epsilon = 0	$\epsilon = 0$	integral with respect to ϵ , then set $\epsilon = 0$ the result is
gilmore-lie-groups_F03329	321	■ 1.000	L[u] =	$L[u] =$	the dynamics of a system is derivable from an action integral of the form $L[u] =$
gilmore-lie-groups_F03330	321	■ 1.000	\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$, and suppose the infinitesimal generators that leave the
gilmore-lie-groups_F03331	321	■ 1.000	d \rho(g)	$d\rho(g)$	12. Representation theory G is a compact Lie group with invariant measure $d\rho(g)$
gilmore-lie-groups_F03332	321	■ 0.995	\operatorname{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)	$\operatorname{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$	and volume $\operatorname{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$ are the irreducible representations of G con-
gilmore-lie-groups_F03333	321	■ 1.000	\phi(g), \psi(g)	$\phi(g), \psi(g)$	structed by reduction of tensor products (Wigner–Stone theorem), and $\phi(g), \psi(g)$

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gilmore-lie-groups-F03334	322	■ 1.000	$\psi(g)=\langle g \mid \psi \rangle$	$\psi(g) = \langle g \mid \psi \rangle$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \mid \psi \rangle$ and $\langle \lambda_{\mu\nu} \mid \psi \rangle =$
gilmore-lie-groups-F03335	322	■ 1.000	$\langle \lambda_{\mu\nu} \mid \psi \rangle =$	$\langle \lambda_{\mu\nu} \mid \psi \rangle =$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \mid \psi \rangle$ and $\langle \lambda_{\mu\nu} \mid \psi \rangle =$
gilmore-lie-groups-F03336	322	■ 1.000	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$ (and similarly for $\phi(g) = \langle g \mid \phi \rangle$) using the Dirac represen-
gilmore-lie-groups-F03337	322	■ 1.000	$\phi(g)=\langle g \mid \phi \rangle$	$\phi(g) = \langle g \mid \phi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} \mid g \rangle \langle g \mid \psi \rangle$ (and similarly for $\phi(g) = \langle g \mid \phi \rangle$) using the Dirac represen-
gilmore-lie-groups-F03338	322	■ 1.000	$\int \phi^*(g) \psi(g) d\rho(g)$	$\int \phi^*(g) \psi(g) d\rho(g)$	tation. Write down the Parseval equality for the inner product $\int \phi^*(g) \psi(g) d\rho(g)$
gilmore-lie-groups-F03339	323	■ 1.000	$\mathrm{He}^{\{+\}}$	He^+	O. BELY (1966), Quantum defect theory III. Electron scattering by He^+ , <i>Proc. Phys. Soc.</i>
gilmore-lie-groups-F03340	324	■ 0.868	$A_{\{n\}}; D_{\{n\}}, B_{\{n\}}, C_{\{n\}}, J$	$A_n; D_n, B_n, C_n, J$	$A_n; D_n, B_n, C_n, J$. <i>Math. Phys.</i> 11 , 513–523.
gilmore-lie-groups-F03341	325	■ 0.866	$\mathrm{mx} / \sim$	$\mathrm{mx} / \sim$	$\mathrm{mx} / \sim$ mcintosh/comun/symm/symm.html.
gilmore-lie-groups-F03342	327	■ 0.995	$A(p\ q),\ 39, 48$	$A(pq), 39, 48$	$A(pq), 39, 48$
gilmore-lie-groups-F03343	327	■ 0.978	$A_{\{1\}},\ 161$	$A_1, 161$	$A_1, 161$
gilmore-lie-groups-F03344	327	■ 0.991	$A_{\{2\}}$	A_2	$A_2, 151, 160$
gilmore-lie-groups-F03345	327	■ 0.865	$A_{\{3\}},\ 46, 161, 162$	$A_3, 46, 161, 162$	$A_3, 46, 161, 162$
gilmore-lie-groups-F03346	327	■ 0.946	$A_{\{n\}},\ 46, 49, 161, 166$	$A_n, 46, 49, 161, 166$	$A_n, 46, 49, 161, 166$
gilmore-lie-groups-F03347	327	■ 0.992	$B_{\{1\}},\ 161$	$B_1, 161$	$B_1, 161$
gilmore-lie-groups-F03348	327	■ 0.749	$B_{\{2\}},\ 151, 160, 161, 164$	$B_2, 151, 160, 161, 164$	$B_2, 151, 160, 161, 164$
gilmore-lie-groups-F03349	327	■ 0.761	$B_{\{3\}},\ 162$	$B_3, 162$	$B_3, 162$
gilmore-lie-groups-F03350	327	■ 0.729	$\boldsymbol{B}_{\{n\}},\ 161, 166, 168$	$\boldsymbol{B}_n, 161, 166, 168$	$B_n, 161, 166, 168$
gilmore-lie-groups-F03351	327	■ 1.000	$C_{\{1\}},\ 161$	$C_1, 161$	$C_1, 161$
gilmore-lie-groups-F03352	327	■ 0.523	$C_{\{2\}},\ 151, 160, 161, 164$	$C_2, 151, 160, 161, 164$	$C_2, 151, 160, 161, 164$
gilmore-lie-groups-F03353	327	■ 0.712	$C_{\{3\}},\ 162$	$C_3, 162$	$C_3, 162$
gilmore-lie-groups-F03354	327	■ 0.997	$C_{\{n\}},\ 161, 166, 168$	$C_n, 161, 166, 168$	$C_n, 161, 166, 168$
gilmore-lie-groups-F03355	327	■ 0.840	$D_{\{2\}},\ 151, 160, 162$	$D_2, 151, 160, 162$	$D_2, 151, 160, 162$
gilmore-lie-groups-F03356	327	■ 0.435	$G\ L(1; \mathbb{Q}),\ 40, 47$	$GL(1; \mathbb{Q}), 40, 47$	$GL(1; \mathbb{Q}), 40, 47$
gilmore-lie-groups-F03357	327	■ 0.979	$G\ L(2; \mathbb{C}),\ 47$	$GL(2; \mathbb{C}), 47$	$GL(2; \mathbb{C}), 47$
gilmore-lie-groups-F03358	327	■ 0.788	$G\ L(2; \mathbb{R})$	$GL(2; \mathbb{R})$	$GL(2; \mathbb{R}), 43$
gilmore-lie-groups-F03359	327	■ 0.731	$G\ L(2; \mathbb{Z}),\ 45, 49$	$GL(2; \mathbb{Z}), 45, 49$	$GL(2; \mathbb{Z}), 45, 49$
gilmore-lie-groups-F03360	327	■ 0.903	$G\ L(3; \mathbb{Z}),\ 45$	$GL(3; \mathbb{Z}), 45$	$GL(3; \mathbb{Z}), 45$
gilmore-lie-groups-F03361	327	■ 0.471	$G\ L(n; \mathbb{F}),\ 34, 36, 74$	$GL(n; \mathbb{F}), 34, 36, 74$	$GL(n; \mathbb{F}), 34, 36, 74$

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gilmore-lie-groups_ F03362	327	0.698	$G_L(n; \mathbb{Q})$	$GL(n; \mathbb{Q})$	$GL(n; \mathbb{Q}), 47$
gilmore-lie-groups_ F03363	327	0.973	$G_2, 151, 160, 162, 165$	$G_2, 151, 160, 162, 165$	$G_2, 151, 160, 162, 165$
gilmore-lie-groups_ F03364	327	0.933	$\operatorname{HT}(p, q), 37, 48$	$HT(p, q), 37, 48$	$HT(p, q), 37, 48$
gilmore-lie-groups_ F03365	327	0.634	$\operatorname{Nil}(n), 38, 48$	$Nil(n), 38, 48$	$Nil(n), 38, 48$
gilmore-lie-groups_ F03366	327	0.788	$O(31), 261$	$O(31), 261$	$O(31), 261$
gilmore-lie-groups_ F03367	327	0.988	$O(3), 40, 78$	$O(3), 40, 78$	$O(3), 40, 78$
gilmore-lie-groups_ F03368	327	1.000	$O(3; \mathbb{Z}), 46$	$O(3; \mathbb{Z}), 46$	$O(3; \mathbb{Z}), 46$
gilmore-lie-groups_ F03369	327	0.991	$O(n), 40, 43, 145$	$O(n), 40, 43, 145$	$O(n), 40, 43, 145$
gilmore-lie-groups_ F03370	327	1.000	$O(n; \mathbb{G}), 41$	$O(n; \mathbb{G}), 41$	$O(n; \mathbb{G}), 41$
gilmore-lie-groups_ F03371	327	1.000	$O(n; \mathbb{Z}), 45, 49$	$O(n; \mathbb{Z}), 45, 49$	$O(n; \mathbb{Z}), 45, 49$
gilmore-lie-groups_ F03372	327	0.737	$O(p, q)$	$O(p, q)$	$O(p, q), 43$
gilmore-lie-groups_ F03373	327	0.939	$P_n, 45$	$P_n, 45$	$P_n, 45$
gilmore-lie-groups_ F03374	327	0.901	$S_L(2; \mathbb{C}), 43$	$SL(2; \mathbb{C}), 43$	$SL(2; \mathbb{C}), 43$
gilmore-lie-groups_ F03375	327	0.985	$S_L(2; \mathbb{R}), 26, 28, 29, 30, 41, 43, 56, 58, 62, 100, 102$	$SL(2; \mathbb{R}), 26, 28, 29, 30, 41, 43, 56, 58, 62, 100, 102$	$SL(2; \mathbb{R}), 26, 28, 29, 30, 41, 43, 56, 58, 62, 100, 102,$
gilmore-lie-groups_ F03376	327	0.956	$S_L(n; \mathbb{R}), 30$	$SL(n; \mathbb{R}), 30$	$SL(n; \mathbb{R}), 30$
gilmore-lie-groups_ F03377	327	0.600	$S_L(n; \mathbb{C}), 43, 47$	$SL(n; \mathbb{C}), 43, 47$	$SL(n; \mathbb{C}), 43, 47$
gilmore-lie-groups_ F03378	327	0.519	$\operatorname{SL}(n; \mathbb{Q})$	$SL(n; \mathbb{Q})$	$SL(n; \mathbb{Q}), 43$
gilmore-lie-groups_ F03379	327	0.747	$\operatorname{SL}(n; \mathbb{R})$	$SL(n; \mathbb{R})$	$SL(n; \mathbb{R}), 43, 47, 164$
gilmore-lie-groups_ F03380	327	0.880	$S_L(n; \mathbb{Z}), 45$	$SL(n; \mathbb{Z}), 45$	$SL(n; \mathbb{Z}), 45$
gilmore-lie-groups_ F03381	327	0.703	$S_{O(2,1)}, 105$	$SO(2, 1), 105$	$SO(2, 1), 105$
gilmore-lie-groups_ F03382	327	0.943	$S_{O(2,1)} / S_{O(2)}, 106$	$SO(2, 1)/SO(2), 106$	$SO(2, 1)/SO(2), 106$
gilmore-lie-groups_ F03383	327	0.820	$\operatorname{SO}(2), 48, 164$	$SO(2), 48, 164$	$SO(2), 48, 164$
gilmore-lie-groups_ F03384	327	0.621	$S_{O(2n)}, 164$	$SO(2n), 164$	$SO(2n), 164$
gilmore-lie-groups_ F03385	327	1.000	$S_{O(2n+1)}, 164$	$SO(2n + 1), 164$	$SO(2n + 1), 164$
gilmore-lie-groups_ F03386	327	0.607	$S_{O(3,1)}, 263$	$SO(3, 1), 263$	$SO(3, 1), 263$
gilmore-lie-groups_ F03387	327	0.979	$\operatorname{SO}(3), 49, 90, 106$	$SO(3), 49, 90, 106$	$SO(3), 49, 90, 106$
gilmore-lie-groups_ F03388	327	0.875	$S_{O(3)} / S_{O(2)}, 107$	$SO(3)/SO(2), 107$	$SO(3)/SO(2), 107$
gilmore-lie-groups_ F03389	327	0.498	$\operatorname{SO}(4,1), 210$	$SO(4, 1), 210$	$SO(4, 1), 210$
gilmore-lie-groups_ F03390	327	0.463	$\operatorname{SO}(n)$	$SO(n)$	$SO(n), 43, 145$

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gilmore-lie-groups_F03391	327	0.637	$\operatorname{SU}(1 ; \mathbb{Q})$, 40,48	$SU(1; \mathbb{Q})$, 40, 48	<i>SU(1; $\mathbb{Q}$), 40, 48</i>
gilmore-lie-groups_F03392	327	0.998	$\operatorname{SU}(1,1)$, 38,43,48,105	$SU(1,1)$, 38, 43, 48, 105	<i>SU(1,1), 38, 43, 48, 105</i>
gilmore-lie-groups_F03393	327	1.000	$S U(1,1) / U(1)$, 106	$SU(1,1)/U(1)$, 106	<i>SU(1,1)/U(1), 106</i>
gilmore-lie-groups_F03394	327	0.606	$\operatorname{SU}(2)$, 48,106	$SU(2)$, 48, 106	<i>SU(2), 48, 106</i>
gilmore-lie-groups_F03395	327	0.999	$S U(2) / U(1)$, 107	$SU(2)/U(1)$, 107	<i>SU(2)/U(1), 107</i>
gilmore-lie-groups_F03396	327	0.983	$\operatorname{SU}(n)$, 43,90,164	$SU(n)$, 43, 90, 164	<i>SU(n), 43, 90, 164</i>
gilmore-lie-groups_F03397	327	0.962	$\operatorname{SU}(p, q)$, 43,164	$SU(p, q)$, 43, 164	<i>SU(p,q), 43, 164</i>
gilmore-lie-groups_F03398	327	0.994	$S_{\{3\}}$, 5,46	S_3 , 5, 46	<i>S₃, 5, 46</i>
gilmore-lie-groups_F03399	328	0.886	$S_{\{n\}}$, 45,49	S_n , 45, 49	<i>S_n, 45, 49</i>
gilmore-lie-groups_F03400	328	0.925	$\operatorname{Sol}(n)$, 38,48	$Sol(n)$, 38, 48	<i>Sol(n), 38, 48</i>
gilmore-lie-groups_F03401	328	0.986	$\operatorname{Sp}(1)$, 40	$Sp(1)$, 40	<i>Sp(1), 40</i>
gilmore-lie-groups_F03402	328	0.995	$\operatorname{Sp}(2 ; \mathbb{R})$, 41	$Sp(2; \mathbb{R})$, 41	<i>Sp(2; $\mathbb{R}$), 41</i>
gilmore-lie-groups_F03403	328	0.883	$\operatorname{Sp}(2 n ; \mathbb{R})$, 41	$Sp(2n; \mathbb{R})$, 41	<i>Sp(2n; $\mathbb{R}$), 41</i>
gilmore-lie-groups_F03404	328	0.739	$\operatorname{Sp}(n)$, 40,164	$Sp(n)$, 40, 164	<i>Sp(n), 40, 164</i>
gilmore-lie-groups_F03405	328	0.999	$\operatorname{Sp}(n ; \mathbb{C})$, 41	$Sp(n; \mathbb{C})$, 41	<i>Sp(n; $\mathbb{C}$), 41</i>
gilmore-lie-groups_F03406	328	0.944	$\operatorname{Sp}(n ; \mathbb{G})$, 41	$Sp(n; \mathbb{G})$, 41	<i>Sp(n; $\mathbb{G}$), 41</i>
gilmore-lie-groups_F03407	328	0.621	$\operatorname{Sp}(n ; \mathbb{R})$, 41	$Sp(n; \mathbb{R})$, 41	<i>Sp(n; $\mathbb{R}$), 41</i>
gilmore-lie-groups_F03408	328	0.903	$\operatorname{Sp}(p, q)$, 164	$Sp(p, q)$, 164	<i>Sp(p,q), 164</i>
gilmore-lie-groups_F03409	328	0.970	$U(1,1)$, 43	$U(1,1)$, 43	<i>U(1,1), 43</i>
gilmore-lie-groups_F03410	328	0.619	$U(2)$, 40,78	$U(2)$, 40, 78	<i>U(2), 40, 78</i>
gilmore-lie-groups_F03411	328	1.000	$U(2 ; \mathbb{Q})$, 164	$U(2; \mathbb{Q})$, 164	<i>U(2; $\mathbb{Q}$), 164</i>
gilmore-lie-groups_F03412	328	0.804	$U(n)$, 40,43,90	$U(n)$, 40, 43, 90	<i>U(n), 40, 43, 90</i>
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gilmore-lie-groups_F03414	328	1.000	$U(p, q)$, 43	$U(p, q)$, 43	<i>U(p,q), 43</i>
gilmore-lie-groups_F03415	328	0.713	$U \operatorname{Sp}(2 n)$, 44	$U Sp(2n)$, 44	<i>U Sp(2n), 44</i>
gilmore-lie-groups_F03416	328	0.397	$\operatorname{UT}(p, q)$, 48	$UT(p, q)$, 48	<i>UT(p,q), 48</i>
gilmore-lie-groups_F03417	328	1.000	$\mathbb{Z}$	$\mathbb{Z}$	<i>$\mathbb{Z}$, integers, 44</i>
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gilmore-lie-groups-F03422	329	■ 0.560	$\overline{S\ U(1,1) / U(1)}$, 108	$\overline{SU(1,1)/U(1)}$, 108	$\overline{SU(1,1)/U(1)}$, 108
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gilmore-lie-groups-F03424	330	■ 0.874	$\mathfrak{gl}(n; \mathbb{F})$, 74, 83	$\mathfrak{gl}(n; \mathbb{F})$, 74, 83	$\mathfrak{gl}(n; \mathbb{F})$, 74, 83
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gilmore-lie-groups-F03429	330	■ 0.997	$\mathfrak{o}(p, q)$, 78	$\mathfrak{o}(p, q)$, 78	$\mathfrak{o}(p, q)$, 78
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gilmore-lie-groups-F03432	330	■ 0.637	$\mathfrak{sl}(n; \mathbb{C})$, 80, 85, 86	$\mathfrak{sl}(n; \mathbb{C})$, 80, 85, 86	$\mathfrak{sl}(n; \mathbb{C})$, 80, 85, 86
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gilmore-lie-groups_F03456	331	0.857	$\mathfrak{s} \mathfrak{u}(n)$, 80, 132, 178	$\mathfrak{su}(n)$, 80, 132, 178	$\mathfrak{su}(n)$, 80, 132, 178
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